Lecture Notes: Linear Algebra I/II - 2025/2026

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Contents

The Fibonacci Sequence (Chapter 1)

Introduction and Generalization

We begin our study with the classical Fibonacci sequence, a sequence of numbers that naturally emerges in various mathematical and biological contexts.

Definition 1.1 (*The Fibonacci Sequence*): The *"Fibonacci sequence"* is an infinite sequence of integers $a_0, a_1, a_2, a_3, \ldots, a_n, \ldots$ defined by the following recurrence relation:

$$\begin{cases} a_0 = 0 \\ a_1 = 1 \\ a_n = a_{n-1} + a_{n-2} \quad \text{for } n \geq 2 \end{cases} \tag{1.1}$$

Remark: This sequence was introduced to Western mathematics by Leonardo Fibonacci, a medieval mathematician from Pisa (1170–1250), to model the growth of rabbit populations. Computing the first few terms, we easily find:

$$a_0 = 0, \quad a_1 = 1, \quad a_2 = 1, \quad a_3 = 2, \quad a_4 = 3, \quad a_5 = 5, \quad a_6 = 8, \quad a_7 = 13, \quad a_8 = 21, \ldots$$

A natural, central question arises: Is there a closed, explicit formula for the general term a_n? To answer this, let us briefly look at other well-known mathematical sequences for comparison:

(a) **Arithmetic Sequence:** Defined by $a_0 = a$ and $a_n = a_{n-1} + d$ for $n \geq 1$. The explicit formula for the general term is given by $a_n = a + n \cdot d$.

(b) **Geometric Sequence:** Defined by $a_0 = a$ and $a_n = a_{n-1} \cdot q$ for $n \geq 1$. The explicit formula is given by $a_n = a \cdot q^n$.

Clearly, the recurrence relation for the Fibonacci sequence is fundamentally different. To find a formula for a_n, it is highly beneficial to step back and generalize the problem.

Definition 1.2 (*Generalized Fibonacci Sequence*): Let $a_0, a_1 \in \mathbb{R}$ be given initial real numbers. We define a sequence $\mathcal{A}$ to be a *"Fibonacci sequence"* if it satisfies the relation:

$$\begin{cases} \alpha_0 = a_0 \\ \alpha_1 = a_1 \\ \alpha_n = \alpha_{n-1} + \alpha_{n-2} \quad \text{for } n \geq 2 \end{cases} \tag{1.2}$$

We denote such a sequence uniquely determined by its starting values as $\mathcal{F}_{a_0,a_1}$. Furthermore, we denote the collection (or space) of all such generalized Fibonacci sequences by Fib.

Notice that our original, classical Fibonacci sequence is precisely $\mathcal{F}_{0,1}$.

The Algebraic Structure of Fib

The collection Fib possesses a very rich internal algebraic structure, which we will exploit to find our explicit formula.

Proposition 1.3 (*Closure Properties of Fib*): The collection of Fibonacci sequences, Fib, is closed under term-wise addition and scalar multiplication. Specifically,

(a) If $\mathcal{A}_1, \mathcal{A}_2 \in \mathrm{Fib}$, then $\mathcal{A}_1 + \mathcal{A}_2 \in \mathrm{Fib}$.

(b) If $\mathcal{A} \in \mathrm{Fib}$ and $\alpha \in \mathbb{R}$, then $\alpha\mathcal{A} \in \mathrm{Fib}$.

Proof

For the first property, let $\mathcal{A}_1 = (a_0, a_1, a_2, \ldots)$ and $\mathcal{A}_2 = (b_0, b_1, b_2, \ldots)$ be two sequences in Fib. We define their sum term-wise as $\mathcal{A}_1 + \mathcal{A}_2 := (c_0, c_1, c_2, \ldots)$, where $c_n = a_n + b_n$. We must verify that the sequence (c_n) obeys the Fibonacci recurrence relation, namely $c_n = c_{n-1}+c_{n-2}$ for $n \geq 2$. By algebraic manipulation, we find:

$$c_{n-1} + c_{n-2} = (a_{n-1} + b_{n-1}) + (a_{n-2} + b_{n-2}) = (a_{n-1} + a_{n-2}) + (b_{n-1} + b_{n-2})$$

Because both $\mathcal{A}_1$ and $\mathcal{A}_2$ are Fibonacci sequences, we can substitute $a_{n-1} + a_{n-2} = a_n$ and $b_{n-1} + b_{n-2} = b_n$. Therefore,

$$c_{n-1} + c_{n-2} = a_n + b_n = c_n$$

This shows that $\mathcal{F}_{a_0,a_1} + \mathcal{F}_{b_0,b_1} = \mathcal{F}_{a_0+b_0,a_1+b_1}$, and therefore, Fib is closed under addition.

For the second property, let $\mathcal{A} = (a_0, a_1, a_2, \dots) \in$ Fib and $\alpha \in \mathbb{R}$. We define $\alpha\mathcal{A} := (\alpha a_0, \alpha a_1, \alpha a_2, \dots)$. We need to check that $\alpha a_n = \alpha a_{n-1} + \alpha a_{n-2}$. This is obviously true because $a_n = a_{n-1}+a_{n-2}$, and we simply multiply both sides by α. In fact, we see that $\alpha\mathcal{F}_{a_0,a_1} = \mathcal{F}_{\alpha a_0,\alpha a_1}$.

Q.E.D.

Corollary 1.4: Linear combinations of Fibonacci sequences are also Fibonacci sequences. That is, for any scalars $\alpha, \beta \in \mathbb{R}$, we have:

$$\alpha\mathcal{F}_{a_0,a_1} + \beta\mathcal{F}_{b_0,b_1} = \mathcal{F}_{\alpha a_0+\beta b_0,\alpha a_1+\beta b_1}$$

Remark: As a direct consequence of this structure, in order to find a general formula for *any* sequence $\mathcal{F}_{a_0,a_1}$, it is enough to find explicit formulas for the two foundational sequences $\mathcal{F}_{1,0}$ and $\mathcal{F}_{0,1}$. This is because any sequence can be decomposed as:

$$\mathcal{F}_{a_0,a_1} = a_0 \cdot \mathcal{F}_{1,0} + a_1 \cdot \mathcal{F}_{0,1}$$

We will see later in the course that Fib represents a 2-dimensional vector space. Every element in this space can be expressed by a linear combination of some two base elements (but 1 element alone is not enough). What precisely constitutes a *"vector space"* and *"dimension"* will be formally defined later in the course.

Deriving the Explicit Formula

Perhaps one of the sequences in Fib matches a form we already know?

First, we can rule out arithmetic sequences. Except for the trivial zero sequence $\mathcal{F}_{0,0}$, no Fibonacci sequence can be arithmetic. The reason is straightforward: for an arithmetic sequence, the difference between consecutive terms must be constant, meaning $a_n - a_{n-1} = d$. However, for a Fibonacci sequence, $a_n - a_{n-1} = a_{n-2}$. This difference grows and cannot remain constant. Thus, arithmetic sequences are unsuitable for this purpose.

Next, we ask: perhaps for some choice of a_0, a_1, the sequence $\mathcal{F}_{a_0,a_1}$ coincides with a geometric sequence? Let us try a geometric sequence of the form $\mathcal{G} = (1, q, q^2, q^3, \dots, q^n, \dots)$.

Lemma 1.5: A geometric sequence $\mathcal{G} = (1, q, q^2, \dots)$ with $q \neq 0$ is a Fibonacci sequence if and only if $q = \frac{1\pm\sqrt{5}}{2}$.

Proof

For $\mathcal{G}$ to be a Fibonacci sequence, it must satisfy the recurrence relation for all $n \geq 2$:

$$q^n = q^{n-1} + q^{n-2}$$

Since $q \neq 0$, we can divide the entire equation by q^{n-2}, which simplifies the condition to a quadratic equation:

$$q^2 = q + 1 \implies q^2 - q - 1 = 0$$

Applying the standard quadratic formula yields two possible solutions:

$$q = \frac{1 \pm \sqrt{5}}{2}$$

We denote these two specific roots as $\varphi := \frac{1+\sqrt{5}}{2} \approx 1.618$ (the golden ratio) and $\psi := \frac{1-\sqrt{5}}{2} \approx -0.618$. Consequently, we have discovered two purely geometric sequences that belong to Fib:

$$\mathcal{F}_{1,\varphi} = (1, \varphi, \varphi^2, \varphi^3, \dots, \varphi^n, \dots)$$
$$\mathcal{F}_{1,\psi} = (1, \psi, \psi^2, \psi^3, \dots, \psi^n, \dots)$$

Q.E.D.

Theorem 1.6 (*Binet's Formula*): The explicit formula for the n-th term of the classical Fibonacci sequence $\mathcal{F}_{0,1}$ is given by *"Binet's Formula"*:

$$a_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$$

Proof

So far, we have found formulas for two specific sequences, $\mathcal{F}_{1,\varphi}$ and $\mathcal{F}_{1,\psi}$. Our original goal is to find a formula for the classical sequence $\mathcal{F}_{0,1}$.

We use the linearity principle established in **??**. We seek scalars $\alpha, \beta \in \mathbb{R}$ such that:

$$\alpha \cdot \mathcal{F}_{1,\varphi} + \beta \cdot \mathcal{F}_{1,\psi} = \mathcal{F}_{0,1}$$

Comparing the first two terms ($n = 0$ and $n = 1$) of these sequences yields a system of linear equations:

$$\begin{cases} \alpha \cdot 1 + \beta \cdot 1 = 0 \\ \alpha \cdot \varphi + \beta \cdot \psi = 1 \end{cases} \tag{1.3}$$

From the first equation, we immediately deduce that $\beta = -\alpha$. Substituting this into the second equation gives:

$$\alpha \cdot \varphi - \alpha \cdot \psi = 1 \implies \alpha(\varphi - \psi) = 1$$

Notice that the difference $\varphi - \psi = \sqrt{5}$. Therefore, $\alpha = 1/\sqrt{5}$ and $\beta = -1/\sqrt{5}$. This leads directly to:

$$a_n = \frac{1}{\sqrt{5}}(\varphi^n - \psi^n) = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$$

Q.E.D.

The Upshot: Asymptotics and Matrix Form

The derivation above yields two profound observations regarding the growth and representation of the Fibonacci sequence.

Important remark: Clearly, as n grows, the term involving ψ becomes negligible. Since $|\psi| = |\frac{1-\sqrt{5}}{2}| \approx 0.618 < 1$, it follows that $\psi^n \to 0$ as $n \to \infty$. Therefore, for large n, we have the approximation $a_n \approx \frac{1}{\sqrt{5}}\varphi^n$. In fact, because $\left|\frac{\psi^n}{\sqrt{5}}\right| < \frac{1}{2}$ for all $n \geq 0$, we arrive at the following precise conclusion:

$$\implies a_n \text{ is the } \textit{closest integer} \text{ to } \frac{1}{\sqrt{5}} \cdot \left(\frac{1+\sqrt{5}}{2}\right)^n$$

Furthermore, this implies that the ratio of consecutive terms satisfies:

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \varphi \tag{1.4}$$

Finally, we can represent the recurrence relation using the language of matrices. Defining a vector $v_n = \binom{a_{n+1}}{a_n}$, the relation $a_{n+1} = a_n + a_{n-1}$ can be written as:

$$\begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_n \\ a_{n-1} \end{pmatrix} \implies v_n = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n \begin{pmatrix} a_1 \\ a_0 \end{pmatrix} \tag{1.5}$$

This formulation shows that the behavior of the sequence is entirely governed by the powers of the matrix $M = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$.

Exercise (*Interesting Exercises*):

(a) **General Formula:** Find an explicit formula for the general Fibonacci sequence $\mathcal{F}_{a_0,a_1}$ that depends only on the initial values a_0, a_1 and the index n.

(b) **The Shift Operator:** Let $S : \text{Fib} \to \text{Fib}$ be the shift map defined by $S(a_0, a_1, \dots) = (a_1, a_2, \dots)$. Formally verify that S is a *linear map*.

(c) **Golden Ratio Continued Fraction:** Consider $\varphi = \frac{1+\sqrt{5}}{2}$.

$$\left. \varphi = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cdots}}}} \right\} \quad \text{what is actually the meaning of this?}$$

(d) **Non-existence of Arithmetic Fibonacci Sequences:** Complete the details to show that no Fibonacci sequence $\mathcal{F}_{a_0,a_1}$ (except for the trivial zero sequence) can be an arithmetic sequence.

(e) **Convergence:** For the classical Fibonacci sequence $a_n = \mathcal{F}_{0,1}$, calculate the limit

$$\lim_{n\to\infty} \frac{a_n}{a_{n-1}}$$

.

(f) **Cassini's Identity:** Show that $a_{n+1}a_{n-1} - a_n^2 = (-1)^n$ for all $n \geq 1$. Hint: Use the determinant of the matrix M^n.

Logic and Mathematics (Chapter 2)

Before advancing into the formal theory of vector spaces, we must establish the rigorous grammar of mathematical reasoning. This chapter provides the logical foundation necessary for constructing and understanding proofs.

Mathematical Statements and Basic Operators

Definition 2.1 (*Mathematical Statement*): A *"mathematical statement"* (or logical statement) is a declarative sentence that is definitively either **True (T)** or **False (F)**. It is a fundamental axiom of logic that a statement cannot be both true and false simultaneously.

Example: Consider the following declarations:

A: *"For every real number x, $x^2 \geq 0$."*

B: *"Every prime number $p \geq 3$ is odd."*

C: *"Every odd number $p \geq 3$ is prime."*

Statements A and B are True. Statement C is False (e.g., 9 is odd but not prime).

Negation, Conjunction, and Disjunction

Definition 2.2 (*Negation*): The *"negation"* of a statement A, denoted $\neg A$ (or sometimes $\sim A$), asserts that A does not hold.

A	$\neg A$
T	F
F	T

Definition 2.3 (*Conjunction / AND*): The statement $A \wedge B$ (read *"A and B"*) is True if and only if both A and B are True.

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

Example: For $x \in \mathbb{R}$, the statement $(x^2 = 1) \wedge (x \geq 0)$ is logically equivalent to $x = 1$.

Definition 2.4 (*Disjunction / OR*): The statement $A \vee B$ (read *"A or B"*) is True if *at least one* of the statements A or B is True. This is an *inclusive* OR.

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

Example: The statement $(x > 1) \vee (x < 2)$ is True for all $x \in \mathbb{R}$. Conversely, $(x < 1) \vee (x > 2)$ is precisely the negation of the statement $1 \leq x \leq 2$.

Implications and Equivalences

Definition 2.5 (*Implication*): The statement $A \implies B$ (*"A implies B"*) is defined to be False only when A is True and B is False. In all other cases, it is True.

A	B	$A \implies B$
T	T	T
T	F	F
F	T	T
F	F	T

Remark: Note that $A \implies B$ is logically equivalent to $\neg A \vee B$. Consequently, if the premise A is False, the implication is *vacuously true* regardless of B. As Prof. Biran noted:

$$(0 = 1) \implies \text{“The Earth is flat”}$$

is a mathematically **True** statement. Causality is irrelevant in formal logic.

Definition 2.6 (*Equivalence*): The statement $A \iff B$ (*"A if and only if B"*) holds when $A \implies B$ and $B \implies A$ are both True. This means A and B share the same truth value.

A	B	$A \iff B$
T	T	T
T	F	F
F	T	F
F	F	T

Exercise: Prove the following via truth tables:

(a) **Contrapositive:** $(A \implies B) \iff (\neg B \implies \neg A)$.

(b) **De Morgan's Laws:** $\neg(A \wedge B) \iff \neg A \vee \neg B$ and $\neg(A \vee B) \iff \neg A \wedge \neg B$.

Predicates and Quantifiers

A *"predicate"* $P(x)$ is a statement whose truth depends on a variable x. We use quantifiers to convert predicates into absolute statements.

Definition 2.7 (*Quantifiers*):

- **Universal ($\forall$):** *"For every x in X, $P(x)$ holds."*
- **Existential ($\exists$):** *"There exists at least one x in X such that $P(x)$ holds."*

Example: Let $P(n)$ be the predicate $n = n^2$ over $\mathbb{Z}$.

- $\forall n \in \mathbb{Z} : P(n)$ is False.
- $\exists n \in \mathbb{Z} : P(n)$ is True (take $n = 1$).

Important remark (*The Order of Quantifiers*): The order of quantifiers is non-commutative. Consider X (ETH students), Y (courses), and $A(x, y) =$ *"student x takes course y"*.

- $\forall x \in X, \exists y \in Y : A(x, y)$ means every student takes at least one course (True).
- $\exists y \in Y, \forall x \in X : A(x, y)$ means there is one specific course taken by *every* student (False).

Negation and Unique Existence

Proposition 2.8 (*Negating Quantifiers*):

$$\neg(\forall x \in X : A(x)) \iff \exists x \in X : \neg A(x)$$
$$\neg(\exists x \in X : A(x)) \iff \forall x \in X : \neg A(x)$$

Remark (*Shorthand Notation*): In practice, we often condense multiple quantifiers of the same type:

- $\exists x \in X, \exists y \in X \implies \exists x, y \in X$
- $\forall x \in X, \forall y \in X \implies \forall x, y \in X$

Definition 2.9 (*Unique Existence*): The notation $\exists! x \in X : P(x)$ means there exists exactly one x satisfying the predicate. Formally:

$$(\exists x \in X : P(x)) \wedge (\forall x, y \in X : (P(x) \wedge P(y) \implies x = y))$$

Example: $\exists! x \in \mathbb{R} : x^2 = 4$ is False (since $x = \pm 2$). However, $\exists! x \in \mathbb{R}_{\geq 0} : x^2 = 4$ is True.

Set Theory (Chapter 3)

In this chapter, we formalize the language of sets, which provides the universal framework for all modern mathematical structures, including the vector spaces we will investigate later.

Basic Definitions and Notation

Definition 3.1 (*Set*): A *“set”* is a collection of distinct objects, which are referred to as the *elements* of the set.

In our notation, we denote sets typically by capital letters. If an object x is an element of a set M, we write $x \in M$. If it is not, we write $x \notin M$.

Remark: Crucially, within a set, the order of elements is irrelevant, and repetitions do not contribute to the set's identity. For example:

$$M = \{1, 3, \text{moon}\} = \{3, \text{moon}, 1\}$$

$$M = \{1, 1, 1\} = \{1\}$$

We frequently define sets using a property or condition $A(x)$ that the elements must satisfy. The standard notation for this *“set-builder”* form is:

$$M = \{x \mid A(x)\} \quad \text{or} \quad M = \{x : A(x)\}$$

Example: The set of integers whose square is less than or equal to 9 is written as:

$$\{x \in \mathbb{Z} \mid x^2 \leq 9\} = \{-3, -2, -1, 0, 1, 2, 3\}$$

Definition 3.2 (*The Empty Set*): The *“empty set”*, denoted by $\emptyset$ or $\{\}$, is the unique set containing no elements.

It is important to note that a set is allowed to contain another set as one of its elements. For instance, consider the set $M = \{1, \text{elephant}, \{a, 5\}\}$. Here, the set $\{a, 5\}$ is a single element of M. Similarly, we can define a set containing only the empty set: $M = \{\emptyset\}$, which is a set containing exactly one element.

Subsets and Inclusion

We now define how sets relate to one another in terms of containment.

Definition 3.3 (*Subset and Proper Subset*): Let P and Q be sets.

(a) We say P is a *“subset”* of Q, denoted $P \subseteq Q$, if every element of P is also an element of Q. Formally: $\forall x \in P : x \in Q$.

(b) We say P is a *“proper subset”* of Q, denoted $P \subsetneq Q$, if $P \subseteq Q$ and $P \neq Q$.

(c) We write $P \nsubseteq Q$ to denote that P is not a subset of Q, meaning $\neg(P \subseteq Q)$.

Remark: Note that some authors use $P \subset Q$ to denote proper subsets, while others use it for generic subsets.

Claim: For any set Q, the empty set is a subset of Q (i.e., $\emptyset \subseteq Q$).

Set Operations

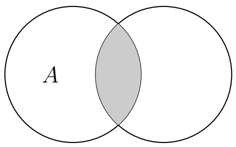
Just as we have logical operators for statements, we have operations to combine or modify sets. Let A and B be subsets of a universal set X.

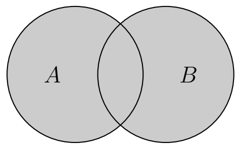
(a) **Intersection:** $A \cap B = \{x \in X \mid x \in A \wedge x \in B\}$.

(b) **Union:** $A \cup B = \{x \in X \mid x \in A \vee x \in B\}$.

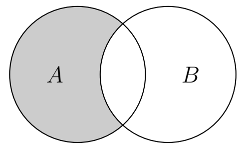
(c) **Difference:** $A \setminus B = \{x \in X \mid x \in A \wedge x \notin B\}$.

(d) **Complement:** The complement of A in X is $A^c = X \setminus A = \{x \in X \mid x \notin A\}$.

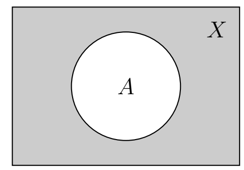




$A \cup B$



$A \setminus B$



A^c

Cartesian Products

To handle ordered pairs and higher-dimensional structures, we introduce the Cartesian product.

Definition 3.4 (*Cartesian Product*): The *"Cartesian product"* of two sets X and Y is the set of all ordered pairs (x, y) where $x \in X$ and $y \in Y$:

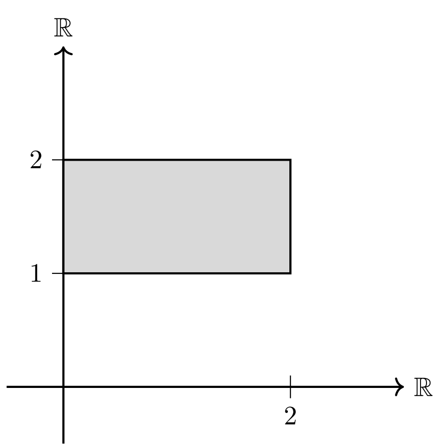
$$X \times Y = \{(x, y) \mid x \in X, y \in Y\}$$

This construction generalizes to n sets. We denote the n-fold product of a set X with itself as X^n:

$$X^n = \underbrace{X \times \cdots \times X}_{n \text{ times}} = \{(x_1, \ldots, x_n) \mid x_i \in X \text{ for } 1 \leq i \leq n\}$$

The elements of X^n are called n-tuples. For example, $X^2 = X \times X$, and $X^3 = X \times X \times X = \{(x_1, x_2, x_3) \mid x_1, x_2, x_3 \in X\}$, and so forth.

Example: If $X = \mathbb{R}$, then $X^2 = \mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ represents the standard Euclidean plane. If we take $X = [0, 2]$ and $Y = [1, 2]$, then $X \times Y$ represents a rectangle in the plane with corners at $(0, 1), (2, 1), (0, 2)$, and $(2, 2)$.



Power Sets and Cardinality

Definition 3.5 (*Power Set*): Let X be a set. The *"power set"* of X, denoted by $\mathcal{P}(X)$ or 2^X, is the set of all subsets of X:

$$\mathcal{P}(X) = \{Q \mid Q \subseteq X\}$$

Example: For $X = \{1, 2, 3\}$, the power set is:

$$\mathcal{P}(X) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$$

Note that $\mathcal{P}(X)$ contains 8 elements.

Definition 3.6 (*Cardinality*): If X is a finite set, meaning a set with finitely many elements $X = \{x_1, \ldots, x_n\}$, we say that the *"cardinality"* of X is the number of elements in X, denoted by $|X| = n$ or $\mathrm{card}(X) = n$.

Exercise: Let A be a finite set. Prove by induction that the cardinality of its power set is given by:

$$|\mathcal{P}(A)| = 2^{|A|}$$

A Note on the Foundations: Russell's Paradox

In the early development of set theory, it was assumed that any property $A(x)$ could define a set (Naive Set Theory). However, Bertrand Russell showed that this leads to a contradiction.

Suppose there exists a *"set of all sets"*, denoted by S. We then define a subset $R \subseteq S$ as follows:

$$R := \{X \in S \mid X \notin X\}$$

Now, we ask: is R an element of itself?

- If $R \in R$, then by the definition of R, it must be that $R \notin R$. This is a contradiction.
- If $R \notin R$, then by the definition of R, it must be that $R \in R$. This is also a contradiction.

Claim: S is not a set.

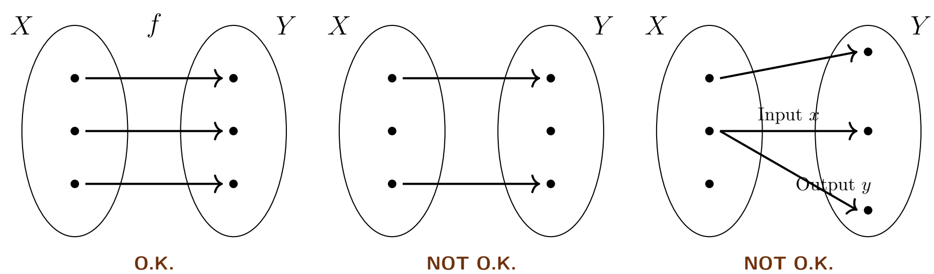
This paradox led to the development of Axiomatic Set Theory (e.g., Zermelo-Fraenkel), which places specific restrictions on how sets can be constructed to ensure mathematical consistency.

Maps (Chapter 4)

Functions and their Domains

Definition 4.1 (*Map / Function*): Let X and Y be sets. A map $f : X \to Y$ (also called a function or transformation) is an assignment that assigns to every element $x \in X$ a uniquely defined element $f(x) \in Y$.

In accordance with this definition, we can visually distinguish between valid and invalid assignments. A function must assign exactly one output to each input in its domain.



X is called the *domain of definition* of f. Y is called the *target* (or target domain) of f.

Definition 4.2 (*Equality of Functions*): Two functions $f : X \to Y$ and $g : X' \to Y'$ are equal if they have the same domain of definition ($X' = X$), the same target ($Y' = Y$), and for all $x \in X$, we have $f(x) = g(x)$.

Sometimes a function is given by a formula, and sometimes it is not. Be careful: providing just a formula is often mathematically insufficient.

Example: Bad notation: $f(x) = x^2$. (What are the domain and target?)
O.K. notation: Let $X := \{x \in \mathbb{R} \mid 0 \leq x\}$. Define $f : X \to \mathbb{R}$ by $f(x) = x^2$.

Example: Let $X = \{x \in \mathbb{Z} \mid x \geq 0\}$ and let Y be the set of family names. We might try to define a map $g : X \to Y$ by setting $g(x)$ to be the family name of the student at ETH whose student number is x.

This g is **not** defined! What is $g(0)$? What is $g(10^{10})$? There are no students with these IDs.

To fix this, we could define $Y' = Y \cup \{\text{ERROR!}\}$ and introduce a rigorous function $h : X \to Y'$ given by:

$$h(x) = \begin{cases} \text{family name} & \text{if a student with student no. } x \text{ exists} \\ \text{ERROR!} & \text{otherwise} \end{cases}$$

Remark: Notation: We frequently use the barred arrow ($\mapsto$) to denote how individual elements are mapped. For example, $f : X \to Y, X \ni x \mapsto f(x) \in Y$. Concretely: $f : \mathbb{R} \to \mathbb{R}, \mathbb{R} \ni x \mapsto x^5 \in \mathbb{R}$.

Images and Restrictions

Definition 4.3 (*Image of a Map*): Let $f : X \to Y$ be a map. The *image* of f is the set:

$$f(X) := \{y \in Y \mid \exists x \in X \text{ s.t. } f(x) = y\}$$

(sometimes written as $\text{image}(f)$). Clearly, $f(X) \subseteq Y$.

Definition 4.4 (*Restriction and Extension*): Let $f : X \to Y$ be a map, and let $A \subseteq X$ be a subset. We can define a new map $f|_A : A \to Y$, called the *restriction* of f to A, simply by $f|_A(x) := f(x)$ for all $x \in A$. Conversely, f is called an *extension* of $f|_A$.

Some Standard Maps

Example:

(a) **Identity Map:** $\mathrm{id} : X \to X$ (sometimes denoted id_X), defined by $\mathrm{id}(x) = x$ for all $x \in X$.

(b) **Constant Map:** $f : X \to Y$. Choose a fixed $y_0 \in Y$. Then $f(x) = y_0$ for all $x \in X$.

(c) **Characteristic Function:** Let $A \subseteq X$. The characteristic function $\chi_A : X \to \{0, 1\}$ is defined as:

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

(d) **Projections:** Given a Cartesian product $X_1 \times X_2$, the standard projection maps are:

$$p_1 : X_1 \times X_2 \to X_1, \quad p_1(x_1, x_2) = x_1$$
$$p_2 : X_1 \times X_2 \to X_2, \quad p_2(x_1, x_2) = x_2$$

Injectivity, Surjectivity, Bijectivity

Definition 4.5 (*Injectivity, Surjectivity, and Bijectivity*)**:** Let $f : X \to Y$ be a map.

(a) We say f is **injective** if $\forall x_1, x_2 \in X$, $f(x_1) = f(x_2) \implies x_1 = x_2$. (Equivalently, by contrapositive: $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$).

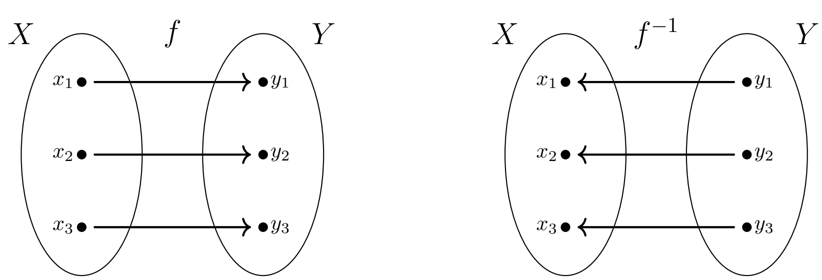
(b) We say f is **surjective** if $\forall y \in Y, \exists x \in X$ such that $f(x) = y$. (Equivalently: $f(X) = Y$).

(c) We say f is **bijective** (or a *bijection*) if f is both injective and surjective. (Equivalently: $\forall y \in Y, \exists! x \in X : f(x) = y$).

Example: Consider variations of the squaring function, which explicitly demonstrate the importance of the chosen domain and target:

- $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$. Not injective (e.g., $f(2) = f(-2)$). Not surjective (no real x maps to -1).
- $f : \mathbb{R}_{\geq 0} \to \mathbb{R}, f(x) = x^2$. Injective, but not surjective.
- $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}, f(x) = x^2$. Bijective.

Definition 4.6 (*Inverse Map*)**:** Let $f : X \to Y$ be a bijection. The *inverse map* $f^{-1} : Y \to X$ is the map that assigns to every $y \in Y$ the unique $x \in X$ such that $f(x) = y$.



Example: For the bijection $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ with $f(x) = x^2$, the inverse map is $f^{-1} : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$, strictly defined by $f^{-1}(y) = \sqrt{y}$.

Composition of Maps

Definition 4.7 (*Composition*)**:** Let $f : X \to Y$ and $g : Y \to Z$ be maps. The composition $g \circ f : X \to Z$ is defined by:

$$(g \circ f)(x) := g(f(x))$$

Example: If $f : X \to Y$ is bijective and $f^{-1} : Y \to X$ is its inverse, then by definition:

$$f^{-1} \circ f = \mathrm{id}_X \quad \text{and} \quad f \circ f^{-1} = \mathrm{id}_Y$$

Remark: If $g \circ f$ is defined, $f \circ g$ is in general NOT defined (unless $Z = X$). But even when $Z = X$, in general $g \circ f \neq f \circ g$. Commutativity is not guaranteed.

Example: Let $f, g : \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$ and $g(x) = x + 1$.

$$(g \circ f)(x) = g(x^2) = x^2 + 1$$
$$(f \circ g)(x) = f(x + 1) = (x + 1)^2 = x^2 + 2x + 1$$

Clearly, $g \circ f \neq f \circ g$.

Lemma 4.8: Let $f : X \to Y$ and $g : Y \to Z$ be maps.

(a) If f and g are injective, then $g \circ f$ is injective.

(b) If f and g are surjective, then $g \circ f$ is surjective.

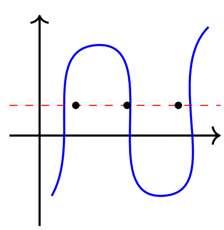
Proof

(a) Assume f and g are injective. Let $x_1, x_2 \in X$ such that $(g \circ f)(x_1) = (g \circ f)(x_2)$. This means $g(f(x_1)) = g(f(x_2))$. Because g is injective, this implies $f(x_1) = f(x_2)$. Because f is injective, this implies $x_1 = x_2$. Thus, $g \circ f$ is injective.

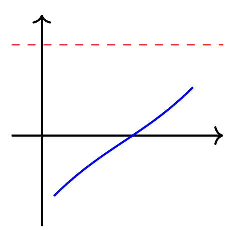
(b) Assume f and g are surjective. Let $z \in Z$. Because g is surjective, there exists $y \in Y$ such that $g(y) = z$. Because f is surjective, there exists $x \in X$ such that $f(x) = y$. Therefore, $g(f(x)) = g(y) = z$, which means $(g \circ f)(x) = z$. Thus, $g \circ f$ is surjective. **Q.E.D.**

Visualizing Properties via Graphs

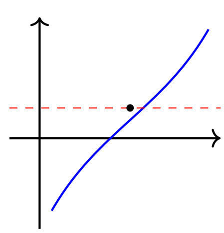
We can often immediately determine the injectivity, surjectivity, and bijectivity of a function—or whether an assignment is a function at all—by examining its Cartesian graph using horizontal and vertical line tests.



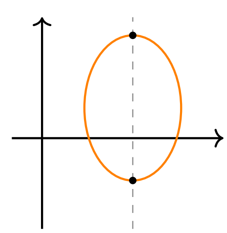
Surjective
(but not injective)



Injective
(but not surjective)



Bijective



Not a Function's Graph
(fails vertical line test)

Image and Inverse Image of Subsets

Definition 4.9 (*Image of a Subset*): Let $f : X \to Y$ be a map, and let $A \subseteq X$. The *image of A under f* is:

$$f(A) := \{y \in Y \mid \exists x \in A \text{ s.t. } f(x) = y\} = \{f(x) \mid x \in A\}$$

Definition 4.10 (*Inverse Image of a Subset*): Let $f : X \to Y$ be a map, and let $B \subseteq Y$. The *inverse image of B under f* is:

$$f^{-1}(B) := \{x \in X \mid f(x) \in B\} \subseteq X$$

Important remark: The notation $f^{-1}(B)$ is somewhat confusing, as this set exists and is perfectly well-defined even if the map f is not invertible!

Example: Consider $f : \mathbb{R} \to \mathbb{R}, f(x) = x^2$ (which is neither injective nor surjective). We can still compute inverse images of subsets without issue:

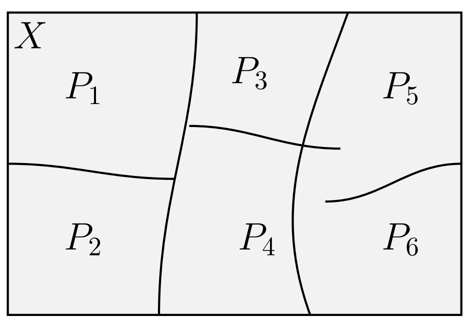
$$\begin{aligned} f^{-1}(\{4\}) &= \{-2, 2\} \\ f^{-1}(\{1, 9\}) &= \{-1, 1, -3, 3\} \\ f^{-1}(\{0\}) &= \{0\} \\ f^{-1}(\{-1\}) &= \emptyset \\ f^{-1}(\{-1, 0\}) &= \{0\} \end{aligned}$$

Partitions

Definition 4.11 (*Partition*): Let X be a set and $\mathcal{P}$ a family of subsets of X, none of which is empty. We say that $\mathcal{P}$ is a *partition* of X if the following hold:

(a) $X = \bigcup_{P \in \mathcal{P}} P$ (the subsets cover X).

(b) For any $P_1, P_2 \in \mathcal{P}$ with $P_1 \neq P_2$, we have $P_1 \cap P_2 = \emptyset$ (the subsets are mutually disjoint).



Fields (Körper) (Chapter 5)

Before we can solve systems of linear equations or define abstract vector spaces, we must answer a more fundamental question: *Where do our numbers come from?*

In linear algebra, we do not restrict ourselves to the real numbers. We can perform our calculations over any algebraic structure where addition, subtraction, multiplication, and division behave exactly as we expect them to. Such a structure is called a **field** (in German: *Körper*).

The Field Axioms (Section 5.a)

To ensure that an algebraic structure behaves nicely, it must satisfy nine rigid rules. These are the Field Axioms.

Definition 5.1.1 (*Field*): A *field* is a tuple $(K, +, \cdot, 0, 1)$ consisting of a set K equipped with two binary operations:

$$+ : K \times K \to K, \quad (x, y) \mapsto x + y \quad \text{(Addition)}$$
$$\cdot : K \times K \to K, \quad (x, y) \mapsto x \cdot y \quad \text{(Multiplication)}$$

and two distinguished, unique elements $0, 1 \in K$, such that the following nine axioms hold:

Axioms of Addition:

K1 $\forall x, y, z \in K : x + (y + z) = (x + y) + z$ *(Associativity of Addition)*

K2 $\forall x, y \in K : x + y = y + x$ *(Commutativity of Addition)*

K3 $\forall x \in K : 0 + x = x$ *(Neutral Element of Addition)*

K4 $\forall x \in K, \exists x' \in K : x + x' = 0$ *(Inverse Element of Addition)*

Axioms of Multiplication:

K5 $\forall x, y, z \in K : x \cdot (y \cdot z) = (x \cdot y) \cdot z$ *(Associativity of Multiplication)*

K6 $\forall x \in K : 1 \cdot x = x$ *(Neutral Element of Multiplication)*

K7 $\forall x \in K \setminus \{0\}, \exists x' \in K : x' \cdot x = 1$ *(Inverse Element of Multiplication)*

Compatibility and Non-Triviality:

K8 $\forall x, y, z \in K : x \cdot (y + z) = x \cdot y + x \cdot z$ *(Distributivity)*

K9 $1 \neq 0$ *(The field has at least two elements)*

Remark (*Notation*): We typically denote the additive inverse x' from (K4) as $-x$, allowing us to define subtraction: $x - y := x + (-y)$. Similarly, we denote the multiplicative inverse x' from (K7) as x^{-1} or $\frac{1}{x}$, allowing us to define division by non-zero elements: $x/y := x \cdot y^{-1}$.

Infinite Fields (Section 5.b)

The most familiar fields are infinite. They form the backbone of calculus and classical geometry.

Example (*Familiar Infinite Fields*):

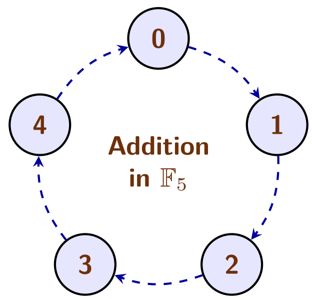
- $\mathbb{Q}$: The rational numbers (fractions). This is the smallest field containing the integers.
- $\mathbb{R}$: The real numbers.
- $\mathbb{C}$: The complex numbers, where we introduce $i^2 = -1$.

Important remark: The integers $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$ do **not** form a field! While they satisfy addition and multiplication axioms perfectly, they completely fail (K7). For instance, there is no integer x' such that $2 \cdot x' = 1$. You cannot divide freely within $\mathbb{Z}$.

Finite Fields and Modular Arithmetic (Section 5.c)

A field does not have to be an infinite, continuous line of numbers. If we change our definitions of "addition" and "multiplication", we can create perfectly valid fields that contain only a finite number of elements.

The secret to building a finite field is *modular arithmetic* (often called "clock arithmetic"). Instead of numbers growing infinitely on a number line, they wrap around a circle.



Definition 5.3.1 (*The Field $\mathbb{F}_p$*): Let p be a prime number. The set of integers modulo p, denoted as $\mathbb{F}_p$ (or sometimes $\mathbb{Z}/p\mathbb{Z}$), contains exactly p elements:

$$\mathbb{F}_p = \{0, 1, 2, \dots, p-1\}$$

Addition and multiplication are performed exactly as regular integers, but if the result exceeds $p-1$, we divide by p and keep only the **remainder**.

Theorem 5.3.2: $\mathbb{F}_p$ is a field if and only if p is a prime number.

Why must p be prime? Consider what happens if we try to build a field modulo 4 (which is not prime). In modulo 4, we have $2 \cdot 2 = 4 \equiv 0 \pmod 4$. We multiplied two non-zero numbers and got zero! In a true field, this is algebraically impossible, because it destroys the ability to divide (axiom K7). Prime numbers beautifully prevent this "zero divisor" problem.

Example (*The Smallest Field: $\mathbb{F}_2$*): The absolute smallest field mathematically possible contains only two elements: $\{0, 1\}$. This is the field of Boolean logic, heavily used in computer science and cryptography. Operations are done modulo 2:

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

Notice the fascinating property of $\mathbb{F}_2$: $1 + 1 = 0$. In this field, every element is its own additive inverse! Subtraction and addition are literally the same operation.

Example (*Calculations in $\mathbb{F}_5$*): Let us explore $\mathbb{F}_5 = \{0, 1, 2, 3, 4\}$.

- **Addition:** $3 + 4 = 7$. But $7 \div 5 = 1$ with a remainder of 2. So, $3 + 4 = 2$ in $\mathbb{F}_5$.
- **Additive Inverses (K4):** What is -2? We need a number that adds to 2 to yield 0. Since $2 + 3 = 5 \equiv 0$, we know that $-2 = 3$.
- **Multiplicative Inverses (K7):** What is the inverse of 3 (i.e., $\frac{1}{3}$)? We need a number x such that $3 \cdot x = 1$. Let's test the options: $3 \cdot 2 = 6 \equiv 1$. Therefore, $3^{-1} = 2$.

Proving Basic Properties from the Axioms (Section 5.d)

The power of an axiomatic definition is that if we can prove a theorem using *only* axioms (K1) through (K9), that theorem is universally true for **every field in existence**—whether it is the infinite space of real numbers $\mathbb{R}$, or the tiny binary world of $\mathbb{F}_2$.

Let us prove two seemingly "obvious" facts that actually require careful algebraic footwork to justify.

Lemma 5.4.1 (*Multiplication by Zero*): For any element x in a field K, we have $0 \cdot x = 0$.

Proof

Let $x \in K$. By the neutral property of zero (K3), we know that $0 = 0 + 0$. Let us multiply both sides of this equation by x:

$$(0+0) \cdot x = 0 \cdot x$$

By the distributivity axiom (K8), we can expand the left side:

$$0 \cdot x + 0 \cdot x = 0 \cdot x$$

Now, by the additive inverse axiom (K4), the element $(0 \cdot x)$ must have an inverse, $-(0 \cdot x)$. We add this inverse to both sides of our equation:

$$\big(0 \cdot x + 0 \cdot x\big) + \big(-(0 \cdot x)\big) = 0 \cdot x + \big(-(0 \cdot x)\big)$$

Using associativity of addition (K1), we regroup the left side:

$$0 \cdot x + \Big(0 \cdot x + \big(-(0 \cdot x)\big)\Big) = 0$$

$$0 \cdot x + 0 = 0$$

Finally, applying (K3) once more, we drop the $+0$ to conclude:

$$0 \cdot x = 0$$

Q.E.D.

Lemma 5.4.2 (*Zero Divisors do not exist*): Let $a, b \in K$. If $a \cdot b = 0$, then either $a = 0$ or $b = 0$.

Proof

Assume $a \cdot b = 0$. We want to show that if $a \neq 0$, then b must be exactly 0.

Suppose $a \neq 0$. Because a is a non-zero element in a field, axiom (K7) guarantees that it has a multiplicative inverse, a^{-1}. Let us multiply both sides of our equation $a \cdot b = 0$ by this inverse:

$$a^{-1} \cdot (a \cdot b) = a^{-1} \cdot 0$$

From our previous lemma, we know that any element multiplied by 0 is 0, so the right side is 0:

$$a^{-1} \cdot (a \cdot b) = 0$$

By associativity of multiplication (K5), we regroup the left side:

$$(a^{-1} \cdot a) \cdot b = 0$$

By the definition of the inverse (K7), $a^{-1} \cdot a = 1$:

$$1 \cdot b = 0$$

Finally, by the neutral element of multiplication (K6), $1 \cdot b = b$. Thus:

$$b = 0$$

We have proven that if the product is zero and a is not zero, b is forced to be zero. **Q.E.D.**

Systems of Linear Equations (Chapter 6)

Fields (Körper)

Before we solve systems of linear equations, we must define the algebraic structures over which these equations are built. Throughout this course, we will work over a base *"field"* (in German: *"Körper"*), usually denoted by K.

A field is a set equipped with two operations (addition $+$ and multiplication $\cdot$) that satisfy a standard set of axioms, including commutativity, associativity, distributivity, and the existence of identities (0 and 1) and inverses.

Example (*Common Fields*):

- $\mathbb{Q} = \left\{\frac{m}{n} \mid m, n \in \mathbb{Z}, n \neq 0\right\}$ (The rational numbers)
- $\mathbb{R}$ (The real numbers)
- $\mathbb{C} = \{a + i \cdot b \mid a, b \in \mathbb{R}, i^2 = -1\}$ (The complex numbers)

Example (*Finite Fields*): Fields do not have to be infinite. We will frequently use finite fields, where arithmetic is performed modulo a prime number.

(a) $\mathbb{F}_2 = \{0, 1\}$. Operations are done modulo 2:

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

(b) $\mathbb{F}_3 = \{0, 1, 2\}$. Operations are done modulo 3:

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\cdot$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

Motivating Example

Let us solve a familiar system of linear equations over $\mathbb{R}$ using elimination:

$$\begin{cases} E_1: & x + 3y + 4z = -1 \\ E_2: & 2x - y + z = 5 \\ E_3: & 3x + 2y + z = 0 \end{cases}$$

We perform operations on the equations to systematically eliminate variables:

$$-2 \cdot E_1 + E_2 \to E_2$$
$$-3 \cdot E_1 + E_3 \to E_3$$

This yields an equivalent, simpler system:

$$\begin{cases} x + 3y + 4z = -1 \\ -7y - 7z = 7 \\ -7y - 11z = 3 \end{cases}$$

Next, we scale the second equation ($-\frac{1}{7} \cdot E_2 \to E_2$) to get:

$$\begin{cases} x + 3y + 4z = -1 \\ y + z = -1 \\ -7y - 11z = 3 \end{cases}$$

Finally, we eliminate y from the third equation ($7 \cdot E_2 + E_3 \to E_3$):

$$\begin{cases} x + 3y + 4z = -1 \\ y + z = -1 \\ -4z = -4 \implies z = 1 \end{cases}$$

By backward substitution:

- $y + 1 = -1 \implies y = -2$
- $x + 3(-2) + 4(1) = -1 \implies x - 2 = -1 \implies x = 1$

Thus, the unique solution is $(x, y, z) = (1, -2, 1)$. We can readily verify this by plugging the values back into the original system.

General Systems of Linear Equations

Definition 6.1 (*Linear Equation*): A *"linear equation"* in n variables $x_1, \ldots, x_n$ over a field K is an equation of the form:

$$a_1 x_1 + \cdots + a_n x_n = b$$

where $a_1, \ldots, a_n \in K$ are the *"coefficients"* and $b \in K$ is the *"constant term"*.

- If $b = 0$, the equation is called *"homogeneous"*.
- If $b \neq 0$, the equation is called *"inhomogeneous"*.

Definition 6.2 (*System of Linear Equations*): A general system of m linear equations in n variables over a field K is written as:

$$\begin{cases} A_{1,1}x_1 + A_{1,2}x_2 + \cdots + A_{1,n}x_n = b_1 \\ A_{2,1}x_1 + A_{2,2}x_2 + \cdots + A_{2,n}x_n = b_2 \\ \vdots \\ A_{m,1}x_1 + A_{m,2}x_2 + \cdots + A_{m,n}x_n = b_m \end{cases} \tag{6.1}$$

Definition 6.3 (*Solution*): A *"solution"* to the system is a sequence of scalars $s_1, \ldots, s_n \in K$ that simultaneously satisfies all m equations when we substitute $x_j = s_j$.

Definition 6.4 (*Equivalence*): Two systems of linear equations (in the same variables) are called *"equivalent"* if they have exactly the same set of solutions.

Matrices

To study systems of linear equations systematically, we strip away the variables and isolate the coefficients into a rectangular array.

Definition 6.5 (*Matrix*): An $m \times n$ *"matrix"* over K is a rectangular array consisting of $m \cdot n$ elements of K, arranged in m rows and n columns.

Notation: We denote a matrix A by its entries:

$$A = (A_{i,j})_{1 \le i \le m, 1 \le j \le n} \quad \text{or simply} \quad (a_{ij})$$

Here, i denotes the row index and j denotes the column index.

We define the set of all such matrices as:

$$\mathcal{M}_{m \times n}(K) := \text{the set of all } m \times n \text{ matrices over } K$$

Definition 6.6 (*Row and Column Vectors*): Special cases of matrices occur when one of the dimensions is 1:

(a) Column Vectors: An $n \times 1$ matrix is called a *"column vector"*. The set of all column vectors of length n is denoted by $K^n := \mathcal{M}_{n\times 1}(K)$.

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in K^n$$

(b) Row Vectors: A $1 \times n$ matrix is called a *"row vector"*. The set of all row vectors of length n is denoted by $K^n_{\text{row}} := \mathcal{M}_{1\times n}(K)$.

$$(x_1, x_2, \ldots, x_n) \in K^n_{\text{row}}$$

Matrix-Vector Multiplication

Definition 6.7 (*Matrix-Vector Multiplication*): Let $A \in \mathcal{M}_{m\times n}(K)$ and $x \in K^n$. We define the product $A \cdot x$ as a new column vector in K^m, computed by taking the dot product of the rows of A with the vector x:

$$\begin{pmatrix} A_{1,1} & A_{1,2} & \dots & A_{1,n} \\ A_{2,1} & A_{2,2} & \dots & A_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m,1} & A_{m,2} & \dots & A_{m,n} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} A_{1,1}x_1 + \cdots + A_{1,n}x_n \\ A_{2,1}x_1 + \cdots + A_{2,n}x_n \\ \vdots \\ A_{m,1}x_1 + \cdots + A_{m,n}x_n \end{pmatrix} \in K^m$$

Using this powerful notation, the massive system of linear equations (**??**) can be written elegantly and compactly as a single matrix equation:

$$Ax = b \tag{6.2}$$

where $A \in \mathcal{M}_{m\times n}(K)$ is the *"coefficient matrix"*, $x \in K^n$ is the vector of variables, and $b \in K^m$ is the *"constant vector"*.

Elementary Row Operations (EROs)

To solve $Ax = b$, we define the *"augmented matrix"* $(A|b) \in \mathcal{M}_{m\times(n+1)}(K)$ by appending b as an extra column to A. We can manipulate this matrix using three basic operations that correspond exactly to operations on the equations themselves.

Definition 6.8 (*Elementary Row Operations*): Let E_i denote the i-th row of the augmented matrix. The three Elementary Row Operations (EROs) are defined and denoted as follows: Let E_i denote the i-th row of the augmented matrix. The three *"Elementary Row Operations"* (EROs) are defined and denoted as follows:

(a) Type 1: Multiply row i by a non-zero scalar $\lambda \in K \setminus \{0\}$.
Notation: $\lambda \cdot E_i \to E_i$

(b) Type 2: Add to row j a scalar multiple of row i.
Notation: $\lambda \cdot E_i + E_j \to E_j$

(c) Type 3: Swap row i and row j.
Notation: $E_i \leftrightarrow E_j$

Theorem 6.9: Performing Elementary Row Operations on the augmented matrix $(A|b)$ does NOT change the set of solutions to the corresponding system of linear equations. In other words, if $(A|b)$ is transformed into $(A'|b')$ via EROs, then $Ax = b$ and $A'x = b'$ are equivalent systems.

Idea of the Proof

Each of the three EROs is strictly reversible. If you apply an operation, you can apply its inverse to get the original system back. Therefore, no solutions are lost, and no false solutions are created.

Q.E.D.

Further Computational Examples

Example (*A system with infinitely many solutions*): Consider the following system:

$$\begin{cases} x_1 - 2x_2 + x_3 + x_4 = 1 \\ 2x_1 - 4x_2 + x_3 + 3x_4 = 2 \\ x_1 - 2x_2 + 2x_3 - x_4 = 1 \end{cases}$$

We write the augmented matrix and apply EROs:

$$\left(\begin{array}{cccc|c} 1 & -2 & 1 & 1 & 1 \\ 2 & -4 & 1 & 3 & 2 \\ 1 & -2 & 2 & -1 & 1 \end{array}\right) \xrightarrow[-1\cdot E_1+E_3\to E_3]{-2\cdot E_1+E_2\to E_2} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 1 & 1 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 \end{array}\right)$$

$$\xrightarrow{1\cdot E_2+E_3\to E_3} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 1 & 1 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 \end{array}\right)$$

From the third row, $-x_4 = 0 \implies x_4 = 0$.
From the second row, $-x_3 + x_4 = 0 \implies x_3 = 0$.
Substituting into the first row gives $x_1 - 2x_2 + 0 + 0 = 1 \implies x_1 = 1 + 2x_2$.

We have a free variable. Let $x_2 = \alpha \in \mathbb{R}$. The general solution is:

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 1+2\alpha \\ \alpha \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

Example (*Conditions for consistency*): Let's see how solutions behave for a general choice of constants $b_1, b_2, b_3 \in \mathbb{R}$:

$$\begin{pmatrix} 1 & -2 & 1 \\ 2 & 1 & 1 \\ 0 & 5 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

We construct the augmented matrix and reduce:

$$\left(\begin{array}{ccc|c} 1 & -2 & 1 & b_1 \\ 2 & 1 & 1 & b_2 \\ 0 & 5 & -1 & b_3 \end{array}\right) \xrightarrow{-2\cdot E_1+E_2\to E_2} \left(\begin{array}{ccc|c} 1 & -2 & 1 & b_1 \\ 0 & 5 & -1 & b_2 - 2b_1 \\ 0 & 5 & -1 & b_3 \end{array}\right)$$

$$\xrightarrow{-1\cdot E_2+E_3\to E_3} \left(\begin{array}{ccc|c} 1 & -2 & 1 & b_1 \\ 0 & 5 & -1 & b_2 - 2b_1 \\ 0 & 0 & 0 & b_3 - b_2 + 2b_1 \end{array}\right)$$

For this system to have *any* solutions, the bottom row cannot result in a contradiction ($0 =$ non-zero). Therefore, the system is consistent if and only if:

$$b_3 - b_2 + 2b_1 = 0$$

Vector Spaces (Chapter 7)

Definition and Axioms

Definition 7.1 (*Vector Space*): A *"vector space"* over a field K is a set V, endowed with two operations:

$$+ : V \times V \longrightarrow V, \quad (v_1, v_2) \mapsto v_1 + v_2 \qquad \text{(Addition of vectors)}$$
$$\cdot : K \times V \longrightarrow V, \quad (\alpha, v) \mapsto \alpha \cdot v \qquad \text{(Scalar multiplication)}$$

such that the following conditions (axioms) hold:

V1 $\forall v_1, v_2, v_3 \in V\colon\ v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$.

V2 $\exists$ an element $0 \in V$ such that $0 + v = v + 0 = v$ for all $v \in V$. (Sometimes denoted 0_V).

V3 $\forall v \in V, \exists v' \in V$ such that $v + v' = 0$.

V4 $\forall v_1, v_2 \in V\colon\ v_1 + v_2 = v_2 + v_1$.

V5 $\forall a, b \in K, \forall v \in V\colon\ a \cdot (b \cdot v) = (a \cdot b) \cdot v$.

V6 $\forall v \in V\colon\ 1 \cdot v = v$.

V7 $\forall \alpha \in K, \forall v_1, v_2 \in V\colon\ \alpha \cdot (v_1 + v_2) = \alpha \cdot v_1 + \alpha \cdot v_2$.

V8 $\forall \alpha_1, \alpha_2 \in K, \forall v \in V\colon\ (\alpha_1 + \alpha_2) \cdot v = \alpha_1 \cdot v + \alpha_2 \cdot v$.

Lemma 7.2: Let V be a vector space over a field K. Then the following properties hold:

(a) The 0 vector from (V2) is unique. To avoid confusion (e.g., with $0 \in K$), we will sometimes denote it by 0_V.

(b) The element v' (corresponding to v) from (V3) is unique. We will denote it by $-v$. We will also write $v - w := v + (-w)$ for all $v, w \in V$.

(c) For all $v \in V$, we have $0 \cdot v = 0_V$ (i.e., $0_K \cdot v = 0_V$).

(d) For all $\alpha \in K$, we have $\alpha \cdot 0_V = 0_V$.

(e) $(-1) \cdot v = -v$ for all $v \in V$.

Proof

(a) Assume there are two such zero elements, 0_1 and 0_2. Then, by applying (V2) to both, we get: $0_1 = 0_1 + 0_2 = 0_2 + 0_1 = 0_2$. Thus, $0_1 = 0_2$.

(b) Assume there are two inverses v_1' and v_2' for an element v. Then we can write:

$$v_1' = v_1' + 0_V = v_1' + (v + v_2') = (v_1' + v) + v_2' = 0_V + v_2' = v_2'$$

Thus, the inverse is unique.

(c) Consider $0 \cdot v = (0+0) \cdot v = 0 \cdot v + 0 \cdot v$. By adding $-(0 \cdot v)$ to both sides, we yield $0_V = 0 \cdot v$.

(d) Similarly, $\alpha \cdot 0_V = \alpha \cdot (0_V + 0_V) = \alpha \cdot 0_V + \alpha \cdot 0_V$. Adding $-(\alpha \cdot 0_V)$ yields $0_V = \alpha \cdot 0_V$.

(e) Observe that $v + (-1) \cdot v = 1 \cdot v + (-1) \cdot v = (1 - 1) \cdot v = 0 \cdot v = 0_V$. Since the inverse is unique by part (b), it must be that $(-1) \cdot v = -v$. **Q.E.D.**

Examples of Vector Spaces

The Space K^n

Example: The set of n-tuples K^n becomes a vector space over K if we endow it with the following operations. Let $v = (a_1, \dots, a_n)$ and $w = (b_1, \dots, b_n) \in K^n$. We define addition in K^n component-wise:

$$v + w = (a_1, \dots, a_n) + (b_1, \dots, b_n) := (a_1 + b_1, \dots, a_n + b_n)$$

For $\alpha \in K$, we define scalar multiplication as:

$$\alpha \cdot v = \alpha \cdot (a_1, \dots, a_n) := (\alpha \cdot a_1, \dots, \alpha \cdot a_n)$$

The zero vector is explicitly defined as $0 := (0, \dots, 0) \in K^n$. (The proof that this constitutes a vector space relies directly on the fact that K itself satisfies the axioms of a field).

Spaces of Matrices

Example: Let $\mathcal{M}_{m\times n}(K)$ be the set of all $m \times n$ matrices with entries in K. We define addition of matrices component-wise: if $A = (A_{i,j})$ and $B = (B_{i,j})$, then $A + B = (A_{i,j} + B_{i,j})$. Similarly, scalar multiplication is defined component-wise: $\alpha \cdot A = (\alpha \cdot A_{i,j})$. With these operations, $\mathcal{M}_{m\times n}(K)$ forms a vector space over K. The zero vector is the $m \times n$ matrix consisting entirely of zeros.

Function Spaces K^S

Example: Let S be an arbitrary set. We define the function space:

$$K^S := \{f : S \to K \mid f \text{ is a function}\}$$

We define addition $+$ and scalar multiplication $\cdot$ on K^S as follows: Let $f, g \in K^S$ and $a \in K$. We define $(f + g) : S \to K$ point-wise by $x \mapsto f(x) + g(x)$. We define $(a \cdot f) : S \to K$ point-wise by $x \mapsto a \cdot f(x)$.

Polynomials $K[x]$

Example: A polynomial over K is an expression of the form:

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

where $a_i \in K$. Two polynomials are considered equal if and only if all their corresponding coefficients are equal. Addition and scalar multiplication are performed exactly as expected by manipulating coefficients.

Important remark: For finite fields, different polynomials may yield the exact same function! For example, if $K = \mathbb{F}_2$, the polynomials $f(x) = x$ and $g(x) = x^2$ produce the same output values for all inputs in $\mathbb{F}_2$. However, **as polynomials**, $f(x)$ and $g(x)$ are strictly different mathematical objects!

Degree of a polynomial: Let $f(x) = a_0 + a_1x + \cdots + a_nx^n$ be a polynomial with $a_n \neq 0$. We say that the *degree* of f is n, and we write $\deg(f) = n \in \mathbb{Z}_{\geq 0}$. Alternatively, if $f(x) \neq 0$:

$$\deg(f) := \max\{k \in \mathbb{Z}_{\geq 0} \mid a_k \neq 0\}$$

If $a_0 = a_1 = \cdots = a_n = 0$, meaning f is the zero polynomial, we set $\deg(f) = -\infty$. (And consequently, $\deg(f) = -\infty \iff f = 0$).

Remark: Every polynomial has a well-defined degree.

Linear Subspaces

Definition 7.3 (*Linear Subspace*): Let V be a vector space over K. A subset $U \subseteq V$ is called a *"linear subspace"* if it satisfies:

(a) $0_V \in U$.

(b) For all $u_1, u_2 \in U$, we have $u_1 + u_2 \in U$ (closed under addition).

(c) For all $\alpha \in K$ and for all $u \in U$, we have $\alpha \cdot u \in U$ (closed under scalar multiplication).

Example: Consider the set $U_b \subseteq K^3$ defined by $U_b = \{(x_1, x_2, x_3) \in K^3 \mid x_1 + x_2 + x_3 = b\}$.
Claim: U_b is a linear subspace of K^3 if and only if $b = 0$.

Proof

Suppose U_b is a valid subspace. Let (x_1, x_2, x_3) and $(y_1, y_2, y_3) \in U_b$. Then by definition, $x_1 + x_2 + x_3 = b$ and $y_1 + y_2 + y_3 = b$. Since U_b is a subspace, it must be closed under addition, meaning $(x_1 + y_1, x_2 + y_2, x_3 + y_3) \in U_b$. This implies:

$$(x_1 + y_1) + (x_2 + y_2) + (x_3 + y_3) = b$$

But rearranging the terms, we also know:

$$(x_1 + x_2 + x_3) + (y_1 + y_2 + y_3) = b + b = 2b$$

Therefore, $2b = b$, which implies $b = 0$. (Conversely, if $b = 0$, it is trivial to check that all three subspace conditions hold). Q.E.D.

Example (*Subspaces of Polynomials*): Fix $d \in \mathbb{Z}_{\geq 0} \cup \{-\infty\}$. We define the set:

$$K[x]_d := \{f \in K[x] \mid \deg(f) \leq d\}$$

(where we agree that $-\infty \leq k$ for all $k \in \mathbb{Z}_{\geq 0}$).

Claim: $K[x]_d \subseteq K[x]$ is a linear subspace of $K[x]$.

Later, we will see that $K[x]_d$ is a subspace of dimension $d + 1$.

Span and Linear Combinations (Chapter 8)

Preparation

Let V be a vector space over a field K. Let $S \subseteq V$ be a subset (which is not necessarily a linear subspace). We aim to answer two fundamental and closely related questions:

(a) What is the *smallest* linear subspace of V that contains S?

(b) What structure do we obtain if we consider the subset of V formed by taking all possible linear combinations of elements from S?

To formalize the concept of a "smallest" subspace, we must first prove that the intersection of any collection of subspaces remains a subspace.

Lemma 8.1: Let V be a vector space over K. Let $\{W_i\}_{i\in I}$ be a collection of linear subspaces $W_i \subseteq V$ of V, indexed by an arbitrary set I. Then their intersection

$$W := \bigcap_{i\in I} W_i$$

is a linear subspace of V.

Proof

First, since every W_i is a subspace, the zero vector 0_V is contained in every W_i. Therefore, $0_V \in \bigcap_{i\in I} W_i = W$.

Next, let $a_1, a_2 \in K$ and $w_1, w_2 \in W$. We must show that $a_1w_1 + a_2w_2 \in W$. Choose an arbitrary index $j \in I$. Since $w_1, w_2 \in \bigcap_{i\in I} W_i$, we trivially have $w_1, w_2 \in W_j$. Because $W_j \subseteq V$ is itself a linear subspace, it is closed under addition and scalar multiplication, which implies:

$$a_1w_1 + a_2w_2 \in W_j$$

Since we have proved that $a_1w_1 + a_2w_2 \in W_j$ holds for every $j \in I$, we conclude that:

$$a_1w_1 + a_2w_2 \in \bigcap_{i\in I} W_i = W$$

This proves that W is a linear subspace. **Q.E.D.**

Definition 8.2 (*Span*): Let $S \subseteq V$ be a subset. We define the *span* of S as the intersection of all linear subspaces of V that contain S:

$$\mathrm{Sp}(S) := \bigcap_{\substack{S\subseteq W\subseteq V \\ W \text{ is a subspace}}} W$$

By convention, the span of the empty set is defined as the zero subspace: $\mathrm{Sp}(\emptyset) := \{0_V\}$.

Definition 8.3 (*Linear Combination*): Let $\alpha_1, \ldots, \alpha_n \in K$ be scalars and $w_1, \ldots, w_n \in V$ be vectors. A *linear combination* of the vectors $w_1, \ldots, w_n$ with coefficients $\alpha_1, \ldots, \alpha_n$ is the vector:

$$\alpha_1w_1 + \cdots + \alpha_nw_n \in V \quad \left(\text{or equivalently, } \sum_{i=1}^{n} \alpha_i w_i\right)$$

Important remark: In this course, we consider **only finitely many elements** in the sums of linear combinations. Infinite sums require topological notions of limits and convergence, which fall outside the scope of general vector spaces.

Lemma 8.4: Let $\emptyset \neq S \subseteq V$. Then $\mathrm{Sp}(S)$ is exactly the set of all possible linear combinations of vectors from S:

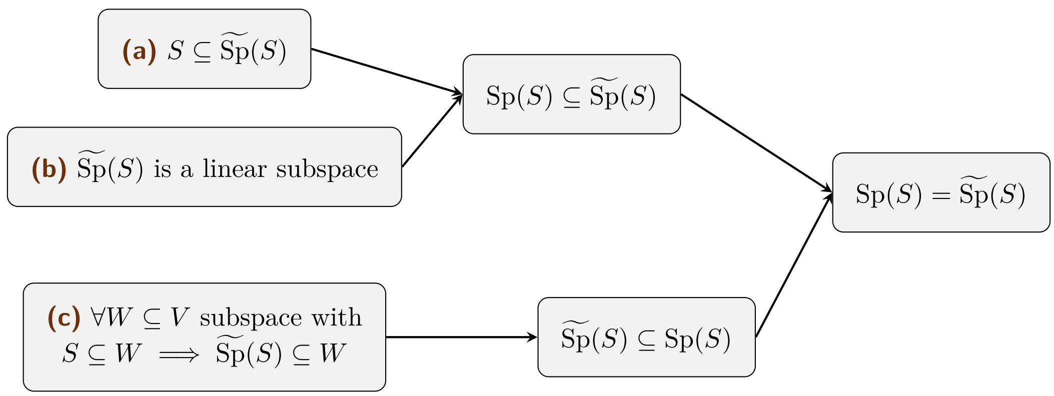
$$\begin{aligned}\mathrm{Sp}(S) &= \{a_1 w_1 + \cdots + a_n w_n \mid n \in \mathbb{N}, a_i \in K, w_i \in S\} \\ &= \text{subset of } V \text{ obtained by taking all possible linear combinations of vectors from } S.\end{aligned}$$

Remark: The set S itself can be finite or infinite. Regardless of the size of S, each individual linear combination uses only a finite number of elements from S.

Proof

Let us denote the set of all linear combinations by $\widetilde{\mathrm{Sp}}(S) = \{a_1 w_1 + \cdots + a_n w_n \mid n \in \mathbb{N}, a_i \in K, w_i \in S\}$.

Before diving into the algebra, let us visualize the logical strategy of the proof. We will establish three facts, which together force $\mathrm{Sp}(S) = \widetilde{\mathrm{Sp}}(S)$:



Note that (a) and (b) imply that $\mathrm{Sp}(S) \subseteq \widetilde{\mathrm{Sp}}(S)$ because $\mathrm{Sp}(S)$ is defined as the intersection of **all** subspaces containing S. Furthermore, (c) implies that $\widetilde{\mathrm{Sp}}(S) \subseteq \mathrm{Sp}(S)$ because $\mathrm{Sp}(S)$ is itself a valid subspace W containing S.

(a) Let $v \in S$. Then we can write $v = 1 \cdot v \in \widetilde{\mathrm{Sp}}(S)$. Thus $S \subseteq \widetilde{\mathrm{Sp}}(S)$.

(b) First, $0_V \in \widetilde{\mathrm{Sp}}(S)$ because for any $w \in S$, $0 \cdot w = 0_V$. Second, let $u, v \in \widetilde{\mathrm{Sp}}(S)$. Then $u = a_1 w_1 + \cdots + a_n w_n$ and $v = b_1 v_1 + \cdots + b_m v_m$. Their sum:

$$u + v = a_1 w_1 + \cdots + a_n w_n + b_1 v_1 + \cdots + b_m v_m \in \widetilde{\mathrm{Sp}}(S)$$

Third, let $c \in K$ and $u \in \widetilde{\mathrm{Sp}}(S)$. Then:

$$c \cdot u = c(a_1 w_1 + \cdots + a_n w_n) = (ca_1) w_1 + \cdots + (ca_n) w_n \in \widetilde{\mathrm{Sp}}(S)$$

Thus $\widetilde{\mathrm{Sp}}(S)$ is closed under addition and scalar multiplication, making it a subspace.

(c) Let $W \subseteq V$ be an arbitrary subspace such that $S \subseteq W$. Let $u \in \widetilde{\mathrm{Sp}}(S)$. Then $u = a_1 w_1 + \cdots + a_n w_n$ with elements $w_i \in S$. Since $S \subseteq W$, we know $w_i \in W$. Since W is a subspace, it is closed under scalar multiplication and finite addition, which guarantees $a_1 w_1 + \cdots + a_n w_n \in W$. So $u \in W$. Thus $\widetilde{\mathrm{Sp}}(S) \subseteq W$. **Q.E.D.**

Checking for Span: The Bridge to Computations

Up to this point, our definition of the span has been highly abstract. We know that if $S = \{w_1, \ldots, w_n\}$ is a finite set, we can drop the set brackets and write its span as the set of all possible linear combinations:

$$\mathrm{Sp}(w_1, \ldots, w_n) = \{x_1 w_1 + \cdots + x_n w_n \mid x_1, \ldots, x_n \in K\}$$

However, as a student of linear algebra, you should immediately ask a practical, computational question: **If I hand you a specific, concrete target vector w, how can we mathematically determine whether or not $w \in \mathrm{Sp}(w_1, \ldots, w_n)$?**

To build our intuition, let us translate this abstract geometric question into algebra. Suppose our vector space is $V = K^m$. This means our vectors are just column vectors with m entries.

If w is indeed in the span of $w_1, \ldots, w_n$, there must exist some scalar "weights" $x_1, \ldots, x_n \in K$ that allow us to build w. In other words, we need to solve the vector equation:

$$x_1 w_1 + x_2 w_2 + \cdots + x_n w_n = w \tag{8.1}$$

Let us write these vectors out explicitly by their components:

$$w_1 = \begin{pmatrix} w_{1,1} \\ w_{2,1} \\ \vdots \\ w_{m,1} \end{pmatrix}, \quad w_2 = \begin{pmatrix} w_{1,2} \\ w_{2,2} \\ \vdots \\ w_{m,2} \end{pmatrix}, \quad \ldots \quad, \quad w_n = \begin{pmatrix} w_{1,n} \\ w_{2,n} \\ \vdots \\ w_{m,n} \end{pmatrix}, \quad \text{and target } w = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{pmatrix}$$

If we substitute these columns into Equation (**??**) and perform the scalar multiplication and vector addition component by component, we get a massive single column vector on the left side:

$$\begin{pmatrix} x_1 w_{1,1} + x_2 w_{1,2} + \cdots + x_n w_{1,n} \\ x_1 w_{2,1} + x_2 w_{2,2} + \cdots + x_n w_{2,n} \\ \vdots \\ x_1 w_{m,1} + x_2 w_{m,2} + \cdots + x_n w_{m,n} \end{pmatrix} = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{pmatrix}$$

Look closely at the left-hand side. This is exactly the definition of matrix-vector multiplication! We can seamlessly separate the known vector components from the unknown scalar weights x_i by factoring this out into a coefficient matrix A multiplied by a variable vector x:

$$\underbrace{\begin{pmatrix} w_{1,1} & w_{1,2} & \ldots & w_{1,n} \\ w_{2,1} & w_{2,2} & \ldots & w_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{m,1} & w_{m,2} & \ldots & w_{m,n} \end{pmatrix}}_{\text{Matrix } A \text{ (whose columns are } w_i)} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}}_{\text{Vector } x} = \underbrace{\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{pmatrix}}_{\text{Target } w}$$

This reveals one of the most profound and useful insights in all of linear algebra: **A matrix-vector product $A \cdot x$ is simply a linear combination of the columns of A, weighted by the entries of x.**

Theorem 8.5: Let $w, w_1, \ldots, w_n \in K^m$. Let $A \in \mathcal{M}_{m \times n}(K)$ be the matrix whose columns are exactly the vectors $w_1, \ldots, w_n$. Then the target vector w is in the span of $w_1, \ldots, w_n$ if and only if the system of linear equations $A \cdot x = w$ has at least one solution $x = (x_1, \ldots, x_n)^T \in K^n$.

$$w \in \mathrm{Sp}(w_1, \ldots, w_n) \iff \text{The system } A \cdot x = w \text{ is consistent.}$$

Remark (*How to check this in practice*): Because of this beautiful equivalence, whenever you are asked "Is w in the span of these vectors?", you do not need to guess the coefficients. You simply construct the augmented matrix $(A \mid w)$ and perform Gaussian elimination. If the resulting system yields a contradiction, the vector is **not** in the span. If the system is solvable, the solutions $x_1, \ldots, x_n$ are exactly the coefficients you need to build w.

Some spaces and how to generate them

To conclude, let us look at some standard vector spaces and identify natural sets of "building block" vectors that span them entirely.

(a) **The Coordinate Space K^n:** Let $e_1 = (1, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$, $\dots$, $e_n = (0, \dots, 0, 1)$ be the standard basis vectors. Every vector $(x_1, \dots, x_n)$ can trivially be written as $x_1e_1 + \cdots + x_ne_n$. Then $K^n = \mathrm{Sp}(e_1, \dots, e_n)$. (**check!**)

(b) **Bounded Polynomials $K[x]_d$:** Let $W = K[x]_d$ be the space of polynomials over K of degree $\leq d$. We can generate this space using the standard monomials:

$$W = \mathrm{Sp}(1, x, x^2, \dots, x^d)$$

(c) **All Polynomials $K[x]$:** Let $V = K[x]$ be the space of *all* polynomials over K. Generating this space requires an infinite set, as there is no maximum degree limit:

$$V = \mathrm{Sp}(\{1, x, x^2, \dots, x^n, \dots\})$$

(d) **Spaces of Matrices $\mathcal{M}_{m\times n}(K)$:** Consider the vector space of all $m \times n$ matrices over K. We can generate this entire space using the *standard matrix basis*. Let $E_{i,j}$ be the $m \times n$ matrix that has a 1 in the i-th row and j-th column, and 0 everywhere else. There are exactly $m \cdot n$ such matrices.

For any arbitrary matrix $A = (A_{i,j}) \in \mathcal{M}_{m\times n}(K)$, we can pull it apart into a linear combination where the scalar weights are simply the matrix entries themselves:

$$A = \sum_{i=1}^{m}\sum_{j=1}^{n} A_{i,j} \cdot E_{i,j}$$

For instance, in $\mathcal{M}_{2\times 2}(K)$, any matrix can be constructed like this:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Thus, a matrix can be viewed as a "vector" built out of simpler matrices! We write:

$$\mathcal{M}_{m\times n}(K) = \mathrm{Sp}(\{E_{i,j} \mid 1 \leq i \leq m, 1 \leq j \leq n\})$$

(**check!!**)

Linear Independence (Chapter 9)

Linear Independence of a Finite List

Let V be a vector space over a field K. In the previous chapter, we studied how a set of vectors can *span* a subspace. We now ask whether a spanning set contains "redundant" vectors. This leads us to the core algebraic concept of linear independence.

Definition 9.1 (*Linear Independence*): Let $v_1, \ldots, v_n$ be a finite list of vectors in V. We say that the list $v_1, \ldots, v_n$ is **linearly independent** if the only linear combination that equals the zero vector 0_V is the trivial combination (where all coefficients are zero).

In formal notation, the vectors are linearly independent if, for all $a_1, \ldots, a_n \in K$:

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n = 0_V \implies a_1 = 0, a_2 = 0, \ldots, a_n = 0$$

Remark (*The Empty Set*): By mathematical convention, the empty list of vectors $\emptyset$ is defined to be linearly independent. But *why* does this make sense?

Recall from Chapter 8 that we defined the span of the empty set as the zero subspace: $\mathrm{Sp}(\emptyset) = \{0_V\}$. If the empty set were to be linearly *dependent*, there would need to exist a linear combination of its elements that non-trivially yields 0_V. However, since $\emptyset$ contains literally no vectors, we cannot even form a linear combination with a non-zero coefficient! Because the condition for dependence is impossible to satisfy, the empty set is vacuously, and logically, independent.

Before looking at examples, we can immediately deduce several structural facts directly from the definition.

Lemma 9.2 (*Basic Properties of Linear Independence*): Let V be a vector space. The following properties hold for any finite list of vectors:

(a) The zero vector 0_V can always be written as a trivial linear combination of any finite list of vectors: $0 \cdot v_1 + \cdots + 0 \cdot v_n = 0_V$.

(b) The order of the vectors in a list does not affect their linear independence.

(c) If a list $v_1, \ldots, v_n$ contains two identical vectors (i.e., $v_k = v_l$ for some $k \neq l$), then the list is NOT linearly independent.

(d) If the zero vector 0_V is an element of the list, then the list is NOT linearly independent.

Proof

Properties (1) and (2) follow trivially from the vector space axioms (specifically the commutativity of vector addition and the property that $0 \cdot v = 0_V$).

Proof of (3): Assume $v_k = v_l$ for some $1 \leq k < l \leq n$. We can construct a specific linear combination where the coefficient for v_k is 1, the coefficient for v_l is -1, and all other coefficients are 0:

$$0 \cdot v_1 + \cdots + 1 \cdot v_k + \cdots + (-1) \cdot v_l + \cdots + 0 \cdot v_n = v_k - v_l = 0_V$$

Because we found a linear combination equal to 0_V where not all coefficients are zero (specifically, $a_k = 1 \neq 0$), the list fails the definition and is therefore not linearly independent.

Proof of (4): Suppose $v_1 = 0_V$. We can assign a non-zero coefficient (e.g., 1) to v_1 and zero to all other vectors:

$$1 \cdot 0_V + 0 \cdot v_2 + \cdots + 0 \cdot v_n = 0_V + 0_V + \cdots + 0_V = 0_V$$

Again, since $a_1 = 1 \neq 0$, we have a non-trivial combination yielding the zero vector. The list is not linearly independent. **Q.E.D.**

Examples of Linear Independence

Let us apply this definition to various vector spaces.

Example (*In $\mathbb{R}^2$*): Consider the vectors $v_1 = \binom{1}{0}$ and $v_2 = \binom{0}{1}$ in $\mathbb{R}^2$.
Assume $a_1 v_1 + a_2 v_2 = 0_{\mathbb{R}^2}$. This means:

$$a_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Thus, $a_1 = 0$ and $a_2 = 0$. The vectors $\{v_1, v_2\}$ are linearly independent.

Example (*Linearly dependent vectors in $\mathbb{R}^2$*): Consider $v_1 = \binom{1}{1}$ and $v_2 = \binom{2}{2}$.
We can easily spot that $2v_1 - v_2 = 0_{\mathbb{R}^2}$. Because we have coefficients $a_1 = 2$ and $a_2 = -1$ (which are non-zero) that yield the zero vector, these vectors are NOT linearly independent.

Example (*The Standard Basis in K^n*): Consider the standard basis vectors in K^n:

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \ldots \quad , \quad e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

Setting $a_1 e_1 + a_2 e_2 + \cdots + a_n e_n = 0_{K^n}$ yields the column vector $(a_1, a_2, \ldots, a_n)^T = (0, 0, \ldots, 0)^T$. This immediately forces $a_i = 0$ for all i. Thus, the set $\{e_1, e_2, \ldots, e_n\}$ is always linearly independent.

Example (*Single Vectors*): Let V be a vector space and $v \in V$. The single-element set $\{v\}$ is linearly independent if and only if $v \neq 0_V$. (If $v = 0_V$, we know from Lemma 9.1 that it is not independent. If $v \neq 0_V$, then $a \cdot v = 0_V \implies a = 0$).

Arbitrary Families and Linear Dependence

As we advance in linear algebra, we frequently need to deal with infinite collections of vectors (such as the set of all polynomials). We must rigorously extend our definition to handle this.

Definition 9.3 (*Linear Independence of an Arbitrary Family*): Let $\mathcal{F} = \{v_i\}_{i \in I}$ be a *family of vectors* in V, indexed by an arbitrary set I (which may be infinite). We say that the family $\mathcal{F}$ is linearly independent if **every finite subset** of $\mathcal{F}$ is linearly independent.

Example (*Families of Polynomials*): In the space of bounded polynomials $K[x]_d$, the set of standard monomials $\{1, x, x^2, \ldots, x^d\}$ is linearly independent. This remains true even for the **infinite family** of all monomials $\mathcal{F} = \{1, x, x^2, \ldots\} \subseteq K[x]$. Any finite subset of this family will just be a finite collection of distinct monomials, which can only sum to the zero polynomial if all their coefficients are strictly zero.

We now formally define the opposite of linear independence, specifically focusing on how it applies to an arbitrary family.

Definition 9.4 (*Linear Dependence*): A family $\mathcal{F} = \{v_i\}_{i \in I}$ of vectors in V is called **linearly dependent** if it is NOT linearly independent.

In other words, this means there exists some finite integer $n \in \mathbb{N}$, a selection of n distinct indices $i_1, \ldots, i_n \in I$, and a set of scalars $a_1, \ldots, a_n \in K$ **where not all a_j are zero**, such that:

$$a_1 v_{i_1} + a_2 v_{i_2} + \cdots + a_n v_{i_n} = 0_V$$

Back to Linear Independence: Redundancy

This definition leads to an incredibly useful geometric intuition: a set of vectors is linearly dependent if and only if at least one vector is "redundant" — meaning it can be built entirely out of the other vectors in the set.

Algebraic Definition
$\{v_1, \ldots, v_n\}$ is
Linearly Dependent
$(\sum a_i v_i = 0_V, \; a_i \neq 0)$

$\xleftrightarrow{\textbf{Proposition 9.1}}$

Geometric Redundancy
At least one vector v_k
is a linear combination
of the others.
$(v_k \in \mathrm{Sp}(v_1, \ldots, \widehat{v_k}, \ldots, v_n))$

Proposition 9.5 (*Equivalence of Dependence and Redundancy*): A list of vectors $v_1, \ldots, v_n$ is linearly dependent if and only if at least one vector in the list can be written as a linear combination of the other vectors.

Proof

We must prove both directions.

Direction 1 ($\Longrightarrow$): Assume the list $v_1, \ldots, v_n$ is linearly dependent. By definition, there exist scalars $a_1, \ldots, a_n \in K$, not all zero, such that:

$$a_1 v_1 + a_2 v_2 + \cdots + a_n v_n = 0_V$$

Since not all coefficients are zero, we can choose an index k such that $a_k \neq 0$. We can isolate $a_k v_k$ on one side of the equation by subtracting all other terms:

$$a_k v_k = -\sum_{\substack{i=1 \\ i \neq k}}^{n} a_i v_i$$

Because $a_k \neq 0$, its multiplicative inverse a_k^{-1} exists in the field K. Multiplying both sides by a_k^{-1} yields:

$$v_k = \sum_{\substack{i=1 \\ i \neq k}}^{n} \left(-\frac{a_i}{a_k} \right) v_i$$

Thus, v_k is a linear combination of the other vectors.

Direction 2 ($\Longleftarrow$): Assume that at least one vector, say v_k, can be written as a linear combination of the others. This means there exist scalars b_i such that:

$$v_k = \sum_{\substack{i=1 \\ i \neq k}}^{n} b_i v_i$$

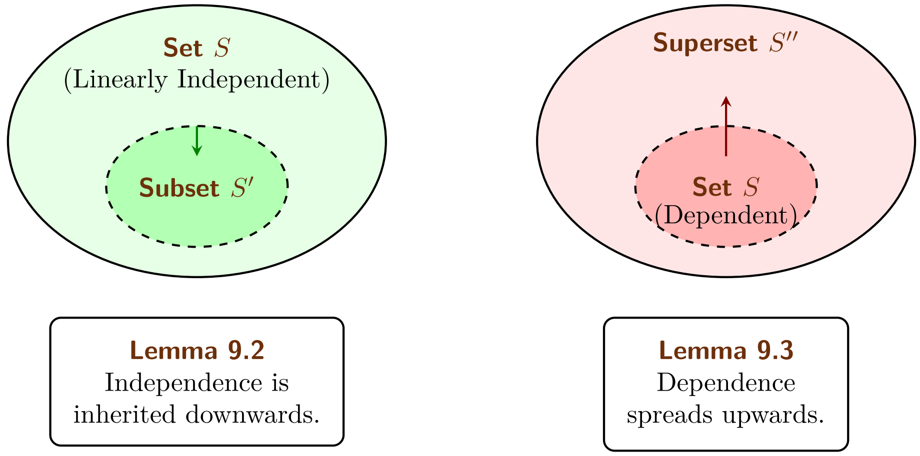
We can subtract v_k from both sides to obtain:

$$0_V = -1 \cdot v_k + \sum_{\substack{i=1 \\ i \neq k}}^{n} b_i v_i$$

This equation is a linear combination of the full list $v_1, \ldots, v_n$ that equals 0_V. Furthermore, the coefficient of v_k is -1, which is non-zero. Since we have found a non-trivial linear combination equaling 0_V, the list is linearly dependent by definition. **Q.E.D.**

Inheritance in Subsets and Supersets

How does linear independence behave when we add or remove vectors from a set? There is a strict, logical hierarchy to how these properties are inherited.



Lemma 9.6 (*Subsets of Independent Sets*): Let $S \subseteq V$ be a linearly independent set. If $S' \subseteq S$ is a subset, then S' is also linearly independent.

Proof

We prove this by contradiction. Suppose S' is NOT linearly independent. Then S' must be linearly dependent. By definition, there exist vectors $v_1, \ldots, v_k \in S'$ and scalars $a_1, \ldots, a_k \in K$, not all zero, such that:

$$a_1 v_1 + \cdots + a_k v_k = 0_V$$

However, since $S' \subseteq S$, every vector v_i is also an element of the larger set S. Therefore, this exact same equation represents a non-trivial linear combination of elements in S that equals 0_V. This would mean S is linearly dependent, which directly contradicts our initial assumption that S is linearly independent. Thus, S' must be linearly independent. Q.E.D.

Lemma 9.7 (*Supersets of Dependent Sets*): Let $S \subseteq V$ be a linearly dependent set. If S'' is a superset containing S (i.e., $S \subseteq S'' \subseteq V$), then S'' is also linearly dependent.

Proof

Because S is linearly dependent, there exist vectors $v_1, \ldots, v_k \in S$ and scalars $a_1, \ldots, a_k \in K$, not all zero, such that:

$$a_1 v_1 + \cdots + a_k v_k = 0_V$$

Since $S \subseteq S''$, the vectors $v_1, \ldots, v_k$ also belong to S''. Therefore, this exact same non-trivial linear combination exists within S'', immediately satisfying the definition of linear dependence for S''. Q.E.D.

Bases of Vector Spaces (Part A) (Chapter 10)

Foundations of Bases and the Exchange Theorem (Section 10.a)

Definition and Unique Representation

We have previously explored two fundamental properties a set of vectors can possess: the ability to *span* the entire space, and the property of being *linearly independent* (having no redundancies). A basis is simply a set that achieves both simultaneously.

Definition 10.1.1 (*Basis*): Let V be a vector space over a field K. A subset $S \subseteq V$ is called a **basis** of V (plural: *"bases"*) if the following two conditions hold:

(a) S is linearly independent.

(b) S generates (or spans) V, meaning $\mathrm{Sp}(S) = V$.

Why is a basis so mathematically desirable? Because it provides a perfect "coordinate system" for our abstract vector space.

Proposition 10.1.2 (*Unique Representation*): A subset $S \subseteq V$ is a basis of V if and only if every vector $v \in V$ can be written in a **unique way** as a linear combination of vectors in S.

To prove this rigorously—especially since S might be an infinite set, like the set of all polynomials—we must first formalize what we mean by "a unique way as a linear combination." In a vector space, we can only ever add a *finite* number of vectors together.

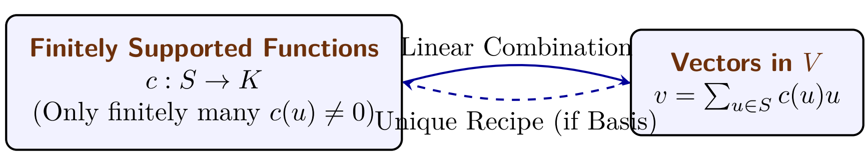
We formalize this using **finitely supported functions**. Let $c : S \to K$ be a function that assigns a scalar coefficient to every vector in S. We require that c is non-zero on only finitely many elements. In other words, the "support" of the function:

$$\mathrm{supp}(c) := \{u \in S \mid c(u) \neq 0\}$$

must be a finite set. We can then define our linear combination as the sum:

$$\sum_{u \in S} c(u) \cdot u$$

Because $\mathrm{supp}(c)$ is finite, the number of non-zero terms in this sum is strictly finite, making it a perfectly valid vector operation. (Furthermore, because vector addition is commutative, the order in which we sum these elements does not matter).



To say that every $v \in V$ can be written in a unique way means: if we have two finitely supported functions $c, c' : S \to K$ such that:

$$\sum_{u \in S} c(u) \cdot u = \sum_{u \in S} c'(u) \cdot u$$

then it must be true that $c(u) = c'(u)$ for all $u \in S$.

Proof of Proposition 10.1.1

We must prove both directions.

Direction 1 ($\implies$): Assume S is a basis. Because S spans V, every $v \in V$ can be written as a linear combination of elements in S. We must show this combination is unique. Suppose there are two combinations yielding the same vector: $\sum_{u\in S} c(u)u = \sum_{u\in S} c'(u)u$. Subtracting one from the other yields:

$$\sum_{u\in S} \big(c(u) - c'(u)\big)u = 0_V$$

Because S is a basis, it is linearly independent. The only way a linear combination of independent vectors can equal 0_V is if all coefficients are zero. Thus, $c(u) - c'(u) = 0 \implies c(u) = c'(u)$ for all $u \in S$. The representation is unique.

Direction 2 ($\Longleftarrow$): Assume every $v \in V$ has a unique representation. Since every vector can be represented, S clearly spans V. We must show S is independent. Consider the linear combination $\sum_{u\in S} c(u)u = 0_V$. We already know one valid representation for the zero vector: assigning 0 to every coefficient. Since the hypothesis guarantees representations are *unique*, the all-zero assignment must be the *only* representation. Thus, S is linearly independent. **Q.E.D.**

Standard Examples of Bases

Let us identify the standard bases for the families of vector spaces we encounter most frequently.

Example (*The Coordinate Space*): In $V = K^n$, the standard basis is $S = \{e_1, e_2, \dots, e_n\}$, where e_i is the column vector with a 1 in the i-th position and 0 elsewhere.

Example (*Bounded Polynomials*): In $V = K[x]_d$, the space of polynomials of degree $\leq d$, the standard basis is $S = \{1, x, x^2, \dots, x^d\}$.

Example (*All Polynomials*): In $V = K[x]$, the space of all polynomials, the standard basis is the infinite set $S = \{1, x, x^2, \dots, x^n, \dots\}$. (Notice how beautifully our "finitely supported function" handles this: any specific polynomial has a finite degree, so it requires only a finite number of non-zero coefficients from this infinite basis).

Example (*Matrices*): In $V = \mathcal{M}_{m\times n}(K)$, the standard basis is $S = \{E_{i,j} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$, where $E_{i,j}$ is the matrix with a 1 in the (i,j) entry and 0 everywhere else.

The Size of a Basis and Exchange Lemmas

Looking at K^n, the standard basis has exactly n elements. Looking at $K[x]_d$, the standard basis has $d+1$ elements. This raises a monumental question: **Are all bases of a given vector space V exactly the same size?**

Yes, they are. Our next major goal is to prove this fact. To do so, we must understand how vectors can be swapped in and out of spanning sets without breaking the span. We develop this through a sequence of four powerful lemmas.

Lemma 10.1.3 (*Lemma A: Simple Exchange*): Let $v_1, \dots, v_n$ be a basis of V. Let $u = a_1v_1 + \cdots + a_nv_n \in V$ be a vector such that $a_1 \neq 0$. Then the modified list $u, v_2, \dots, v_n$ is also a basis of V.

Proof

We must show the new list is independent and spans V. **Independence:** Assume $b_1u + b_2v_2 + \cdots + b_nv_n = 0_V$. Substitute the definition of u:

$$b_1(a_1v_1 + a_2v_2 + \cdots + a_nv_n) + b_2v_2 + \cdots + b_nv_n = 0_V$$

Regrouping the terms by the basis vectors v_i:

$$(b_1a_1)v_1 + (b_1a_2 + b_2)v_2 + \cdots + (b_1a_n + b_n)v_n = 0_V$$

Since $v_1, \dots, v_n$ are linearly independent, all coefficients must be zero. In particular, $b_1a_1 = 0$. Because we assumed $a_1 \neq 0$, we are forced to conclude $b_1 = 0$. Once $b_1 = 0$, the equation reduces to $b_2v_2 + \cdots + b_nv_n = 0_V$, which forces $b_2 = \cdots = b_n = 0$. Thus, the new list is independent.

Spanning: We can rearrange $u = a_1v_1 + a_2v_2 + \cdots + a_nv_n$ to solve for v_1:

$$v_1 = a_1^{-1}u - a_1^{-1}a_2v_2 - \cdots - a_1^{-1}a_nv_n$$

This shows $v_1 \in \mathrm{Sp}(u, v_2, \ldots, v_n)$. Since all the other basis vectors $v_2, \ldots, v_n$ are trivially in this span, all vectors of the original basis are contained within it. Thus $V = \mathrm{Sp}(v_1, \ldots, v_n) \subseteq \mathrm{Sp}(u, v_2, \ldots, v_n)$. **Q.E.D.**

Lemma 10.1.4 (*Lemma B: Targeted Redundancy*): Let $v_1, \ldots, v_n$ be a list of vectors. Suppose it is linearly dependent. Then there exists an index $1 \leq j \leq n$ such that $v_j \in \mathrm{Sp}(v_1, \ldots, v_{j-1})$. Moreover, if we drop v_j from the list, the span of the remaining vectors is exactly the same as the original span.

Proof

Because the list is dependent, there exist coefficients not all zero such that $\sum_{i=1}^{n} a_iv_i = 0_V$. Let j be the *largest* index such that $a_j \neq 0$. Then our equation is actually:

$$a_1v_1 + a_2v_2 + \cdots + a_jv_j = 0_V$$

Because $a_j \neq 0$, we can solve for v_j:

$$v_j = -\frac{a_1}{a_j}v_1 - \cdots - \frac{a_{j-1}}{a_j}v_{j-1}$$

This proves $v_j \in \mathrm{Sp}(v_1, \ldots, v_{j-1})$. Because v_j can be synthesized entirely from the vectors strictly preceding it, dropping v_j does not shrink the span. **Q.E.D.**

Lemma 10.1.5 (*Lemma C: Span Overcrowding*): If $\mathrm{Sp}(v_1, \ldots, v_n) = V$ and $u \in V$, then the list $u, v_1, \ldots, v_n$ is linearly dependent.

Proof

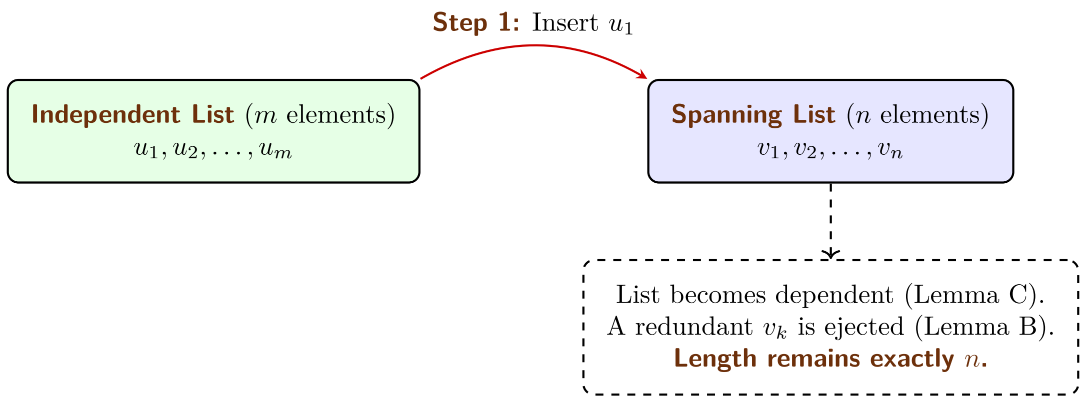
Because the v_i's span V, we can write $u = a_1v_1 + \cdots + a_nv_n$. Rearranging this gives:

$$1 \cdot u - a_1v_1 - \cdots - a_nv_n = 0_V$$

Since the coefficient of u is 1 (which is non-zero), this is a non-trivial linear combination that equals zero. The list is dependent. **Q.E.D.**

We now arrive at the crescendo of this chapter: The Steinitz Exchange Lemma. This theorem rigorously proves that an independent set can never outnumber a spanning set.

Theorem 10.1.6 (*Lemma D: Steinitz Exchange Theorem*): Let $u_1, \ldots, u_m$ be a linearly independent list in V. Let $v_1, \ldots, v_n$ be a list that spans V (i.e., $\mathrm{Sp}(v_1, \ldots, v_n) = V$). Then $m \leq n$.



Proof

We proceed by injecting the independent u vectors into the spanning list one by one, forcefully replacing v vectors to maintain a constant list length of n.

Step 1: Add u_1 to the front of the spanning list: $u_1, v_1, \ldots, v_n$. Because the v_i's span V, Lemma C dictates this new list is dependent. By Lemma B, there is a redundant vector in this list that can be dropped without altering the span. Could the redundant vector be u_1? No. Lemma B specifies the redundant vector is in the span of the vectors *preceding* it. u_1 is at the very front (its preceding span is $\emptyset$), and $u_1 \neq 0_V$ because it is part of an independent set. Thus, the redundant vector must be one of the v_i's, say v_k. We drop v_k. Our list is now $u_1, v_1, \ldots, \widehat{v_k}, \ldots, v_n$. It still spans V, and it has exactly n elements.

Step 2: Add u_2 to the front: $u_2, u_1, \ldots, \widehat{v_k}, \ldots, v_n$. By Lemma C, this is dependent. By Lemma B, a redundant vector exists. It cannot be u_2 (front of the line). It cannot be u_1, because u_1 cannot be in the span of u_2 (since the u's are independent). Therefore, the redundant vector *must* be another v. We drop it. Length remains n.

The Contradiction: Assume, for the sake of contradiction, that $m > n$. We can continue the above process exactly n times. After step n, we will have systematically ejected every single v_i from the list, leaving us with a list consisting purely of u-vectors: $u_n, u_{n-1}, \ldots, u_1$. Crucially, because we only ever used Lemma B to drop vectors, this list of u-vectors **still spans** V.

Because $m > n$, we still have at least one vector left in our independent set: u_{n+1}. Let us add it to the front of our current spanning list:

$$u_{n+1}, u_n, u_{n-1}, \ldots, u_1$$

Because the set $u_n, \ldots, u_1$ spans V, Lemma C states that this new combined list must be linearly dependent. However, this list is just a subset of our original $u_1, \ldots, u_m$, which was explicitly assumed to be linearly independent! This is a fatal mathematical contradiction.

Therefore, our assumption that $m > n$ must be false. We conclude that $m \leq n$. **Q.E.D.**

Bases of a Vector Space (Part B) (Chapter 11)

The Dimension Theorem (Section 11.a)

The Strong Exchange Lemma

Before we establish the main theorem of this chapter, we formalize a stronger version of the Steinitz Exchange Theorem (Lemma D from the previous section). We proved that an independent list cannot be larger than a spanning list. However, the exact mechanism of the proof actually guarantees something even more powerful: we can precisely substitute the independent vectors into the spanning set.

Lemma 11.1.1 (*Lemma D': Strong Steinitz Exchange*): Let V be a vector space over K, and suppose that $V = \mathrm{Sp}(v_1, \dots, v_n)$ for some list of vectors $v_1, \dots, v_n \in V$. Let $u_1, \dots, u_m \in V$ be a linearly independent list.

Then $m \leq n$, and moreover, one can delete exactly m vectors from the spanning list $v_1, \dots, v_n$ such that the remaining $n - m$ vectors, together with all the vectors $u_1, \dots, u_m$, still span V.

$$\mathrm{Sp}(u_1, \dots, u_m, v'_1, \dots, v'_{n-m}) = V$$

(where the v' vectors are the survivors of the original spanning list).

Proof

The fact that $m \leq n$ was already proven in Lemma D. To prove the "moreover" part, we simply apply the exact step-by-step substitution procedure described in the proof of Lemma D. For steps $j = 1, 2, \dots, m$, we sequentially add u_j to the front of the list. At each step, the list becomes dependent (Lemma C), and we forcefully eject a redundant v_k vector (Lemma B) to restore independence while perfectly preserving the span. After m steps, we have injected all m independent vectors and deleted exactly m of the original spanning vectors. The resulting mixed list still spans V.

Q.E.D.

The Existence and Uniqueness of Dimension

We are now ready to state and prove the foundational theorem of finite-dimensional vector spaces. It guarantees that bases always exist and that they provide an absolute, rigid measure of the "size" of a vector space.

Theorem 11.1.2 (*The Basis Theorem*): Let V be a finite-dimensional vector space over K. Then V has a basis with finitely many elements.

Moreover, **every** basis of V is finite, and any two bases of V have exactly the same number of elements.

Remark (*Infinite Dimensional Spaces*): This theorem is actually true even if V is not finite-dimensional (this is known as the "Hamel Basis Theorem" and requires Zorn's Lemma). However, in this course, we will only prove the theorem for finite-dimensional spaces. The infinite-dimensional case is rigorously covered in advanced algebra courses (e.g., *Grundstrukturen* in the 2nd semester).

Proof of the Basis Theorem

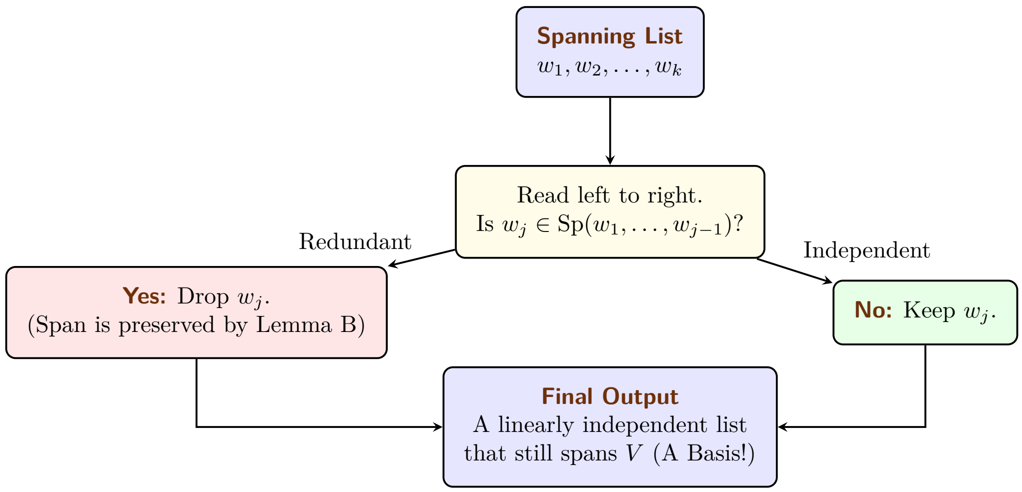
Because this theorem is so monumental, we will break its proof down into three distinct, manageable statements.

Statement 1: V has a finite basis.

Proof

Because we assumed V is finite-dimensional, there must exist some finite list of vectors that spans V. Let us call this spanning set $S = \{w_1, \dots, w_k\}$. If S is already linearly independent, it is a basis, and we are done.

If S is linearly dependent, we can extract a basis from it using the following systematic reduction algorithm (often called a "Redundancy Filter"):



Specifically: 1. If the zero vector 0_V is in the list, drop it. 2. Read the list from left to right. If a vector w_j is a linear combination of its strictly preceding vectors (i.e., $w_j \in \mathrm{Sp}(w_1, \dots, w_{j-1})$), drop it. By Lemma B, dropping this redundant vector does not change the span of the list. We repeat this until we have checked every vector. The remaining list still spans V. Furthermore, by Proposition 9.1, it must be linearly independent, because we systematically deleted every vector that could be written as a combination of the others. Thus, the remaining list is a finite basis.

Q.E.D.

Statement 2: Every basis of V is finite.

Proof

We prove this by contradiction. Suppose there exists a basis $\mathcal{A}$ of V that is *not* a finite set.

From Statement 1, we know that V definitely possesses at least one finite basis. Let us call this finite basis $S = \{v_1, \dots, v_n\}$. The size of this finite basis is exactly n.

Because $\mathcal{A}$ is infinite, it contains infinitely many linearly independent vectors. Therefore, we can arbitrarily select $n + 100$ distinct vectors from $\mathcal{A}$. Let us call them $u_1, \dots, u_{n+100}$. Because they belong to a basis, this chosen list of $n + 100$ vectors is strictly linearly independent.

We now have two lists:

- A spanning list S containing exactly n elements.
- An independent list containing $n + 100$ elements.

By the Steinitz Exchange Theorem (Lemma D), the number of elements in any independent list must be less than or equal to the number of elements in any spanning list. Therefore, we must have:

$$n + 100 \leq n$$

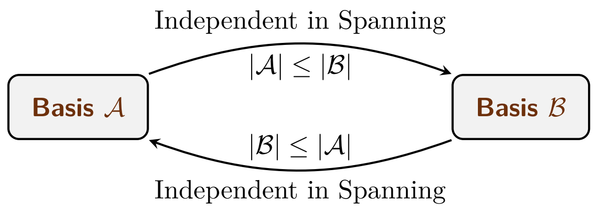
This implies $100 \leq 0$, which is a blatant mathematical contradiction! Thus, our initial assumption must be wrong. Every basis of V must be finite. Q.E.D.

Statement 3: Any two bases have the same cardinality.

Proof

Let $\mathcal{A}$ and $\mathcal{B}$ be two arbitrary bases of V. By Statement 2, we know that both $\mathcal{A}$ and $\mathcal{B}$ must be finite sets. Let $|\mathcal{A}|$ denote the number of elements in $\mathcal{A}$, and $|\mathcal{B}|$ denote the number of elements in $\mathcal{B}$.

Because they are bases, both sets are simultaneously independent and spanning. This allows us to apply Lemma D twice in alternating directions:



(a) Because $\mathcal{A}$ is linearly independent and $\mathcal{B}$ spans V, Lemma D dictates that $|\mathcal{A}| \leq |\mathcal{B}|$.

(b) Reversing their roles, because $\mathcal{B}$ is linearly independent and $\mathcal{A}$ spans V, Lemma D dictates that $|\mathcal{B}| \leq |\mathcal{A}|$.

Since $|\mathcal{A}| \leq |\mathcal{B}|$ and $|\mathcal{B}| \leq |\mathcal{A}|$, it must be that $|\mathcal{A}| = |\mathcal{B}|$. All bases contain exactly the same number of elements. **Q.E.D.**

The Concept of Dimension

Because the number of elements in a basis is a rigid, absolute invariant of the vector space, we can finally define the space's dimension.

Definition 11.1.3 (*Dimension*): Let V be a finite-dimensional vector space over K. We define the **dimension** of V to be the number of elements in any basis of V. We denote this integer by $\dim V = n \in \mathbb{Z}_{\geq 0}$.

If V is not finite-dimensional, we write $\dim V = \infty$.

Remark (*The Zero Space*): Consider the zero vector space $V = \{0_V\}$. Its only spanning set is $\{0_V\}$, which is linearly dependent. The only linearly independent subset that spans it is the empty set $\emptyset$. Therefore, the basis of the zero space is $\emptyset$. Because the empty set contains 0 elements, we have:

$$\dim\{0_V\} = 0$$

Examples of Dimensions

Now that we have formally defined dimension, we can readily determine the dimensions of our familiar vector spaces by simply counting the elements in their standard bases.

Example (*The Coordinate Space*): The standard basis for K^n is $\{e_1, \ldots, e_n\}$. Because this set contains exactly n elements, we have:

$$\dim K^n = n$$

Example (*Bounded Polynomials*): The standard basis for the space of polynomials of degree $\leq d$ is $\{1, x, x^2, \ldots, x^d\}$. Be careful when counting here! Because we start at $x^0 = 1$, this set contains $d+1$ elements. Thus:

$$\dim K[x]_d = d + 1$$

Example (*Matrices*): The standard basis for the matrix space $\mathcal{M}_{m\times n}(K)$ consists of the individual single-entry matrices $E_{i,j}$. Since there are m rows and n columns, there are exactly $m \cdot n$ such matrices. Thus:

$$\dim \mathcal{M}_{m\times n}(K) = m \cdot n$$

Example (*Function Spaces*): Let S be an arbitrary set. Recall that the space of all functions from S to K, denoted as $K^S := \{f \mid f : S \to K\}$, forms a vector space over K.

The dimension of this space matches the cardinality of the underlying set S:

$$\dim K^S = |S|$$

Exercise: If S is a finite set, say $S = \{s_1, \dots, s_n\}$, can you construct an explicit basis for K^S? *(Hint: Try to define a family of functions f_i where each function outputs 1 for exactly one specific element s_i, and 0 for all other elements).*

Remark: When the domain S is an infinite set, the space K^S is infinite-dimensional. In this case, we unfortunately do not have a "nice" or easily describable basis for it.

Dimension and Subspaces (Chapter 12)

In this chapter, we synthesize the fundamental results regarding finite-dimensional vector spaces and explore the relationship between a space and its subspaces through the lens of dimensionality.

Summary (*Properties of Finite-Dimensional Spaces*): Let V be a finite-dimensional vector space over a field K. The following structural principles hold:

(a) Every finite list of vectors that spans V contains a sublist which constitutes a *"basis"* of V. Formally, if $(v_1, \dots, v_n)$ spans V, then there exists a sublist $(v_{i_1}, \dots, v_{i_d})$ with $1 \le i_1 < i_2 < \dots < i_d \le n$ that is a basis for V, where $d = \dim V$.

(b) Every linearly independent list of vectors in V can be extended to a basis of V. If the list $(u_1, \dots, u_m)$ is linearly independent, there exist vectors $w_1, \dots, w_{d-m}$ such that

$$(u_1, \dots, u_m, w_1, \dots, w_{d-m})$$

is a basis of V, where $d = \dim V$.

Theorem 12.1 (*Equivalent Characterizations of a Basis*): Let V be a vector space with $\dim V = n \in \mathbb{Z}_{\ge 0}$. Let $v_1, \dots, v_n \in V$ be a list of n vectors. Then the following statements are equivalent:

(a) The vectors $v_1, \dots, v_n$ are linearly independent.

(b) The vectors $v_1, \dots, v_n$ generate V, i.e., $\mathrm{Sp}(v_1, \dots, v_n) = V$.

(c) The vectors $v_1, \dots, v_n$ constitute a basis for V.

Proof

The equivalence is established through the following implications:

(a) $\implies$ (c): Suppose that **(a)** holds, so the list $v_1, \dots, v_n$ is linearly independent. According to the property that every linearly independent list can be extended to a basis, and observing that the list already possesses length $n = \dim V$, the extension requires no additional vectors. Thus, the list is already a basis for V, satisfying **(c)**.

(c) $\implies$ (a) & (b): This follows directly from the definition of a basis, which by requirement must satisfy both the condition of linear independence in **(a)** and the spanning property in **(b)**.

(b) $\implies$ (c): If the spanning property in **(b)** holds, the list $v_1, \dots, v_n$ generates V. By the properties of finite-dimensional spaces, this list must contain a sublist that is a basis. Since $\dim V = n$, any basis must have exactly length n. Therefore, the sublist must be the entire list itself, making $v_1, \dots, v_n$ a basis as required by **(c)**. Q.E.D.

Proposition 12.2 (*Dimensional Constraints on Subspaces*): Let V be a finite-dimensional vector space over K. Then every linear subspace $U \subseteq V$ is also finite-dimensional. Moreover, $\dim U \le \dim V$, with the equality $\dim U = \dim V$ holding if and only if $U = V$.

Proof

Let $n = \dim V$. We first establish that U possesses a finite basis. If $U = \{0\}$, the empty set serves as the basis and we are done. Assume $U \neq \{0\}$. We select a non-zero vector $v_1 \in U$. If $\mathrm{Sp}(v_1) = U$, we have found our basis.

If $\mathrm{Sp}(v_1) \neq U$, we choose a second vector $v_2 \in U \setminus \mathrm{Sp}(v_1)$. It follows from the properties of linear independence that the list $\{v_1, v_2\}$ is linearly independent. We proceed iteratively: after j steps, we obtain a linearly independent list $\{v_1, \dots, v_j\} \subseteq U$. If $\mathrm{Sp}(v_1, \dots, v_j) = U$, the process terminates. Otherwise, we choose $v_{j+1} \in U \setminus \mathrm{Sp}(v_1, \dots, v_j)$. By the principle that adding a vector outside the span of a linearly independent set preserves independence, the expanded list $\{v_1, \dots, v_{j+1}\}$ remains linearly independent.

Crucially, in a space V of dimension n, no linearly independent list can exceed length n. Consequently, this algorithmic procedure must terminate in k steps, where $k \le n$. At this stage, we have a list $\{v_1, \ldots, v_k\}$ that is linearly independent and spans U, thereby constituting a basis for U. This implies U is finite-dimensional and $\dim U = k \le n = \dim V$.

Furthermore, if $\dim U = \dim V = n$, then U contains a linearly independent list of length n. By **??**, such a list satisfies property **(a)** and is therefore also a basis for V. This implies $\mathrm{Sp}(v_1, \ldots, v_n) = V$. Since $U = \mathrm{Sp}(v_1, \ldots, v_n)$ by the construction of the basis, we conclude $U = V$. **Q.E.D.**

Row and Column Spaces (Chapter 13)

In this chapter, we bridge the abstract theory of vector spaces with the algorithmic power of matrix theory. We use Gauss elimination as a scalpel to dissect the structure of coordinate spaces.

Linear Combinations and Matrix Equations

Consider the vector space K^m over a field K. Let $w_1, \ldots, w_n \in K^m$ be a collection of vectors. We begin by addressing three fundamental questions that govern our understanding of these spaces.

Question (*The Three Fundamental Questions*):

(a) How can we systematically determine if a given vector $w \in K^m$ belongs to the span $\mathrm{Sp}(w_1, \ldots, w_n)$?

(b) Can we describe the subspace $\mathrm{Sp}(w_1, \ldots, w_n)$ precisely using algebraic conditions?

(c) How can we determine whether the vectors $w_1, \ldots, w_n$ are linearly independent?

Answer (*To Questions 1 and 2*): By definition, $w \in \mathrm{Sp}(w_1, \ldots, w_n)$ if and only if there exist scalars $x_1, \ldots, x_n \in K$ such that $w = x_1 w_1 + \cdots + x_n w_n$. We can assemble the vectors w_j as columns of an $m \times n$ matrix A. This abstract linear combination is equivalent to the matrix equation:

$$\underbrace{\begin{pmatrix} | & & | \\ w_1 & \ldots & w_n \\ | & & | \end{pmatrix}}_{A} \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} | \\ w \\ | \end{pmatrix}$$

Thus, w is in the span if and only if the linear system $Ax = w$ is *"consistent"* (i.e., it possesses at least one valid solution vector x). We analyze this by applying Gauss elimination on the *"augmented matrix"* $(A \mid w)$.

To answer the second question and describe the entire span precisely, we replace the specific target w with a general, variable vector $b = (b_1, \ldots, b_m)^T$ and reduce the augmented matrix $(A \mid b)$ to a row-reduced echelon form $(A' \mid b')$.

This row operation procedure transforms our original problem into a much simpler, equivalent linear system $A'x = b'$. As shown in Prof. Biran's notes, the resulting matrix generally has the following characteristic block structure:

$$(A' \mid b') = \left(\begin{array}{ccc|c} * & \cdots & * & b'_1 \\ \vdots & \text{Rows with pivots} & \vdots & \vdots \\ * & \cdots & * & b'_r \\ \hline 0 & \cdots & 0 & b'_{r+1} \\ \vdots & \text{Zero rows} & \vdots & \vdots \\ 0 & \cdots & 0 & b'_m \end{array}\right) \tag{13.1}$$

In this reduced block structure, any row of A' consisting entirely of zeros translates to an algebraic equation of the form $0 = b'_i$. For the overall system to be consistent (meaning a solution actually exists), it is an absolute requirement that the right-hand side of these specific equations also evaluates to zero.

In other words, the system is consistent if and only if the entries in b' corresponding to the zero-rows of A' strictly vanish. Therefore, we can concisely describe the span as the set of all vectors satisfying these constraints:

$$\mathrm{Sp}(w_1, \ldots, w_n) = \left\{ b \in K^m \;\middle|\; b'_{r+1} = \cdots = b'_m = 0 \right\}.$$

Example (*Calculating a Specific Span*): Let $v_1 = (1,1,1)^T$ and $v_2 = (1,0,-1)^T \in K^3$. To describe $\mathrm{Sp}(v_1, v_2)$, we set up the augmented matrix with a general vector b and row-reduce:

$$\left(\begin{array}{cc|c} 1 & 1 & b_1 \\ 1 & 0 & b_2 \\ 1 & -1 & b_3 \end{array}\right) \xrightarrow{R_2-R_1, R_3-R_1} \left(\begin{array}{cc|c} 1 & 1 & b_1 \\ 0 & -1 & b_2-b_1 \\ 0 & -2 & b_3-b_1 \end{array}\right) \xrightarrow{R_3-2R_2} \left(\begin{array}{cc|c} 1 & 1 & b_1 \\ 0 & -1 & b_2-b_1 \\ \hline 0 & 0 & b_1-2b_2+b_3 \end{array}\right)$$

The system is consistent if and only if the expression corresponding to the zero-row evaluates to zero. Thus, the span is perfectly described by that single plane equation:

$$\mathrm{Sp}(v_1, v_2) = \{(b_1, b_2, b_3)^T \in K^3 \mid b_1 - 2b_2 + b_3 = 0\}.$$

Answer (*To Question 3*): The vectors are linearly independent if and only if the homogeneous equation $Ax = 0$ has only the trivial solution (i.e., $x_1 = \cdots = x_n = 0$).

This occurs if and only if the row-reduced echelon form A' has a pivot in every column. If even a single column lacks a pivot, the corresponding variable becomes *"free"* (e.g., it can be parameterized to take on any scalar value). The existence of a free variable immediately allows for non-trivial solutions, which by definition means the vectors are linearly dependent.

Theorem 13.1 (*Fundamental Constraints on Coordinate Vectors*): Let $v_1, \ldots, v_n \in K^m$.

(a) If $n > m$, then the vectors $v_1, \ldots, v_n$ are linearly dependent.

(b) If $n < m$, then the vectors $v_1, \ldots, v_n$ cannot span K^m.

Proof

(a) Let $A = (v_1 \mid \cdots \mid v_n)$. The number of pivots in A' is bounded by the number of rows m. If $n > m$, at least $n - m$ columns are forced to be without pivots. These free variables guarantee non-trivial solutions to $Ax = 0$, hence dependence.

(b) The number of pivots is bounded by the number of columns n. If $n < m$, there are at least $m - n$ zero-rows in A'. By the logic of equation (**??**), we can easily choose a vector b such that the resulting b' has non-zero entries in those bottom rows, making the system inconsistent.

Q.E.D.

Row and Column Spaces

Definition 13.2 (*Subspaces of a Matrix*): Let $A \in \mathcal{M}_{m\times n}(K)$.

(a) The *"column space"* $\mathrm{Cols}(A) \subseteq K^m$ is the span of the columns of A.

(b) The *"row space"* $\mathrm{Rows}(A) \subseteq K^n$ is the span of the rows of A.

Proposition 13.3 (*Invariance of Row Space*): Elementary row operations do not change the row space: $\mathrm{Rows}(A) = \mathrm{Rows}(A')$.

Proof

New rows are, simply, linear combinations of the old rows, and because elementary operations are reversible, the spans remain identical. Q.E.D.

Important remark: Row operations **do** change the column space, as seen by $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$.

However, they **preserve** the linear relations between columns: $\sum \alpha_i w_i = 0 \iff \sum \alpha_i w_i' = 0$.

Constructing Bases

Theorem 13.4 (*Basis for the Column Space*): The columns of A that correspond to the *"pivot columns"* of its row-reduced echelon form B constitute a basis for $\mathrm{Cols}(A)$.

Proof

By the principle that row operations preserve the linear dependence relations between columns, we have:

$$\sum_{i=1}^{n} \alpha_i w_i = 0 \iff \sum_{i=1}^{n} \alpha_i w_i' = 0.$$

Thus, a set of columns in A is linearly independent (or spans the column space) if and only if the corresponding set of columns in B possesses the same property in $\mathrm{Cols}(B)$.

Consider the row-reduced echelon matrix $B \in \mathcal{M}_{m\times n}(K)$ with r pivots. We visualize its block structure and the specific indices of its pivot columns $j_1, j_2, \dots, j_r$:

$$B = \begin{array}{cc} \begin{array}{cccccccc} & \text{col } j_1 & & \text{col } j_2 & & \text{col } j_r & & \\ & \downarrow & & \downarrow & & \downarrow & & \end{array} & \\ \left(\begin{array}{cccccccc} \cdots & \mathbf{1} & * & \mathbf{0} & * & \mathbf{0} & * & \cdots \\ \cdots & \mathbf{0} & \cdots & \mathbf{1} & \cdots & \mathbf{0} & * & \cdots \\ & \vdots & & \vdots & \ddots & \vdots & \vdots & \\ \cdots & \mathbf{0} & \cdots & \mathbf{0} & \cdots & \mathbf{1} & * & \cdots \\ \hline & 0 & & 0 & & 0 & 0 & \cdots \\ & \vdots & & \vdots & & \vdots & \vdots & \ddots \\ & 0 & & 0 & & 0 & 0 & \cdots \end{array}\right) & \begin{array}{c} \leftarrow 1 \\ \leftarrow 2 \\ \vdots \\ \leftarrow r \\ \\ \\ \\ \end{array} \end{array} \tag{13.2}$$

In this echelon form, the pivot columns are precisely the first r standard basis vectors of K^m:

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad e_r = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

where for each e_i, the entry 1 is located at the i-th row. Because B is in row-reduced echelon form, every entry in every column w_j' that lies below the r-th row must be zero. This implies that:

$$w_j' \in K^r \times \{0\}^{m-r} = \left\{(x_1, \dots, x_r, 0, \dots, 0)^T \in K^m\right\}.$$

Since the standard basis vectors $e_1, \dots, e_r$ are clearly linearly independent and they span the subspace $K^r \times \{0\}^{m-r}$, we establish the following chain of inclusions:

$$K^r \times \{0\}^{m-r} = \mathrm{Sp}(e_1, \dots, e_r) \subseteq \mathrm{Cols}(B) \subseteq K^r \times \{0\}^{m-r}.$$

This double inclusion forces the equality $\mathrm{Cols}(B) = \mathrm{Sp}(e_1, \dots, e_r) = K^r \times \{0\}^{m-r}$. Consequently, the pivot columns of B form a basis for its column space. By the preservation of linear relations, the corresponding columns in the original matrix A form a basis for $\mathrm{Cols}(A)$. **Q.E.D.**

Theorem 13.5 (*Basis for the Row Space*): The non-zero rows of the row-reduced echelon matrix B form a basis for $\mathrm{Rows}(A)$.

Proof

By Proposition **??**, $\mathrm{Rows}(A) = \mathrm{Rows}(B)$. Let r be the number of pivots in B. The non-zero rows of B are $w_1', \dots, w_r'$. By definition, they span $\mathrm{Rows}(B)$.

To show linear independence, consider the indices of the pivot columns $j_1, \dots, j_r$. For each row $i \in \{1, \dots, r\}$, the entry in the j_i-th column is 1, while in all other rows $k \neq i$, the entry in the j_i-th column is 0. Suppose $\sum_{i=1}^r \alpha_i w_i' = 0$. Looking at the j_k-th component of this vector sum, we have:

$$\sum_{i=1}^{r} \alpha_i B_{i,j_k} = \alpha_k \cdot 1 + \sum_{i \neq k} \alpha_i \cdot 0 = \alpha_k = 0.$$

Since this holds for all k, all coefficients must be zero. Thus, the non-zero rows are linearly independent and form a basis. Q.E.D.

Corollary 13.6 (*Rank Invariance*): The number of pivots is independent of the sequence of row operations, as it always equals $\dim \mathrm{Rows}(A)$.

Proof

By Theorem **??**, the non-zero rows of any row-reduced echelon form B of A form a basis for $\mathrm{Rows}(A)$. The number of such rows is exactly the number of pivots. Since the dimension of a vector space is a fixed property, the number of pivots must be the same regardless of the specific row-reduction process used to reach an echelon form. Q.E.D.

Theorem 13.7 (*The Rank Equality*): For any $A \in \mathcal{M}_{m\times n}(K)$, $\dim \mathrm{Rows}(A) = \dim \mathrm{Cols}(A)$. This shared dimension is the *"rank"* of A, denoted $\mathrm{rank}(A)$.

Proof

Let r be the number of pivots in the RREF of A. By Theorem **??**, $\dim \mathrm{Cols}(A) = r$. By Theorem **??**, $\dim \mathrm{Rows}(A) = r$. Thus, the dimensions are exactly equal. Q.E.D.

Sums of Vector Spaces (Chapter 14)

Definition 14.1 (*Sum of Subspaces*): Let V be a vector space over K, and let $U, W \subseteq V$ be linear subspaces. We define the *"sum"* of U and W, denoted by $U + W$, as the set:

$$U + W := \{u + w \mid u \in U, w \in W\}.$$

In a similar way, we can define sums of more than two subspaces. If $U_1, \ldots, U_r \subseteq V$ are subspaces, we define their total sum as:

$$U_1 + \cdots + U_r := \{u_1 + \cdots + u_r \mid u_1 \in U_1, \ldots, u_r \in U_r\}.$$

Proposition 14.2 (*Properties of Sums*): Let $U, W \subseteq V$ be subspaces. Then, the following properties hold:

(a) $U + W = \mathrm{Sp}(U \cup W)$. In particular, $U + W \subseteq V$ is a subspace, and U and W are themselves subspaces of $U + W$.

(b) If U and W are finite-dimensional, then so is $U + W$. Moreover, one can systematically find a basis for $U + W$ by following these sequential steps: first, choose a basis $(v_1, \ldots, v_k)$ for $U \cap W$ (where $k = \dim(U \cap W)$). Second, extend this to a basis $(v_1, \ldots, v_k, u_1, \ldots, u_{l-k})$ for U (where $l = \dim U$). Third, extend the initial intersection basis to a basis $(v_1, \ldots, v_k, w_1, \ldots, w_{m-k})$ for W (where $m = \dim W$). Finally, the combined elements

$$(v_1, \ldots, v_k, u_1, \ldots, u_{\ell-k}, w_1, \ldots, w_{m-k})$$

form a basis for $U + W$.

Proof

We will explicitly prove the second statement. By definition, the subspace $U + W$ is generated by the combined vectors $(v_1, \ldots, v_k, u_1, \ldots, u_{l-k}, w_1, \ldots, w_{m-k})$. It remains to show that these vectors are linearly independent.

Suppose there exists a linear combination equal to the zero vector:

$$\alpha_1 v_1 + \cdots + \alpha_k v_k + \beta_1 u_1 + \cdots + \beta_{\ell-k} u_{\ell-k} + \gamma_1 w_1 + \cdots + \gamma_{m-k} w_{m-k} = 0. \tag{14.1}$$

We can rearrange this equation to isolate the terms belonging to U and W:

$$\underbrace{\beta_1 u_1 + \cdots + \beta_{\ell-k} u_{\ell-k}}_{\in U} = \underbrace{-\alpha_1 v_1 - \cdots - \alpha_k v_k - \gamma_1 w_1 - \cdots - \gamma_{m-k} w_{m-k}}_{\in W}.$$

Let us denote this common vector by x. Since the left side is entirely in U and the right side is entirely in W, we have $x \in U$ and $x \in W$, which implies $x \in U \cap W$.

Because $(v_1, \ldots, v_k)$ is a basis for $U \cap W$, there must exist scalars $\delta_1, \ldots, \delta_k \in K$ such that:

$$x = \delta_1 v_1 + \cdots + \delta_k v_k.$$

Equating this with our original expression for x in U, we obtain:

$$\beta_1 u_1 + \cdots + \beta_{\ell-k} u_{\ell-k} = \delta_1 v_1 + \cdots + \delta_k v_k \implies \sum_{i=1}^{\ell-k} \beta_i u_i - \sum_{j=1}^{k} \delta_j v_j = 0.$$

However, the vectors $(v_1, \ldots, v_k, u_1, \ldots, u_{\ell-k})$ form a basis for U, meaning they are strictly linearly independent. Consequently, all coefficients in this sum must be zero. In particular, $\beta_1 = \cdots = \beta_{\ell-k} = 0$.

Substituting these zero values back into our initial equation (**??**), the terms involving u_i vanish, leaving:

$$\alpha_1 v_1 + \cdots + \alpha_k v_k + \gamma_1 w_1 + \cdots + \gamma_{m-k} w_{m-k} = 0.$$

Since the vectors $(v_1, \ldots, v_k, w_1, \ldots, w_{m-k})$ form a basis for W, they are linearly independent. This immediately implies that $\alpha_1 = \cdots = \alpha_k = 0$ and $\gamma_1 = \cdots = \gamma_{m-k} = 0$.

Because all coefficients α_i, β_j, and γ_k have been forced to zero, the combined family of vectors is linearly independent, and therefore constitutes a valid basis for $U+W$. **Q.E.D.**

Theorem 14.3 (*Dimension Formula for Sums*): Let V be a vector space, and let $U,W \subseteq V$ be finite-dimensional subspaces. Then, the following dimension formula holds:

$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W).$$

Proof

Using the basis constructed in Proposition **??**, we can directly count the number of basis vectors for $U+W$:

$$\begin{aligned}\dim(U+W) &= k + (\ell - k) + (m - k) \\ &= \ell + m - k \\ &= \dim U + \dim W - \dim(U \cap W).\end{aligned}$$

Q.E.D.

Corollary 14.4 (*Equivalent Conditions for Trivial Intersection*): Let V be a vector space, and let $U,W \subseteq V$ be subspaces. The following statements are equivalent:

(a) $U \cap W = \{0\}$.

(b) For every $v \in U+W$, there exist unique vectors $u \in U$ and $w \in W$ such that $v = u + w$.

(c) Every $x \in U \cap W$ can be uniquely written as $x = u + w$ with $u \in U$ and $w \in W$.

(d) The equality $0 = u + w$ with $u \in U$ and $w \in W$ implies that $u = 0$ and $w = 0$.

Proof

We prove the main equivalences by showing the cycle of implications **(a)** $\implies$ **(b)** $\implies$ **(d)** $\implies$ **(a)**. (The equivalence of **(c)** follows naturally from the same uniqueness constraints).

(a) $\implies$ **(b):** Suppose that $v \in U+W$ can be written in two ways: $v = u + w = u' + w'$, where $u, u' \in U$ and $w, w' \in W$. Rearranging this equation yields $u - u' = w' - w$. The left side is in U and the right side is in W, which means the vector $u - u'$ resides in the intersection $U \cap W$. Since we assume $U \cap W = \{0\}$, it follows that $u - u' = 0$ and $w' - w = 0$. Therefore, $u = u'$ and $w = w'$, proving uniqueness.

(b) $\implies$ **(d):** Consider the zero vector, which can obviously be written as $0 = 0_U + 0_W$. If we also have $0 = u + w$ for some $u \in U$ and $w \in W$, the assumed uniqueness in statement **(b)** immediately forces $u = 0$ and $w = 0$.

(d) $\implies$ **(a):** Let $x \in U \cap W$. We can express the zero vector as $0 = x + (-x)$. Since $x \in U \cap W$, we know $x \in U$ and $-x \in W$. If statement **(d)** holds, this equality strictly implies $x = 0$. Consequently, the only element in the intersection is the zero vector, so $U \cap W = \{0\}$. **Q.E.D.**

Definition 14.4 (*Complement of a Subspace*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. A subspace $W \subseteq V$ is called a *"complement"* of U if it satisfies the following two conditions:

$$U + W = V \quad \text{and} \quad U \cap W = \{0\}.$$

Remark: Note that if W is a complement of U, then U and W satisfy all of the five equivalent statements derived from Corollary **??**.

Proposition 14.5 (*Existence of a Complement*): Let V be a finite-dimensional vector space, and let $U \subseteq V$ be a subspace. Then, there exists a subspace $W \subseteq V$ which is a complement to U.

Proof

Choose a basis $\mathcal{B}_U = (u_1, \dots, u_\ell)$ for U, where $\ell = \dim U$. By the Steinitz Exchange Lemma, we can extend this to a full basis for V:

$$\mathcal{B}_V = (u_1, \dots, u_\ell, w_1, \dots, w_m),$$

where $\ell + m = \dim V$. We define $W := \mathrm{Sp}(w_1, \dots, w_m)$.

We claim that W is a valid complement of U. Indeed, since the combined basis spans the entirety of V, it is clear that $U + W = V$. To see that $U \cap W = \{0\}$, let $x \in U \cap W$. Because $x \in U$, we can write $x = \sum_{i=1}^{l} a_i u_i$. Because $x \in W$, we can also write $x = \sum_{j=1}^{m} b_j w_j$. Equating these expressions yields:

$$\sum_{i=1}^{\ell} a_i u_i = \sum_{j=1}^{m} b_j w_j \implies \sum_{i=1}^{\ell} a_i u_i - \sum_{j=1}^{m} b_j w_j = 0.$$

Since $\mathcal{B}_V$ is a basis for V, all of its constituent vectors are strictly linearly independent. This forces all scalar coefficients to be zero: $a_i = 0$ for all i and $b_j = 0$ for all j. Thus, $x = 0$, proving that $U \cap W = \{0\}$. Q.E.D.

Linear Maps (Chapter 15)

Introduction to Linear Maps (Section 15.a)

Linear maps, also known as linear transformations, are homomorphisms of vector spaces. They are the primary objects of study in linear algebra, as they preserve the underlying algebraic structure of vector spaces.

Definition 15.a.1 (*Linear Map*): Let V and W be vector spaces over a field K. A map $T : V \to W$ is called a *"linear map"* if it satisfies the following two conditions:

(a) For all $v_1, v_2 \in V$, $T(v_1 + v_2) = T(v_1) + T(v_2)$.

(b) For all $\alpha \in K$ and $v \in V$, $T(\alpha \cdot v) = \alpha \cdot T(v)$.

Remark (*Notation and Terminology*): In many contexts, we omit the parentheses and write Tv instead of $T(v)$. The conditions outlined in **??** essentially state that $T : V \to W$ is a map that respects the operations defined on the vector spaces; specifically, the addition and scalar multiplication in V are mapped perfectly to the corresponding operations in W.

We denote the set of all linear maps from V to W by $\operatorname{Hom}_K(V, W)$, or simply $\operatorname{Hom}(V, W)$. While some authors use alternative notations such as $\operatorname{Lin}(V, W)$ or $\hom_K(V, W)$, $\operatorname{Hom}_K(V, W)$ remains the standard academic convention.

Exercise (*Linearity Criterion*): Prove that a map $T : V \to W$ is linear if and only if for all $\alpha_1, \alpha_2 \in K$ and $v_1, v_2 \in V$:

$$T(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 T(v_1) + \alpha_2 T(v_2).$$

Example (*Fundamental Examples of Linear Maps*):

(a) The identity map $\mathrm{id}_V : V \to V$, defined by $\mathrm{id}_V(v) := v$ for all $v \in V$.

(b) The zero map $0 : V \to W$, defined by $0(v) := 0_W$ for all $v \in V$.

Clearly, these initial examples are elementary, yet they serve as vital benchmarks.

(c) The derivative map $D : K[x] \to K[x]$. For any polynomial $p(x) = \sum_{k=0}^{n} a_k x^k$, we define its image as:

$$D(p(x)) := \sum_{k=1}^{n} k a_k x^{k-1}.$$

This map is linear, as the derivative of a linear combination of polynomials is simply the same linear combination of their derivatives.

(d) Consider the vector space $C([a, b])$ consisting of continuous functions $f : [a, b] \to \mathbb{R}$. The definite integral defines a linear map $T : C([a, b]) \to \mathbb{R}$, given by:

$$T(f) := \int_a^b f(x)\, dx.$$

(e) The shift map $S : K^\infty \to K^\infty$, defined on infinite sequences as:

$$S((a_1, a_2, a_3, \dots)) := (a_2, a_3, a_4, \dots).$$

Example (*Matrix Transformation*): This is a central example in finite-dimensional linear algebra. Let $A \in \mathcal{M}_{m \times n}(K)$ be a matrix. We define a map $T_A : K^n \to K^m$ via matrix-vector multiplication:

$$T_A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} := A \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Recall from matrix arithmetic that $A\cdot(\alpha\cdot x) = \alpha\cdot(A\cdot x)$ and $A\cdot(x+y) = A\cdot x + A\cdot y$. Consequently, T_A is a linear map.

Proposition 15.a.2 (*Basic Properties of Linear Maps*)**:** Let V and W be vector spaces over a field K, and let $T : V \to W$ be a linear map. Then, the following properties hold:

(a) For all $n \in \mathbb{N}$, $\alpha_1, \dots, \alpha_n \in K$, and $v_1, \dots, v_n \in V$, we have:

$$T\left(\sum_{i=1}^{n} \alpha_i v_i\right) = \sum_{i=1}^{n} \alpha_i T(v_i).$$

(b) $T(0_V) = 0_W$.

Proof

(a) The first statement follows directly from the definition of linearity (**??**) by applying induction on the number of vectors n.

(b) To prove the second statement, we evaluate the image of the zero vector:

$$0_W = T(0_V) - T(0_V) = T(0_V - 0_V) = T(0_V).$$

Therefore, any linear map must map the zero vector of the domain directly to the zero vector of the codomain.

Q.E.D.

Theorem 15.a.3 (*Existence and Uniqueness of Linear Maps on a Basis*)**:** Let V be a finite-dimensional vector space, and let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V. Given any arbitrary sequence of vectors $w_1, \dots, w_n \in W$, there exists a unique linear map $T : V \to W$ such that:

$$T(v_i) = w_i, \quad \forall\, 1 \le i \le n.$$

Proof

Existence: Let $v \in V$ be an arbitrary vector. Since $\mathcal{B}$ is a basis, there exist unique scalars $a_1, \dots, a_n \in K$ such that $v = \sum_{i=1}^{n} a_i v_i$. We define the map T by setting:

$$T(v) := \sum_{i=1}^{n} a_i w_i. \tag{15.1}$$

Because the coordinates a_i are uniquely determined for each v, the map T is well-defined. By verifying the conditions of **??**, it is straightforward to show that T is linear. Furthermore, for each basis vector v_j, the only non-zero coordinate is $a_j = 1$, which naturally implies $T(v_j) = w_j$.

Uniqueness: Suppose there exists another linear map $S : V \to W$ such that $S(v_i) = w_i$ for all $1 \le i \le n$. For any $v = \sum_{i=1}^{n} a_i v_i \in V$, the linearity of S dictates:

$$S(v) = S\left(\sum_{i=1}^{n} a_i v_i\right) = \sum_{i=1}^{n} a_i S(v_i) = \sum_{i=1}^{n} a_i w_i = T(v).$$

Since $S(v) = T(v)$ for all $v \in V$, the maps are functionally identical, establishing that $S = T$.

Q.E.D.

Example (*Important Example 15.a.4: Linear Maps on Coordinate Spaces*)**:** Consider the coordinate space K^n equipped with the standard basis $\mathcal{E} = (e_1, \dots, e_n)$. Let $w_1, \dots, w_n \in K^m$ be any vectors. **??** guarantees a unique linear map $T : K^n \to K^m$ such that $T(e_i) = w_i$ for all $1 \le i \le n$.

Question: Can we provide an explicit matrix description for this map T?

Answer: Let $A \in \mathcal{M}_{m\times n}(K)$ be the matrix whose j-th column is precisely the vector w_j:

$$A = \begin{pmatrix} | & & | \\ w_1 & \dots & w_n \\ | & & | \end{pmatrix}.$$

The linear transformation $T_A : K^n \to K^m$ defined by $v \mapsto A \cdot v$ easily satisfies $T_A(e_j) = w_j$ for all j. Due to the absolute uniqueness property established in **??**, we immediately conclude that $T = T_A$.

Lemma 15.a.5 (*Matrix Representation of Linear Maps*): For every linear transformation $T : K^n \to K^m$, there exists a unique matrix $A \in \mathcal{M}_{m\times n}(K)$ such that $T = T_A$.

Proof

Let $w_i = T(e_i)$ for each $i \in \{1, \dots, n\}$. As demonstrated in the preceding example, the matrix A formed by using these specific vectors as columns defines a linear map T_A that perfectly agrees with T on the standard basis $\mathcal{E}$. Because $\mathcal{E}$ is a basis for K^n, **??** strictly implies that T and T_A must be the exact same map. **Q.E.D.**

Kernel, Image, and the Rank-Nullity Theorem (Section 15.b)

In this section, we explore two fundamental subspaces associated with every linear map: the kernel and the image. These subspaces provide deep insight into the structure of the transformation and ultimately lead us to one of the most celebrated results in linear algebra, the Rank-Nullity Theorem.

Proposition 15.b.1 (*Kernel and Image*): Let V and W be vector spaces over a field K, and let $T : V \to W$ be a linear map.

(a) The set $\ker(T) := \{v \in V \mid T(v) = 0_W\} \subseteq V$ is a linear subspace of V. It is called the *"kernel"* of T.

(b) The set $\operatorname{Im}(T) := \{T(v) \mid v \in V\} \subseteq W$ is a linear subspace of W. It is called the *"image"* of T.

Remark (*Set-Theoretic Perspective*): Using standard notation from set theory, we can concisely write the kernel as the fiber over the zero vector, $\ker(T) = T^{-1}(\{0_W\})$, and the image as the range of the entire domain, $\operatorname{Im}(T) = T(V)$. (Note: The image is sometimes denoted as $\operatorname{Image}(T)$).

Proof of ??

(a) To show that $\ker(T) \subseteq V$ is a subspace, let $a, b \in K$, and let $v_1, v_2 \in \ker(T)$. By the linearity of T, we have:

$$T(av_1 + bv_2) = aT(v_1) + bT(v_2) = a \cdot 0_W + b \cdot 0_W = 0_W.$$

This implies that $av_1 + bv_2 \in \ker(T)$. Furthermore, since $T(0_V) = 0_W$, the zero vector $0_V \in \ker(T)$. Thus, the kernel is a valid subspace.

(b) To show that $\operatorname{Im}(T) \subseteq W$ is a subspace, let $a, b \in K$, and let $w_1, w_2 \in \operatorname{Im}(T)$. By definition, there exist vectors $v_1, v_2 \in V$ such that $T(v_1) = w_1$ and $T(v_2) = w_2$. Therefore, we can write:

$$aw_1 + bw_2 = aT(v_1) + bT(v_2) = T(av_1 + bv_2).$$

Since $av_1 + bv_2 \in V$, it follows immediately that $aw_1 + bw_2 \in \operatorname{Im}(T)$. Moreover, since $T(0_V) = 0_W$, we have $0_W \in \operatorname{Im}(T)$. Thus, the image is a subspace.

Q.E.D.

Proposition 15.b.2 (*Injectivity and the Kernel*)**:** A linear map $T : V \to W$ is injective if and only if $\ker(T) = \{0_V\}$.

Proof

First, suppose T is injective. If $v \in \ker(T)$, then $T(v) = 0_W$. Since we also know that $T(0_V) = 0_W$, the injectivity of T forces $v = 0_V$. Thus, $\ker(T) = \{0_V\}$.

Conversely, suppose that $\ker(T) = \{0_V\}$. Let $v_1, v_2 \in V$ such that $T(v_1) = T(v_2)$. By linearity, we obtain:

$$T(v_1) - T(v_2) = 0_W \implies T(v_1 - v_2) = 0_W.$$

This implies that $v_1 - v_2 \in \ker(T)$. By our assumption, the only element in the kernel is the zero vector, and therefore, $v_1 - v_2 = 0_V$, which yields $v_1 = v_2$. Consequently, T is injective. Q.E.D.

Definition 15.b.3 (*Rank of a Linear Map*)**:** Let $T :\in \operatorname{Hom}(V, W)$ be a linear map such that $\dim \operatorname{Im}(T) < \infty$. We define the *"rank"* of T as the dimension of its image:

$$\operatorname{rank}(T) := \dim \operatorname{Im}(T).$$

Theorem 15.b.4 (*Rank-Nullity Theorem*)**:** Let V be a finite-dimensional vector space, and let $T : V \to W$ be a linear map. Then, the following dimension formula holds:

$$\dim V = \dim \ker(T) + \dim \operatorname{Im}(T).$$

Proof

Let $k = \dim \ker(T)$ and $n = \dim V$. Let $(u_1, \dots, u_k)$ be a basis for $\ker(T)$. By the Steinitz Exchange Lemma, we can extend this linearly independent set to a full basis for V, denoted by $\mathcal{B} = (u_1, \dots, u_k, v_1, \dots, v_{n-k})$. We claim that the family $\mathcal{C} = (T(v_1), \dots, T(v_{n-k}))$ forms a basis for $\operatorname{Im}(T)$.

(a) **Spanning:** Let $w \in \operatorname{Im}(T)$. By definition, there exists some $v \in V$ such that $w = T(v)$. We can express v uniquely as a linear combination of the basis vectors in $\mathcal{B}$:

$$v = \sum_{i=1}^{k} a_i u_i + \sum_{j=1}^{n-k} b_j v_j.$$

Applying the linear map T, and recalling that $T(u_i) = 0_W$ for all i, we find:

$$\begin{aligned} w &= T\left(\sum_{i=1}^{k} a_i u_i + \sum_{j=1}^{n-k} b_j v_j\right) \\ &= \sum_{i=1}^{k} a_i T(u_i) + \sum_{j=1}^{n-k} b_j T(v_j) \\ &= 0_W + \sum_{j=1}^{n-k} b_j T(v_j). \end{aligned}$$

Thus, the vectors in $\mathcal{C}$ span $\operatorname{Im}(T)$.

(b) **Linear Independence:** Suppose there exist scalars $b_j \in K$ such that:

$$\sum_{j=1}^{n-k} b_j T(v_j) = 0_W.$$

By the linearity of T, this is equivalent to $T\left(\sum_{j=1}^{n-k} b_j v_j\right) = 0_W$, which implies that the vector $\sum_{j=1}^{n-k} b_j v_j$ lies in $\ker(T)$. Because $(u_1, \dots, u_k)$ is a basis for the kernel, there must exist scalars $a_i \in K$ such that:

$$\sum_{j=1}^{n-k} b_j v_j = \sum_{i=1}^{k} a_i u_i \implies \sum_{j=1}^{n-k} b_j v_j - \sum_{i=1}^{k} a_i u_i = 0_V .$$

However, since $\mathcal{B}$ is a basis for V, all of its constituent vectors are linearly independent. This forces all coefficients to be zero, specifically $b_j = 0$ for all $1 \leq j \leq n-k$.

Therefore, $\mathcal{C}$ is linearly independent and spans the image, making it a basis for $\operatorname{Im}(T)$. Consequently, $\dim \operatorname{Im}(T) = n - k = \dim V - \dim \ker(T)$.

Q.E.D.

Corollary 15.b.5 (*Equivalence of Injectivity and Surjectivity*): Let V and W be finite-dimensional vector spaces such that $\dim V = \dim W$. For any linear map $T : V \to W$, the following statements are equivalent:

(a) T is injective.

(b) T is surjective.

(c) T is an isomorphism.

Proof

By the Rank-Nullity Theorem (**??**), we know that $\dim \ker(T) = \dim V - \dim \operatorname{Im}(T)$. Since we are given that $\dim V = \dim W$, we can analyze the implications:

- T is injective $\iff$ $\ker(T) = \{0_V\}$ $\iff$ $\dim \ker(T) = 0$ $\iff$ $\dim \operatorname{Im}(T) = \dim V = \dim W$ $\iff$ T is surjective.

Because injectivity implies surjectivity (and vice versa) in this specific dimension constraint, either condition guarantees that the map is bijective, and therefore, an isomorphism. Q.E.D.

Isomorphisms and Vector Space Equivalence

Isomorphisms allow us to classify vector spaces and identify when two ostensibly different spaces are essentially the *"same"* in terms of their linear algebraic structure.

Theorem 15.b.12 (*Isomorphisms Preserve Bases*): Let $T : V \to W$ be an isomorphism between two vector spaces V and W. Let $\mathcal{S} = \{v_i\}$ be a family of vectors in V. We define its image family as $T(\mathcal{S}) = \{T(v_i)\}_{v_i \in \mathcal{S}} \subseteq W$. Then, the following hold:

(a) $\mathcal{S}$ is linearly independent in V if and only if $T(\mathcal{S})$ is linearly independent in W.

(b) $\mathcal{S}$ spans V if and only if $T(\mathcal{S})$ spans W.

(c) $\mathcal{S}$ is a basis of V if and only if $T(\mathcal{S})$ is a basis of W.

Definition 15.b.13 (*Isomorphic Spaces*): Two vector spaces V and W are defined to be *"isomorphic"*, denoted as $V \cong W$, if there exists an isomorphism between them. Formally, this means there exist linear maps $T : V \to W$ and $S : W \to V$ such that:

$$T \circ S = \mathrm{id}_W, \quad \text{and} \quad S \circ T = \mathrm{id}_V .$$

$$\begin{tikzcd} V \arrow[r, shift left, "T"] & W \arrow[l, shift left, "S"] \end{tikzcd}$$

Remark (*The Significance of Isomorphism*): If two vector spaces are isomorphic ($V \cong W$), they possess identical linear algebraic properties, such as having the exact same dimension. We can conceptually think of V and W as the exact same space, just represented in two different ways.

However, it is crucial to note that identifying V with W requires the explicit choice of an isomorphism $T : V \to W$. Because there is usually no preferred (or *"canonical"*) choice for this map, it is generally better practice not to forcefully identify different vector spaces simply because they are isomorphic. We will explore the nuances of canonical choices later in the course.

Linear Maps and Bases (Chapter 16)

Writing Linear Maps Using Bases and Coordinates

In this lecture, we will assume all vector spaces to be *"finite-dimensional"* unless otherwise stated. We will write bases of a vector space V as an ordered list (or tuple) $\mathcal{B} = (v_1, \dots, v_n)$.

Definition 16.1 (*Coordinate Vector*): Let V be a vector space over K, and let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis of V. Let $v \in V$. We define the *"coordinate-vector"* of v with respect to the basis $\mathcal{B}$ as

$$[v]_{\mathcal{B}} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in K^n,$$

where $a_1, \dots, a_n \in K$ are the unique scalars in K such that $v = a_1 v_1 + \cdots + a_n v_n$.

Remark: Note that $[v]_{\mathcal{B}}$ is uniquely defined, since every $v \in V$ has a unique representation as a linear combination of the basis vectors $v_1, \dots, v_n$.

Define a map $\Phi_{\mathcal{B}} : V \to K^n$ given by:

$$\Phi_{\mathcal{B}}(v) := [v]_{\mathcal{B}} \in K^n.$$

Proposition 16.2 (*Isomorphism of the Coordinate Map*): The map $\Phi_{\mathcal{B}}$ is an isomorphism.

Proof

We first show that $\Phi_{\mathcal{B}}$ is a linear map. Let $a, b \in K$, and let $v, v' \in V$. Write v and v' as linear combinations of the elements of $\mathcal{B}$:

$$v = a_1 v_1 + \cdots + a_n v_n, \quad \text{and} \quad v' = b_1 v_1 + \cdots + b_n v_n,$$

with $a_i, b_i \in K$. By definition, we have:

$$[v]_{\mathcal{B}} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}, \quad \text{and} \quad [v']_{\mathcal{B}} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

Now, $av + bv' = (aa_1 + bb_1)v_1 + \cdots + (aa_n + bb_n)v_n$, and hence:

$$\Phi_{\mathcal{B}}(av + bv') = [av + bv']_{\mathcal{B}} = \begin{pmatrix} aa_1 + bb_1 \\ \vdots \\ aa_n + bb_n \end{pmatrix} = a \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} + b \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = a\Phi_{\mathcal{B}}(v) + b\Phi_{\mathcal{B}}(v').$$

This proves that $\Phi_{\mathcal{B}}$ is linear.

We now claim that $\Phi_{\mathcal{B}}$ is an isomorphism. By previous results, it is enough to show $\Phi_{\mathcal{B}}$ is bijective. Indeed, if $v \in \ker(\Phi_{\mathcal{B}})$, then $v = 0v_1 + \cdots + 0v_n = 0$, which implies $\ker(\Phi_{\mathcal{B}}) = \{0\}$. This shows that $\Phi_{\mathcal{B}}$ is injective.

For the surjectivity of $\Phi_{\mathcal{B}}$, let $(a_1, \dots, a_n)^T \in K^n$. Put $v = a_1 v_1 + \cdots + a_n v_n \in V$. Then, $\Phi_{\mathcal{B}}(v) = (a_1, \dots, a_n)^T$, which implies $\mathrm{Im}(\Phi_{\mathcal{B}}) = K^n$. Q.E.D.

Remark: The inverse map to $\Phi_{\mathcal{B}}$ is given by:

$$\Phi_{\mathcal{B}}^{-1} : K^n \to V, \quad \Phi_{\mathcal{B}}^{-1} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} := a_1 v_1 + \cdots + a_n v_n.$$

Remark: The previous **??** gives yet another proof for the following Corollary.

Corollary 16.3 (*Classification of Finite-Dimensional Vector Spaces*): Let V be a vector space over K with $\dim V = n \in \mathbb{Z}_{\geq 0}$. Then, $V \cong K^n$.

Proof

Choose a basis $\mathcal{B}$ for V. Then, $\Phi_{\mathcal{B}} : V \to K^n$ is an isomorphism. **Q.E.D.**

Remark:

- The Corollary does NOT hold for $\dim V = \infty$, at least not as stated.
- We have seen that $V \cong K^n$ where $n = \dim V < \infty$, but in general, there does not exist a *"canonical"* (or preferred) isomorphism $V \to K^n$. For example, $\Phi_{\mathcal{B}}$ depends on a choice of a basis $\mathcal{B}$.

Example (*Coordinate Vector Calculation*): Consider $V = \mathbb{R}^2$, let $\mathcal{E} = \left(\binom{1}{0}, \binom{0}{1}\right)$ be the standard basis, and let $\mathcal{B} = \left(\binom{1}{1}, \binom{1}{-1}\right)$ be another basis (check that $\mathcal{B}$ is indeed a basis!). We have

$$\left[\begin{pmatrix} x \\ y \end{pmatrix}\right]_{\mathcal{E}} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \text{and we want to find} \quad \left[\begin{pmatrix} x \\ y \end{pmatrix}\right]_{\mathcal{B}} = \begin{pmatrix} a \\ b \end{pmatrix} =?$$

Let us first try to determine $\left[\binom{1}{0}\right]_{\mathcal{B}}$ and $\left[\binom{0}{1}\right]_{\mathcal{B}}$. We set $\left[\binom{1}{0}\right]_{\mathcal{B}} = \binom{a_1}{a_2}$, where

$$a_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + a_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

This leads to the system of linear equations:

$$\begin{cases} a_1 + a_2 = 1 \\ a_1 - a_2 = 0 \end{cases}$$

Solving this system yields $a_1 = a_2 = \frac{1}{2}$, and therefore, $\left[\binom{1}{0}\right]_{\mathcal{B}} = \binom{1/2}{1/2}$.

Similarly, for the second standard basis vector, we set

$$\left[\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]_{\mathcal{B}} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \quad \text{where} \quad b_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + b_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The resulting system is:

$$\begin{cases} b_1 + b_2 = 0 \\ b_1 - b_2 = 1 \end{cases}$$

Solving this yields $b_1 = \frac{1}{2}$ and $b_2 = -\frac{1}{2}$, so $\left[\binom{0}{1}\right]_{\mathcal{B}} = \binom{1/2}{-1/2}$.

Finally, using the fact that $\Phi_{\mathcal{B}}$ is linear, we calculate:

$$\left[\begin{pmatrix} x \\ y \end{pmatrix}\right]_{\mathcal{B}} = x \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right]_{\mathcal{B}} + y \left[\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]_{\mathcal{B}} = \begin{pmatrix} x/2 \\ x/2 \end{pmatrix} + \begin{pmatrix} y/2 \\ -y/2 \end{pmatrix} = \begin{pmatrix} \frac{x+y}{2} \\ \frac{x-y}{2} \end{pmatrix}.$$

Representation Matrices

Now we arrive at the main question. Let V and W be vector spaces over K, let $n := \dim V$ and $m := \dim W$, and let $T : V \to W$ be a linear map. Fix a basis $\mathcal{B}$ for V and a basis $\mathcal{C}$ for W. Consider the following commutative diagram test:

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "\Phi_{\mathcal{B}}"'] & W \arrow[d, "\Phi_{\mathcal{C}}"] \\ K^n \arrow[r, "\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}"'] & K^m \end{tikzcd}$$

Since $\Phi_{\mathcal{B}}^{-1}$, T, and $\Phi_{\mathcal{C}}$ are all linear maps, the map $\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1} : K^n \to K^m$ is also linear. By previous results, any linear map $K^n \to K^m$ can be written using a matrix $A \in \mathcal{M}_{m\times n}(K)$. In other words, there exists a unique matrix $A \in \mathcal{M}_{m\times n}(K)$ such that $\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1} = T_A$.

The matrix A is called the *"representation matrix"* of T with respect to the bases $\mathcal{B}$ and $\mathcal{C}$. We also say that A represents T in the bases $\mathcal{B}$ and $\mathcal{C}$. We denote this matrix by $[T]_{\mathcal{C}}^{\mathcal{B}} \in \mathcal{M}_{m\times n}(K)$.

Question:

(a) How can we calculate the entries of A?

(b) How does A depend on different choices of the bases $\mathcal{B}$ and $\mathcal{C}$?

(c) If we have three vector spaces V, W, U with bases $\mathcal{B}, \mathcal{C}, \mathcal{D}$ respectively, and $T : V \to W$ and $S : W \to U$ are linear maps where T is represented by the matrix M and S by the matrix N, then what is the matrix representing $S \circ T : V \to U$ with respect to the bases $\mathcal{B}$ and $\mathcal{D}$?

Let $T : V \to W$ be a linear map, let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis for V, and let $\mathcal{C} = (w_1, \ldots, w_m)$ be a basis for W. For any $v_i \in V$, its image is $T(v_i) \in W$. Under the map T_A, the standard basis vector $e_i \in K^n$ maps to K^m:

$$T_A(e_i) = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}(e_i) = \Phi_{\mathcal{C}}(T(v_i)) = [T(v_i)]_{\mathcal{C}}.$$

But these are precisely the columns of A. In other words:

$$A = [T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \ldots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix} \in \mathcal{M}_{m\times n}(K).$$

Another way to describe this matrix is $A = (a_{ij}) \in \mathcal{M}_{m\times n}(K)$, where the entries a_{ij} are defined uniquely by:

$$T(v_j) = a_{1j}w_1 + \cdots + a_{mj}w_m = \sum_{i=1}^{m} a_{ij}w_i.$$

Proposition 16.4 (*Evaluation via Representation Matrix*): Let $T : V \to W$ be a linear map between two finite-dimensional vector spaces V and W. Let $\mathcal{B}$ and $\mathcal{C}$ be bases for V and W, respectively. Then, for all $v \in V$:

$$[T]_{\mathcal{C}}^{\mathcal{B}} \cdot [v]_{\mathcal{B}} = [T(v)]_{\mathcal{C}}.$$

Proof

The equality $[T]_{\mathcal{C}}^{\mathcal{B}} \cdot [v]_{\mathcal{B}} = [T(v)]_{\mathcal{C}}$ is equivalent to evaluating the linear transformation defined by the matrix $[T]_{\mathcal{C}}^{\mathcal{B}}$ at the vector $[v]_{\mathcal{B}} \in K^n$:

$$T_{[T]_{\mathcal{C}}^{\mathcal{B}}}([v]_{\mathcal{B}}) = [T(v)]_{\mathcal{C}}.$$

By definition, $T_{[T]_{\mathcal{C}}^{\mathcal{B}}} = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}$. But this is precisely how we defined the matrix $[T]_{\mathcal{C}}^{\mathcal{B}}$.

Q.E.D.

Example:

(a) Let $V = K^n$ and $W = K^m$, and let $T : K^n \to K^m$ be given by $T = T_A$ for $A \in \mathcal{M}_{m\times n}(K)$, so $T(v) = A \cdot v$. Take $\mathcal{B} = (e_1, \ldots, e_n)$ and $\mathcal{C} = (e_1, \ldots, e_m)$ to be the standard bases. Then, $[T]_{\mathcal{C}}^{\mathcal{B}} = A$.

(b) Let $V = K[x]_3$ and $W = K[x]_2$, and let $D : V \to W$ be the derivative map defined by $D(p(x)) = p'(x)$. Let $\mathcal{B} = (1, x, x^2, x^3)$ and $\mathcal{C} = (1, x, x^2)$. We calculate:

$$\begin{aligned} D(1) &= 0 = 0 \cdot 1 + 0 \cdot x + 0 \cdot x^2 \\ D(x) &= 1 = 1 \cdot 1 + 0 \cdot x + 0 \cdot x^2 \\ D(x^2) &= 2x = 0 \cdot 1 + 2 \cdot x + 0 \cdot x^2 \\ D(x^3) &= 3x^2 = 0 \cdot 1 + 0 \cdot x + 3 \cdot x^2 \end{aligned}$$

Therefore, the representation matrix is:

$$[D]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

(b) Suppose we view D as a linear map $D : K[x]_3 \to K[x]_3$. Then,

$$[D]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Representing Composition of Linear Maps by Matrices

Let $T : V \to W$ and $S : W \to U$ be linear maps. Let $\mathcal{A} = (v_1, \ldots, v_n)$ be a basis for V, let $\mathcal{B} = (w_1, \ldots, w_m)$ be a basis for W, and let $\mathcal{C} = (u_1, \ldots, u_p)$ be a basis for U. The dimensions are thus $n = \dim V$, $m = \dim W$, and $p = \dim U$.

Consider the composition $S \circ T : V \longrightarrow U$.

Question: What is the matrix $[S \circ T]_{\mathcal{C}}^{\mathcal{A}}$? How is it related to the matrices $[T]_{\mathcal{B}}^{\mathcal{A}}$ and $[S]_{\mathcal{C}}^{\mathcal{B}}$?

To answer this, let us write the representation matrices as follows:

$$[T]_{\mathcal{B}}^{\mathcal{A}} = (a_{ij}), \quad [S]_{\mathcal{C}}^{\mathcal{B}} = (b_{ij}), \quad [S \circ T]_{\mathcal{C}}^{\mathcal{A}} = (c_{ij}).$$

By definition, the images of the basis vectors are given by:

$$T(v_j) = \sum_{k=1}^{m} a_{kj} w_k, \quad \forall\, 1 \le j \le n, \tag{16.1}$$

$$S(w_k) = \sum_{i=1}^{p} b_{ik} u_i, \quad \forall\, 1 \le k \le m, \tag{16.2}$$

$$(S \circ T)(v_j) = \sum_{i=1}^{p} c_{ij} u_i, \quad \forall\, 1 \le j \le n. \tag{16.3}$$

But we can also calculate $(S \circ T)(v_j)$ using (**??**) and (**??**):

$$\begin{aligned}(S \circ T)(v_j) &= S\left(\sum_{k=1}^{m} a_{kj} w_k\right)\\ &= \sum_{k=1}^{m} a_{kj} S(w_k)\\ &= \sum_{k=1}^{m} a_{kj}\left(\sum_{i=1}^{p} b_{ik} u_i\right)\\ &= \sum_{k=1}^{m}\sum_{i=1}^{p} a_{kj} b_{ik} u_i = \sum_{i=1}^{p}\left(\sum_{k=1}^{m} b_{ik} a_{kj}\right) \cdot u_i.\end{aligned}$$

Comparing this with (**??**), we find that $c_{ij} = \sum_{k=1}^{m} b_{ik} a_{kj}$. In other words, row i of $[S]_{\mathcal{C}}^{\mathcal{B}}$ multiplied by column j of $[T]_{\mathcal{B}}^{\mathcal{A}}$ yields the entry c_{ij}:

$$\begin{pmatrix} b_{i1} & \dots & b_{im} \end{pmatrix} \cdot \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix} = c_{ij}.$$

This reveals that:

$$[S]_{\mathcal{C}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{A}} = [S \circ T]_{\mathcal{C}}^{\mathcal{A}},$$

where the matrix dimensions follow $(p \times m) \cdot (m \times n) = (p \times n)$.

In order to obtain $[S \circ T]_{\mathcal{C}}^{\mathcal{A}}$ from $[T]_{\mathcal{B}}^{\mathcal{A}}$ and $[S]_{\mathcal{C}}^{\mathcal{B}}$, we need to use a new operation on matrices. It is called *"matrix multiplication"*. But is this operation really new? - Yes, but not entirely...

Matrices (Chapter 17)

Matrix Multiplication (Section 17.a)

Definition 17.a.1 (*Matrix Multiplication*): Let $A \in \mathcal{M}_{m\times n}(K)$ and $B \in \mathcal{M}_{n\times p}(K)$. We define a new matrix, which is called the *"product"* (or *"multiplication"*) of A and B, denoted by $A \cdot B \in \mathcal{M}_{m\times p}(K)$.

Write $A = (a_{ij})_{\substack{1\le i\le m\\ 1\le j\le n}}$ and $B = (b_{ij})_{\substack{1\le i\le n\\ 1\le j\le p}}$. The matrix $A\cdot B$ has entries $(c_{ij})_{\substack{1\le i\le m\\ 1\le j\le p}}$, where

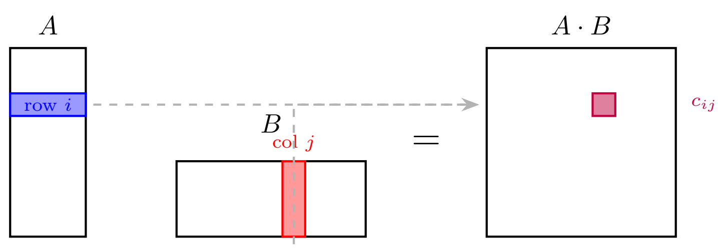
$$c_{ij} := a_{i1}b_{1j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}b_{kj}$$

We often write AB instead of $A\cdot B$.

Important remark: The product AB is defined only when the number of columns of A equals the number of rows of B.

An alternative description of $A\cdot B$ can be visualized by taking the dot product of the i-th row of A with the j-th column of B:

$$c_{ij} = (\text{row } i \text{ of } A)\cdot(\text{column } j \text{ of } B) = a_{i1}b_{1j} + \cdots + a_{ik}b_{kj} + \cdots + a_{in}b_{nj}$$



Another way to describe it is by considering the columns of B, denoted as $w_1, \ldots, w_p$. The matrix $A\cdot B$ is then formed by multiplying A with each column vector w_j:

$$A \cdot \begin{pmatrix} | & & | \\ w_1 & \ldots & w_p \\ | & & | \end{pmatrix} = \begin{pmatrix} | & & | \\ Aw_1 & \ldots & Aw_p \\ | & & | \end{pmatrix}$$

Example: We compute the product of a 2×3 matrix and a 3×2 matrix:

$$\begin{pmatrix} 3 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 4 & 0 \end{pmatrix} = \begin{pmatrix} 3\cdot 1 + 2\cdot 0 + 1\cdot 4 & 3\cdot 2 + 2\cdot 1 + 1\cdot 0 \\ 1\cdot 1 + 0\cdot 0 + 2\cdot 4 & 1\cdot 2 + 0\cdot 1 + 2\cdot 0 \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 9 & 2 \end{pmatrix}$$

Conversely, multiplying the 3×2 matrix by the 2×3 matrix yields a 3×3 matrix:

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 4 & 0 \end{pmatrix} \cdot \begin{pmatrix} 3 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 5 & 2 & 5 \\ 1 & 0 & 2 \\ 12 & 8 & 4 \end{pmatrix}$$

If $A \in \mathcal{M}_{m\times n}(K)$ and $B \in \mathcal{M}_{n\times p}(K)$, then $A\cdot B$ is defined, but $B\cdot A$ is not defined, unless $p = m$.

Some Special Cases

(a) If $A \in \mathcal{M}_{m\times n}(K)$ and $v \in K^n_{\text{col}}$, we have previously defined the matrix-vector product $A \cdot v \in K^m_{\text{col}}$. However, we can also identify $K^n_{\text{col}} = \mathcal{M}_{n\times 1}(K)$. If we view v as an $n \times 1$ matrix, then the matrix multiplication of A and v yields the exact same result as applying A to the vector v.

(b) If $A \in \mathcal{M}_{m\times n}(K)$ and $w \in K^m_{\text{row}} = \mathcal{M}_{1\times m}(K)$, then we can define the left-multiplication $w \cdot A \in \mathcal{M}_{1\times n}(K) = K^n_{\text{row}}$. If A is represented by its row vectors $v_1, \dots, v_m$, this operation scales and sums the rows of A according to the entries of w.

(c) Let $v = (a_1, \dots, a_n) \in K^n_{\text{row}} = \mathcal{M}_{1\times n}(K)$, and let $w = (b_1, \dots, b_n)^T \in K^n_{\text{col}} = \mathcal{M}_{n\times 1}(K)$. Then the product $v \cdot w \in \mathcal{M}_{1\times 1}(K) = K$ is simply the standard scalar product:

$$v \cdot w = a_1 b_1 + \cdots + a_n b_n$$

Connections to Linear Maps

As a conclusion from our previous discussions on linear maps, let $T : V \to W$ and $S : W \to U$ be linear maps between finite-dimensional vector spaces. Let $\mathcal{B}$, $\mathcal{C}$, and $\mathcal{E}$ be bases for V, W, and U, respectively. The matrix representation of the composition $S \circ T$ is given by the product of their respective representation matrices:

$$[S \circ T]^{\mathcal{B}}_{\mathcal{E}} = [S]^{\mathcal{C}}_{\mathcal{E}} \cdot [T]^{\mathcal{B}}_{\mathcal{C}}$$

$$\begin{tikzcd}
V \arrow[r, "T"] \arrow[rr, bend left, "S\circ T"] & W \arrow[r, "S"] & U \\
\mathrm{Sp}(\mathcal{B}) \arrow[u, phantom, "\in"] & \mathrm{Sp}(\mathcal{C}) \arrow[u, phantom, "\in"] & \mathrm{Sp}(\mathcal{E}) \arrow[u, phantom, "\in"]
\end{tikzcd}$$

Notation and The Identity Matrix

Let $\mathcal{I}$ be an index set. For $i, j \in \mathcal{I}$, we define the **Kronecker-delta** as:

$$\delta_{ij} := \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Using this, we define the *"identity matrix"* $I_n \in \mathcal{M}_{n\times n}(K)$ as $I_n = (\delta_{ij})_{\substack{1\le i\le n \\ 1\le j\le n}}$. Explicitly, this is:

$$I_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

Remark: Let V be a finite-dimensional vector space, and let $n := \dim V$. Let $\mathrm{id}_V : V \to V$ be the identity map, and let $\mathcal{B}$ be any basis for V. Then, $[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}} = I_n$. In other words, the matrix representing the identity map with respect to the basis $\mathcal{B}$ is precisely the identity matrix I_n.

Exercise: Prove the preceding remark.

Properties of Matrix Multiplication

Proposition 17.a.2 (*Algebraic Properties of Matrix Multiplication*): Matrix multiplication satisfies the following algebraic properties:

Associativity: For all $A \in \mathcal{M}_{m\times n}(K)$, $B \in \mathcal{M}_{n\times p}(K)$, and $C \in \mathcal{M}_{p\times q}(K)$, we have:

$$(AB)C = A(BC) \in \mathcal{M}_{m\times q}(K)$$

Distributivity: For all $A, B \in \mathcal{M}_{m\times n}(K)$, $C \in \mathcal{M}_{n\times p}(K)$, and $C' \in \mathcal{M}_{q\times m}(K)$, we have:

$$(A+B)C = AC + BC \in \mathcal{M}_{m\times p}(K)$$
$$C'(A+B) = C'A + C'B \in \mathcal{M}_{q\times n}(K)$$

Identity: For all $A \in \mathcal{M}_{m\times n}(K)$, we have $I_m A = A$ and $AI_n = A$.

Scalar Multiplication: For all $\alpha \in K$, $A \in \mathcal{M}_{m\times n}(K)$, and $B \in \mathcal{M}_{n\times p}(K)$, we have:

$$\alpha(AB) = (\alpha A)B = A(\alpha B)$$

Proof

Let $T_A : K^n \to K^m$, $T_B : K^p \to K^n$, and $T_C : K^q \to K^p$ be the linear maps defined by left-multiplication with the matrices A, B, and C, respectively (i.e., $T_A(v) = A\cdot v$ for all $v \in K^n$, $T_B(w) = B\cdot w$ for all $w \in K^p$, and $T_C(u) = C\cdot u$ for all $u \in K^q$). For any dimension r, we denote the standard basis of K^r by $\mathcal{E}_r$. We have already established that:

$$[T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} = A, \quad [T_B]_{\mathcal{E}_p}^{\mathcal{E}_n} = B, \quad \text{and} \quad [T_C]_{\mathcal{E}_q}^{\mathcal{E}_p} = C$$

Proof of (a): We rely on the associativity of function composition, which states that $(T_A \circ T_B) \circ T_C = T_A \circ (T_B \circ T_C)$. These are both maps from K^q to K^m. We now pass to the matrix representation of these maps with respect to the standard bases $\mathcal{E}_q$ and $\mathcal{E}_m$:

$$[(T_A \circ T_B) \circ T_C]_{\mathcal{E}_q}^{\mathcal{E}_m} = [T_A \circ (T_B \circ T_C)]_{\mathcal{E}_q}^{\mathcal{E}_m}$$

By applying the rule for the matrix representation of composed maps, we get:

$$[T_A \circ T_B]_{\mathcal{E}_p}^{\mathcal{E}_m} \cdot [T_C]_{\mathcal{E}_q}^{\mathcal{E}_p} = [T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} \cdot [T_B \circ T_C]_{\mathcal{E}_q}^{\mathcal{E}_n}$$

Expanding further, we obtain:

$$\left([T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} \cdot [T_B]_{\mathcal{E}_p}^{\mathcal{E}_n}\right) \cdot [T_C]_{\mathcal{E}_q}^{\mathcal{E}_p} = [T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} \cdot \left([T_B]_{\mathcal{E}_p}^{\mathcal{E}_n} \cdot [T_C]_{\mathcal{E}_q}^{\mathcal{E}_p}\right)$$

Substituting the matrices back in yields $(A\cdot B)\cdot C = A\cdot(B\cdot C)$.

Proof of (b): From the properties of linear maps, we know that $(T_A + T_B) \circ T_C = T_A \circ T_C + T_B \circ T_C$. Passing to the matrix representations, we have:

$$[(T_A + T_B) \circ T_C]_{\mathcal{E}_q}^{\mathcal{E}_m} = [T_A \circ T_C + T_B \circ T_C]_{\mathcal{E}_q}^{\mathcal{E}_m}$$

This implies that:

$$[T_A + T_B]_{\mathcal{E}_n}^{\mathcal{E}_m} \cdot [T_C]_{\mathcal{E}_q}^{\mathcal{E}_n} = [T_A \circ T_C]_{\mathcal{E}_q}^{\mathcal{E}_m} + [T_B \circ T_C]_{\mathcal{E}_q}^{\mathcal{E}_m}$$

Which directly gives us $(A+B)\cdot C = A\cdot C + B\cdot C$. The other distributive identity in (b) is proven similarly.

Proof of (c): The identity matrix I_m represents the identity map: $I_m = [\mathrm{id}_{K^m}]_{\mathcal{E}_m}^{\mathcal{E}_m}$. Since composing with the identity map does not change a function, we have $\mathrm{id}_{K^m} \circ T_A = T_A$. Therefore:

$$[\mathrm{id}_{K^m} \circ T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} = [T_A]_{\mathcal{E}_n}^{\mathcal{E}_m} = A$$

This implies that $I_m \cdot A = A$. The proof for $AI_n = A$ follows the same logic.

Proof of (d): Let $T_A : K^n \to K^m$ and $T_B : K^p \to K^n$ be the linear maps associated with A and B, respectively. We want to show that $\alpha(AB) = (\alpha A)B = A(\alpha B)$. This is equivalent to showing that the corresponding linear maps are equal.

First, consider the map $Q : K^n \to K^n$ defined by $Q(v) := \alpha v$. This map is linear. If $\mathcal{E}_n$ is the standard basis for K^n, then for any $e_j \in \mathcal{E}_n$, $Q(e_j) = \alpha e_j$. Thus, the matrix representation of Q with respect to $\mathcal{E}_n$ is $[Q]_{\mathcal{E}_n}^{\mathcal{E}_n} = \alpha I_n$.

Now, let's look at the compositions:

- The linear map corresponding to $(\alpha A)B$ is $T_{\alpha A} \circ T_B$. We know that $T_{\alpha A}(v) = (\alpha A)v = \alpha(Av) = \alpha T_A(v)$, so $T_{\alpha A} = \alpha T_A$. Thus, $T_{(\alpha A)B} = (\alpha T_A) \circ T_B$.

- The linear map corresponding to $A(\alpha B)$ is $T_A \circ T_{\alpha B}$. Similarly, $T_{\alpha B} = \alpha T_B$. Thus, $T_{A(\alpha B)} = T_A \circ (\alpha T_B)$.

We need to show that $(\alpha T_A) \circ T_B = T_A \circ (\alpha T_B)$. For any $v \in K^p$:

$$\begin{aligned} ((\alpha T_A) \circ T_B)(v) &= (\alpha T_A)(T_B(v)) \\ &= \alpha(T_A(T_B(v))) \quad \text{(by definition of scalar multiplication for maps)} \end{aligned}$$

And for the other side:

$$\begin{aligned} (T_A \circ (\alpha T_B))(v) &= T_A((\alpha T_B)(v)) \\ &= T_A(\alpha(T_B(v))) \quad \text{(by definition of scalar multiplication for maps)} \\ &= \alpha(T_A(T_B(v))) \quad \text{(by linearity of } T_A\text{)} \end{aligned}$$

Since both compositions yield the same result, the corresponding matrices are equal: $(\alpha A)B = A(\alpha B)$.

Finally, to show $\alpha(AB) = (\alpha A)B$: The matrix $\alpha(AB)$ corresponds to the linear map $\alpha(T_A \circ T_B)$. The matrix $(\alpha A)B$ corresponds to the linear map $(\alpha T_A) \circ T_B$. We have already shown that $(\alpha T_A) \circ T_B = \alpha(T_A \circ T_B)$ (from the first part of the derivation above). Therefore, $\alpha(AB) = (\alpha A)B$.

Q.E.D.

Exercise: The proof for statement **(d)** of **??** relies on the properties of scalar multiplication of linear maps. You are encouraged to verify these properties.

Remark: In the space $\mathcal{M}_{n\times n}(K)$, we can add and subtract matrices, and we can also multiply matrices. There is additionally a multiplicative unit I_n. Because these operations satisfy associativity, distributivity, and other relevant axioms, it follows that $\mathcal{M}_{n\times n}(K)$ forms a (non-commutative) ring with a unity.

Important remark: Matrix multiplication is, in general, **not commutative**. There exist matrices A and B such that $A \cdot B \neq B \cdot A$. For example:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$
$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Remark: In the space $\mathcal{M}_{n\times n}(K)$, we can add and subtract matrices, and we can also multiply matrices. There is additionally a multiplicative unit I_n. Because these operations satisfy associativity, distributivity, and other relevant axioms, it follows that $\mathcal{M}_{n\times n}(K)$ forms a (non-commutative) ring with a unity.

Invertible Matrices

Definition 17.a.3 (*Invertibility*): An $n \times n$ matrix $A \in \mathcal{M}_{n\times n}(K)$ is called *"invertible"* if there exists a matrix $B \in \mathcal{M}_{n\times n}(K)$ such that $A \cdot B = B \cdot A = I_n$.

Remark: If A is invertible, then there exists **only one** matrix B such that $A \cdot B = B \cdot A = I_n$.

Proof

Assume there are two matrices B and C such that $AB = BA = I_n$ and $AC = CA = I_n$. Then, we can calculate:

$$B = B \cdot I_n = B \cdot (A \cdot C) = (B \cdot A) \cdot C = I_n \cdot C = C$$

This proves the uniqueness. **Q.E.D.**

Definition 17.a.4 (*Inverse Matrix and the General Linear Group*): If A is invertible, the unique matrix B such that $AB = BA = I_n$ is called the *"inverse"* of A and is denoted by A^{-1}. We have:

$$A \cdot A^{-1} = A^{-1} \cdot A = I_n$$

"The set of all invertible $n \times n$ matrices over K" is denoted by:

$$\mathrm{GL}_n(K) = \mathrm{GL}(n, K) := \{A \in \mathcal{M}_{n\times n}(K) \mid A \text{ is invertible}\}$$

Definition 17.a.5 (*Group*): Let G be a set and $\cdot : G \times G \to G$, $(a, b) \mapsto a \cdot b \in G$, be a binary operation. We say that $(G, \cdot)$ is a **group** if the following axioms hold: Let G be a set and $\cdot : G \times G \to G$, $(a, b) \mapsto a \cdot b \in G$, be a binary operation. We say that $(G, \cdot)$ is a *"group"* if the following axioms hold:

(a) **Associativity:** For all $g_1, g_2, g_3 \in G$, we have $g_1 \cdot (g_2 \cdot g_3) = (g_1 \cdot g_2) \cdot g_3$.

(b) **Identity Element:** There exists an element $e \in G$ such that $e \cdot g = g \cdot e = g$ for all $g \in G$. The element e is called the *"unit"* (or *"identity"*) of G.

(c) **Inverse Element:** For every $g \in G$, there exists an element $h \in G$ such that $g \cdot h = h \cdot g = e$. The element h is called the *"inverse"* of g.

Furthermore, if G satisfies $g_1 \cdot g_2 = g_2 \cdot g_1$ for all $g_1, g_2 \in G$, we say that G is a *"commutative group"* (or *"Abelian group"*).

Proposition 17.a.6 (*Group Properties*): Let G be a group. Then:

(a) The unit $e \in G$ is unique. (The unit e is sometimes denoted as $1 \in G$.)

(b) For every $g \in G$, its inverse is uniquely defined. It is denoted by g^{-1} (so that $g \cdot g^{-1} = g^{-1} \cdot g = e$).

(c) For every $g \in G$, we have $(g^{-1})^{-1} = g$.

(d) For all $g_1, g_2 \in G$, we have $(g_1 \cdot g_2)^{-1} = g_2^{-1} \cdot g_1^{-1}$.

Proposition 17.a.7 (*The General Linear Group*): The set $\mathrm{GL}_n(K)$, endowed with matrix multiplication $(A, B) \mapsto A \cdot B$, is a *"group"*. The unity is the identity matrix I_n. The inverse of a matrix $A \in \mathrm{GL}_n(K)$ is A^{-1}. Moreover, for all $A, B \in \mathrm{GL}_n(K)$, we have:

$$(A^{-1})^{-1} = A \quad \text{and} \quad (AB)^{-1} = B^{-1}A^{-1}$$

Proof

Let $A, B \in \mathrm{GL}_n(K)$. To establish that $\mathrm{GL}_n(K)$ forms a group, we first need to show closure, namely that $AB \in \mathrm{GL}_n(K)$ as well. Indeed, we can verify this by checking the inverse property:

$$(B^{-1} \cdot A^{-1})(AB) = B^{-1}A^{-1}AB = B^{-1}I_nB = B^{-1}B = I_n$$

and, multiplying from the right,

$$(AB)(B^{-1}A^{-1}) = ABB^{-1}A^{-1} = AI_nA^{-1} = AA^{-1} = I_n$$

This proves that AB is invertible with inverse $B^{-1}A^{-1}$, and therefore, $AB \in \mathrm{GL}_n(K)$. The other statements in the proposition follow immediately from what we have proved earlier regarding the uniqueness of inverses. Q.E.D.

Corollary 17.a.8 (*Unique Solution via Matrix Inverse*): Let $A \in \mathrm{GL}_n(K)$. Then, for all $b \in K^n$, the linear system of equations $A \cdot x = b$ has a unique solution, namely $x = A^{-1} \cdot b$.

Proof

If $Ax = b$, then multiplying both sides from the left by A^{-1} yields:

$$A^{-1}(Ax) = A^{-1}b \implies I_nx = A^{-1}b \implies x = A^{-1}b$$

Conversely, if we substitute $x = A^{-1}b$ into the equation, we obtain:

$$A(A^{-1}b) = (AA^{-1})b = I_n b = b$$

This confirms both the existence and uniqueness of the solution. **Q.E.D.**

Proposition 17.a.9 (*Equivalence of Invertibility and Isomorphism*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then, $A \in \mathrm{GL}_n(K)$ (i.e., A is invertible) if and only if the associated linear map $T_A : K^n \to K^n$ is an isomorphism. Moreover, in this case, $(T_A)^{-1} = T_{A^{-1}}$.

Proof

First, assume that A is invertible. We claim that $T_{A^{-1}} \circ T_A = \mathrm{id}_{K^n}$ and $T_A \circ T_{A^{-1}} = \mathrm{id}_{K^n}$, which will show that T_A is an isomorphism. Indeed, for all $v \in K^n$:

$$T_{A^{-1}} \circ T_A(v) = A^{-1} \cdot (T_A(v)) = A^{-1} \cdot A \cdot v = I_n \cdot v = v$$

Similarly, one shows that for all $v \in K^n$:

$$T_A \circ T_{A^{-1}}(v) = I_n \cdot v = v$$

This proves that T_A is an isomorphism.

Conversely, assume that $T_A : K^n \to K^n$ is an isomorphism. Let $S := (T_A)^{-1}$. By a previous result, we know that any linear map from K^n to K^n is given by left-multiplication with a matrix, so $S = T_B$ for some $B \in M_{n\times n}(K)$. Now, for all $w \in K^n$:

$$w = S \circ T_A(w) = T_B(A \cdot w) = B \cdot A \cdot w = (BA) \cdot w$$

We apply this equality for the standard basis vectors $w = e_1, w = e_2, \ldots, w = e_n$ and obtain that $BA = I_n$. Similarly, using $T_A \circ S = \mathrm{id}_{K^n}$, one shows that $AB = I_n$ too. Thus, A is invertible. **Q.E.D.**

Exercise:

(a) Let $A \in \mathcal{M}_{m\times n}(K)$ and $B \in \mathcal{M}_{n\times p}(K)$. Prove that $(AB)^T = B^T \cdot A^T$.
(*Hint: Proving this by direct calculation is not so bad.*)

Proof Sketch

Let $A = (a_{ij})$ and $B = (b_{jk})$. Let $C = AB = (c_{ik})$. Then $c_{ik} = \sum_{\ell=1}^n a_{i\ell} b_{\ell k}$. The (i,k)-th entry of $(AB)^T$ is $c_{ki} = \sum_{\ell=1}^n a_{k\ell} b_{\ell i}$. Now consider $B^T = (b'_{k\ell})$ where $b'_{k\ell} = b_{\ell k}$, and $A^T = (a'_{\ell i})$ where $a'_{\ell i} = a_{i\ell}$. The (i,k)-th entry of $B^T A^T$ is $\sum_{\ell=1}^n b'_{i\ell} a'_{\ell k} = \sum_{\ell=1}^n b_{\ell i} a_{k\ell}$. This is exactly c_{ki}, thus $(AB)^T = B^T A^T$. **Q.E.D.**

(b) Let $A \in \mathrm{GL}_n(K)$. Prove that $A^T \in \mathrm{GL}_n(K)$ and that $(A^T)^{-1} = (A^{-1})^T$.

Proof Sketch

For the first part, recall that $\det(A^T) = \det(A)$. Since $A \in \mathrm{GL}_n(K)$, we know $\det(A) \neq 0$, which implies $\det(A^T) \neq 0$, and thus $A^T \in \mathrm{GL}_n(K)$. For the second part, we need to show that $A^T(A^{-1})^T = I_n$ and $(A^{-1})^T A^T = I_n$. Using the result from part (1):

$$A^T(A^{-1})^T = (A^{-1}A)^T = I_n^T = I_n.$$

Similarly,

$$(A^{-1})^T A^T = (AA^{-1})^T = I_n^T = I_n.$$

This confirms that $(A^{-1})^T$ is indeed the inverse of A^T. **Q.E.D.**

Definition 17.a.10 (*Triangular and Diagonal Matrices*): A matrix $A \in \mathcal{M}_{n\times n}(K)$, with entries $A = (a_{ij})_{\substack{1\le i\le n \\ 1\le j\le n}}$, is called:

- *"Upper triangular"* if $a_{ij} = 0$ for all $i > j$.
- *"Lower triangular"* if $a_{ij} = 0$ for all $i < j$.
- *"Diagonal"* if $a_{ij} = 0$ for all $i \neq j$. A diagonal matrix has the form:

$$\begin{pmatrix} * & & 0 \\ & \ddots & \\ 0 & & * \end{pmatrix}$$

Lemma 17.a.11 (*Closure of Triangular Matrices*): Let A and B be two diagonal matrices (respectively, both upper triangular, or both lower triangular). Then, their product $A \cdot B$ is of the same type.

Proof

Exercise: Let $A = (a_{ij})$ and $B = (b_{ij})$ be two upper triangular matrices. This means $a_{ij} = 0$ for $i > j$ and $b_{ij} = 0$ for $i > j$. Let $C = AB = (c_{ij})$. We want to show that $c_{ij} = 0$ for $i > j$. The entry c_{ij} is given by the sum $c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$.

Consider c_{ij} where $i > j$. For each term $a_{ik} b_{kj}$ in the sum:

- If $k < i$: For a_{ik} to be non-zero, k must be less than or equal to i. If $k > j$, then b_{kj} must be zero (since B is upper triangular).
- If $k \geq i$: Since $i > j$, we have $k > j$. This implies $b_{kj} = 0$ (since B is upper triangular).

More precisely, for c_{ij} with $i > j$, we have $a_{ik} = 0$ for all $k < i$ such that $k < i$ is false (i.e., $k \geq i$), and $b_{kj} = 0$ for all $k > j$. Thus, in the sum $\sum_{k=1}^{n} a_{ik} b_{kj}$, if $k < i$, then a_{ik} can be non-zero. But if $k > j$, b_{kj} is zero. So we only need to consider $k \leq j$. However, if $k \leq j$ and $i > j$, then $i > k$. This means $a_{ik} = 0$ for all $k \leq j$. Therefore, every term $a_{ik} b_{kj}$ in the sum must be zero when $i > j$. Hence, $c_{ij} = 0$ for $i > j$, and C is upper triangular. The proofs for diagonal and lower triangular matrices follow similar logic. **Q.E.D.**

Change of Basis (Section 17.b)

Question: How does the representation matrix $[T]$ of a linear map $T : V \to W$ depend on the choices of bases $\mathcal{B}$ and $\mathcal{C}$?

Corollary 17.b.1 (*Change of Basis Formula*): Let V and W be finite-dimensional vector spaces over K. Let $n := \dim V$ and $m := \dim W$. Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases for V, and let $\mathcal{C}$ and $\mathcal{C}'$ be two bases for W. Then, $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'} \in \mathrm{GL}_n(K)$ and $[\mathrm{id}_W]_{\mathcal{C}}^{\mathcal{C}'} \in \mathrm{GL}_m(K)$, with inverses given by:

$$[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'} = ([\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}})^{-1} \quad \text{and} \quad [\mathrm{id}_W]_{\mathcal{C}}^{\mathcal{C}'} = ([\mathrm{id}_W]_{\mathcal{C}'}^{\mathcal{C}})^{-1}$$

Let $T : V \to W$ be a linear map. Then, its representation matrix changes according to the formula:

$$[T]_{\mathcal{C}'}^{\mathcal{B}'} = [\mathrm{id}_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$$

Proof

We can write T as the composition $T = \mathrm{id}_W \circ T \circ \mathrm{id}_V = \mathrm{id}_W \circ (T \circ \mathrm{id}_V)$. Passing to the matrix representations, this yields:

$$[T]_{\mathcal{C}'}^{\mathcal{B}'} = [\mathrm{id}_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T \circ \mathrm{id}_V]_{\mathcal{C}}^{\mathcal{B}'} = [\mathrm{id}_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$$

Also, we have the identity $\mathrm{id}_V = \mathrm{id}_V \circ \mathrm{id}_V$. This implies that:

$$I_n = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'} \cdot [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}}$$

and similarly:

$$I_n = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}'} = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$$

This demonstrates that $[\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$ is invertible and that its inverse is indeed $([\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}})^{-1}$. The proof for $[\mathrm{id}_W]_{\mathcal{C}}^{\mathcal{C}'}$ follows the exact same logic. **Q.E.D.**

Definition 17.b.2 (*Change of Basis Matrix*): Let V be a finite-dimensional vector space over K, and let $n := \dim V$. Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases of V. The matrix $[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} \in \mathrm{GL}_n(K)$ is called the *"change of basis matrix"* between $\mathcal{B}$ and $\mathcal{B}'$, or the *"transition matrix"* from basis $\mathcal{B}$ to basis $\mathcal{B}'$.

Remark: If $\mathcal{B} = (w_1, \dots, w_n)$ and $\mathcal{B}' = (w_1', \dots, w_n')$, then the transition matrix is explicitly constructed by placing the coordinate vectors of the old basis elements with respect to the new basis into the columns:

$$[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [w_1]_{\mathcal{B}'} & \dots & [w_n]_{\mathcal{B}'} \\ | & & | \end{pmatrix}$$

Proposition 17.b.3 (*Representation of Isomorphisms*): Let $T : V \to W$ be an isomorphism between two finite-dimensional vector spaces. Let $\mathcal{B}$ and $\mathcal{C}$ be bases for V and W, respectively. Then, the matrix $[T]_{\mathcal{C}}^{\mathcal{B}}$ is an invertible matrix, and furthermore:

$$[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} = ([T]_{\mathcal{C}}^{\mathcal{B}})^{-1}$$

Proof

Let $n := \dim V = \dim W$. By the properties of composed linear maps and representation matrices, we have:

$$[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = I_n$$

and conversely:

$$[T]_{\mathcal{C}}^{\mathcal{B}} \cdot [T^{-1}]_{\mathcal{B}}^{\mathcal{C}} = [\mathrm{id}_W]_{\mathcal{C}}^{\mathcal{C}} = I_n$$

This proves the claim. **Q.E.D.**

Equivalence and Similarity of Matrices

Question: Suppose $T : V \to V$ is an endomorphism. Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases for V. What is the relation between $[T]_{\mathcal{B}'}^{\mathcal{B}'}$ and $[T]_{\mathcal{B}}^{\mathcal{B}}$?

Corollary 17.b.4 (*Similarity of Representation Matrices*): Let V be a finite-dimensional vector space, and let $T : V \to V$ be a linear map. Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases for V. Then:

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'} = ([\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$$

Definition 17.b.5 (*Equivalence and Similarity*): We define two types of relations between matrices:

(a) Let $A, B \in \mathcal{M}_{m\times n}(K)$. We say that A and B are *"equivalent"* if there exist matrices $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$ such that $B = PAQ$.

(b) Let $A, B \in \mathcal{M}_{n\times n}(K)$. We say that A and B are *"similar"* if there exists a matrix $P \in \mathrm{GL}_n(K)$ such that $B = P^{-1}AP$.

Exercise:

(a) Show that *"equivalence"* between $m\times n$ matrices defines an equivalence relation on $\mathcal{M}_{m\times n}(K)$.

(b) Show that two matrices $A, B \in \mathcal{M}_{m\times n}(K)$ are equivalent if and only if there exist two vector spaces V and W with $\dim V = n$ and $\dim W = m$, two bases $\mathcal{B}$ and $\mathcal{B}'$ of V, and two bases $\mathcal{C}$ and $\mathcal{C}'$ of W such that:

$$A = [T]_{\mathcal{C}}^{\mathcal{B}} \quad \text{and} \quad B = [T]_{\mathcal{C}'}^{\mathcal{B}'}$$

In other words, A is equivalent to B if and only if A and B are matrix representations of the exact same linear map T, but with respect to different choices of bases for V and W.

(c) Show that *"similarity"* between $n \times n$ matrices defines an equivalence relation on $\mathcal{M}_{n\times n}(K)$.

(d) Show that two square matrices $M, N \in \mathcal{M}_{n\times n}(K)$ are similar if and only if there exists an n-dimensional space V, an endomorphism $T : V \to V$, and two bases $\mathcal{B}$ and $\mathcal{B}'$ of V such that $M = [T]_{\mathcal{B}}^{\mathcal{B}}$ and $N = [T]_{\mathcal{B}'}^{\mathcal{B}'}$.

Proposition 17.b.6 (*Normal Form for Linear Maps*): Let V and W be finite-dimensional vector spaces over K, with dimensions $n = \dim V$ and $m = \dim W$. Let $T : V \to W$ be a linear map with $\operatorname{rank}(T) = r$. (Recall that $\operatorname{rank}(T) := \dim \operatorname{Im}(T)$). Then, there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ of V and a basis $\mathcal{C} = (w_1, \dots, w_m)$ of W such that:

$$[T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} I_r & O_{r,n-r} \\ O_{m-r,r} & O_{m-r,n-r} \end{pmatrix}$$

where $O_{p,q}$ stands for the zero matrix in $M_{p\times q}(K)$.

Proof

Put $\ell := \dim\ker(T)$. By the rank theorem, we know that $r = n - \ell$. Let $v_1, \dots, v_\ell$ be a basis of $\ker(T)$, and extend it to a full basis $(v_1, \dots, v_\ell, v_{\ell+1}, \dots, v_n)$ of V. It will be useful to work below with the rearranged basis:

$$\mathcal{B} = (v_{\ell+1}, \dots, v_n, v_1, \dots, v_\ell)$$

In the proof of the rank theorem, we have seen that $T(v_{\ell+1}), \dots, T(v_n)$ form a basis for $\operatorname{Im}(T)$. Define $w_i := T(v_{\ell+i})$ for $i = 1, \dots, r$ (i.e., $w_1 = T(v_{\ell+1}), \dots, w_r = T(v_{\ell+r})$), and extend this list to a basis of W:

$$\mathcal{C} := (w_1, \dots, w_r, w_{r+1}, \dots, w_m)$$

It now directly follows from these definitions that the representation matrix $[T]_{\mathcal{C}}^{\mathcal{B}}$ takes the exact block form described in the proposition. **Q.E.D.**

Corollary 17.b.7 (*Matrix Equivalence and Normal Form*): Let $A \in \mathcal{M}_{m\times n}(K)$. Then, there exist invertible matrices $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$ such that:

$$PAQ = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix}$$

where $r = \operatorname{rank}_{\mathrm{col}}(A)$. In particular, $\operatorname{rank}_{\mathrm{col}}(A) \le \min\{m, n\}$.

Proof

Consider the linear map $T_A : K^n \to K^m$ defined by left-multiplication, $T_A(v) = A \cdot v$. Then, $r = \operatorname{rank}_{\mathrm{col}}(A) = \dim \operatorname{Im}(T_A)$, because we have previously seen that

$$\operatorname{Im}(T_A) = \operatorname{Sp}(T_A(e_1), \dots, T_A(e_n)) = \operatorname{Sp}(\text{columns of } A).$$

Now, we apply the previous proposition to T_A, yielding that:

$$[T_A]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix}$$

for some basis $\mathcal{B}$ of K^n and some basis $\mathcal{C}$ of K^m. Using the change of basis formula, we can rewrite this as:

$$\underbrace{[\mathrm{id}_{K^m}]_{\mathcal{E}_m}^{\mathcal{C}}}_{=:P} \cdot \underbrace{[T_A]_{\mathcal{E}_m}^{\mathcal{E}_n}}_{=A} \cdot \underbrace{[\mathrm{id}_{K^n}]_{\mathcal{E}_n}^{\mathcal{B}}}_{=:Q} = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix}$$

where $\mathcal{E}_n$ and $\mathcal{E}_m$ are the standard bases of K^n and K^m, respectively. Setting $P = [\mathrm{id}_{K^m}]^{\mathcal{C}}_{\mathcal{E}_m}$ and $Q = [\mathrm{id}_{K^n}]^{\mathcal{B}}_{\mathcal{E}_n}$ concludes the proof. Q.E.D.

Remark: The fact that $\mathrm{rank}_{\mathrm{col}}(A) \leq \min\{m,n\}$ also follows from observing the linear map $T_A : K^n \to K^m$. We have:

$$\mathrm{rank}_{\mathrm{col}}(A) = \dim \mathrm{Im}(T_A) = n - \dim \ker(T_A) \leq n$$

and simultaneously,

$$\mathrm{rank}_{\mathrm{col}}(A) = \dim \mathrm{Im}(T_A) \leq \dim K^m = m$$

Question: Recall that we have an equivalence relation on $M_{m\times n}(K)$: $A \sim B$ if there exist $P \in GL_m(K)$ and $Q \in GL_n(K)$ such that $B = PAQ$. Can we describe the equivalence classes of this equivalence relation?

Corollary 17.b.8 (*Classification of Matrices by Rank*): Let $A, B \in \mathcal{M}_{m\times n}(K)$. Then, $A \sim B$ if and only if $\mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}_{\mathrm{col}}(B)$. Moreover, for every $0 \leq r \leq \min\{m,n\}$, there is precisely one equivalence class of matrices A with $\mathrm{rank}_{\mathrm{col}} = r$.

Proof

Let $A, B \in \mathcal{M}_{m\times n}(K)$. First, assume that $\mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}_{\mathrm{col}}(B)$, and denote this common column rank by r. By **??**, we know that $A \sim \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ and $B \sim \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$. Since $\sim$ is an equivalence relation, it naturally follows that $A \sim B$.

Conversely, assume that $A \sim B$. This implies that there exist matrices $P \in GL_m(K)$ and $Q \in GL_n(K)$ such that $B = PAQ$. Transitioning to the associated linear maps, this means $T_B = T_P \circ T_A \circ T_Q$. We analyze the image of T_B:

$$\mathrm{Im}(T_B) = T_B(K^n) = T_P \circ T_A(T_Q(K^n)) = T_P \circ T_A(K^n) = T_P(\mathrm{Im}(T_A))$$

Note that $T_Q(K^n) = K^n$ because T_Q is an isomorphism. Consequently, the dimension of the image is:

$$\mathrm{rank}_{\mathrm{col}}(B) = \dim \mathrm{Im}(T_B) = \dim T_P(\mathrm{Im}(T_A)) = \dim \mathrm{Im}(T_A) = \mathrm{rank}_{\mathrm{col}}(A)$$

where the second to last equality holds because T_P is an isomorphism, preserving the dimension of subspaces. Q.E.D.

Column Rank and Row Rank (Section 17.c)

Theorem 17.c.1 (*Equality of Column and Row Rank*): Let $A \in \mathcal{M}_{m\times n}(K)$. Then, $\mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}_{\mathrm{row}}(A)$.

Remark (*Notation and Properties of Rank*):

(a) Because of this theorem, we will from now on denote $\mathrm{rank}(A) := \mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}_{\mathrm{row}}(A)$.

(b) Suppose A' is a row-reduced echelon matrix that is row-equivalent to A. Then, we have:

$$\mathrm{rank}(A) = \mathrm{rank}_{\mathrm{row}}(A) = \mathrm{rank}_{\mathrm{row}}(A') = \text{number of pivots in } A'$$

(c) Recall that for a linear map $T : V \to W$, we defined $\mathrm{rank}(T) := \dim \mathrm{Im}(T)$. Moreover, if $A \in \mathcal{M}_{m\times n}(K)$, then $\mathrm{rank}(T_A) = \dim \mathrm{Im}(T_A) = \mathrm{rank}_{\mathrm{col}}(A)$.

Preparation for the Proof

Lemma 17.c.2 (*Rank Invariance under Composition with Isomorphisms*): Let $T : V \to W$ be a linear map, and let $S : W \to W'$ and $L : P \to V$ be isomorphisms. Then:

(a) $\operatorname{rank}(S \circ T) = \operatorname{rank}(T)$.

(b) $\operatorname{rank}(T \circ L) = \operatorname{rank}(T)$.

Proof

Proof of (a): By definition, $\operatorname{rank}(S \circ T) = \dim(S(T(V)))$. Since S is an isomorphism, its restriction to the subspace $T(V) \subseteq W$ is injective. An injective linear map preserves the dimension of a vector space, and therefore:

$$\dim(S(T(V))) = \dim(T(V)) = \operatorname{rank}(T)$$

Proof of (b): By definition, $\operatorname{rank}(T \circ L) = \dim(T(L(P)))$. Since L is an isomorphism, it is surjective, which implies that $L(P) = V$. Consequently:

$$\dim(T(L(P))) = \dim(T(V)) = \operatorname{rank}(T)$$

This completes the proof. **Q.E.D.**

Lemma 17.c.3 (*Invertibility of the Transpose*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then, $A \in \mathrm{GL}_n(K)$ if and only if $A^T \in \mathrm{GL}_n(K)$.

Proof

We know that A is invertible if and only if the associated linear map $T_A : K^n \to K^n$ is an isomorphism. This is true if and only if $\operatorname{rank}(T_A) = n$, which is equivalent to $\operatorname{rank}_{\mathrm{col}}(A) = n$.

We also know that A is row-equivalent to some row-reduced echelon matrix A'. Since row-equivalence corresponds to multiplying from the left by invertible matrices, we have that $A \in GL_n(K)$ if and only if $A' \in GL_n(K)$. Because A' is an $n \times n$ row-reduced echelon matrix, A' is invertible if and only if it has no zero rows. This holds if and only if A' has n pivots, which is equivalent to $\operatorname{rank}_{\mathrm{row}}(A') = n$. Since row operations preserve the row space, this means $\operatorname{rank}_{\mathrm{row}}(A) = n$.

In summary, we have established that $A \in \mathrm{GL}_n(K)$ if and only if $\operatorname{rank}_{\mathrm{row}}(A) = n$. But by definition, the row rank of A is exactly the column rank of its transpose, so $\operatorname{rank}_{\mathrm{row}}(A) = \operatorname{rank}_{\mathrm{col}}(A^T)$. Therefore, $A \in \mathrm{GL}_n(K)$ if and only if $\operatorname{rank}_{\mathrm{col}}(A^T) = n$, which means A^T is invertible. **Q.E.D.**

Lemma 17.c.4 (*Rank Invariance under Matrix Equivalence*): Let $A, B \in \mathcal{M}_{m\times n}(K)$ such that $B = PAQ$, where $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$. Then:

(a) $\operatorname{rank}_{\mathrm{col}}(A) = \operatorname{rank}_{\mathrm{col}}(B)$.

(b) $\operatorname{rank}_{\mathrm{row}}(A) = \operatorname{rank}_{\mathrm{row}}(B)$.

Proof

(a) Transitioning to the associated linear maps, we have $T_B = T_P \circ T_A \circ T_Q$. Since P and Q are invertible matrices, both T_P and T_Q are isomorphisms. Applying **??**, we deduce that $\operatorname{rank}(T_B) = \operatorname{rank}(T_A)$. Recalling that the rank of a linear map given by a matrix matches the column rank of that matrix, we obtain $\operatorname{rank}_{\mathrm{col}}(B) = \operatorname{rank}_{\mathrm{col}}(A)$.

(b) Taking the transpose of the equation $B = PAQ$ yields $B^T = Q^T A^T P^T$. Since P and Q are invertible matrices, their transposes P^T and Q^T are also invertible (we have previously seen that $(P^T)^{-1} = (P^{-1})^T$). We can now apply part (a) of this **??**, which we have already proven, to the matrices A^T and B^T. This gives us $\operatorname{rank}_{\mathrm{col}}(B^T) = \operatorname{rank}_{\mathrm{col}}(A^T)$. By the definition of the transpose, this immediately translates to $\operatorname{rank}_{\mathrm{row}}(B) = \operatorname{rank}_{\mathrm{row}}(A)$. **Q.E.D.**

Proof of the Main Theorem

We are now ready for the proof of **??**.

Proof of ??

Let $r := \mathrm{rank}_{\mathrm{col}}(A)$. By a previous proposition regarding the equivalence of matrices, we know that there exist invertible matrices $P \in GL_m(K)$ and $Q \in GL_n(K)$ such that:

$$PAQ = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix} =: B$$

By **??**, we know that $\mathrm{rank}_{\mathrm{row}}(A) = \mathrm{rank}_{\mathrm{row}}(B)$. Thus, it suffices to determine the row rank of B.

The matrix B is of the block form $\left(\begin{smallmatrix} I_r & O \\ O & O \end{smallmatrix}\right)$. The non-zero rows of B are precisely the first r standard basis vectors $e_1^T, e_2^T, \ldots, e_r^T \in K^n_{\mathrm{row}}$. Since these vectors form a linearly independent set, the dimension of the row space of B is exactly r. Consequently, $\mathrm{rank}_{\mathrm{row}}(B) = r$.

Therefore, we conclude that $\mathrm{rank}_{\mathrm{row}}(A) = r = \mathrm{rank}_{\mathrm{col}}(A)$, completing the proof. **Q.E.D.**

Important remark (*Column Space versus Row Space*): Although $\mathrm{rank}_{\mathrm{col}}(A) = \mathrm{rank}_{\mathrm{row}}(A)$, the column space and the row space are, in general, different vector spaces. First of all, if $A \in \mathcal{M}_{m\times n}(K)$, then the column space is a subspace of K^m, whereas the row space is a subspace of K^n_{row} (which is isomorphic to K^n). But even in the case where $m = n$ (i.e., $A \in \mathcal{M}_{n\times n}(K)$), these two spaces are generally distinct! They share the exact same dimension, but they are not the same space.

More on Matrices and Linear Equations (Section 17.d)

Let $A \in \mathcal{M}_{m\times n}(K)$, and consider the homogeneous system of linear equations $A \cdot x = 0$. We denote the space of solutions of this system by:

$$\mathrm{Sol}(A, 0) := \{x \in K^n \mid Ax = 0\} = \ker(T_A)$$

Note that by the Rank-Nullity Theorem, we have

$$\dim \mathrm{Sol}(A, 0) = n - \mathrm{rank}(A),$$

because $\dim \ker(T_A) = n - \dim \mathrm{Im}(T_A)$.

Let A' be a row-reduced echelon matrix that is row-equivalent to A (for example, A' is obtained from A by Gauss elimination). We then know that the number of pivots of A' equals $\mathrm{rank}(A)$, and the number of free variables is $n - \mathrm{rank}(A)$.

We can visualize the structure of A' and its corresponding variables $x_1, \ldots, x_n$ as follows. The columns containing the leading 1's (pivots) correspond to the pivot variables, while the remaining columns correspond to the free variables. Clearly, the solution spaces are identical:

$$\mathrm{Sol}(A, 0) = \mathrm{Sol}(A', 0)$$

Denote by $x_{i_1}, \ldots, x_{i_\ell}$ the free variables, where $1 \le i_1 < \cdots < i_\ell \le n$ and $\ell = n - \mathrm{rank}(A)$. Similarly, denote by $x_{j_1}, \ldots, x_{j_r}$ the pivot variables, where $1 \le j_1 < \cdots < j_r \le n$ and $r = \mathrm{rank}(A)$.

For each choice of values for the free variables $x_{i_1}, \ldots, x_{i_\ell} \in K$, the pivot variables $x_{j_1}, \ldots, x_{j_r}$ are uniquely determined. So, we obtain a map $\Phi : K^\ell \to \mathrm{Sol}(A, 0)$, where $\ell = n - \mathrm{rank}(A)$, defined by:

$$\Phi(\lambda_1, \ldots, \lambda_\ell) = \text{the unique solution of } Ax = 0 \text{ (which is the solution of } A'x = 0)$$

with $x_{i_1} = \lambda_1, \ldots, x_{i_k} = \lambda_k, \ldots, x_{i_\ell} = \lambda_\ell$.

Proposition 17.d.1 (*Linearity and Isomorphism of the Solution Map*): The map Φ is linear. Moreover, it is an isomorphism.

Proof

Linearity of Φ: The k-th non-zero row of A' gives the following equation:

$$x_{j_k} + (\text{linear combination of the free variables that come after } j_k) = 0$$

This allows us to isolate the pivot variable x_{j_k}:

$$x_{j_k} = -\sum_{\{q \mid 1 \le q \le \ell,\, i_q > j_k\}} a'_{k,i_q} \cdot x_{i_q}$$

This explicitly shows that the pivot variables $x_{j_1}, \dots, x_{j_r}$ depend linearly on the free variables $x_{i_1}, \dots, x_{i_\ell}$. This proves the linearity of Φ.

Isomorphism: To prove that Φ is an isomorphism, it is enough to show that $\ker(\Phi) = \{0\}$, because $\Phi : K^\ell \to \mathrm{Sol}(A, 0)$ is a linear map between two spaces of the exact same dimension ℓ.

But this clearly holds: if $\lambda = (\lambda_1, \dots, \lambda_\ell) \in \ker(\Phi)$, which means $\Phi(\lambda_1, \dots, \lambda_\ell) = 0$, then by the definition of Φ, we must have $\lambda_1 = \cdots = \lambda_\ell = 0$. This is simply because each λ_k is an actual entry in the output vector $\Phi(\lambda_1, \dots, \lambda_\ell)$.

Furthermore, we can explicitly state the inverse map Φ^{-1}. For any $x \in \mathrm{Sol}(A, 0)$, we extract the entries corresponding to the free variables:

$$\Phi^{-1}(x) = \begin{pmatrix} x_{i_1} \\ \vdots \\ x_{i_\ell} \end{pmatrix}$$

This completes the proof. **Q.E.D.**

Corollary 17.d.2 (*Structure of Non-Homogeneous Solutions*): Let $b \in K^m$, and $A \in \mathcal{M}_{m \times n}(K)$. Then:

(a) $\mathrm{Sol}(A, b) := \{x \in K^n \mid A \cdot x = b\} \neq \emptyset$ if and only if $b \in \mathrm{ColS}(A)$.

(b) If $\mathrm{Sol}(A, b) \neq \emptyset$ and $y \in \mathrm{Sol}(A, b)$, then:

$$\mathrm{Sol}(A, b) = y + \mathrm{Sol}(A, 0) := \{y + x \mid x \in \mathrm{Sol}(A, 0)\}$$

Proof

This is left as an exercise.

Hints: 1) Write the matrix A in terms of its column vectors $v_1, \dots, v_n$:

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$$

For any vector $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in K^n$, the matrix-vector product is a linear combination of the columns:

$$A \cdot x = x_1 \begin{pmatrix} | \\ v_1 \\ | \end{pmatrix} + \cdots + x_n \begin{pmatrix} | \\ v_n \\ | \end{pmatrix}$$

2) Let $y \in \mathrm{Sol}(A, b)$. First, assume $x \in \mathrm{Sol}(A, 0)$. Then $A \cdot x = 0$ and $A \cdot y = b$. Adding these together yields:

$$A(y + x) = A \cdot y + A \cdot x = b + 0 = b$$

Conversely, if $z \in \mathrm{Sol}(A,b)$, meaning $A \cdot z = b$, then define $x := z - y$. We obviously have $z = y + x$, and checking the multiplication gives:

$$A \cdot x = A \cdot (z - y) = A \cdot z - A \cdot y = b - b = 0$$

This shows that any solution z can be written as $y + x$ for some $x \in \mathrm{Sol}(A, 0)$. **Q.E.D.**

Proposition 17.d.3 (*Rank Criterion for Solvability*): Let $A \in \mathcal{M}_{m\times n}(K)$ and $b \in K^m$. The following statements are equivalent:

(a) $\mathrm{rank}(A|b) = \mathrm{rank}(A)$.

(b) The system of equations $Ax = b$ has a solution.

Proof

Write A in terms of its columns, $A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$. We have the following chain of equivalences:

$$\begin{aligned} \mathrm{rank}(A|b) = \mathrm{rank}(A) &\iff \text{col-rank}(A|b) = \text{col-rank}(A) \\ &\iff \dim \mathrm{Sp}(v_1, \dots, v_n, b) = \dim \mathrm{Sp}(v_1, \dots, v_n) \\ &\iff b \in \mathrm{Sp}(v_1, \dots, v_n) \\ &\iff b \in \mathrm{ColS}(A) \\ &\iff \mathrm{Sol}(A,b) \neq \emptyset \end{aligned}$$

The last step holds directly by **??**, thus finishing the proof. **Q.E.D.**

Proposition 17.d.4 (*Criterion for a Unique Solution*): Let $A \in \mathcal{M}_{m\times n}(K)$ and $b \in K^m$. The following statements are equivalent:

(a) $\mathrm{rank}(A) = \mathrm{rank}(A|b) = n$.

(b) The system $Ax = b$ has a unique solution.

Proof

This is left as an exercise.

Hints: Write $A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$. Note that:

$$\mathrm{rank}(A) = n \iff \dim \mathrm{Sp}(v_1, \dots, v_n) = n$$

Since there are n column vectors, this means that $v_1, \dots, v_n$ are linearly independent, and therefore, they form a basis for $\mathrm{ColS}(A)$. **Q.E.D.**

Elementary Row Operations Revisited (Section 17.e)

Let $A \in \mathcal{M}_{n\times p}(K)$. We have previously learned about elementary row operations, which are utilized as an integral part of the Gaussian elimination algorithm. The main point of this section is to demonstrate that all of these operations can be expressed through matrix multiplications.
Notation: Let $n \in \mathbb{Z}_{\geq 1}$, and let $1 \leq i \leq n$, $1 \leq j \leq n$. We denote by $R_{ij} \in \mathcal{M}_{n\times n}(K)$ the matrix whose entries are all zero, except for the entry in the i-th row and j-th column, which is equal to 1.

We will now define three types of elementary matrices in $\mathcal{M}_{n\times n}(K)$.

Definition 17.e.1 (*Elementary Matrices*): We define the following three types of elementary matrices:

(a) **Type I:** Let $1 \leq i \leq n$, $1 \leq j \leq n$ with $i \neq j$, and let $\alpha \in K$. We define $Q_{ij}(\alpha)$ to be the matrix obtained from I_n by applying the row operation $R_i + \alpha R_j \longrightarrow R_i$ (i.e., replacing the i-th row by the sum of the i-th row and α times the j-th row). Explicitly, we have:

$$Q_{ij}(\alpha) = I_n + \alpha \cdot R_{ij}$$

(b) **Type II:** Let $1 \leq i \leq n$, $1 \leq j \leq n$ with $i \neq j$. We define P_{ij} to be the matrix obtained from I_n by permuting (exchanging) the i-th row and the j-th row, denoted by $R_i \leftrightarrow R_j$.

(c) **Type III:** Let $1 \leq i \leq n$, and let $0 \neq \alpha \in K$. We denote by $S_i(\alpha)$ the matrix obtained from I_n by applying the row operation $\alpha R_i \longrightarrow R_i$ (i.e., multiplying the i-th row by the non-zero scalar α).

Exercise: Show that the Type II elementary matrix can be written explicitly as $P_{ij} = I_n - R_{ii} - R_{jj} + R_{ij} + R_{ji}$.

Example: Consider the space $\mathcal{M}_{4\times 4}(K)$. Examples of the three types of elementary matrices are:

$$Q_{1,3}(\alpha) = \begin{pmatrix} 1 & 0 & \alpha & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$P_{3,4} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$S_3(\alpha) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \alpha & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Lemma 17.e.2 (*Matrix Multiplication and Row Operations*): Let $A \in \mathcal{M}_{n\times p}(K)$. Multiplying A from the left by an elementary matrix performs the corresponding elementary row operation on A:

(a) Multiplying A from the left with $Q_{ij}(\alpha)$ results in the row operation $R_i + \alpha R_j \longrightarrow R_i$:

$$A \xrightarrow{R_i + \alpha R_j \longrightarrow R_i} Q_{ij}(\alpha) \cdot A$$

(b) Multiplying A from the left with P_{ij} results in the row operation $R_i \leftrightarrow R_j$:

$$A \xrightarrow{R_i \leftrightarrow R_j} P_{ij} \cdot A$$

(c) Multiplying A from the left with $S_i(\alpha)$ results in the row operation $\alpha R_i \longrightarrow R_i$:

$$A \xrightarrow{\alpha R_i \longrightarrow R_i} S_i(\alpha) \cdot A$$

Important remark (*Column Operations*): The same principle holds true for elementary column operations, but via right-multiplication:

(a') Multiplying A from the right with $Q_{ij}(\alpha)$ results in the column operation $C_j + \alpha C_i \longrightarrow C_j$:

$$A \xrightarrow{C_j + \alpha C_i \longrightarrow C_j} A \cdot Q_{ij}(\alpha)$$

(b') Multiplying A from the right with P_{ij} results in the column operation $C_i \leftrightarrow C_j$:

$$A \xrightarrow{C_i \leftrightarrow C_j} A \cdot P_{ij}$$

(c') Multiplying A from the right with $S_i(\alpha)$ results in the column operation $\alpha C_i \longrightarrow C_i$:

$$A \xrightarrow{\alpha C_i \longrightarrow C_i} A \cdot S_i(\alpha)$$

Proof

Statements (a), (b), and (c) can be proven directly by calculating the matrix multiplications (you are encouraged to check this!).

Statements (a'), (b'), and (c') can be deduced from the row operation versions by passing to the transposed matrix. For example, to prove (a'):

$$(A \cdot Q_{ij}(\alpha))^T = Q_{ij}(\alpha)^T \cdot A^T = Q_{ji}(\alpha) \cdot A^T$$

By part (a), this is exactly the matrix obtained from A^T after applying to it the row operation $R_j + \alpha R_i \longrightarrow R_j$. But taking the transpose again, this is equal to:

$$(\text{matrix obtained from } A \text{ after applying } C_j + \alpha C_i \longrightarrow C_j)^T$$

Taking the transpose of both sides, it follows that $A \cdot Q_{ij}(\alpha)$ is the matrix obtained from A after the column operation $C_j + \alpha C_i \longrightarrow C_j$. The proofs for (b') and (c') follow a similar logic. **Q.E.D.**

Lemma 17.e.3 (*Inverses of Elementary Matrices*): Every elementary matrix is invertible, and the inverse of each is an elementary matrix of the exact same type:

$$Q_{ij}(\alpha)^{-1} = Q_{ij}(-\alpha), \quad P_{ij}^{-1} = P_{ij}, \quad \text{and} \quad S_i(\alpha)^{-1} = S_i(\alpha^{-1})$$

Proof

Exercise. (*Hint:* Note that $P_{ij} \cdot P_{ij}$ is the matrix obtained from P_{ij} by exchanging rows i and j, which brings us immediately back to I_n. Apply similar logical steps for the other types.)

Q.E.D.

Theorem 17.e.4 (*Decomposition into Elementary Matrices*): For every invertible matrix $A \in \mathrm{GL}_n(K)$, there exists an integer $k \geq 1$ and a sequence of elementary matrices $T_1, \ldots, T_k$ such that:

$$T_k \cdot T_{k-1} \ldots T_1 \cdot A = I_n$$

Furthermore, we have $A = T_1^{-1} \ldots T_k^{-1}$ and $A^{-1} = T_k \ldots T_1$. In particular, any invertible matrix A can be written as a finite product of elementary matrices.

Proof

We already know that every matrix $A \in \mathcal{M}_{n \times n}(K)$ can be brought to a row-reduced echelon matrix, which we will call A', using the Gaussian elimination algorithm.

In our case, A is invertible, and hence $\operatorname{rank}(A) = n$. Because row operations do not change the rank, we have $\operatorname{rank}(A') = \operatorname{rank}(A)$, which implies that $\operatorname{rank}(A') = n$. The only $n \times n$ row-reduced echelon matrix with rank n is the identity matrix, meaning $A' = I_n$.

Since each step in the Gaussian elimination algorithm corresponds to an elementary operation, we can deduce from Lemma **??** that there exist elementary matrices $T_1, \ldots, T_k$ (of types I, II, and III) such that:

$$T_k \cdot T_{k-1} \ldots T_1 \cdot A = I_n$$

By left-multiplying successively with the inverses of these elementary matrices, we isolate A and deduce that $A = T_1^{-1} \ldots T_k^{-1}$. Taking the inverse of A yields $A^{-1} = T_k \ldots T_1$. **Q.E.D.**

Theorem 17.e.5 (*Matrix Inversion Algorithm*): Let $A \in \mathcal{M}_{n \times n}(K)$. Construct the augmented matrix by placing A alongside I_n, denoted as $(A \mid I_n)$. Then, A is invertible if and only if $(A \mid I_n)$ can be brought, via a sequence of elementary row operations, to the form $(I_n \mid B)$ for some matrix $B \in \mathcal{M}_{n \times n}(K)$. Furthermore, if this is the case, we have $A^{-1} = B$.

Before proving this theorem, we require an intermediate technical result.

Lemma 17.e.6 (*Block Matrix Multiplication*): Let $B, C, D \in \mathcal{M}_{n\times n}(K)$. Then:

$$B \cdot (C \mid D) = (B \cdot C \mid B \cdot D)$$

Proof

This follows directly from the definition of matrix multiplication. Check the explicit details as an exercise! This follows directly from the definition of matrix multiplication. (Hint: Consider the (k,l)-th entry of $B \cdot (C \mid D)$ and show it matches the (k,l)-th entry of $(B \cdot C \mid B \cdot D)$.) Q.E.D.

Proof of ??

Assume that A is invertible. By **??**, there exist elementary matrices $T_1, \dots, T_k$ such that $T_k \dots T_1 \cdot A = I_n$. Using **??**, we can multiply the augmented matrix $(A \mid I_n)$ from the left by this chain of elementary matrices:

$$T_k \dots T_1 \cdot (A \mid I_n) = (T_k \dots T_1 \cdot A \mid T_k \dots T_1 \cdot I_n) = (I_n \mid T_k \dots T_1)$$

Let us define $B := T_k \dots T_1$. We can clearly see that:

$$B \cdot A = (T_k \dots T_1) \cdot A = I_n$$

and using the decomposition of A provided by **??**,

$$A \cdot B = (T_1^{-1} \dots T_k^{-1}) \cdot (T_k \dots T_1) = I_n$$

This explicitly implies that $A^{-1} = B$.

Conversely, assume that $(A \mid I_n)$ can be brought to $(I_n \mid B)$ by a sequence of elementary row operations, for some matrix $B \in M_{n\times n}(K)$. This directly implies that the matrix A itself is row-equivalent to I_n. Consequently, A must have rank n, which inherently means that A is invertible. This concludes the proof. Q.E.D.

Example: Let us find the inverse of the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in \mathcal{M}_{2\times 2}(\mathbb{Q})$. We place A and I_2 together in an augmented matrix and apply Gaussian elimination:

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right) \xrightarrow{R_2 - 3R_1 \longrightarrow R_2} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array}\right)$$

Next, we scale the second row to create a pivot of 1:

$$\xrightarrow{-\frac{1}{2}R_2 \longrightarrow R_2} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right)$$

Finally, we eliminate the entry above the second pivot:

$$\xrightarrow{R_1 - 2R_2 \longrightarrow R_1} \left(\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right)$$

The left side is now the identity matrix I_2, which means the right side is our inverse matrix. We conclude that:

$$A^{-1} = \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$$

Constructing New Vector Spaces Out of Old Ones (Chapter 18)

The Homomorphism Space, The Dual Space And The Dual Map (Section 18.a)

The Homomorphism Space

Proposition 18.a.1 (*The Vector Space Structure of the Homomorphism Space*): Let V and W be two vector spaces over K. Then, $\mathrm{Hom}(V, W)$ has the structure of a vector space over K, if we endow it with the following operations:

- For all $T_1, T_2 \in \mathrm{Hom}(V, W)$, $(T_1 + T_2)(v) := T_1(v) + T_2(v)$ for all $v \in V$.
- For all $T \in \mathrm{Hom}(V, W)$ and $\alpha \in K$, $(\alpha \cdot T)(v) := \alpha \cdot T(v)$ for all $v \in V$.

Proof

We claim that $T_1 + T_2 \in \mathrm{Hom}(V, W)$, meaning $T_1 + T_2$ is a linear map. Indeed, for a scalar a and a vector v, we have

$$\begin{aligned}(T_1 + T_2)(a \cdot v) &= T_1(a \cdot v) + T_2(a \cdot v) = a \cdot T_1(v) + a \cdot T_2(v) \\ &= a \cdot (T_1(v) + T_2(v)) = a \cdot (T_1 + T_2)(v),\end{aligned}$$

and for vectors v_1, v_2, we have

$$\begin{aligned}(T_1 + T_2)(v_1 + v_2) &= T_1(v_1 + v_2) + T_2(v_1 + v_2) = T_1(v_1) + T_1(v_2) + T_2(v_1) + T_2(v_2) \\ &= (T_1 + T_2)(v_1) + (T_1 + T_2)(v_2).\end{aligned}$$

Consequently, $T_1 + T_2$ is a linear map. The proof that $\alpha \cdot T$ is linear is analogous.

So far, we have defined $+\colon \mathrm{Hom}(V, W) \times \mathrm{Hom}(V, W) \to \mathrm{Hom}(V, W)$ and $\cdot\colon K \times \mathrm{Hom}(V, W) \to \mathrm{Hom}(V, W)$.

Regarding the neutral element, the map $0\colon V \to W$ defined by $v \mapsto 0$ for all $v \in V$ is linear. We claim that $0 + T = T$. Indeed,

$$(0 + T)(v) = 0(v) + T(v) = 0 + T(v) = T(v)$$

for all $v \in V$. Similarly, one shows that $T + 0 = T$. **Q.E.D.**

Exercise: Complete the proof that $\mathrm{Hom}(V, W)$ is a vector space by checking that it satisfies all the axioms of a vector space, using the zero map $0\colon V \to W$ as the additive identity.

For example, consider the distributivity axiom. For all $\alpha \in K$ and $T_1, T_2 \in \mathrm{Hom}(V, W)$, we have $\alpha \cdot (T_1 + T_2) = \alpha \cdot T_1 + \alpha \cdot T_2$. Indeed, for $v \in V$,

$$\begin{aligned}(\alpha \cdot (T_1 + T_2))(v) &= \alpha \cdot (T_1 + T_2)(v) = \alpha(T_1(v) + T_2(v)) \\ &= \alpha \cdot T_1(v) + \alpha \cdot T_2(v) = (\alpha \cdot T_1)(v) + (\alpha \cdot T_2)(v) \\ &= (\alpha \cdot T_1 + \alpha \cdot T_2)(v).\end{aligned}$$

Therefore, $\alpha \cdot (T_1 + T_2) = \alpha \cdot T_1 + \alpha \cdot T_2$.

Theorem 18.a.2 (*The Canonical Isomorphism from the Homomorphism Space* $\mathrm{Hom}(V, W)$ *to the Matrix Space* $\mathcal{M}_{m\times n}(K)$): Let V and W be finite-dimensional vector spaces over K, let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis for V, and let $\mathcal{C} = (w_1, \ldots, w_m)$ be a basis for W. Then, $\Psi_{\mathcal{C}}^{\mathcal{B}}\colon \mathrm{Hom}(V, W) \to$

$\mathcal{M}_{m\times n}(K)$, defined by

$$\Psi^{\mathcal{B}}_{\mathcal{C}}(T) := [T]^{\mathcal{B}}_{\mathcal{C}} = \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \dots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix},$$

is a linear map, and moreover, an isomorphism.

Proof

Recall that $T_{[T]^{\mathcal{B}}_{\mathcal{C}}} = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}$. Recall also that $\Phi_{\mathcal{B}}(v) = [v]_{\mathcal{B}} \in K^n$ for all $v \in V$. Furthermore, $\Phi_{\mathcal{C}}(w) = [w]_{\mathcal{C}} \in K^m$ for all $w \in W$. We have

$$\Phi_{\mathcal{B}}^{-1}\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = a_1 v_1 + \dots + a_n v_n, \quad \Phi_{\mathcal{C}}^{-1}\begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} = b_1 w_1 + \dots + b_m w_m.$$

We claim that the map $\Psi^{\mathcal{B}}_{\mathcal{C}}$ is bijective. Indeed, define $\Theta\colon \mathcal{M}_{m\times n}(K) \to \mathrm{Hom}(V,W)$ by $\Theta(A) := \Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}} \in \mathrm{Hom}(V,W)$ for $A \in \mathcal{M}_{m\times n}(K)$. We have:

$$\begin{aligned} \Psi^{\mathcal{B}}_{\mathcal{C}} \circ \Theta(A) &= \Psi^{\mathcal{B}}_{\mathcal{C}}(\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}}) \\ &= \big([\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}}(v_1)]_{\mathcal{C}} \dots [\Phi_{\mathcal{C}}^{-1} \circ T_A \circ \Phi_{\mathcal{B}}(v_n)]_{\mathcal{C}}\big) \\ &= \begin{pmatrix} | & & | \\ T_A\Phi_{\mathcal{B}}(v_1) & \dots & T_A\Phi_{\mathcal{B}}(v_n) \\ | & & | \end{pmatrix} \\ &= \begin{pmatrix} | & & | \\ T_A(e_1) & \dots & T_A(e_n) \\ | & & | \end{pmatrix} = A. \end{aligned}$$

This uses the facts that $[\Phi_{\mathcal{C}}^{-1}(b)]_{\mathcal{C}} = b$ for all $b \in K^m$, and $\Phi_{\mathcal{B}}(v_k) = e_k$. This proves that $\Psi^{\mathcal{B}}_{\mathcal{C}} \circ \Theta = \mathrm{id}_{\mathcal{M}_{m\times n}(K)}$.

Now, by definition, $\Theta \circ \Psi^{\mathcal{B}}_{\mathcal{C}}(T) = \Phi_{\mathcal{C}}^{-1} \circ T_{\Psi^{\mathcal{B}}_{\mathcal{C}}(T)} \circ \Phi_{\mathcal{B}} = \Phi_{\mathcal{C}}^{-1} \circ T_{[T]^{\mathcal{B}}_{\mathcal{C}}} \circ \Phi_{\mathcal{B}} = T$. This shows that $\Theta \circ \Psi^{\mathcal{B}}_{\mathcal{C}} = \mathrm{id}_{\mathrm{Hom}(V,W)}$. Therefore, $\Psi^{\mathcal{B}}_{\mathcal{C}}$ is bijective.

It remains to prove that $\Psi^{\mathcal{B}}_{\mathcal{C}}$ is linear. Indeed, let $\alpha \in K$, and let $T \in \mathrm{Hom}(V,W)$. Then,

$$\begin{aligned} \Psi^{\mathcal{B}}_{\mathcal{C}}(\alpha \cdot T) &= ([(\alpha T)(v_1)]_{\mathcal{C}} \dots [(\alpha T)(v_n)]_{\mathcal{C}}) \\ &= ([\alpha T(v_1)]_{\mathcal{C}} \dots [\alpha T(v_n)]_{\mathcal{C}}) \\ &= (\alpha[T(v_1)]_{\mathcal{C}} \dots \alpha[T(v_n)]_{\mathcal{C}}) \\ &= \alpha \cdot \begin{pmatrix} | & & | \\ [T(v_1)]_{\mathcal{C}} & \dots & [T(v_n)]_{\mathcal{C}} \\ | & & | \end{pmatrix} \\ &= \alpha \cdot \Psi^{\mathcal{B}}_{\mathcal{C}}(T). \end{aligned}$$

Similarly, one shows that $\Psi^{\mathcal{B}}_{\mathcal{C}}(T_1 + T_2) = \Psi^{\mathcal{B}}_{\mathcal{C}}(T_1) + \Psi^{\mathcal{B}}_{\mathcal{C}}(T_2)$. Q.E.D.

Corollary 18.a.3 (*Dimensionality of the Homomorphism Space*): If V and W are finite-dimensional vector spaces, then the homomorphism space $\mathrm{Hom}(V,W)$ is also finite-dimensional, and $\dim \mathrm{Hom}(V,W) = \dim V \cdot \dim W$.

Proof

Let $n := \dim V$ and $m := \dim W$. By **??**, we have an isomorphism

$$\mathrm{Hom}(V,W) \cong \mathcal{M}_{m\times n}(K).$$

This means $\operatorname{Hom}(V, W)$ is a finite-dimensional space. Furthermore,

$$\dim \operatorname{Hom}(V, W) = \dim \mathcal{M}_{m\times n}(K) = m \cdot n.$$

Q.E.D.

Rings of Endomorphisms

Example:

(a) Every field K is also a commutative ring.

(b) $\mathbb{Z}$ is a commutative ring.

(c) $\mathcal{M}_{n\times n}(K)$ is a ring, where $A \cdot B$ is matrix multiplication, and the unity is I_n. Clearly, it is not commutative.

Corollary 18.a.4 (*The Ring of Endomorphisms*): Let V be a vector space over K. Then, $\operatorname{End}(V) := \operatorname{Hom}(V, V)$ is a ring with a unity. This ring is, in general, not commutative. The multiplication in $\operatorname{End}(V)$ is given by composition $T_1 \cdot T_2 := T_1 \circ T_2$ for all $T_1, T_2 \in \operatorname{Hom}(V, V)$, and the unity is id_V. Moreover, if V is finite-dimensional with $\dim V = n$, and $\mathcal{B} = (w_1, \ldots, w_n)$ is a basis for V, then $\Psi^{\mathcal{B}}_{\mathcal{B}} \colon \operatorname{Hom}(V, V) \to \mathcal{M}_{n\times n}(K)$ is an isomorphism of rings.

Proof (sketch)

The multiplication $T_1 \cdot T_2 := T_1 \circ T_2$ is associative, since $(T_1 \cdot T_2) \cdot T_3 = (T_1 \circ T_2) \circ T_3 = T_1 \circ (T_2 \circ T_3) = T_1 \cdot (T_2 \cdot T_3)$. Also, $T \cdot \mathrm{id}_V = T \circ \mathrm{id}_V = T$, and $\mathrm{id}_V \cdot T = \mathrm{id}_V \circ T = T$. Furthermore, $T \cdot (T_1 + T_2) = T \circ (T_1 + T_2) = T \circ T_1 + T \circ T_2$.

Now, if V is finite-dimensional, and $\mathcal{B}$ is a basis for V, then

$$\Psi^{\mathcal{B}}_{\mathcal{B}}(T_1 \circ T_2) = [T_1 \circ T_2]^{\mathcal{B}}_{\mathcal{B}} = [T_1]^{\mathcal{B}}_{\mathcal{B}} \cdot [T_2]^{\mathcal{B}}_{\mathcal{B}} = \Psi^{\mathcal{B}}_{\mathcal{B}}(T_1) \cdot \Psi^{\mathcal{B}}_{\mathcal{B}}(T_2)$$

via matrix multiplication. Q.E.D.

Proof

First, we verify that $(\operatorname{End}(V), +, \circ)$ satisfies the ring axioms. We already know that $(\operatorname{End}(V), +)$ is an abelian group from the proposition regarding the vector space structure of $\operatorname{Hom}(V, W)$. We now verify the properties related to multiplication (composition):

(a) **Associativity of Multiplication**: For any $T_1, T_2, T_3 \in \operatorname{End}(V)$, function composition is inherently associative. That is, $(T_1 \circ T_2) \circ T_3 = T_1 \circ (T_2 \circ T_3)$.

(b) **Existence of Multiplicative Identity**: The identity map $\mathrm{id}_V \colon V \to V$ is an endomorphism. For any $T \in \operatorname{End}(V)$, we have $T \circ \mathrm{id}_V = T$ and $\mathrm{id}_V \circ T = T$. Thus, id_V is the multiplicative identity.

(c) **Distributivity**: For any $T, T_1, T_2 \in \operatorname{End}(V)$, we must show both left and right distributivity.

- **Left Distributivity**: For any $v \in V$,

  $$(T \circ (T_1 + T_2))(v) = T((T_1 + T_2)(v)) = T(T_1(v) + T_2(v)).$$

  By the linearity of T, this equals $T(T_1(v)) + T(T_2(v)) = (T \circ T_1)(v) + (T \circ T_2)(v) = (T \circ T_1 + T \circ T_2)(v)$. Hence, $T \circ (T_1 + T_2) = T \circ T_1 + T \circ T_2$.

- **Right Distributivity**: For any $v \in V$,

  $$\begin{aligned} ((T_1 + T_2) \circ T)(v) &= (T_1 + T_2)(T(v)) \\ &= T_1(T(v)) + T_2(T(v)) \\ &= (T_1 \circ T)(v) + (T_2 \circ T)(v) \\ &= (T_1 \circ T + T_2 \circ T)(v). \end{aligned}$$

  Hence, $(T_1 + T_2) \circ T = T_1 \circ T + T_2 \circ T$.

Thus, $\mathrm{End}(V)$ is a ring with unity id_V. It is generally not commutative, as demonstrated by matrix multiplication.

Now, assume V is finite-dimensional with $\dim V = n$, and let $\mathcal{B}$ be a basis for V. We have already established that $\Psi_{\mathcal{B}}^{\mathcal{B}}\colon \mathrm{End}(V) \to \mathcal{M}_{n\times n}(K)$ is a linear isomorphism. To prove it is a ring isomorphism, we must show it preserves the multiplicative structure and the multiplicative identity.

(a) **Preservation of Multiplication**: For any $T_1, T_2 \in \mathrm{End}(V)$, a fundamental result from Linear Algebra I states that the matrix representation of a composition of linear maps is the product of their matrix representations. That is,
$$\Psi_{\mathcal{B}}^{\mathcal{B}}(T_1 \circ T_2) = [T_1 \circ T_2]_{\mathcal{B}}^{\mathcal{B}} = [T_1]_{\mathcal{B}}^{\mathcal{B}} \cdot [T_2]_{\mathcal{B}}^{\mathcal{B}} = \Psi_{\mathcal{B}}^{\mathcal{B}}(T_1) \cdot \Psi_{\mathcal{B}}^{\mathcal{B}}(T_2).$$

(b) **Preservation of Multiplicative Identity**: The matrix representation of the identity endomorphism with respect to any basis $\mathcal{B}$ is the identity matrix I_n.
$$\Psi_{\mathcal{B}}^{\mathcal{B}}(\mathrm{id}_V) = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = I_n.$$

Since $\Psi_{\mathcal{B}}^{\mathcal{B}}$ is a bijective linear map that preserves both multiplication and the multiplicative identity, it is an isomorphism of rings. Q.E.D.

The Dual Space

An important special case of $\mathrm{Hom}(V, W)$ is when $W = K$.

Definition 18.a.5 (*The Dual Space and Linear Functionals*)**:** Let V be a vector space over K. The *"dual space"* of V is defined to be the vector space $V^* := \mathrm{Hom}(V, K)$. The elements of V^* are linear maps $\ell\colon V \to K$, and are sometimes called *"functionals"*.

Corollary 18.a.6 (*Dimensionality of the Dual Space*)**:** Let V be a finite-dimensional vector space over K. Then, $\dim V^* = \dim V$.

Proof

We have $\dim V^* = \dim \mathrm{Hom}(V, K) = \dim V \cdot \dim K = \dim V \cdot 1 = \dim V$. Q.E.D.

Let $V = K^n_{\mathrm{col}}$. Then, we can identify V^* with K^n_{row} as follows: If $\ell \in K^n_{\mathrm{row}}$, then ℓ defines a functional $f_\ell\colon K^n_{\mathrm{col}} \to K$ by $f_\ell(v) := \ell \cdot v \in \mathcal{M}_{1\times 1}(K) \cong K$. The map $D\colon K^n_{\mathrm{row}} \to (K^n_{\mathrm{col}})^*$ defined by $D(\ell) := f_\ell$ is a linear map, and therefore, it is an isomorphism.

Let V be a finite-dimensional vector space over K, and let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V. For $1 \le i \le n$, define a functional $v_i^* \in V^*$ as follows: $v_i^*(v_j) := \delta_{ij}$. This means that $v_i^*(v_i) = 1$, and $v_i^*(v_j) = 0$ for all $j \ne i$. Since $v_1, \dots, v_n$ form a basis for V, our definition determines uniquely a well-defined functional $v_i^*\colon V \to K$.

Lemma 18.a.7 (*Construction of the Dual Basis*)**:** The elements $v_1^*, \dots, v_n^* \in V^*$ form a basis for V^*. This basis is called the *"dual basis"* to $\mathcal{B}$. We denote this basis by $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$. Moreover, for all $\ell \in V^*$, $\ell = \sum_{i=1}^n \ell(v_i) \cdot v_i^*$.

Proof

Let $\ell \in V^*$. We claim that $\ell = \sum_{i=1}^n \ell(v_i) \cdot v_i^*$, or in other words, we claim that $\ell(v) = \sum_{i=1}^n \ell(v_i) \cdot v_i^*(v)$. Indeed, both sides of the equation are linear maps $V \to K$, and therefore, it is enough to check that it holds for $v = v_1, v = v_2, \dots, v = v_n$, because $v_1, \dots, v_n$ form a basis for V. And indeed, for $v = v_k$, we have:
$$\sum_{i=1}^n \ell(v_i) \cdot v_i^*(v_k) = \sum_{i=1}^n \ell(v_i) \cdot \delta_{ik} = \ell(v_k).$$
The claim just proven shows that $v_1^*, \dots, v_n^*$ span V^*. Since $\dim V^* = \dim V = n$, it follows that $v_1^*, \dots, v_n^*$ form a basis for V^*. Q.E.D.

Question: Let V be a finite-dimensional vector space over K, and let $\mathcal{B}$ and $\mathcal{C}$ be two bases for V. $\mathcal{B}^*$ and $\mathcal{C}^*$ are two bases for V^*. What is the relation between $[\mathrm{id}_{V^*}]^{\mathcal{C}^*}_{\mathcal{B}^*}$ and $[\mathrm{id}_V]^{\mathcal{C}}_{\mathcal{B}}$?

Proposition 18.a.8 (*Dual Change of Basis and Transposition*): The relationship between transition matrices in the dual space and the primal space is characterized by transposition and inversion. Specifically, the transition matrix between dual bases $\mathcal{C}^*$ and $\mathcal{B}^*$ is the transpose of the inverse transition matrix between the original bases $\mathcal{C}$ and $\mathcal{B}$:

$$[\mathrm{id}_{V^*}]^{\mathcal{C}^*}_{\mathcal{B}^*} = \left([\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{C}}\right)^T = \left(\left([\mathrm{id}_V]^{\mathcal{C}}_{\mathcal{B}}\right)^{-1}\right)^T.$$

Thus, the coordinate transformation in the dual space corresponds to the transpose of the inverse transformation in the primal space.

Proof

Let $\mathcal{B} = (v_1, \dots, v_n)$, and let $\mathcal{C} = (w_1, \dots, w_n)$. We have

$$[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{C}} = \begin{pmatrix} | & & | \\ [v_1]_{\mathcal{C}} & \dots & [v_n]_{\mathcal{C}} \\ | & & | \end{pmatrix}.$$

Let $A := [\mathrm{id}_V]$ and write $A = (a_{ij})$. By definition, $v_i = \sum_{k=1}^n a_{ki} w_k$. Let $\mathcal{C}^* = (w_1^*, \dots, w_n^*)$ be the dual basis of $\mathcal{C}$. We apply the previous lemma to $\ell = w_j^*$ and the basis $\mathcal{B}$:

$$w_j^* = \sum_{i=1}^n w_j^*(v_i) \cdot v_i^* = \sum_{i=1}^n w_j^* \left(\sum_{k=1}^n a_{ki} w_k\right) \cdot v_i^* = \sum_{i=1}^n a_{ji} v_i^*.$$

Here, only $k = j$ remains from the inner sum. This implies that $[\mathrm{id}_{V^*}]^{\mathcal{C}^*}_{\mathcal{B}^*} = A^T = ([\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{C}})^T$.

Q.E.D.

The Dual Map

Lemma 18.a.9 (*Definition of the Dual Map*): Let V and W be vector spaces over K, and let $T\colon V \to W$ be a linear map. Define a new map $T^*\colon W^* \to V^*$ by $T^*(\ell) := \ell \circ T$ for all $\ell \in W^*$. Then, T^* is a linear map. It is called the *"dual map"* to T.

Proof

Let $\ell \in W^*$, meaning $\ell\colon W \to K$ is linear. Clearly, $\ell \circ T\colon V \to K$ is a linear map because both T and ℓ are linear, so $T^*(\ell) := \ell \circ T \in V^*$. This shows that $T^*\colon W^* \to V^*$ has its target indeed in V^*, as claimed.

We will show now that T^* is a linear map. Let $\ell \in W^*$, and let $\alpha \in K$. We have that $T^*(\alpha\ell) \in V^*$ is the map:

$$T^*(\alpha\ell)(v) = ((\alpha\ell) \circ T)(v) = (\alpha\ell)(T(v)) = \alpha\ell(T(v)) = \alpha \cdot (\ell \circ T)(v) = \alpha(T^*\ell)(v).$$

So, $T^*(\alpha\ell) = \alpha \cdot T^*(\ell)$. Similarly, one shows that $T^*(\ell_1 + \ell_2) = T^*(\ell_1) + T^*(\ell_2)$.

Q.E.D.

Proposition 18.a.10 (*Functoriality and Composition of Duals*): Let U, V, W be vector spaces over K, and let $T\colon V \to W$ and $S\colon W \to U$ be linear maps. Consider the dual maps $T^*\colon W^* \to V^*$ and $S^*\colon U^* \to W^*$. Then, $(S \circ T)^* = T^* \circ S^*$.

Proof

Exercise.

Q.E.D.

Exercise: Let V be a vector space over K. Show that:

(a) $(\mathrm{id}_V)^* = \mathrm{id}_{V^*}$.

(b) $(0: V \to V)^* = (0: V^* \to V^*)$.

Summary: In summary, applying $(-)^*$, or $\mathrm{Hom}(-, K)$, takes every vector space V and gives us a new vector space V^*. It also takes any linear map $T: V \to W$ and gives us a new map T^* between the duals of V and W, but the arrow is reversed: $T^*: W^* \to V^*$. This is an example of a *"contravariant functor"*.

Lemma 18.a.11 (*Transposition of the Representation Matrix*): Let $S: V \to W$ be a linear map between two finite-dimensional vector spaces V and W over K. Let $\mathcal{B}$ be a basis for V, and let $\mathcal{C}$ be a basis for W. Consider $S^*: W^* \to V^*$ and the bases $\mathcal{C}^*, \mathcal{B}^*$ for W^*, V^*, respectively. Then, $[S^*]^{\mathcal{C}^*}_{\mathcal{B}^*} = ([S]^{\mathcal{B}}_{\mathcal{C}})^T$.

Proof

Write $\mathcal{B} = (v_1, \ldots, v_n)$, $\mathcal{C} = (w_1, \ldots, w_m)$, and $A := [S]^{\mathcal{B}}_{\mathcal{C}} = (a_{ij})$. The matrix A satisfies $S(v_j) = \sum_{i=1}^m a_{ij} w_i$. Now, for $w_j^* \in \mathcal{C}^*$, we have

$$S^*(w_j^*) = w_j^* \circ S = \sum_{i=1}^n (w_j^* \circ S)(v_i) \cdot v_i^* = \sum_{i=1}^n w_j^*(S(v_i)) \cdot v_i^*,$$

where the second equality follows from the Lemma on the Dual Basis. Substituting $S(v_i)$, we get

$$\sum_{i=1}^n w_j^* \left(\sum_{k=1}^m a_{ki} w_k \right) \cdot v_i^* = \sum_{i=1}^n a_{ji} v_i^*.$$

The coordinates of $S^*(w_j^*)$ in the basis $\mathcal{B}^* = (v_1^*, \ldots, v_n^*)$ are $\begin{pmatrix} a_{j1} \\ \vdots \\ a_{jn} \end{pmatrix}$, which corresponds to row j in A, represented as a column (i.e., column j in A^T). **Q.E.D.**

Reflexivity

Let V be a vector space over K. We have V^*, but also $V^{**} := (V^*)^*$, obtained by applying the *"dual"* operation one more time to V^*. V^{**} is sometimes called the *"bidual space"* of V. Is there any relation between V^{**} and V?

Theorem 18.a.12 (*The Canonical Isomorphism to the Bidual Space*): Let V be a finite-dimensional vector space over K. Then, there is a canonical isomorphism $\tau: V \to (V^*)^*$ given by: for $v \in V$, $\tau(v): V^* \to K$ is the map $\tau(v)(\ell) := \ell(v)$ for all $\ell \in V^*$.

Remark: The map $\tau: V \to (V^*)^*$ exists and is defined in the same way as above regardless of V being finite- or infinite-dimensional. Moreover, τ is always injective. However, when V is infinite-dimensional, τ fails to be surjective.

Proof

We claim that τ is a linear map. Indeed, if $v \in V$ and $\alpha \in K$, we have

$$\begin{aligned} \tau(\alpha v) &= (V^* \ni \ell \mapsto \ell(\alpha v) \in K) \\ &= (V^* \ni \ell \mapsto \alpha \ell(v) \in K) \\ &= \alpha \cdot (V^* \ni \ell \mapsto \ell(v) \in K) = \alpha \tau(v). \end{aligned}$$

Similarly, one shows that $\tau(v' + v'') = \tau(v') + \tau(v'')$.

We now claim that τ is injective. To see this, we will show that $\ker(\tau) = 0$. Indeed, let $v \in \ker(\tau)$, meaning $\tau(v) = 0$. Then, $\ell(v) = 0$ for all $\ell \in V^*$. Choose a basis $\mathcal{B} = (v_1, \ldots, v_n)$ for V. Write $v = c_1 v_1 + \cdots + c_n v_n$. Apply v_k^* to v, and we obtain $v_k^*(c_1 v_1 + \cdots + c_n v_n) = 0$. But

$v_k^*(c_1v_1 + \cdots + c_nv_n) = c_k$. Therefore, $c_k = 0$, and this holds for $1 \le k \le n$, which implies that $v = 0$. This proves that τ is injective.

To finish the proof, note that $\dim(V^*)^* = \dim V^* = \dim V$. Therefore, since $\tau\colon V \to (V^*)^*$ is injective, it must be an isomorphism. Q.E.D.

Remark: **??** is not true for infinite-dimensional vector spaces. More specifically, $\tau\colon V \to (V^*)^*$ is injective, but not surjective. In fact, when V is infinite-dimensional, $(V^*)^*$ is not isomorphic to V.

Naturality

The map $\tau\colon V \to (V^*)^*$ exists for every space V, and it is defined in the same way. To distinguish τ's for different spaces, we denote it by $\tau_V\colon V \to (V^*)^*$. Let V and W be two vector spaces over K. Then, for any linear map $T\colon V \to W$, we have the following naturality condition, represented by a commutative diagram:

$$\begin{tikzcd} V \arrow[r, "\tau_V"] \arrow[d, "T"'] & (V^*)^* \arrow[d, "(T^*)^*"] \\ W \arrow[r, "\tau_W"'] & (W^*)^* \end{tikzcd}$$

Exercise: Let V be a finite-dimensional vector space over K, and let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis for V. Let $\mathcal{B}^* = (v_1^*, \ldots, v_n^*)$ be the dual basis, and let $(\mathcal{B}^*)^* = ((v_1^*)^*, \ldots, (v_n^*)^*)$ be the basis for $(V^*)^*$ which is dual to $\mathcal{B}^*$. Show that $\tau(\mathcal{B}) = (\mathcal{B}^*)^*$, meaning $\tau(v_i) = (v_i^*)^*$ for all $1 \le i \le n$.

Direct Sums (Section 18.b)

The External Direct Sum

Proposition 18.b.1 (*Vector Space Structure of the Cartesian Product*): Let V and W be two vector spaces over K. Then, the Cartesian product $V \times W$ becomes a vector space over K, provided we endow it with the following operations:

- $(v_1, w_1) + (v_2, w_2) := (v_1 + v_2, w_1 + w_2)$ for all $(v_1, w_1), (v_2, w_2) \in V \times W$.
- $\alpha \cdot (v, w) := (\alpha v, \alpha w)$ for all $\alpha \in K$ and $(v, w) \in V \times W$.

The zero element of this space is given by $0_{V\times W} := (0_V, 0_W)$. This new vector space $V \times W$ is usually denoted by $V \oplus W$ and is called the direct sum of V and W.

Proof

Exercise. One must verify that all vector space axioms are satisfied under these componentwise operations. Q.E.D.

Remark (*Canonical Maps associated with Direct Sums*): The direct sum $V \oplus W$ is naturally equipped with several canonical linear maps:

(a) **Canonical Embeddings**:

- $i_V\colon V \to V \oplus W$ defined by $i_V(v) := (v, 0)$.
- $i_W\colon W \to V \oplus W$ defined by $i_W(w) := (0, w)$.

(b) **Canonical Projections**:

- $p_V : V \oplus W \to V$ defined by $p_V(v,w) := v$.
- $p_W : V \oplus W \to W$ defined by $p_W(v,w) := w$.

Clearly, all these maps are linear. Furthermore, the projections p_V and p_W are surjective, while the embeddings i_V and i_W are injective. By virtue of these embeddings, we can often identify V and W as subspaces of $V \oplus W$ through the identifications $v \leftrightarrow (v,0)$ and $w \leftrightarrow (0,w)$.

Internal Direct Sums and Complements

Proposition 18.b.2 (*Isomorphism to a Complementary Subspace*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. Suppose $U' \subseteq V$ is another subspace that acts as a complement of U in V, which implies that $U + U' = V$ and $U \cap U' = \{0\}$. Then, the map

$$\varphi\colon U \oplus U' \to V, \quad \varphi(u,u') := u + u'$$

is a linear isomorphism.

Proof

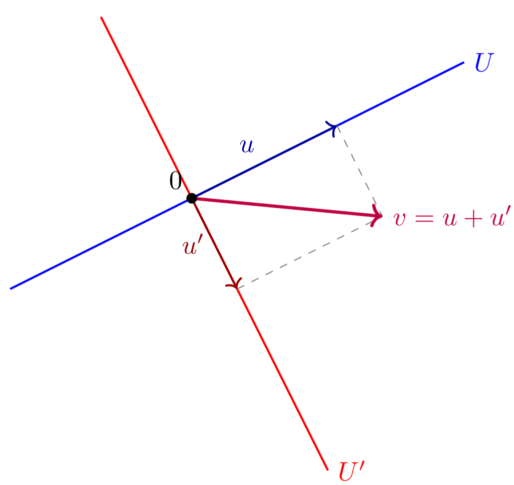
The linearity of φ follows immediately from the definitions of the operations in $U \oplus U'$ and the linearity of addition in V.

To establish injectivity, suppose $\varphi(u,u') = 0$, which implies $u + u' = 0$, and therefore $u = -u'$. Since $u \in U$ and $u' \in U'$, it follows that $u = -u' \in U \cap U'$. However, as U' is a complement of U, we have $U \cap U' = \{0\}$ by definition. Consequently, $u = u' = 0$, which shows that $\ker(\varphi) = \{0_{U \oplus U'}\}$. Thus, φ is injective.

Regarding surjectivity, we observe that since U' is a complement of U in V, we have $U + U' = V$ by definition. Thus, for any $v \in V$, there exist $u \in U$ and $u' \in U'$ such that $u + u' = v$. This implies $\varphi(u,u') = v$, and therefore $\mathrm{Im}(\varphi) = V$. This proves that φ is surjective, and consequently, an isomorphism. Q.E.D.

Remark: While $U \oplus U'$ is formally a different vector space from V, the existence of the canonical isomorphism φ allows us to treat V as the direct sum of its internal subspaces whenever a complement is specified.

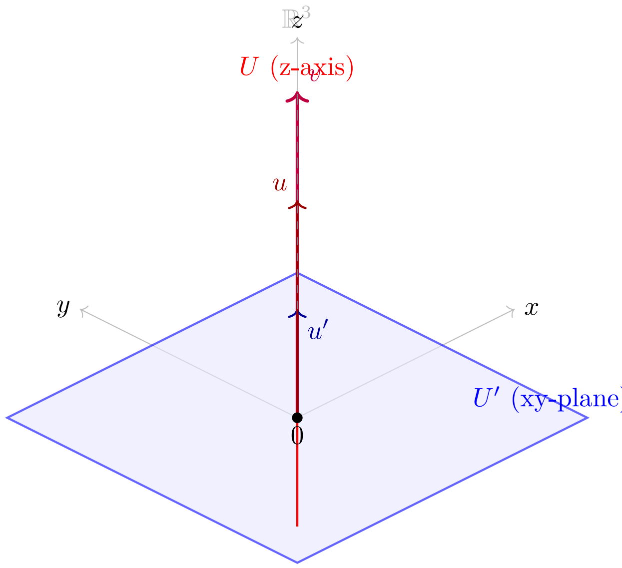
Geometrically, in $\mathbb{R}^2$, this represents the decomposition of any vector v uniquely into a sum of a vector $u \in U$ and a vector $u' \in U'$, constructed via a parallelogram parallel to the complementary subspaces.



Similarly, we can extend this intuition to $\mathbb{R}^3$. Consider U to be the z-axis (a line through the origin) and U' to be the xy-plane (a plane through the origin). Then U and U' are subspaces of $\mathbb{R}^3$.

- $U = \mathrm{Sp}(e_3) = \{(0,0,z) \mid z \in \mathbb{R}\}$
- $U' = \mathrm{Sp}(e_1, e_2) = \{(x,y,0) \mid x, y \in \mathbb{R}\}$

Any vector $v = (x,y,z) \in \mathbb{R}^3$ can be uniquely written as a sum of a vector $u \in U$ and a vector $u' \in U'$: $v = (0,0,z) + (x,y,0)$. Here, $u = (0,0,z) \in U$ and $u' = (x,y,0) \in U'$. Furthermore, $U \cap U' = \{(0,0,0)\}$, which is the zero vector. Thus, $\mathbb{R}^3 = U \oplus U'$. Geometrically, this means any point in 3D space can be reached by first moving along the z-axis and then moving in the xy-plane.



Generalization to Multiple Summands

The concept of a direct sum can be extended to any finite collection of vector spaces $U_1, \ldots, U_m$ over K. We define their direct sum as:

$$U_1 \oplus \cdots \oplus U_m := U_1 \times \cdots \times U_m,$$

endowed with coordinatewise operations.

For an infinite family of vector spaces $\{U_i\}_{i \in I}$, we define the direct sum $\bigoplus_{i \in I} U_i$ as the set of functions $t \colon I \to \bigcup_{i \in I} U_i$ such that $t(i) \in U_i$ for all $i \in I$, and $t(i) \neq 0$ for at most finitely many indices $i \in I$. This ensures that any element of the direct sum can be represented as a finite formal sum of components.

Quotient Spaces (Section 18.c)

Equivalence Relation and the Quotient Space

Let V be a vector space over K, and let $U \subseteq V$ be a subspace. We will define a new vector space over K, denoted by V/U, together with a linear map $\pi \colon V \to V/U$ such that π is surjective and $\ker(\pi) = U$. The space V/U will have some other good properties.

Definition (*The Quotient V/U And The Quotient Map π*): The space V/U is called the *"quotient"* of V by U, or V modulo U. The map π is called the *"quotient map"*, or the *"projection onto the quotient space"*.

We begin by defining a relation on V. For $v_1, v_2 \in V$, we define $v_1 \sim v_2$ if and only if $v_1 - v_2 \in U$.

Lemma (*Equivalence Relation on V*): The relation $\sim$ defines an equivalence relation on V.

Proof

Reflexivity: For all $v \in V$, $v \sim v$ because $v - v = 0 \in U$.

Symmetry: If $v_1 \sim v_2$, then $v_1 - v_2 \in U$. Since U is a subspace, $-(v_1 - v_2) = v_2 - v_1 \in U$, which implies $v_2 \sim v_1$.

Transitivity: If $v_1 \sim v_2$ and $v_2 \sim v_3$, then $v_1 - v_2 \in U$ and $v_2 - v_3 \in U$. Because U is a subspace, their sum is in U:

$$(v_1 - v_2) + (v_2 - v_3) = v_1 - v_3 \in U,$$

which means $v_1 \sim v_3$. **Q.E.D.**

We denote the equivalence class of $v \in V$ by $[v]$. Thus,

$$[v] = v + U = \{v + u \mid u \in U\}.$$

Definition 18.c.1 (*The Quotient Space*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. We define $V/U := V/\sim$ to be the set of equivalence classes of the equivalence relation $\sim$. So, the elements of V/U are $[v]$, where $v \in V$.

Alternatively, each element in V/U is a set of the form $v + U$, where $v \in V$. The space V/U is called the quotient of V by U (or modulo U).

Geometrically, the nature of the quotient space $\mathbb{R}^2/U$ depends on the dimension of U:

- If $U = \{0\}$ is the trivial subspace, then each equivalence class $[v] = v + \{0\} = \{v\}$ is simply a single **point**. In this case, no points are *"collapsed"* together, and therefore $\mathbb{R}^2/\{0\} \cong \mathbb{R}^2$.
- If U is a **line** through the origin, the equivalence classes $[v]$ forming the space $\mathbb{R}^2/U$ are the lines parallel to U, as seen in Figure **??**.

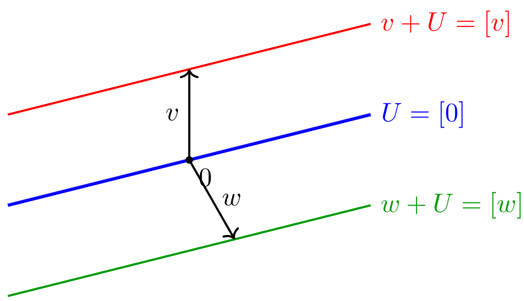


Figure 18.1: Geometric interpretation of the quotient space $\mathbb{R}^2/U$.

Similarly, we can extend this intuition to $\mathbb{R}^3$, as illustrated in Figure **??**:

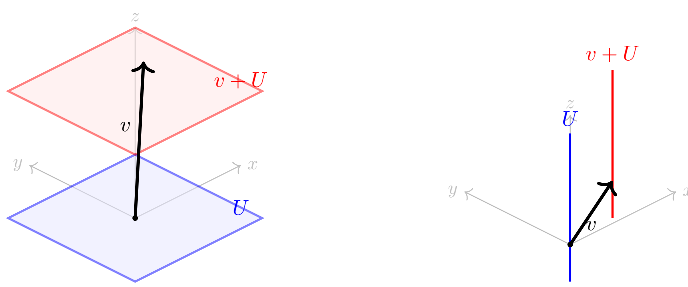
- If $U = \{0\}$ is the trivial subspace, then each equivalence class $[v] = \{v\}$ is simply a **point**. Since no two distinct vectors are identified, the quotient space $\mathbb{R}^3/\{0\}$ is isomorphic to $\mathbb{R}^3$ itself.
- If U is a **plane** through the origin, then $\mathbb{R}^3/U$ is 1-dimensional (by Proposition **??**) and therefore isomorphic to a line. Each element $[v]$ is a plane parallel to U. The quotient space can be viewed as the set of all such parallel planes *"stacked"* along a transversal direction (see Figure **??** and Proposition **??**).
- If U is a **line** through the origin, then $\mathbb{R}^3/U$ is 2-dimensional (by Proposition **??**) and therefore isomorphic to a plane. Each element $[v]$ is a line parallel to U (see Figure **??** and Proposition **??**).
- If $U = \mathbb{R}^3$ is the entire space, then there is only one equivalence class, $[v] = \mathbb{R}^3$. The quotient space $\mathbb{R}^3/\mathbb{R}^3$ is the zero vector space, with dimension 0.

Remark: The crucial insight is that an equivalence class $[v] = v + U = \{v + u \mid u \in U\}$ is not merely a single vector, but a translate of the subspace U. Consequently, the "shape" or dimension of $[v]$ is identical to the dimension of U. Specifically:

- If U is a line, then every $[v]$ is a line parallel to U.
- If U is a plane, then every $[v]$ is a plane parallel to U.

The dimension of the quotient space V/U represents the number of independent directions available to shift these objects. For instance, in $\mathbb{R}^3$, if U is a line (1-dimensional), there are 2 independent directions in which one can move this line to obtain a distinct parallel line (effectively moving the line across a 2-dimensional plane).

Thus, the space of all such lines is 2-dimensional. This geometric intuition aligns perfectly with the dimension formula established in Proposition **??**. In the case where $U = \{0\}$, the equivalence class is a point (0-dimensional), and we retain all 3 degrees of freedom of the ambient space.



(a) Plane case ($1D$-quotient) (b) Line case ($2D$-quotient)

Figure 18.2: Geometric interpretation of the quotient space $\mathbb{R}^3/U$.

The Vector Space Structure of V/U

We want to turn V/U into a vector space over K. We define addition as $[v_1] + [v_2] := [v_1 + v_2]$ for all $[v_1], [v_2] \in V/U$. Alternatively, this can be written as $(v_1 + U) + (v_2 + U) := (v_1 + v_2) + U$. For scalar multiplication, let $\alpha \in K$ and $[v] \in V/U$. We define $\alpha \cdot [v] := [\alpha v]$. The zero element is defined as $0_{V/U} := [0_V] = U$.

Proposition 18.c.2 (*Well-definedness of Operations in V/U*): The operations defined above are well-defined. Moreover, they give V/U the structure of a vector space over K.

Proof

To establish well-definedness, we must show that the operations do not depend on the choice of representatives. For scalar multiplication, suppose $[v] = [v']$, and let $\alpha \in K$. We need to show that $[\alpha v] = [\alpha v']$. Indeed, since $[v] = [v']$, we have $v - v' \in U$. Because U is a subspace, $\alpha(v - v') = \alpha v - \alpha v' \in U$. Consequently, $[\alpha v] = [\alpha v']$.

Similarly, for addition, suppose $[v_1] = [v_1']$ and $[v_2] = [v_2']$. We need to show that $[v_1 + v_2] = [v_1' + v_2']$. By assumption, $v_1 - v_1' \in U$ and $v_2 - v_2' \in U$. Adding these yields:

$$(v_1 + v_2) - (v_1' + v_2') = (v_1 - v_1') + (v_2 - v_2') \in U.$$

Therefore, $v_1 + v_2 \sim v_1' + v_2'$, which implies $[v_1 + v_2] = [v_1' + v_2']$. This proves that the operations are well-defined.

It remains to show that they satisfy the axioms of a vector space. These follow directly from the corresponding axioms in V. (Exercise: complete the proof!) **Q.E.D.**

We now define a map $\pi \colon V \to V/U$ by $\pi(v) := [v] \in V/U$ for all $v \in V$.

Proposition 18.c.3 (*The Quotient Map*): The map $\pi\colon V \to V/U$ is a linear map. Moreover, it is surjective (i.e., $\operatorname{Im}(\pi) = V/U$), and $\ker(\pi) = U$.

Proof

Linearity: For $v_1, v_2 \in V$, $\pi(v_1 + v_2) = [v_1 + v_2] = [v_1] + [v_2] = \pi(v_1) + \pi(v_2)$. For $\alpha \in K$ and $v \in V$, $\pi(\alpha v) = [\alpha v] = \alpha[v] = \alpha\pi(v)$.

Surjectivity: Let $x \in V/U$. By definition, x is an equivalence class $[v]$ of some element $v \in V$. Thus, $x = \pi(v)$.

Kernel: We claim that $\ker(\pi) = U$. First, let $v \in \ker(\pi)$, meaning $\pi(v) = 0$. Then, $[v] = [0]$, which implies $v = v - 0 \in U$. Hence, $\ker(\pi) \subseteq U$. Conversely, let $u \in U$. Then, $u \sim 0$ because $u - 0 \in U$. Therefore, $\pi(u) = [u] = [0] = 0$, so $u \in \ker(\pi)$. This shows that $U \subseteq \ker(\pi)$, completing the proof. **Q.E.D.**

Proposition 18.c.4 (*Dimension of the Quotient Space*): Suppose that V is a finite-dimensional vector space. Then, V/U is also finite-dimensional, and $\dim V/U = \dim V - \dim U$.

Proof

We have seen that $\pi\colon V \to V/U$ is a surjective linear map. This guarantees that V/U is finite-dimensional. (Generally, if $T\colon W' \to W''$ is a surjective linear map and W' is finite-dimensional, then W'' is also finite-dimensional, because the images of a basis of W' span W'').

We now apply the Rank-Nullity Theorem to π:

$$\dim V = \dim\ker(\pi) + \dim\operatorname{Im}(\pi).$$

Since $\ker(\pi) = U$ and $\operatorname{Im}(\pi) = V/U$, we obtain $\dim V/U = \dim V - \dim U$. **Q.E.D.**

The Isomorphism Theorem and Universal Property

Theorem 18.c.5 (*The First Isomorphism Theorem*): Let V and W be vector spaces over K, and let $T\colon V \to W$ be a linear map. We define a new map $\overline{T}\colon V/\ker(T) \to \operatorname{Im}(T)$ as follows: For $x \in V/\ker(T)$, choose a representative $v \in V$ such that $x = [v]$. Define $\overline{T}(x) := T(v)$. Then:

(a) $\overline{T}$ is well-defined and is a linear map.

(b) $\overline{T}\colon V/\ker(T) \to \operatorname{Im}(T)$ is an isomorphism.

(c) The following diagram commutes: $T = i \circ \overline{T} \circ \pi$, where $i\colon \operatorname{Im}(T) \to W$ is the inclusion map.

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "\pi"'] & W \\ V/\ker(T) \arrow[r, "\overline{T}\ (\cong)"] & \operatorname{Im}(T) \arrow[u, "i"'] \end{tikzcd}$$

Proof

(a) To show that $\overline{T}$ is well-defined, suppose $x = [v] = [v']$. We must show that $T(v) = T(v')$. Indeed, $[v] = [v']$ implies that $v - v' \in \ker(T)$, meaning $T(v - v') = 0$. By linearity, $T(v) = T(v')$.

The linearity of $\overline{T}$ is straightforward: $\overline{T}(\alpha[v]) = \overline{T}([\alpha v]) = T(\alpha v) = \alpha T(v) = \alpha\overline{T}([v])$. Also, $\overline{T}([v_1] + [v_2]) = \overline{T}([v_1 + v_2]) = T(v_1 + v_2) = T(v_1) + T(v_2) = \overline{T}([v_1]) + \overline{T}([v_2])$.

(b) To show that $\overline{T}$ is injective, let $x \in \ker(\overline{T})$, and write $x = [v]$ for some $v \in V$. Then, $\overline{T}(x) = T(v) = 0$, meaning $v \in \ker(T)$. This implies that $x = [v] = 0_{V/\ker(T)}$.

To show that $\overline{T}$ is surjective, let $w \in \mathrm{Im}(T)$. Then, there exists $v \in V$ such that $T(v) = w$. Consequently, $\overline{T}([v]) = T(v) = w$, proving surjectivity.

(c) The commutativity of the diagram follows immediately from the definitions. (Exercise: check this!)

Q.E.D.

Theorem 18.c.6 (*Universal Property of Quotient Spaces*): Let V be a vector space over K, and let $U \subseteq V$ be a subspace. The quotient space V/U has the following universal property: For every vector space W and every linear map $T\colon V \to W$ for which $T(U) = 0$ (i.e., for which $\ker(T) \supseteq U$), there exists a unique linear map $T'\colon V/U \to W$ such that $T = T' \circ \pi$. Moreover, $\ker(T') = \ker(T)/U$.

Remark: In algebraic jargon, every linear map $T\colon V \to W$ that maps U to 0 factors through V/U via $T = T' \circ \pi$, and moreover, in a unique way. We say that T induces the map T', or that T descends to a map $T'\colon V/U \to W$.

Proof

We define $T'\colon V/U \to W$ as follows: let $x \in V/U$. Choose $v \in V$ such that $x = [v]$. Define $T'(x) := T(v)$.

Exercise:

(a) Show that T' is well-defined (i.e., if $[v] = [v']$, then $T(v) = T(v')$).

(b) Show that T' is linear.

The diagram commutes by construction, since for all $v \in V$, $T' \circ \pi(v) = T'([v]) = T(v)$.

For uniqueness, suppose $T = T'' \circ \pi$ is another factorization. We will show that $T'' = T'$. Let $x \in V/U$. Since π is surjective, there exists $v \in V$ such that $x = \pi(v)$. We then have $T(v) = T'(\pi(v))$ and $T(v) = T''(\pi(v))$, which immediately yields $T'(x) = T''(x)$. Q.E.D.

Exercise: Prove that $\ker(T') = \ker(T)/U$.

Quotients, Complements, and Dual Spaces

Let V be a vector space over K, and let $U \subseteq V$ be a subspace. Let $W \subseteq V$ be a complement of U (i.e., W is a subspace such that $U \cap W = \{0\}$ and $U + W = V$).

Proposition 18.c.7 (*Canonical Isomorphism from a Complement*): Under the assumptions above, there is a canonical isomorphism $Q\colon W \to V/U$, defined by $Q(w) := [w] \in V/U$ for all $w \in W$.

Remark: The isomorphism Q is canonical only once the complement W is specified. In general, there are many choices of complements to U, and there is no preferred choice.

Proof

Consider the inclusion map $i\colon W \to V$ and the projection map $\pi\colon V \to V/U$. Since $Q = \pi \circ i$, Q is a linear map.

We claim that $\ker(Q) = \{0\}$. Indeed, if $w \in \ker(Q)$, then $[w] = 0$, implying that $w \in U$. However, since $w \in W$, we must have $w \in U \cap W = \{0\}$, and therefore $w = 0$. This shows that $\ker(Q) = \{0\}$.

We also claim that Q is surjective. Let $x \in V/U$ and choose $v \in V$ such that $x = [v]$. Since $V = U + W$, there exist $u \in U$ and $w \in W$ such that $v = w + u$. It follows that $[v] = [w+u] = [w]$, so $Q(w) = x$. This proves that Q is surjective, and consequently, an isomorphism. **Q.E.D.**

Exercise: Describe the inverse of Q, $Q^{-1} \colon V/U \xrightarrow{\cong} W$. (Hint: ...)

Quotients, Complements, and Annihilators

Proposition 18.c.8 (*Annihilators and Dual Quotients*): Let $U \subseteq V$ be a subspace, and define the *"annihilator"* $U^\perp := \{\ell \in V^* \mid \ell|_U = 0\}$. Then:

(a) $U^\perp \subseteq V^*$ is a linear subspace.

(b) There is a canonical isomorphism $(V/U)^* \cong U^\perp$.

(c) If V is finite-dimensional, then $U^* \cong V^*/U^\perp$.

Proof

(a) Clearly, the zero functional vanishes on U. For $\ell_1, \ell_2 \in U^\perp$ and $\alpha, \beta \in K$, we have $(\alpha\ell_1 + \beta\ell_2)(u) = \alpha\ell_1(u) + \beta\ell_2(u) = 0$ for all $u \in U$, establishing that $U^\perp$ is a subspace.

(b) We define a map $L \colon (V/U)^* \to U^\perp$ as follows: Let $S \in (V/U)^*$, i.e., $S \colon V/U \to K$. Define $L(S) \in V^*$ to be $L(S) := S \circ \pi$.

Exercise:

(a) Show that for all $S \in (V/U)^*$, $S \circ \pi \in U^\perp$.

(b) Show that L is a linear map.

We claim that L is an isomorphism. To see this, we define a map $P \colon U^\perp \to (V/U)^*$ as follows. Let $T \in U^\perp$, i.e., $T \colon V \to K$ such that $T|_U = 0$. By the Universal Property of Quotients, there exists a unique $T' \colon V/U \to K$ such that $T = T' \circ \pi$. Define $P(T) := T'$.

We claim that $L \circ P = \mathrm{id}_{U^\perp}$ and $P \circ L = \mathrm{id}_{(V/U)^*}$. Indeed, for all $S \colon V/U \to K$, we have $P \circ L(S) = P(S \circ \pi) = S$, because of the uniqueness statement in the theorem about the universal property of quotient spaces. Also, for all $T \in U^\perp$, $L \circ P(T) = L(T') = T' \circ \pi = T$. This proves that L is bijective, and since we know L is linear, it is an isomorphism. It follows that P is also a linear isomorphism.

(c) Assume V is finite-dimensional. Consider the restriction map $R \colon V^* \to U^*$ defined by $R(\varphi) := \varphi|_U$ for all $\varphi \in V^*$.

Exercise: Show that R is a linear map.

We claim that R is surjective. Indeed, let $\psi \in U^*$. We need to show that there exists $\varphi \in V^*$ such that $\varphi|_U = \psi$. The finite-dimensionality of V is important here. Pick a basis $u_1, \dots, u_k$ for U, and extend it to a basis $u_1, \dots, u_k, v_{k+1}, \dots, v_n$ of V. Define a linear map $\varphi \colon V \to K$ by $\varphi(u_i) := \psi(u_i)$ for all $1 \le i \le k$, and $\varphi(v_j) := 0$ for all $k < j \le n$. Clearly, $\varphi|_U = \psi$ because φ and ψ coincide on the elements of a basis of U. This proves $\mathrm{Im}(R) = U^*$.

We claim that $\ker(R) = U^\perp$. Indeed, this follows directly from definitions: $\ker(R) = \{\varphi \in V^* \mid \varphi|_U = 0\} = U^\perp$. By the First Isomorphism Theorem, R induces an isomorphism $\overline{R} \colon V^*/U^\perp \xrightarrow{\cong} U^*$.

Q.E.D.

Exercise: Let $T \colon V \to W$ be a linear map between vector spaces. Prove that the annihilator of the image of T is equal to the kernel of its dual map:

$$(\operatorname{Im} T)^\perp = \ker(T^*).$$

Miscellaneous Topics and Motivation (Chapter 19)

Geometric Motivation: Why Abstraction Matters (Section 19.a)

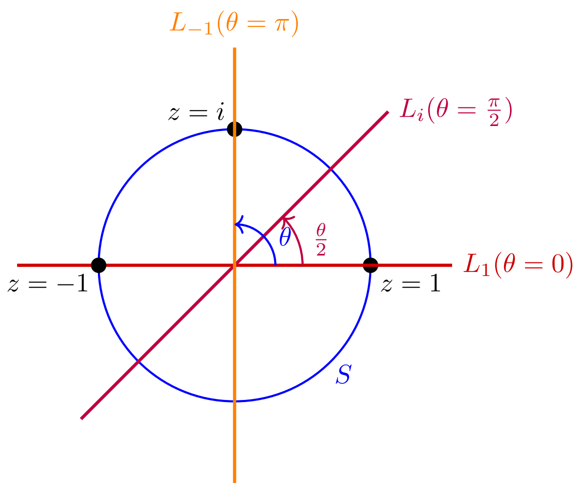
It is a common temptation to simply identify every n-dimensional vector space over K with the standard coordinate space K^n. However, while they are algebraically isomorphic for any fixed parameter, they may not be "continuously" isomorphic when the vector spaces depend dynamically on parameters.

The Möbius Vector Bundle

Definition 19.a.1 (*A Continuous Family of Subspaces*): Let $S \subseteq \mathbb{C}$ be the unit circle defined by $S = \{e^{i\theta} \mid \theta \in [0, 2\pi]\}$. For each $z = e^{i\theta} \in S$, we define a 1-dimensional subspace $L_z \subseteq \mathbb{R}^2$ as follows:

$$L_z := \mathrm{Sp}\left(\cos\frac{\theta}{2}, \sin\frac{\theta}{2}\right).$$

Notice a topological anomaly: $z = 1$ corresponds to both $\theta = 0$ and $\theta = 2\pi$. In the first case ($\theta = 0$), the spanning vector is $(1, 0)$, so $L_1 = \mathrm{Sp}(1, 0)$. In the second case ($\theta = 2\pi$), the spanning vector is $(-1, 0)$, which gives the exact same subspace $L_1 = \mathrm{Sp}(-1, 0)$. The line L_z rotates at exactly half the speed of the point z.



Proposition 19.a.2 (*Non-triviality of the Möbius Bundle*): There does not exist a family of linear isomorphisms $T_z \colon L_z \to \mathbb{R}$ that depends continuously on the parameter $z \in S$.

Proof Sketch

Consider the disjoint union of all these lines parameterized by z, which forms a space $M = \{(z, v) \mid z \in S, v \in L_z\}$. Topologically, this forms a vector bundle over S^1 known as the Möbius strip.

If such a continuous family of isomorphisms T_z existed, it would allow us to construct a global homeomorphism $\Phi \colon M \to S \times \mathbb{R}$. Removing the zero-section (the center circle where $v = 0$) from both spaces would imply that $M \setminus \{0\} \cong S \times (\mathbb{R} \setminus \{0\})$.

However, $S \times (\mathbb{R} \setminus \{0\})$ consists of two disconnected cylinders (an "upper" and "lower" half). In contrast, if you cut a Möbius strip down its center line, it remains a single connected piece. This topological contradiction demonstrates that we cannot consistently and continuously identify all L_z with $\mathbb{R}$. Abstraction is therefore necessary to handle "twisted" families of spaces. **Q.E.D.**

Affine Geometry (Section 19.b)

In our strict definition of a vector space, there is always a distinguished origin 0. Affine geometry studies spaces that "look like" vector spaces locally, but have forgotten their origin.

Definition 19.b.1 (*Affine Space*): A subset $X \subseteq V$ is called an *"affine space"* if there exists a linear subspace $U \subseteq V$ and a translation vector $x_0 \in V$ such that $X = x_0 + U$.

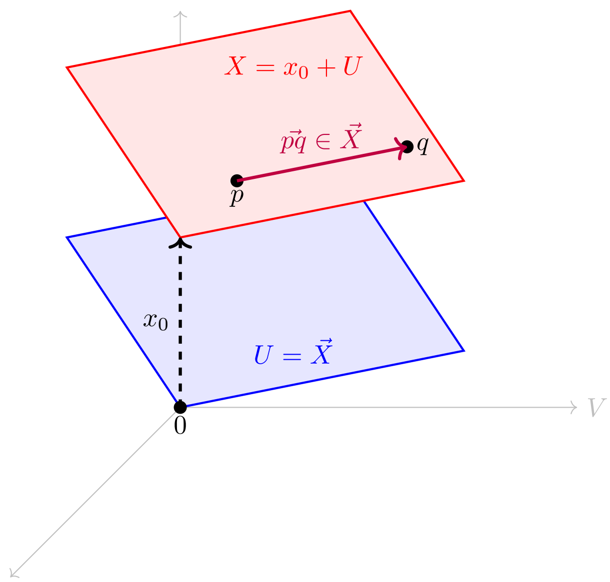
We say that X is modeled on the subspace U, and we write the *"direction space"* as $\vec{X} := U$. The elements of X are called *"points"*, while the elements of $\vec{X}$ are called *"vectors"*.

Proposition 19.b.2 (*Algebra of Affine Spaces*): Let X be an affine space modeled on $\vec{X}$. Then:

(a) We cannot inherently add two points $p, q \in X$, as $p + q \notin X$ in general.

(b) We can translate a point $p \in X$ by a vector $v \in \vec{X}$ to obtain a new point $p + v \in X$.

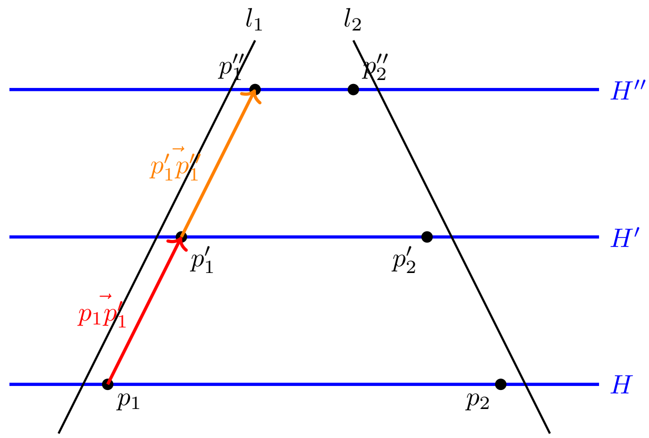
(c) For any two points $p, q \in X$, their difference defines a unique vector $\vec{pq} := q - p \in \vec{X}$.



Theorem 19.b.3 (*Thales' Theorem in Affine Geometry*): Let H, H', H'' be three distinct parallel affine planes in an ambient space V. Let l_1, l_2 be two affine lines intersecting these planes at the points p_1, p_1', p_1'' and p_2, p_2', p_2'' respectively. Then:

$$\frac{\vec{p_1 p_1''}}{\vec{p_1 p_1'}} = \frac{\vec{p_2 p_2''}}{\vec{p_2 p_2'}}.$$

Crucially, the ratio of these vectors does not depend on the specific transversals chosen, but only on the relative distances between the parallel planes.



Introduction to Eigenvalues: Fibonacci Sequences (Section 19.c)

The Vector Space of Fibonacci Sequences

Consider the set $\mathrm{Fib} \subseteq \mathbb{R}^\infty$ of all infinite real sequences $(a_0, a_1, a_2, \dots)$ that satisfy the linear recurrence relation $a_n = a_{n-1} + a_{n-2}$ for all $n \geq 2$.

Proposition 19.c.1 (*Structure of Fib*): The set Fib is a 2-dimensional subspace of $\mathbb{R}^\infty$. The evaluation map $F\colon \mathrm{Fib} \to \mathbb{R}^2$ sending a sequence to its first two terms, $(a_n) \mapsto (a_0, a_1)$, is a linear isomorphism.

To find a closed-form formula for the Fibonacci numbers (like Binet's formula), we study the **shift map** $S\colon \mathbb{R}^\infty \to \mathbb{R}^\infty$ defined by shifting the sequence one position to the left:

$$S(a_0, a_1, a_2, \dots) := (a_1, a_2, a_3, \dots).$$

Since the shifted sequence also satisfies the Fibonacci recurrence, this map restricts to a linear endomorphism $S'\colon \mathrm{Fib} \to \mathrm{Fib}$.

By utilizing the isomorphism F, we can translate the abstract shift operator S' into concrete matrix multiplication on $\mathbb{R}^2$. The sequence (a_0, a_1) shifts to $(a_1, a_2) = (a_1, a_0 + a_1)$, which corresponds to multiplication by the matrix $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$.

$$\begin{tikzcd}
\mathrm{Fib} \arrow[r, "S'"] \arrow[d, "F"', "\cong"] & \mathrm{Fib} \arrow[d, "\cong"', "F"] \\
\mathbb{R}^2 \arrow[r, "T_A"'] & \mathbb{R}^2
\end{tikzcd}$$

Eigenvalues and Eigenvectors

The problem of finding a geometric sequence $(1, \lambda, \lambda^2, \dots)$ inside Fib is equivalent to finding sequences that only scale by λ when shifted. That is, we seek vectors $\mathcal{F}$ such that $S'(\mathcal{F}) = \lambda\mathcal{F}$.

Definition 19.c.2 (*Eigenvalues and Eigenvectors*): Let V be a vector space and $T \in \mathrm{End}(V)$. A scalar $\lambda \in K$ is called an *"eigenvalue"* of T if there exists a non-zero vector $v \in V$ such that:

$$T(v) = \lambda v.$$

The vector v is called an *"eigenvector"* associated with the eigenvalue λ.

Determinants (Chapter 20)

Introduction and Motivation (Section 20.a)

The special case of 2×2 matrices

In linear algebra, determining whether a matrix is invertible is a fundamental problem. Let us begin by analyzing the familiar 2×2 case. Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathcal{M}_{2\times 2}(K)$. There is a very simple, well-known algebraic criterion to check whether A is invertible.

Proposition 20.a.1 (*Invertibility of 2×2 Matrices*): The matrix A is invertible if and only if $ad-bc \neq 0$. Moreover, if A is invertible, then its inverse is given by the explicit formula:

$$A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The scalar quantity $ad - bc$ is called the *"determinant"* of A, denoted $\det(A)$.

Proof

First, we show that A is not invertible if and only if its rank is strictly less than 2. Because A is a 2×2 matrix, having a rank less than 2 implies that its column vectors are linearly dependent.

Remark (*Exercise*): Check that the columns being linearly dependent implies that either the first column is the zero vector (i.e., $\binom{a}{c} = \binom{0}{0}$) or the second column is a scalar multiple of the first (i.e., $\binom{b}{d} = \tau\binom{a}{c}$ for some scalar $\tau \in K$).

If $\binom{a}{c} = \lambda\binom{b}{d}$, then $a = \lambda b$ and $c = \lambda d$. Consequently, $ad - bc = (\lambda b)d - b(\lambda d) = 0$. If, on the other hand, $\binom{b}{d} = \tau\binom{a}{c}$, then $b = \tau a$ and $d = \tau c$. Consequently, $ad - bc = a(\tau c) - (\tau a)c = 0$. Thus, we have proved that if A is not invertible, then $ad - bc = 0$.

Conversely, assume $ad - bc = 0$. If $a = 0$, then $bc = 0$, so either $b = 0$ or $c = 0$. This means either $A = \left(\begin{smallmatrix} 0 & 0 \\ c & d \end{smallmatrix}\right)$ or $A = \left(\begin{smallmatrix} 0 & b \\ 0 & d \end{smallmatrix}\right)$. In both of these cases, A, clearly, has linearly dependent columns and is therefore not invertible.

Now, assume $a \neq 0$. This implies $d = \frac{bc}{a}$, and thus we can write $A = \begin{pmatrix} a & b \\ c & \frac{bc}{a} \end{pmatrix}$. We can observe that the second column of A is exactly a multiple of the first column because $\binom{b}{\frac{bc}{a}} = \frac{b}{a}\binom{a}{c}$. This implies that the columns of A are linearly dependent. Hence, $\operatorname{rank}(A) < 2$, which implies A is not invertible. **Q.E.D.**

To confirm the sufficiency of this condition and to find the explicit inverse, the formula for A^{-1} can be verified by direct matrix multiplication:

$$\frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{ad-bc}\begin{pmatrix} ad-bc & bd-bd \\ -ac+ac & -bc+ad \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Goals: While the 2×2 formula is easy to memorize, it does not immediately suggest how to handle larger matrices. Our main goals now are to make the determinant conceptually rigorous and to generalize the determinant function to arbitrary $n \times n$ matrices.

Multilinear and Alternating Functions

We wish to generalize the determinant to arbitrary dimensions. To this end, we should view an $n \times n$ matrix not just as a grid of numbers, but as a collection of n row vectors.

Definition 20.a.2 (*n-Linearity*): Let $D : \mathcal{M}_{n\times n}(K) \to K$ be a function. We say that D is *"n-linear"* if for every $1 \le i \le n$, D is linear as a function of the i-th row when all the other $n-1$ rows are held fixed.

We can think of the matrix space $\mathcal{M}_{n\times n}(K)$ as a Cartesian product of row vectors: $K^n_{\text{row}} \times \cdots \times K^n_{\text{row}}$. If A has rows $\alpha_1, \dots, \alpha_n$, we write $D(A) = D(\alpha_1, \dots, \alpha_n)$. Saying that D is n-linear means that for any chosen row index i, the following two linearity conditions hold:

(a) $D(\alpha_1, \dots, \alpha_i + \alpha_i', \dots, \alpha_n) = D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) + D(\alpha_1, \dots, \alpha_i', \dots, \alpha_n)$.

(b) $D(\alpha_1, \dots, c \cdot \alpha_i, \dots, \alpha_n) = c \cdot D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n)$ for all scalars $c \in K$.

Example: For a matrix $A \in \mathcal{M}_{n\times n}(K)$, write $A(i,j)$ for its entry at place i,j. Let $k_1, \dots, k_n$ be a fixed sequence of integers such that $1 \le k_i \le n$ for all i. Fix also a scalar $c \in K$. Consider $D : \mathcal{M}_{n\times n}(K) \to K$ defined by the product of one entry from each row:

$$D(A) := c \cdot A(1,k_1) \cdot A(2,k_2) \cdots A(n,k_n).$$

We claim that D is an n-linear function.

To see this, consider $D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) = b \cdot A(i,k_i)$, where $b \in K$ is the product of all the fixed terms. Notice that b depends only on the rows $\alpha_1, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_n$ and does not depend on row i. Let $\alpha_i' = (\alpha_{i,1}', \dots, \alpha_{i,n}')$ be another possible i-th row. Then, by standard distribution:

$$D(\dots, \alpha_i + \alpha_i', \dots) = b \cdot (A(i,k_i) + \alpha_{i,k_i}') = D(\dots, \alpha_i, \dots) + D(\dots, \alpha_i', \dots).$$

Similarly, for any scalar $\lambda \in K$:

$$D(\dots, \lambda \cdot \alpha_i, \dots) = b \cdot (\lambda A(i,k_i)) = \lambda \cdot D(\dots, \alpha_i, \dots).$$

Therefore, D is indeed n-linear.

Example (*Diagonal Product*): A more concrete example: $D(A) := A_{11} \cdot A_{22} \cdot \dots \cdot A_{nn}$ (i.e., the product of the terms on the main diagonal) is an n-linear function.

Now, let us try to find all 2-linear functions $D : \mathcal{M}_{2\times 2}(K) \to K$. Write the identity matrix as $I = \left(\begin{smallmatrix} -\epsilon_1- \\ -\epsilon_2- \end{smallmatrix}\right)$, where $\epsilon_1 = (1,0)$ and $\epsilon_2 = (0,1)$.

If D is 2-linear, then we have:

$$\begin{aligned} D(A) &= D(A_{11}\epsilon_1 + A_{12}\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11} D(\epsilon_1, A_{21}\epsilon_1 + A_{22}\epsilon_2) + A_{12} D(\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11}A_{21} D(\epsilon_1, \epsilon_1) + A_{11}A_{22} D(\epsilon_1, \epsilon_2) + A_{12}A_{21} D(\epsilon_2, \epsilon_1) + A_{12}A_{22} D(\epsilon_2, \epsilon_2). \end{aligned}$$

Therefore, D is completely determined by its values on the following four pairings: $D(\epsilon_1, \epsilon_1)$, $D(\epsilon_1, \epsilon_2)$, $D(\epsilon_2, \epsilon_1)$, and $D(\epsilon_2, \epsilon_2)$.

Lemma 20.a.3 (*Closure of n-Linear Functions*): Any linear combination of n-linear functions is again an n-linear function.

Proof

It is enough to prove this for a linear combination of two n-linear functions. Let $D, E : \mathcal{M}_{n\times n}(K) \to K$ be n-linear functions and $a, b \in K$. We define the combined function as $(aD + bE)(A) := aD(A) + bE(A)$. By expanding $(aD + bE)(\dots, \alpha_i + \alpha_i', \dots)$ and $(aD + bE)(\dots, \lambda\alpha_i, \dots)$ using the individual linearities of D and E, we see the result naturally holds. **Q.E.D.**

Example: Consider $D : \mathcal{M}_{2\times 2}(K) \to K$ defined by $D(A) := A_{11}A_{22} - A_{12}A_{21}$. The function D is 2-linear because it can be written as $D = D_1 + D_2$, where $D_1(A) = A_{11}A_{22}$ and $D_2(A) = -A_{12}A_{21}$, both of which are 2-linear by our previous example. Furthermore, $D(I) = 1$, and if A' is obtained from A by interchanging the rows, then $D(A') = -D(A)$.

Indeed, $D(A') = D\left(\begin{smallmatrix} A_{21} & A_{22} \\ A_{11} & A_{12} \end{smallmatrix}\right) = A_{21} \cdot A_{12} - A_{22}A_{11} = -(A_{11}A_{22} - A_{12}A_{21}) = -D(A)$.

Definition 20.a.4 (*Alternating Property*)**:** Let $D : \mathcal{M}_{n\times n}(K) \to K$ be an n-linear function. We say that D is *"alternating"* if for every matrix A in which some two rows are equal, we have $D(A) = 0$.

Proposition 20.a.5 (*Adjacent Swap Property*)**:** Let D be an n-linear function. Assume that D has the property that whenever A has two equal adjacent rows, then $D(A) = 0$. Then the following are equivalent:

(a) D is alternating.

(b) For any matrix B, if B' is obtained from B by interchanging two of its rows, then $D(B') = -D(B)$.

Proof

We begin with the proof of **(b)**. Assume first that B' is obtained from B by interchanging two adjacent rows (say, row k and row $k+1$). Consider evaluating D on a matrix where both the k-th and $(k+1)$-th rows are equal to $\beta_k + \beta_{k+1}$. By our assumption, this yields 0. Expanding this via n-linearity gives:

$$D(\dots, \beta_k + \beta_{k+1}, \beta_k + \beta_{k+1}, \dots) = D(\dots, \beta_k, \beta_k, \dots) + D(\dots, \beta_k, \beta_{k+1}, \dots)$$
$$+D(\dots, \beta_{k+1}, \beta_k, \dots) + D(\dots, \beta_{k+1}, \beta_{k+1}, \dots) = 0.$$

By assumption, the terms with adjacent equal rows are zero, leaving $D(B) + D(B') = 0$, which naturally implies $D(B') = -D(B)$.

Now suppose B' is obtained from B by interchanging rows k and ℓ where $k < \ell$. We can obtain B' by doing a sequence of adjacent row interchanges. Moving row ℓ to position k requires $r = \ell - k$ adjacent interchanges. Next, moving the original row k down to position ℓ requires $r - 1$ adjacent interchanges. The total number of interchanges is $2r - 1$. Since this is always an odd number, we flip the sign an odd number of times. Thus, $D(B') = (-1)^{2r-1} D(B) = -D(B)$.

We now prove **(a)**. Let A be a matrix in which row i equals row j where $i < j$. If $j = i + 1$, then by assumption $D(A) = 0$. If $j > i + 1$, we interchange row j with row $i + 1$ to obtain a matrix A' with two equal adjacent rows. By assumption $D(A') = 0$, but by **(b)**, $D(A') = -D(A)$ (since we performed an odd number of adjacent swaps), which implies $D(A) = 0$. **Q.E.D.**

The Determinant Function

Definition 20.a.6 (*Determinant Function*)**:** A function $D : \mathcal{M}_{n\times n}(K) \to K$ is called a *"determinant function"* if it is n-linear, it is alternating, and it normalizes the identity matrix such that $D(I) = 1$.

Example (*Warmup*)**:** Let us find all determinant functions on $M_{2\times 2}(K)$. Write $I = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} -\epsilon_1- \\ -\epsilon_2- \end{smallmatrix}\right)$, meaning $\epsilon_1 = (1, 0)$ and $\epsilon_2 = (0, 1)$. If D is a 2-linear function, we can expand it algebraically:

$$\begin{aligned} D(A) &= D(A_{11}\epsilon_1 + A_{12}\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11} D(\epsilon_1, A_{21}\epsilon_1 + A_{22}\epsilon_2) + A_{12} D(\epsilon_2, A_{21}\epsilon_1 + A_{22}\epsilon_2) \\ &= A_{11}A_{21} D(\epsilon_1, \epsilon_1) + A_{11}A_{22} D(\epsilon_1, \epsilon_2) \\ &\quad + A_{12}A_{21} D(\epsilon_2, \epsilon_1) + A_{12}A_{22} D(\epsilon_2, \epsilon_2). \end{aligned}$$

Consequently, D is completely determined by its values on these standard basis row pairings.

If D is alternating, then evaluating equal rows gives zero, meaning $D(\epsilon_1, \epsilon_1) = D(\epsilon_2, \epsilon_2) = 0$. Additionally, swapping rows flips the sign, so $D(\epsilon_2, \epsilon_1) = -D(\epsilon_1, \epsilon_2) = -D(I)$. Finally, if D is a determinant function, $D(I) = 1$. Substituting these values back into our expansion, we perfectly recover the familiar formula:

$$D(A) = A_{11}A_{22}(1) + A_{12}A_{21}(-1) = A_{11}A_{22} - A_{12}A_{21}.$$

Therefore, the 2×2 determinant function is completely unique!

Definition 20.a.7 (*Submatrix*): Let $A \in \mathcal{M}_{n\times n}(K)$, $n \geq 2$, and $1 \leq i \leq n$, $1 \leq j \leq n$. We denote by $A(i|j) \in \mathcal{M}_{(n-1)\times(n-1)}(K)$ the *"submatrix"* obtained from A by deleting row i and column j. If D is an $(n-1)$-linear function, we intuitively define $D_{ij}(A) := D(A(i|j))$.

Theorem 20.a.8 (*Recursive Construction of Determinants*): Let $n \geq 2$, and let D be an $(n-1)$-linear function which is alternating. Let $1 \leq j \leq n$. Define $E_j : \mathcal{M}_{n\times n}(K) \to K$ by

$$E_j(A) := \sum_{i=1}^{n} (-1)^{i+j} A_{ij} D_{ij}(A).$$

Then, E_j is an n-linear function and it is alternating. Furthermore, if D is a determinant function, then so is E_j.

Proof

Clearly, $D_{ij}(A)$ is independent of the i-th row of A because that row is deleted in the construction of $A(i|j)$. Since D is $(n-1)$-linear, the map $A \mapsto D_{ij}(A)$ is linear when viewed as a function of each of its remaining rows. The function $A \mapsto A_{ij}D_{ij}(A)$ is therefore n-linear. Since linear combinations of n-linear functions are n-linear, $E_j(A)$ is n-linear too.

We show that E_j is alternating. By **??**, it is enough to show $E_j(A) = 0$ whenever A has two equal adjacent rows. Assume $\alpha_k = \alpha_{k+1}$ for some index k. For any $i \neq k, k+1$, the submatrix $A(i|j)$ still has two equal rows, hence $D_{ij}(A) = 0$. This leaves only two non-zero terms in the sum:

$$E_j(A) = (-1)^{k+j} A_{kj} D_{kj}(A) + (-1)^{k+1+j} A_{(k+1)j} D_{(k+1)j}(A).$$

But, since $\alpha_k = \alpha_{k+1}$, we have $A_{kj} = A_{(k+1)j}$ and $A(k|j) = A(k+1|j)$, meaning $D_{kj}(A) = D_{(k+1)j}(A)$. Thus, the two terms perfectly cancel out, yielding $E_j(A) = 0$. This proves E_j is alternating.

Finally, assume D is a determinant function, meaning $D(I_{n-1}) = 1$. Note that $I_n(j|j) = I_{n-1}$ for all j. When computing $E_j(I_n)$, the term $(I_n)_{ij}$ is 1 if $i = j$, and 0 if $i \neq j$. Therefore, the sum collapses to a single term:

$$E_j(I_n) = (-1)^{j+j} D_{jj}(I_n) = D(I_n(j|j)) = 1.$$

This shows that E_j is a valid determinant function. **Q.E.D.**

Corollary 20.a.9 (*Existence of Determinants*): For all $n \geq 1$, at least one determinant function exists.

Example: Consider $A \in \mathcal{M}_{3\times 3}(K)$. Using the uniquely defined 2×2 determinant function $|B| := B_{11}B_{22} - B_{12}B_{21}$, we can explicitly write out $E_1(A)$ (expanding along the first column) as:

$$E_1(A) = A_{11} \begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} - A_{21} \begin{vmatrix} A_{12} & A_{13} \\ A_{32} & A_{33} \end{vmatrix} + A_{31} \begin{vmatrix} A_{12} & A_{13} \\ A_{22} & A_{23} \end{vmatrix}.$$

We can similarly construct $E_2(A)$ and $E_3(A)$ along the other columns, and all of these form valid determinant functions on $M_{3\times 3}(K)$.

Uniqueness of the Determinant (Section 20.b)

We have shown that a determinant function exists. Now, we will prove that it is absolutely unique. Let $D : \mathcal{M}_{n\times n}(K) \to K$ be an n-linear alternating function. Let $A \in \mathcal{M}_{n\times n}(K)$ be a matrix with rows $\alpha_1, \ldots, \alpha_n$. We denote the identity matrix as I, whose rows are the standard basis vectors $\epsilon_1, \ldots, \epsilon_n$ of K^n_{row}.

Clearly, we can express each row of A in terms of the standard basis: $\alpha_i = \sum_{j=1}^n A(i,j)\epsilon_j$ for $1 \le i \le n$. Using the n-linearity of D, we can expand $D(A)$ as follows:

$$\begin{aligned}
D(A) &= D(\alpha_1, \ldots, \alpha_n) \\
&= D\left(\sum_{j=1}^n A(1,j)\epsilon_j, \alpha_2, \ldots, \alpha_n\right) \\
&= \sum_{j=1}^n A(1,j) \cdot D(\epsilon_j, \alpha_2, \ldots, \alpha_n) \\
&= \sum_{j=1}^n A(1,j) \cdot D\left(\epsilon_j, \sum_{k=1}^n A(2,k)\epsilon_k, \ldots, \alpha_n\right) \\
&= \sum_{j,k} A(1,j) \cdot A(2,k) \cdot D(\epsilon_j, \epsilon_k, \ldots, \alpha_n)
\end{aligned}$$

where both j and k run between 1 and n. Continuing this expansion for all n rows yields an enormous sum:

$$D(A) = \sum_{k_1,\ldots,k_n} A(1,k_1) \cdot A(2,k_2) \cdots A(n,k_n) \cdot D(\epsilon_{k_1}, \epsilon_{k_2}, \ldots, \epsilon_{k_n})$$

where each of the indices $k_1, \ldots, k_n$ runs between 1 and n.

Thus far, we have not used the fact that D is alternating. Because D is alternating, the term $D(\epsilon_{k_1}, \ldots, \epsilon_{k_n}) = 0$ whenever two of the indices $k_1, \ldots, k_n$ are equal. Thus, we are interested only in ordered sequences $(k_1, \ldots, k_n)$ in which $k_i \in \{1, \ldots, n\}$ for all i, and $k_i \neq k_j$ for all $i \neq j$.

Such a sequence is called a *"permutation"* of $\{1, \ldots, n\}$, or a permutation of degree n. Alternatively, we can think of a permutation of degree n as a bijective function $\sigma : \{1, \ldots, n\} \to \{1, \ldots, n\}$. Then, the sequence $(k_1, \ldots, k_n) = (\sigma(1), \sigma(2), \ldots, \sigma(n))$ is a permutation as defined earlier. In other words, a permutation of degree n is simply a re-ordering of a list of n elements: $(1, \ldots, n) \mapsto (\sigma(1), \ldots, \sigma(n))$.

We can nicely rewrite our expansion of $D(A)$ as:

$$D(A) = \sum_{\sigma} A(1,\sigma(1)) \cdot A(2,\sigma(2)) \cdots A(n,\sigma(n)) \cdot D(\epsilon_{\sigma(1)}, \epsilon_{\sigma(2)}, \ldots, \epsilon_{\sigma(n)})$$

where σ runs over all possible permutations of degree n.

Properties of Permutations

An important property about permutations is that if σ is a permutation of degree n, one can pass from the sequence $(1, \ldots, n)$ to $(\sigma(1), \ldots, \sigma(n))$ by performing a finite sequence of interchanges of pairs.

More precisely, a *"transposition"* is a permutation $\theta : \{1, \ldots, n\} \to \{1, \ldots, n\}$ for which there exist $i, j \in \{1, \ldots, n\}$ with $i \neq j$ such that $\theta(i) = j$, $\theta(j) = i$, and $\theta(k) = k$ for all $k \neq i, j$.

Every permutation σ can be written as a composition of transpositions:

$$\sigma = \theta_1 \circ \cdots \circ \theta_m \tag{20.1}$$

where $\theta_1, \ldots, \theta_m$ are transpositions. However, these θ_i's and the total number m are not unique. The same permutation σ can be written as $\sigma = \theta_1 \circ \cdots \circ \theta_m = \theta'_1 \circ \cdots \circ \theta'_{m'}$.

Remark: Why is the decomposition (**??**) always possible? Start with the sorted list $(1, \ldots, n)$. If $\sigma(1) \neq 1$, then interchange 1 and $\sigma(1)$. We now have $(\sigma(1), \ldots, 1, \ldots, n)$. Next, look at the second position. If it is not $\sigma(2)$, interchange it with $\sigma(2)$, and proceed similarly down the line. Of course, there are other ways to do it. For example, the permutation $(3,2,1)$ can be obtained in 3 steps:

$$(1,2,3) \to (2,1,3) \to (2,3,1) \to (3,2,1)$$

But, it can also be obtained in just one step: $(1,2,3) \to (3,2,1)$.

Lemma 20.b.1 (*Parity of Transpositions*): If $(\sigma(1),\ldots,\sigma(n))$ is obtained from $(1,\ldots,n)$ in two different ways—one time by a sequence of m transpositions, and another time by a sequence of m' transpositions—then m and m' must have the same parity. That is, m' is even if and only if m is even, and m' is odd if and only if m is odd.

In other words, $(-1)^{m'} = (-1)^m$. The parity depends only on the permutation σ, and not on the particular sequence of transpositions used.

A permutation σ is called *"even"* if m is even, and *"odd"* if m is odd. The number $\operatorname{sgn}(\sigma) := (-1)^m \in \{-1,1\}$ is called the *"sign"* of σ.

Proof of Lemma

We know from previous results that determinant functions $E : \mathcal{M}_{n\times n}(K) \to K$ exist for $n \geq 1$. Let us fix one such determinant function E (we denote it by E to distinguish it from D in the previous discussion).

Consider the evaluation $E(\epsilon_{\sigma(1)},\ldots,\epsilon_{\sigma(n)})$. If $(\sigma(1),\ldots,\sigma(n))$ can be obtained from $(1,\ldots,n)$ by a sequence of m transpositions, then the matrix with rows $\epsilon_{\sigma(1)},\ldots,\epsilon_{\sigma(n)}$ can be obtained from the identity matrix I by a sequence of m interchanges of pairs of rows.

Because E is alternating, swapping two rows flips the sign of the output. Therefore:

$$E(\epsilon_{\sigma(1)},\ldots,\epsilon_{\sigma(n)}) = (-1)^m E(\epsilon_1,\ldots,\epsilon_n) = (-1)^m E(I) = (-1)^m.$$

But, this exact same logic applies to the sequence of m' transpositions. Thus:

$$E(\epsilon_{\sigma(1)},\ldots,\epsilon_{\sigma(n)}) = (-1)^{m'} E(\epsilon_1,\ldots,\epsilon_n) = (-1)^{m'} E(I) = (-1)^{m'}.$$

This identically implies that $(-1)^m = (-1)^{m'}$, proving the lemma. Q.E.D.

Returning to our original function $D(A)$, we have seen that:

$$D(A) = \sum_\sigma A(1,\sigma(1))\cdots A(n,\sigma(n)) \cdot D(\epsilon_{\sigma(1)},\ldots,\epsilon_{\sigma(n)}).$$

By the alternating property of D, performing the transpositions necessary to sort $(\sigma(1),\ldots,\sigma(n))$ back to $(1,\ldots,n)$ yields:

$$D(\epsilon_{\sigma(1)},\ldots,\epsilon_{\sigma(n)}) = \operatorname{sgn}(\sigma)\cdot D(\epsilon_1,\ldots,\epsilon_n) = \operatorname{sgn}(\sigma)\cdot D(I). \tag{20.2}$$

Theorem 20.b.2 (*Leibniz Formula & Uniqueness*): For all $n \geq 1$, there exists a unique determinant function on $\mathcal{M}_{n\times n}(K)$. We will denote it by $\det$. It is mathematically given by the formula:

$$\det(A) = \sum_\sigma \operatorname{sgn}(\sigma)\cdot A(1,\sigma(1))\cdot A(2,\sigma(2))\cdots A(n,\sigma(n)).$$

Proof

This follows immediately from (**??**). Indeed, if D is a determinant function, then $D(I) = 1$. Substituting this into (**??**) yields the exact formula for $D(A)$ stated in the theorem, proving that the function is entirely unique. Q.E.D.

Corollary 20.b.3 (*Scaling of Alternating Functions*): For any n-linear alternating function $D : \mathcal{M}_{n\times n}(K) \to K$, we have $D(A) = \det(A)\cdot D(I)$ for all $A \in \mathcal{M}_{n\times n}(K)$.

The Symmetric Group

Denote by S_n the set of permutations $\sigma : \{1,\ldots,n\} \to \{1,\ldots,n\}$. The set S_n has the algebraic structure of a group under composition: if $\sigma,\tau \in S_n$, their composition is $(\sigma\circ\tau)(x) = \sigma(\tau(x))$. The identity function id is the neutral element. The inverse σ^{-1} exists because σ is bijective. Furthermore, the size of the symmetric group is $|S_n| = n!$ (Exercise).

Lemma 20.b.4 (*Multiplicativity of Signum*): For all $\sigma, \tau \in S_n$, we have $\operatorname{sgn}(\sigma \circ \tau) = \operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\tau)$.

Proof

Suppose $\sigma = \theta_1 \circ \cdots \circ \theta_m$ and $\tau = \lambda_1 \circ \cdots \circ \lambda_\ell$, where the θ_i's and λ_j's are all transpositions. Then, their composition is simply $\sigma \circ \tau = \theta_1 \circ \cdots \circ \theta_m \circ \lambda_1 \circ \cdots \circ \lambda_\ell$, which explicitly consists of $m + \ell$ transpositions. Therefore:

$$\operatorname{sgn}(\sigma \circ \tau) = (-1)^{m+\ell} = (-1)^m \cdot (-1)^\ell = \operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\tau).$$

Q.E.D.

Multiplicativity of the Determinant

Theorem 20.b.5 (*Multiplicativity of the Determinant*): For any matrices $A, B \in M_{n\times n}(K)$, we have $\det(A \cdot B) = \det(A) \cdot \det(B)$.

Proof

Fix a matrix $B \in \mathcal{M}_{n\times n}(K)$, and consider a function $D : \mathcal{M}_{n\times n}(K) \to K$ defined by $D(A) := \det(A \cdot B)$.

Claim: D is n-linear and alternating.

Proof of claim: Let the rows of A be $\alpha_1, \ldots, \alpha_n$. If we view α_i as a $1 \times n$ matrix, then the matrix product $\alpha_i \cdot B$ is also $1 \times n$. In fact, the rows of the product matrix $A \cdot B$ are exactly $\alpha_1 \cdot B, \ldots, \alpha_n \cdot B$. So, $D(A) = \det(\alpha_1 \cdot B, \ldots, \alpha_n \cdot B)$.

Clearly, for two row vectors α_i and α_i', we have $(\alpha_i + \alpha_i') \cdot B = \alpha_i \cdot B + \alpha_i' \cdot B$. Also, for any scalar $c \in K$, $(c\alpha_i) \cdot B = c(\alpha_i \cdot B)$. Since the determinant function det is n-linear with respect to its rows, it heavily follows that D is also n-linear.

Furthermore, if two rows of A are equal (i.e., $\alpha_i = \alpha_j$), then the corresponding rows of $A \cdot B$ are also equal ($\alpha_i \cdot B = \alpha_j \cdot B$). Because det is alternating, $D(A) = \det(\alpha_1 \cdot B, \ldots, \alpha_n \cdot B) = 0$. This rigorously proves the claim that D is alternating.

We continue with the proof of **??**. By **??**, any n-linear alternating function satisfies $D(A) = \det(A) \cdot D(I)$. We compute $D(I)$ using our definition: $D(I) = \det(I \cdot B) = \det(B)$. Substituting this back yields $D(A) = \det(A) \cdot \det(B)$, completing the proof. Q.E.D.

Corollary 20.b.6 (*Determinant of an Inverse*): If A is invertible, then $\det(A) \neq 0$. Moreover, $\det(A^{-1}) = \frac{1}{\det(A)}$.

Proof

Assume A is invertible, and denote its inverse by A^{-1}. Since $A \cdot A^{-1} = I$, taking the determinant of both sides and applying **??** yields $\det(A) \cdot \det(A^{-1}) = \det(I) = 1$. This automatically implies that $\det(A)$ cannot be zero, and dividing by $\det(A)$ yields the expected formula. Q.E.D.

More on Determinants: Properties of Determinants and the Transpose (Section 20.c)

Recall that for any matrix $M \in M_{m\times n}(K)$, we have the transposed matrix $M^T \in M_{n\times m}(K)$, where the entries are given by $M^T(i,j) = M(j,i)$.

Theorem 20.c.1 (*Determinant of the Transpose*): For all $A \in \mathcal{M}_{n\times n}(K)$, we have $\det(A^T) = \det(A)$.

Proof

If σ is a permutation of degree n, then by definition of the transpose, $A^T(i, \sigma(i)) = A(\sigma(i), i)$. Therefore, expanding the determinant definition gives:

$$\det(A^T) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \cdot A(\sigma(1), 1) \cdots A(\sigma(n), n).$$

If we let $j = \sigma(i)$, then $i = \sigma^{-1}(j)$, and consequently $A(\sigma(i), i) = A(j, \sigma^{-1}(j))$. Hence, we can rearrange the product completely:

$$A(\sigma(1), 1) \cdots A(\sigma(n), n) = A(1, \sigma^{-1}(1)) \cdots A(n, \sigma^{-1}(n)) \tag{20.3}$$

This holds because every factor on the left-hand side of (**??**) will appear exactly once on the right-hand side, and vice-versa.

Now, since $\sigma \circ \sigma^{-1} = \mathrm{id}$, we have $\operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\mathrm{id}) = 1$, which implies $\operatorname{sgn}(\sigma) = \operatorname{sgn}(\sigma^{-1})$. When σ runs over S_n (all possible permutations of degree n), its inverse σ^{-1} also naturally runs over the entirety of S_n.

This implies:

$$\begin{aligned}
\det(A^T) &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) A(\sigma(1), 1) \cdots A(\sigma(n), n) \\
&= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma^{-1}) A(1, \sigma^{-1}(1)) \cdots A(n, \sigma^{-1}(n)) \\
&= \sum_{\tau \in S_n} \operatorname{sgn}(\tau) A(1, \tau(1)) \cdots A(n, \tau(n)) \\
&= \det(A).
\end{aligned}$$

Q.E.D.

Example: Let $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$. Its transpose is $A^T = \left(\begin{smallmatrix} 1 & 3 \\ 2 & 4 \end{smallmatrix}\right)$. We have $\det(A) = (1)(4) - (2)(3) = -2$, and $\det(A^T) = (1)(4) - (3)(2) = -2$. Thus, the theorem holds.

Corollary 20.c.2 (*Column Alternating Property*): If $D : \mathcal{M}_{n \times n}(K) \to K$ is an alternating n-linear function of the rows of the matrix, then D is also an alternating n-linear function of the columns of the matrix. In fact, $D(A^T) = D(A)$ for all $A \in \mathcal{M}_{n \times n}(K)$. In particular, this holds for $D = \det$.

Proof

For any $A \in M_{n \times n}(K)$, we have previously established $D(A) = \det(A) \cdot D(I)$. Applying this to A^T gives $D(A^T) = \det(A^T) \cdot D(I)$. Since $\det(A^T) = \det(A)$, this evaluates exactly to $\det(A) \cdot D(I) = D(A)$. **Q.E.D.**

Because the determinant acts symmetrically on rows and columns, we can perform column operations just as we do row operations.

Theorem 20.c.3 (*Invariance Under Row Operations*): Let $A \in \mathcal{M}_{n \times n}(K)$.

(a) If B is obtained from A by adding a multiple of one row of A to another row (i.e., $c \cdot E_j + E_i \to E_i$ for $i \neq j$), or by doing the analogous operation on columns, then $\det(B) = \det(A)$.

(b) If B is obtained from A by interchanging two rows (i.e., $E_i \leftrightarrow E_j$ for $i \neq j$), or columns, then $\det(B) = -\det(A)$.

(c) If B is obtained from A by multiplying a row by a scalar (i.e., $c \cdot E_i \to E_i$), then $\det(B) = c \det(A)$.

Proof

Statement **(b)** follows directly from the fact that the determinant is an alternating function. Statement **(c)** follows from the n-linearity of the determinant.

To prove **(a)**, let A have rows $\alpha_1, \dots, \alpha_n$. Assume $j < i$. Expanding by linearity on the i-th row gives:

$$\begin{aligned}\det(B) &= \det(\alpha_1, \dots, \alpha_i + c\alpha_j, \dots, \alpha_j, \dots, \alpha_n) \\ &= \det(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) + c\det(\alpha_1, \dots, \alpha_j, \dots, \alpha_j, \dots, \alpha_n) \\ &= \det(A) + 0 = \det(A).\end{aligned}$$

The second term identically vanishes because two rows are equal (α_j appears twice). If $j > i$, the proof is similar. **Q.E.D.**

Determinants of Block Matrices

Calculating the determinant of a large matrix by expanding all permutations is computationally expensive. Block matrices provide a powerful shortcut. Consider a block matrix M of the form $M = \left(\begin{smallmatrix} A & B \\ 0 & C \end{smallmatrix}\right)$, where $A \in \mathcal{M}_{r\times r}(K)$, $C \in \mathcal{M}_{s\times s}(K)$, and $B \in \mathcal{M}_{r\times s}(K)$.

Proposition 20.c.4 (*Determinant of Block Matrices*): For a block matrix M defined as above, $\det(M) = \det(A) \cdot \det(C)$, where $A \in \mathcal{M}_{r\times r}(K)$, $C \in \mathcal{M}_{s\times s}(K)$, and $B \in \mathcal{M}_{r\times s}(K)$.

Proof

Define a helper function $D(A, B, C) := \det\left(\begin{smallmatrix} A & B \\ 0 & C \end{smallmatrix}\right)$ as a function $D : \mathcal{M}_{r\times r}(K) \times \mathcal{M}_{r\times s}(K) \times \mathcal{M}_{s\times s}(K) \to K$.

If we fix the first two matrices A and B, then the function mapping $C \mapsto D(A, B, C)$ is s-linear and alternating. By **??**, any such function satisfies:

$$D(A, B, C) = \det(C) \cdot D(A, B, I_s). \tag{20.4}$$

Let us calculate $D(A, B, I_s) = \det\left(\begin{smallmatrix} A & B \\ 0 & I_s \end{smallmatrix}\right)$. By doing row operations of the type $R_i + cR_j \to R_i$ (where $j \neq i$), we can use the last s rows of the identity matrix I_s to eliminate the entries in B entirely. This transforms the matrix into $\left(\begin{smallmatrix} A & 0 \\ 0 & I_s \end{smallmatrix}\right)$.

Since these operations do not change the determinant, we have:

$$\det\begin{pmatrix} A & B \\ 0 & I_s \end{pmatrix} = \det\begin{pmatrix} A & 0 \\ 0 & I_s \end{pmatrix}. \tag{20.5}$$

The function $A \mapsto \det\left(\begin{smallmatrix} A & 0 \\ 0 & I_s \end{smallmatrix}\right)$ is r-linear and alternating. By **??** again, it evaluates to $\det(A) \cdot \det\left(\begin{smallmatrix} I_r & 0 \\ 0 & I_s \end{smallmatrix}\right) = \det(A) \cdot \det(I_{r+s}) = \det(A) \cdot 1 = \det(A)$.

Going back through (**??**) and then to (**??**), we finally get that $D(A, B, C) = \det(A) \cdot \det(C)$. **Q.E.D.**

Remark (*Exercise*): Show that $\det\left(\begin{smallmatrix} A & 0 \\ B & C \end{smallmatrix}\right) = \det(A) \cdot \det(C)$, where $A \in \mathcal{M}_{r\times r}(K)$, $C \in \mathcal{M}_{s\times s}(K)$, and $B \in \mathcal{M}_{s\times r}(K)$. (Hint: Use column operations to reduce B to zero, or consider the transpose of the matrix and apply **??**.)

Example (*Calculation of a 4×4 Determinant*): Let $A = \left(\begin{smallmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 1 & -1 & -1 \\ 1 & 2 & 3 & 0 \end{smallmatrix}\right) \in \mathcal{M}_{4\times 4}(\mathbb{R})$. We begin by performing row operations $R_2 - 2R_1 \to R_2$, $R_3 - 4R_1 \to R_3$, and $R_4 - R_1 \to R_4$. This transforms A into:

$$\begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ 0 & 5 & -9 & -13 \\ 0 & 3 & 1 & -3 \end{pmatrix}$$

Now, we clean the second column below the diagonal using $R_3 - \frac{5}{4}R_2 \to R_3$ and $R_4 - \frac{3}{4}R_2 \to R_4$. This reveals a beautiful block structure:

$$\left(\begin{array}{cc|cc} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ \hline 0 & 0 & -4 & -8 \\ 0 & 0 & 4 & 0 \end{array}\right)$$

By applying **??** to this partitioned form, we can see that the determinant of the 4×4 matrix is simply the product of the determinants of the 2×2 diagonal blocks:

$$\det(A) = \det \begin{pmatrix} 1 & -1 \\ 0 & 4 \end{pmatrix} \cdot \det \begin{pmatrix} -4 & -8 \\ 4 & 0 \end{pmatrix}$$

Consequently, we evaluate the individual pieces:

$$\det(A) = (4) \cdot (32) = 128.$$

Cofactor Expansion and the Classical Adjoint

Recall that if D is a determinant function on $\mathcal{M}_{(n-1)\times(n-1)}(K)$, then for any fixed $j \in \{1, \ldots, n\}$, the function $E_j(A) := \sum_{i=1}^{n} (-1)^{i+j} A_{ij} D(A(i|j))$ is a determinant function as well. But, we have seen that on $\mathcal{M}_{n\times n}(K)$ there is a strictly unique determinant function, $\det$.

Corollary 20.c.5 (*Laplace Expansion*): For all $A \in \mathcal{M}_{n\times n}(K)$, we have the following n different formulas for $\det(A)$ (expanding along column j):

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+j} A_{ij} \det A(i|j), \quad \text{for any } j = 1, \ldots, n.$$

The scalars $C_{ij} := (-1)^{i+j} \det A(i|j)$ are called the i, j **cofactor of** A. We can succinctly write the expansion of $\det(A)$ by the cofactors of the j-th column as $\det(A) = \sum_{i=1}^{n} A_{ij} \cdot C_{ij}$.

Lemma (*Orthogonality of Cofactors*): If we replace A_{ij} in the formula by A_{ik} (where k is fixed), we always get 0. In other words, for all $A \in \mathcal{M}_{n\times n}(K)$ and for $j \neq k$, we have $\sum_{i=1}^{n} A_{ik} \cdot C_{ij} = 0$.

Proof

Let $B \in \mathcal{M}_{n\times n}(K)$ be the matrix obtained from A by replacing (not exchanging!) column j of A by column k of A. So, in B, column j equals column k. Clearly, $\det(B) = 0$ since B has two equal columns.

Expanding $\det(B)$ along the j-th column gives:

$$0 = \det(B) = \sum_{i=1}^{n} (-1)^{i+j} B_{ij} \det B(i|j).$$

By construction, $B_{ij} = A_{ik}$ for all i. Furthermore, since we only modified column j, deleting column j from both matrices leaves identical submatrices: $B(i|j) = A(i|j)$ for all i. Thus:

$$0 = \sum_{i=1}^{n} (-1)^{i+j} A_{ik} \det A(i|j) = \sum_{i=1}^{n} A_{ik} \cdot C_{ij}.$$

Q.E.D.

Summary: For $A \in \mathcal{M}_{n\times n}(K)$, we have $\sum_{i=1}^{n} A_{ik} \cdot C_{ij} = \delta_{jk} \cdot \det(A)$, where δ_{jk} is the Kronecker delta (which is 1 if $j = k$, and 0 otherwise).

Definition (*Classical Adjoint*): The transpose of the matrix of cofactors, $(C_{ij})^T$, is called the *"classical adjoint"* (or *"adjugate"*) of A, denoted $\operatorname{adj} A \in \mathcal{M}_{n\times n}(K)$. Explicitly, $(\operatorname{adj} A)_{ij} = C_{ji} = (-1)^{i+j} \det A(j|i)$.

Our summary formula above immediately implies the matrix equation:

$$(\operatorname{adj} A) \cdot A = (\det A) \cdot I \tag{20.6}$$

Lemma (*Left Adjugate Identity*): It is also true that $A \cdot (\operatorname{adj} A) = (\det A) \cdot I$.

Proof

Since $A^T(i|j) = (A(j|i))^T$, we have

$$(\operatorname{adj} A^T)_{ij} = (-1)^{i+j} \det A^T(j|i) = (-1)^{i+j} \det(A(i|j)^T) = (-1)^{i+j} \det A(i|j).$$

This perfectly matches $(\operatorname{adj} A)_{ji}$, which is exactly $((\operatorname{adj} A)^T)_{ij}$. Thus, $\operatorname{adj} A^T = (\operatorname{adj} A)^T$. Applying (**??**) to the matrix A^T gives $(\operatorname{adj} A^T) \cdot A^T = (\det A^T) \cdot I$. Using our transpose identities, this becomes $(\operatorname{adj} A)^T \cdot A^T = (\det A) \cdot I$. Taking the transpose of both sides finally yields $A \cdot (\operatorname{adj} A) = (\det A) \cdot I$. **Q.E.D.**

Theorem 20.c.6 (*Inverse via Adjugate*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then, A is invertible if and only if $\det A \neq 0$. When A is invertible, its inverse is given by $A^{-1} = \frac{1}{\det A} \cdot (\operatorname{adj} A)$.

Cramer's Rule

Let $A \in \mathcal{M}_{n\times n}(K)$. We would like an explicit formula to solve the system of linear equations $A \cdot x = b$, where $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ are the unknowns and $b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \in K^n$ is a given vector.

Assume $A \cdot x = b$. Multiplying both sides by the adjoint gives $(\operatorname{adj} A) \cdot A \cdot x = (\operatorname{adj} A) \cdot b$. This directly implies $(\det A) \cdot x = (\operatorname{adj} A) \cdot b$. Looking at the j-th row of this vector equation, we get:

$$\begin{aligned} (\det A) \cdot x_j &= \sum_{i=1}^{n} (\operatorname{adj} A)_{ji} \cdot b_i \\ &= \sum_{i=1}^{n} (-1)^{i+j} b_i \det A(i|j) \\ &= \det(A(j,b)) \end{aligned}$$

where $A(j, b)$ is the matrix obtained from A by replacing column j with the vector b.

Theorem 20.c.7 (*Cramer's Rule*): Let $A \in \mathcal{M}_{n\times n}(K)$ with $\det A \neq 0$. Let $b = (b_1, \ldots, b_n)^T \in K^n$. Then, the system $Ax = b$, where $x = (x_1, \ldots, x_n)^T$, has a unique solution, and it is given by the formula:

$$x_j = \frac{\det A(j,b)}{\det A} \quad \text{for all } 1 \leq j \leq n,$$

where $A(j, b)$ is the matrix obtained from A by replacing column j with b.

Proof of ??

We have seen previously in Linear Algebra I that if A is invertible, then $Ax = b$ has a unique solution. We have just shown above that $(\det A) \cdot x_j = \det A(j, b)$. Since $\det A \neq 0$, we can safely divide to obtain $x_j = \frac{\det A(j,b)}{\det A}$. **Q.E.D.**

Example: Consider the system $\left(\begin{smallmatrix}2&1\\1&3\end{smallmatrix}\right)\left(\begin{smallmatrix}x_1\\x_2\end{smallmatrix}\right)=\left(\begin{smallmatrix}5\\5\end{smallmatrix}\right)$. We have $\det(A)=2(3)-1(1)=5$. Using Cramer's rule:

$$x_1=\frac{\det\begin{pmatrix}5&1\\5&3\end{pmatrix}}{5}=\frac{15-5}{5}=2,\quad x_2=\frac{\det\begin{pmatrix}2&5\\1&5\end{pmatrix}}{5}=\frac{10-5}{5}=1.$$

Thus, the solution is $x=\binom{2}{1}$.

Determinants of Endomorphisms

Warm up: Let $A\in\mathcal{M}_{n\times n}(K)$, and let $P\in\mathrm{GL}(n,K)$ (meaning P is an invertible $n\times n$ matrix). Then, by multiplicativity (**??**):

$$\det(P^{-1}\cdot A\cdot P)=(\det P^{-1})\cdot(\det A)\cdot(\det P)=\frac{1}{\det P}\cdot\det A\cdot\det P=\det A.$$

In other words, similar matrices mathematically have the exact same determinant.

Let V be a finite-dimensional vector space over K, and put $n:=\dim V$. Let $T:V\to V$ be a linear map (an endomorphism). Pick a basis $\mathcal{B}$ for V. We have a matrix representation $[T]^{\mathcal{B}}_{\mathcal{B}}\in\mathcal{M}_{n\times n}(K)$, and we can consider its determinant $\det[T]^{\mathcal{B}}_{\mathcal{B}}\in K$.

Question: Does $\det[T]^{\mathcal{B}}_{\mathcal{B}}$ depend on the specific choice of the basis $\mathcal{B}$?

Answer: Let $\mathcal{B}'$ be another arbitrary basis for V. Denote by $P:=[\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}}$ the transition matrix between the basis $\mathcal{B}'$ and $\mathcal{B}$. Recall that $[T]^{\mathcal{B}'}_{\mathcal{B}'}=[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}'}\cdot[T]^{\mathcal{B}}_{\mathcal{B}}\cdot[\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}}$. We also formally have $[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}'}=([\mathrm{id}_V]^{\mathcal{B}'}_{\mathcal{B}})^{-1}=P^{-1}$. So, $[T]^{\mathcal{B}'}_{\mathcal{B}'}=P^{-1}\cdot[T]^{\mathcal{B}}_{\mathcal{B}}\cdot P$. This implies $\det[T]^{\mathcal{B}'}_{\mathcal{B}'}=\det[T]^{\mathcal{B}}_{\mathcal{B}}$.

Recall also that if $T,S:V\to V$ are two linear maps, then $[S\circ T]^{\mathcal{B}}_{\mathcal{B}}=[S]^{\mathcal{B}}_{\mathcal{B}}\cdot[T]^{\mathcal{B}}_{\mathcal{B}}$. If T is an isomorphism, then $[T^{-1}]^{\mathcal{B}}_{\mathcal{B}}=([T]^{\mathcal{B}}_{\mathcal{B}})^{-1}$. Finally, $[\mathrm{id}_V]^{\mathcal{B}}_{\mathcal{B}}=I$.

Corollary 20.c.8 (*Determinant of an Endomorphism*): Let V be a finite-dimensional vector space over K, and let $T:V\to V$ be a linear map. Then, $\det([T]^{\mathcal{B}}_{\mathcal{B}})$ does NOT depend on the choice of the basis $\mathcal{B}$ of V. We denote this intrinsic scalar simply by $\det(T)\in K$.

The function $\det:\mathrm{End}(V)\to K, T\mapsto\det(T)$ has the following elegant properties:

(a) $\det(S\circ T)=\det(S)\cdot\det(T)$ for all $S,T\in\mathrm{End}(V)$.

(b) T is an isomorphism if and only if $\det T\neq 0$. If T is an isomorphism, then $\det(T^{-1})=\frac{1}{\det T}$.

(c) $\det(\mathrm{id}_V)=1$.

(d) $\det(0)=0\in K$ (where 0 is the zero linear map).

Important remark: In contrast to an endomorphism $T:V\to V$, if we take a linear map $T:V\to W$ between two vector spaces of the same dimension and bases $\mathcal{B},\mathcal{C}$ of V,W, then $\det[T]^{\mathcal{B}}_{\mathcal{C}}$ generally depends on the choice of $\mathcal{B}$ and $\mathcal{C}$.

Eigenvalues & Eigenvectors (Chapter 21)

Motivation (Section 21.a)

Fibonacci revisited

Consider the sequence $f_0, f_1, f_2, f_3, \dots, f_n, \dots$ defined by $f_0 = 0$, $f_1 = 1$, and $f_n := f_{n-1} + f_{n-2}$ for $n \geq 2$. This yields the sequence: $0, 1, 1, 2, 3, 5, 8, 13, 21, \dots$

We found a formula for f_n:

$$f_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right) \quad \forall n \geq 0.$$

We defined Fib := the set of all sequences $(a_0, a_1, a_2, \dots, a_n, \dots)$ that satisfy $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$.

Claim: $\mathrm{Fib} \subseteq \mathbb{R}^\infty$ is a 2-dimensional vector subspace of the space of all infinite sequences

$$(x_0, x_1, x_2, \dots, x_n, \dots).$$

In fact, the map $F : \mathrm{Fib} \longrightarrow \mathbb{R}^2$ given by $F(a_0, a_1, \dots, a_n, \dots) := (a_0, a_1) \in \mathbb{R}^2$ is an isomorphism.

Claim: Put $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi := \frac{1-\sqrt{5}}{2}$. Then the two geometric sequences $(1, \varphi, \varphi^2, \dots, \varphi^n, \dots)$ and $(1, \psi, \psi^2, \dots, \psi^n, \dots)$ belong to Fib and, moreover, they form a basis for Fib.

We also saw that using these two sequences, we could easily find an explicit formula for all Fibonacci sequences. But how did we find these two special sequences?

The shift map $S : \mathrm{Fib} \longrightarrow \mathrm{Fib}$ defined by $S(a_0, a_1, \dots, a_n, \dots) := (a_1, a_2, \dots, a_{n+1}, \dots)$ is a linear map. Let us try to find a Fibonacci sequence $\mathcal{F} \neq 0$ such that $S(\mathcal{F}) = \lambda \cdot \mathcal{F}$ for some λ.

$$S(\mathcal{F}) = \lambda \cdot \mathcal{F} \iff \lambda = \frac{1 \pm \sqrt{5}}{2} \text{ and } \mathcal{F} = (a, a\lambda, a\lambda^2, \dots, a\lambda^n, \dots)$$

Definition 21.a.1 (*Eigenvalue and Eigenvector*): Let V be a vector space over K and $T : V \to V$ a linear map (an endomorphism of V). We say that $\lambda \in K$ is an *"eigenvalue"* (a.k.a. *"characteristic value"*) of T if there exists a vector $v \neq 0$ such that $Tv = \lambda v$. Any vector $v \neq 0$ with the property that $Tv = \lambda v$ is called an *"eigenvector"* (of T) for the eigenvalue λ.

Definition 21.a.2 (*Diagonalizable Endomorphism*):

$$[T]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix} \tag{21.1}$$

Lemma 21.a.3 (*Diagonalizable Matrix Criterion*): T is diagonalizable if and only if there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ consisting of eigenvectors of T (i.e., $\forall i$, $Tv_i = \lambda_i \cdot v_i$ for some $\lambda_i \in K$). Moreover, if for some basis $\mathcal{B} = (v_1, \dots, v_n)$ the matrix $[T]^{\mathcal{B}}_{\mathcal{B}}$ has the shape (**??**), then for all i, λ_i is an eigenvalue of T and v_i is an eigenvector for λ_i.

Proof

Recall that $[T]^{\mathcal{B}}_{\mathcal{B}} = ([Tv_1]_{\mathcal{B}} \dots [Tv_n]_{\mathcal{B}})$ and $[Tv_j]_{\mathcal{B}} = \begin{pmatrix} \alpha_{1j} \\ \vdots \\ \alpha_{nj} \end{pmatrix}$, where $Tv_j = \alpha_{1j} v_1 + \dots + \alpha_{nj} v_n$. The matrix $[T]^{\mathcal{B}}_{\mathcal{B}}$ has the shape (**??**) if and only if $\alpha_{ij} = 0$ for all $i \neq j$, and $\alpha_{kk} = \lambda_k \iff Tv_k = \lambda_k \cdot v_k$.

Q.E.D.

Remark: We saw that for all $A \in \mathcal{M}_{n\times n}(K)$, we have a linear map $T_A : K^n \to K^n$, $T_A(v) := A \cdot v$. We say that a matrix $A \in \mathcal{M}_{n\times n}(K)$ is diagonalizable if the linear map T_A is diagonalizable.

Example: Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and $K = \mathbb{R}$. $T_A\binom{1}{1} = \binom{3}{3} = 3\binom{1}{1}$. Thus, $\lambda = 3$ is an eigenvalue and $v = \binom{1}{1}$ is a corresponding eigenvector. Furthermore, $T_A\binom{1}{-1} = \binom{1}{-1}$. This implies that $\lambda = 1$ is an eigenvalue and $v = \binom{1}{-1}$ is a corresponding eigenvector. Geometrically, this represents a stretching by a factor of 3 in $\mathbb{R}^2$ along one axis and the identity map along the other axis.

Example: Let $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Let us find all possible eigenvalues and eigenvectors. We are looking for $A \cdot v = \lambda v$ with $\lambda \in \mathbb{R}$ and $0 \neq v \in \mathbb{R}^2$. This gives $(A - \lambda I)v = 0$. This implies that $A - \lambda I$ is not invertible, which in turn means $\det(A - \lambda I) = 0$ (which leads to a quadratic equation).

Proposition 21.a.4 (*Linear Independence of Eigenvectors*): Let V be a vector space over K and $T : V \to V$ a linear map. Suppose that $\lambda_1, \dots, \lambda_n \in K$ are eigenvalues of T which are pairwise distinct (i.e., $\lambda_i \neq \lambda_j$ for all $i \neq j$). Let $v_1, \dots, v_n$ be eigenvectors for $\lambda_1, \dots, \lambda_n$ respectively. Then $v_1, \dots, v_n$ are linearly independent. (Put shortly: eigenvectors corresponding to a list of distinct eigenvalues are always linearly independent).

Proof

We proceed by induction on n. Base case $n = 1$: v_1 is an eigenvector for λ_1. By **??**, $v_1 \neq 0$, which implies that v_1 is linearly independent.

Inductive step for $n \geq 1$: Suppose the proposition holds for any list of n eigenvalues and eigenvectors. We will prove now that the proposition holds also for a list of $n+1$ eigenvalues and eigenvectors. Let $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$ be pairwise distinct eigenvalues of T, and $v_1, \dots, v_n, v_{n+1}$ a given list of eigenvectors of $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$ respectively. We need to prove that $v_1, \dots, v_n, v_{n+1}$ are linearly independent. Assume that

$$a_1 v_1 + \cdots + a_n v_n + a_{n+1} v_{n+1} = 0. \tag{21.2}$$

We need to show that $a_1 = \cdots = a_n = a_{n+1} = 0$. Apply the transformation T to (**??**):

$$a_1 T(v_1) + \cdots + a_n T(v_n) + a_{n+1} T(v_{n+1}) = 0.$$

But since $T(v_i) = \lambda_i v_i$, we have:

$$a_1 \lambda_1 v_1 + \cdots + a_n \lambda_n v_n + a_{n+1} \lambda_{n+1} v_{n+1} = 0. \tag{21.3}$$

Consider now the combination (**??**) $- \lambda_{n+1} \cdot$ (**??**):

$$a_1(\lambda_1 - \lambda_{n+1})v_1 + \cdots + a_n(\lambda_n - \lambda_{n+1})v_n + a_{n+1}(\lambda_{n+1} - \lambda_{n+1})v_{n+1} = 0.$$

By the induction hypothesis, $v_1, \dots, v_n$ are linearly independent. Therefore, $a_1(\lambda_1 - \lambda_{n+1}) = \cdots = a_n(\lambda_n - \lambda_{n+1}) = 0$. By assumption, the eigenvalues are distinct, so $\lambda_i \neq \lambda_{n+1}$. Consequently, $a_1 = \cdots = a_n = 0$. Going back to (**??**), we get $a_{n+1} v_{n+1} = 0$. But since $v_{n+1} \neq 0$ (because v_{n+1} is an eigenvector), this implies that $a_{n+1} = 0$. **Q.E.D.**

The Characteristic Polynomial

Proposition 21.a.5 (*Eigenvalue Injective Criterion*): Let V be a vector space over K and $T \in \mathrm{End}(V)$. Then λ is an eigenvalue of T if and only if the linear map $T - \lambda \cdot \mathrm{id}_V$ is not injective.

Proof

The scalar λ is an eigenvalue if and only if there exists a non-zero vector $v \in V$ such that $Tv - \lambda v = 0$. This is equivalent to stating that $(T - \lambda \cdot \mathrm{id}_V)v = 0$, which implies that $\ker(T - \lambda \cdot \mathrm{id}_V) \neq \{0\}$. This implies that $T - \lambda \cdot \mathrm{id}_V$ is not injective. **Q.E.D.**

Definition 21.a.6 (*Characteristic Polynomial*): Let V be a finite-dimensional vector space over K and $T \in \operatorname{End}(V)$. We define the *"characteristic polynomial"* of T as $P_T(x) := \det(T - x \cdot \mathrm{id}_V)$, where x is a formal variable. We define the *"characteristic polynomial"* of a matrix $A \in \mathcal{M}_{n\times n}(K)$ in the same way: $P_A(x) := \det(A - x \cdot I_n)$.

Definition 21.a.7 (*Eigenspace*): Let V be a vector space over K and $T \in \operatorname{End}(V)$. Let λ be an eigenvalue of T. The *"eigenspace"* corresponding to λ is defined as:

$$\operatorname{Eig}_T(\lambda) := \{v \in V \mid Tv = \lambda v\} = \ker(T - \lambda \cdot \mathrm{id}_V) = \{v \in V \mid v \text{ is an eigenvector of } \lambda\} \cup \{0\}.$$

Clearly (by this definition), $\operatorname{Eig}_T(\lambda) \subseteq V$ is a linear subspace.

Lemma 21.a.8 (*Properties of the Characteristic Polynomial*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then:

(a) $P_A(x)$ is a polynomial of degree n with leading coefficient $(-1)^n$ (i.e., $P_A(x) = c_n x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + \cdots + c_0$, with $c_i \in K$ for all i, and $c_n = (-1)^n$).

(b) $P_{A^T}(x) = P_A(x)$.

(c) If $A = \begin{pmatrix} \lambda_1 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix}$ has upper triangular form, then $P_A(x) = (\lambda_1 - x) \cdot \dots \cdot (\lambda_n - x)$.

(d) If $A = \left(\begin{array}{c|c} B_1 & * \\ \hline 0 & B_2 \end{array}\right)$ is a block matrix, then $P_A(x) = P_{B_1}(x) \cdot P_{B_2}(x)$.

(e) If A and B are similar matrices (i.e., $B = P^{-1}AP$ for some invertible matrix P), then $P_A(x) = P_B(x)$.

Proof

(a) Let

$$P_A(x) = \det(A - x \cdot I_n) = \det \begin{pmatrix} A_{11} - x & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} - x & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nn} - x \end{pmatrix}.$$

Denote this matrix (inside the determinant) by B. This determinant equals

$$\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \cdot B_{1,\sigma 1} \dots B_{n,\sigma n} = (A_{11} - x) \cdot (A_{22} - x) \dots (A_{nn} - x)$$
$$+ (\text{products of degree} \le n-1).$$
$$+ (\text{terms from other permutations, of degree} \le n-2).$$

If $\sigma \neq \mathrm{id} = \left(\begin{smallmatrix} 1 & 2 & \cdots & n \\ 1 & 2 & \cdots & n \end{smallmatrix}\right)$, then at least one factor in the product $B_{1,\sigma 1} \dots B_{n,\sigma n}$ does not contain x. $= (-1)^n x^n + (\text{terms of degree} \le n-1)$.

(b) $P_{A^T}(x) = \det(A^T - x \cdot I_n) = \det((A - x \cdot I_n)^T) = \det(A - x \cdot I_n) = P_A(x)$.

(c)+(d) These properties follow directly from the definition of the determinant and its behavior with block matrices, as established in Chapter 20 (Determinants). For example, for (c), the determinant of an upper triangular matrix is the product of its diagonal entries. For (d), the determinant of a block triangular matrix is the product of the determinants of its diagonal blocks.

(e) $P_B(x) = \det(B - x \cdot I_n) = \det(P^{-1}AP - x \cdot I_n) = \det(P^{-1}(A - x \cdot I_n)P) = \det(A - x \cdot I_n) = P_A(x)$. **Q.E.D.**

Remark: If V is an n-dimensional vector space over K and $T \in \operatorname{End}(V)$, then $P_T(x)$ is a polynomial of degree n with leading coefficient $(-1)^n$.

Proof

Let $\mathcal{B}$ be a basis for V. Then $[T - x \cdot \mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = [T]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I_n$. This implies that $P_T(x) = \det([T - x \cdot \mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}}) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I_n) = P_{[T]_{\mathcal{B}}^{\mathcal{B}}}(x)$. The claim of the remark now follows from **(a)** of **??**. **Q.E.D.**

Definition 21.a.9 (*Trace*): Let $A \in \mathcal{M}_{n\times n}(K)$. Define the *"trace"* (German: *"Spur"*) of A, denoted $\mathrm{Tr}(A) := \sum_{i=1}^n A_{ii} \in K$ (which is the sum of the entries along the diagonal of A). If V is a finite-dimensional vector space over K and $T \in \mathrm{End}(V)$, define $\mathrm{Tr}(T) := \mathrm{Tr}([T]_{\mathcal{B}}^{\mathcal{B}})$, where $\mathcal{B}$ is some basis for V. We will see soon that $\mathrm{Tr}(T)$ is well defined (i.e., it does not depend on the choice of $\mathcal{B}$).

Lemma 21.a.10 (*Properties of Trace*): For all $A, B \in \mathcal{M}_{n\times n}(K)$, we have:

(a) $\mathrm{Tr}(A \cdot B) = \mathrm{Tr}(B \cdot A)$.

(b) If A and B are similar matrices, then $\mathrm{Tr}(A) = \mathrm{Tr}(B)$.

In particular, if $\mathcal{B}$ and $\mathcal{B}'$ are two bases for V, then $\mathrm{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}) = \mathrm{Tr}([T]_{\mathcal{B}'}^{\mathcal{B}'})$. Thus, $\mathrm{Tr}(T)$ is well defined.

Proof

(a) $\mathrm{Tr}(A \cdot B) = \sum_{i=1}^n (A \cdot B)_{ii} = \sum_{i=1}^n \sum_{j=1}^n A_{ij} \cdot B_{ji} = \sum_{i,j} A_{ij} B_{ji} = \sum_{i,j} A_{ji} B_{ij} = \sum_{i=1}^n \sum_{j=1}^n B_{ij} A_{ji} = \sum_{i=1}^n (B \cdot A)_{ii} = \mathrm{Tr}(B \cdot A)$.

(b) If $B = P^{-1}AP$, then $\mathrm{Tr}(B) = \mathrm{Tr}(P^{-1}AP) = \mathrm{Tr}(P^{-1} \cdot (AP)) = \mathrm{Tr}(AP \cdot P^{-1}) = \mathrm{Tr}(A)$.

Regarding the claim about $\mathrm{Tr}(T)$, recall that $[T]_{\mathcal{B}'}^{\mathcal{B}'} = [\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'}$, and $[\mathrm{id}_V]_{\mathcal{B}'}^{\mathcal{B}} = ([\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1}$. Use **(b)** to conclude the proof. **Q.E.D.**

Why is the trace interesting? Let $A \in \mathcal{M}_{n\times n}(K)$ and $P_A(x) = c_n x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0$ its characteristic polynomial. We have seen that $c_n = (-1)^n$.

Lemma 21.a.11 (*Coefficients of the Characteristic Polynomial*): $c_{n-1} = (-1)^{n-1} \cdot \mathrm{Tr}(A)$ and $c_0 = \det(A)$. In other words, $P_A(x) = (-1)^n x^n + (-1)^{n-1}\mathrm{Tr}(A) \cdot x^{n-1} + c_{n-2}x^{n-2} + \cdots + c_1 x + \det(A)$.

Proof

$$P_A(x) = \det \begin{pmatrix} A_{11} - x & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} - x & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \dots & A_{nn} - x \end{pmatrix}.$$ Denote this matrix by B.

$$P_A(x) = (A_{11} - x) \cdot (A_{22} - x) \dots (A_{nn} - x) + \sum_{\sigma \in S_n, \sigma \neq \mathrm{id}} \mathrm{sgn}(\sigma) B_{1,\sigma 1} \dots B_{n,\sigma n}. \tag{21.4}$$

If $\sigma \neq \mathrm{id}$, then there exist at least two indices $i \neq j$ such that $i \neq \sigma i$ and $j \neq \sigma j$. Thus, the product contains at least two factors ($B_{i,\sigma i}$ and $B_{j,\sigma j}$) that are not from the diagonal of B, and so do not contain x. Therefore, the degree of the sum in (**??**) is $\leq n-2$. $P_A(x) = (A_{11}-x)(A_{22}-x)\dots(A_{nn}-x) +$ (another polynomial of degree $\leq n-2$). $= (-1)^n x^n + \left[A_{11}(-x)^{n-1} + \cdots + A_{nn}(-x)^{n-1}\right] + \dots$ Expanding the product $(A_{11} - x)(A_{22} - x)\dots(A_{nn} - x)$ yields $(-1)^n x^n + (-1)^{n-1}(A_{11} + \cdots + A_{nn})x^{n-1} + \dots$. $= (-1)^n x^n + (-1)^{n-1} \cdot \mathrm{Tr}(A)x^{n-1} + \cdots + c_1 x + c_0$.

Finally, $c_0 = P_A(0) = \det(A - 0 \cdot I_n) = \det(A)$ (obtained by substituting $x = 0$). **Q.E.D.**

Definition 21.a.12 (*Direct Sum*): Let V be a vector space over K and $U_1, \dots, U_r \subseteq V$ subspaces. Denote $W := U_1 + \cdots + U_r \subseteq V$. We say that W is a *"direct sum"* of $U_1, \dots, U_r$ (or that $U_1 + \cdots + U_r$ is a *"direct sum"*) if every $w \in W$ has a unique representation as $w = u_1 + \cdots + u_r$, with $u_i \in U_i$ for all i.

Exercise: $U_1 + \cdots + U_r$ is a direct sum if and only if any of the following conditions hold:

(a) The only way to write $0 = u_1 + \cdots + u_r$, with $u_i \in U_i$ for all i, is with $u_1 = \cdots = u_r = 0$.

(b) For all $2 \leq j \leq r$, we have $U_j \cap (U_1 + \cdots + U_{j-1}) = \{0\}$.

Under the additional assumption that $U_1, \ldots, U_r$ are all finite-dimensional, $U_1 + \cdots + U_r$ is a direct sum if and only if whenever $\mathcal{B}_1, \ldots, \mathcal{B}_r$ are bases for $U_1, \ldots, U_r$ respectively, then $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$ forms a basis for $U_1 + \cdots + U_r$.

Exercise: If $U_1 + \cdots + U_r$ is a direct sum, then there exists a canonical isomorphism $U_1 \oplus \cdots \oplus U_r \xrightarrow{\sim} U_1 + \cdots + U_r$, given by $(u_1, \ldots, u_r) \mapsto u_1 + \cdots + u_r$. So when $U_1 + \cdots + U_r$ is a direct sum, we can identify $U_1 + \cdots + U_r$ with $U_1 \oplus \cdots \oplus U_r$.

Proposition 21.a.13 (*Direct Sum of Eigenspaces*): Let V be a vector space over K and $T \in \mathrm{End}(V)$. Let $\lambda_1, \ldots, \lambda_r \in K$ be pairwise distinct eigenvalues of T. Then the sum $\mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_r)$ is a direct sum.

Proof

Assume $u_1 + \cdots + u_r = u_1' + \cdots + u_r'$, where $u_i, u_i' \in \mathrm{Eig}_T(\lambda_i)$ for all $1 \leq i \leq r$. We need to show that $u_i = u_i'$ for all $i \iff (u_i - u_i' = 0 \text{ for all } i)$. We have:

$$(u_1 - u_1') + \cdots + (u_r - u_r') = 0. \tag{21.5}$$

Assume by contradiction that there exists an i such that $u_i - u_i' \neq 0$. Remove from the list $u_1 - u_1', \ldots, u_r - u_r'$ all those vectors that are 0. We get a non-empty list $u_{i_1} - u_{i_1}', \ldots, u_{i_\ell} - u_{i_\ell}'$ of non-zero vectors, and by (**??**), we still have

$$(u_{i_1} - u_{i_1}') + \cdots + (u_{i_\ell} - u_{i_\ell}') = 0. \tag{21.6}$$

where $u_{i_k} - u_{i_k}' \in \mathrm{Eig}_T(\lambda_{i_k})$ Now for all k, $u_{i_k} - u_{i_k}'$ is either an eigenvector of λ_{i_k} or 0, but by our assumptions, $u_{i_k} - u_{i_k}' \neq 0$. Recall from **??** that eigenvectors corresponding to pairwise distinct eigenvalues are linearly independent. $u_{i_1} - u_{i_1}', \ldots, u_{i_\ell} - u_{i_\ell}'$ are linearly independent vectors, and by (**??**), their sum is 0. This is a contradiction. **Q.E.D.**

Corollary 21.a.14 (*Diagonalizability via Eigenspaces*): Let V be finite-dimensional (over K) and $T \in \mathrm{End}(V)$ with distinct eigenvalues $\lambda_1, \ldots, \lambda_r \in K$. Then, T is diagonalizable if and only if the dimensions of its eigenspaces sum to $\dim V$:

$$\sum_{i=1}^{r} \dim \mathrm{Eig}_T(\lambda_i) = \dim V.$$

Exercise: If V is finite-dimensional, then T can have only finitely many eigenvalues. (Of course, it is implicitly assumed $\lambda_j \neq \lambda_k$ for all $j \neq k$).

Proof

Choose a basis $\mathcal{B}_i$ for $\mathrm{Eig}_T(\lambda_i)$. Recall that $W := \mathrm{Eig}_T(\lambda_1) + \cdots + \mathrm{Eig}_T(\lambda_r)$ is a direct sum. This implies that $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$ is a basis for W. Now $W \subseteq V$ is a linear subspace, and $\dim W = \sum_{i=1}^r |\mathcal{B}_i| = \sum_{i=1}^r \dim \mathrm{Eig}_T(\lambda_i) \leq \dim V$. This implies $W = V$ if and only if $\sum_{i=1}^r \dim \mathrm{Eig}_T(\lambda_i) = \dim V$. Consequently, V has a basis (namely $(\mathcal{B}_1, \ldots, \mathcal{B}_r)$) whose elements are eigenvectors of T if and only if this dimension equality holds. This implies that T is diagonalizable. **Q.E.D.**

Multiplicities of Eigenvalues (Section 21.b)

Example: Consider the matrices $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. We have that their characteristic polynomials are identical: $P_A(x) = P_B(x) = (1-x)^2 = (x-1)^2$.

Example (*Jordan Block*): Consider the matrix $J_{\lambda,n} \in \mathcal{M}_{n\times n}(K)$, for $\lambda \in K$.

$$J := J_{\lambda,n} := \begin{pmatrix} \lambda & 1 & 0 & \dots & 0 \\ 0 & \lambda & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & 0 & \lambda \end{pmatrix} \tag{21.7}$$

This is called a *"Jordan block"*. It features λ's along the main diagonal, 1's along the "diagonal" just above the main diagonal, and 0's everywhere else. The characteristic polynomial is $P_{J_{\lambda,n}}(x) = (\lambda - x)^n$. This implies that J has only one eigenvalue, namely λ.

Let us find the eigenvectors for the eigenvalue λ:

$$J - \lambda \cdot I_n = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

This matrix clearly has $n-1$ linearly independent columns. Therefore, the rank is $\operatorname{rank}(J-\lambda\cdot I_n) = n-1$. By the rank-nullity theorem, $\dim\ker(J-\lambda\cdot I_n) = 1$. Consequently, the eigenspace $\operatorname{Eig}_J(\lambda)$ is 1-dimensional. Clearly, $e_1 = (1,0,\dots,0)^T$ is an eigenvector for λ. Thus, $\operatorname{Eig}_J(\lambda) = \operatorname{Sp}(e_1)$.

Important remark: There is an important difference between polynomials $p(x) \in K[x]$ and the functions they define $K \to K$. For example, if $K = \mathbb{Z}_2$, then $p(x) = x$ and $q(x) = x^2$ are different polynomials, but they define the same function $\mathbb{Z}_2 \to \mathbb{Z}_2$. (What happens for $K = \mathbb{Q}, \mathbb{R}, \mathbb{C}, \dots$?)

Lemma 21.b.1 (*Decomposition of Diagonalizable Matrices*): Let $A \in M_{n\times n}(K)$ (or $T \in \operatorname{End}(V)$) be diagonalizable. Then $P_A(x)$ decomposes as a product of linear (i.e., degree 1) factors:

$$P_A(x) = (\lambda_1 - x)\cdot\dots\cdot(\lambda_n - x),$$

where $\lambda_1,\dots,\lambda_n$ are the eigenvalues of A (but there might be repetitions among the λ_i's).

Proof

By assumption, there exists an invertible matrix P such that:

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix} =: D. \tag{21.8}$$

(i.e., A is similar to D). Since similar matrices have the same characteristic polynomial, we have:

$$P_A(x) = P_D(x) = \det \begin{pmatrix} \lambda_1 - x & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n - x \end{pmatrix} = (\lambda_1 - x)\dots(\lambda_n - x).$$

Q.E.D.

Example: Let $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ over $K = \mathbb{R}$. The associated linear map $T_A : \mathbb{R}^2 \to \mathbb{R}^2$ represents a counter-clockwise rotation. The characteristic polynomial is $P_A(x) = \det\left(\begin{smallmatrix} -x & -1 \\ 1 & -x \end{smallmatrix}\right) = x^2 + 1$. This polynomial has no roots in $\mathbb{R}$. And, of course, A has no real eigenvalues.

A quick digression about polynomials...

Let $0 \neq p(x) \in K[x]$ and $\lambda \in K$. By dividing $p(x)$ by $(x-\lambda)$, we can write:

$$p(x) = (x-\lambda) \cdot q(x) + R, \tag{21.9}$$

where $\deg q(x) = \deg p(x) - 1$, and $R \in K$ is the remainder. If we substitute $x = \lambda$ into (??), we obtain $p(\lambda) = 0 \cdot q(\lambda) + R = R$. Thus, $p(x) = (x-\lambda) \cdot q(x) + p(\lambda)$. Conclusion: $p(\lambda) = 0 \iff p(x) = (x-\lambda) \cdot q(x)$ for some $q(x) \in K[x]$. We say that λ is a *"zero"* of $p(x)$ (De: Nullstelle), or a *"root"* of $p(x)$. Sometimes, λ is also a zero of $q(x)$, so $q(x) = (x-\lambda) \cdot h(x)$, which implies $p(x) = (x-\lambda)^2 \cdot h(x)$, etc.

Definition 21.b.2 (*Multiplicity of a Zero*): We say that λ is a zero of *"multiplicity"* $m \in \mathbb{N}_{>0}$ if $p(x) = (x-\lambda)^m \cdot g(x)$, where $g(x) \in K[x]$ and $g(\lambda) \neq 0$ (i.e., λ is not a zero of $g(x)$). If $m = 1$, we say λ is a *"simple zero"* of $p(x)$. If $m \geq 2$, we say λ is a *"multiple zero"* of $p(x)$.

Geometric & Algebraic Multiplicity of Eigenvalues

Definition 21.b.3 (*Geometric and Algebraic Multiplicity*): Let V be a vector space over K and $T \in \mathrm{End}(V)$. Let λ be an eigenvalue of T. We define the *"geometric multiplicity"* of λ to be:

$$m_g(T;\lambda) := \dim \mathrm{Eig}_T(\lambda).$$

Assume now, in addition, that V is finite-dimensional. We define the *"algebraic multiplicity"* of λ to be the multiplicity of the zero λ in the characteristic polynomial $P_T(x)$ of T. We denote it by $m_a(T;\lambda)$.

Proposition 21.b.4 (*Multiplicity Inequality*): For any eigenvalue λ, the geometric multiplicity is less than or equal to the algebraic multiplicity:

$$m_g(T;\lambda) \leq m_a(T;\lambda).$$

Proof

Let $v_1, \dots, v_k$ be a basis for the eigenspace $\mathrm{Eig}_T(\lambda)$. We extend this basis to a basis $\mathcal{B} = (v_1, \dots, v_k, w_{k+1}, \dots, w_n)$ of V. Then the matrix of T with respect to the basis $\mathcal{B}$ has the block form:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = ([Tv_1]_{\mathcal{B}} \dots [Tv_k]_{\mathcal{B}} \mid [Tw_{k+1}]_{\mathcal{B}} \dots [Tw_n]_{\mathcal{B}})$$

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{c|c} \lambda I_k & C_1 \\ \hline 0 & C \end{array}\right) \tag{21.10}$$

where $C \in \mathcal{M}_{(n-k)\times(n-k)}(K)$. We compute the characteristic polynomial by utilizing the determinant property for block matrices (as established in previous sections):

$$P_T(x) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I_n) = (\lambda - x)^k \cdot \det(C - xI_{n-k})$$

$$P_T(x) = (\lambda - x)^k \cdot P_C(x) = (-1)^k (x-\lambda)^k \cdot P_C(x).$$

This factorization directly shows that the multiplicity of λ as a zero of $P_T(x)$ is at least k. Therefore, we conclude that $m_a(T;\lambda) \geq k = m_g(T;\lambda)$. **Q.E.D.**

Example: Revisiting the Jordan block $J := J_{\lambda,n}$ defined in (??): We previously saw that $P_J(x) = (\lambda - x)^n$, which directly implies that $m_a(J;\lambda) = n$. We also found that $\mathrm{Eig}_J(\lambda)$ is a 1-dimensional space, which means $m_g(J;\lambda) = 1$.

Diagonalizability (Section 21.c)

Theorem 21.c.1 (*Equivalent Conditions for Diagonalizability*): Let V be an n-dimensional vector space over K and $T \in \mathrm{End}(V)$. The following statements are equivalent:

(a) T is diagonalizable.

(b) There exists a basis for V whose elements are eigenvectors of T.

(c) The characteristic polynomial $P_T(x)$ splits into a product of linear factors and, for every eigenvalue of T, the geometric and algebraic multiplicities are equal.

(d) $\sum_{i=1}^{k} \dim \mathrm{Eig}_T(\lambda_i) = \dim V$, where $\lambda_1, \dots, \lambda_k \in K$ are all the distinct eigenvalues of T.

(e) $V \cong \bigoplus_{i=1}^{k} \mathrm{Eig}_T(\lambda_i)$.

Proof

(a) $\iff$ **(b)** has already been proved.

(a) $\implies$ **(c)**: Let $\mathcal{B}$ be a basis in which $[T]^{\mathcal{B}}_{\mathcal{B}}$ has diagonal form. By changing the order of the elements in $\mathcal{B}$, we can arrange that $[T]^{\mathcal{B}}_{\mathcal{B}}$ has the following block-diagonal shape:

$$[T]^{\mathcal{B}}_{\mathcal{B}} = \left(\begin{array}{c|c|c} \begin{matrix}\lambda_1 & \dots & 0\\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_1\end{matrix} & 0 & 0 \\ \hline 0 & \ddots & 0 \\ \hline 0 & 0 & \begin{matrix}\lambda_k & \dots & 0\\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_k\end{matrix}\end{array}\right) \tag{21.11}$$

where $\mathcal{B} = (v_1^{(1)}, \dots, v_{\ell_1}^{(1)}, v_1^{(2)}, \dots, v_{\ell_2}^{(2)}, \dots, v_1^{(k)}, \dots, v_{\ell_k}^{(k)})$. Here, $\lambda_1, \dots, \lambda_k$ are the eigenvalues of T (with $\lambda_i \neq \lambda_j$ for all $i \neq j$) and $v_1^{(i)}, \dots, v_{\ell_i}^{(i)}$ are eigenvectors of λ_i for all i. The characteristic polynomial is:

$$P_T(x) = \det([T]^{\mathcal{B}}_{\mathcal{B}} - x \cdot I_n) = (\lambda_1 - x)^{\ell_1} \cdot \dots \cdot (\lambda_k - x)^{\ell_k}. \tag{21.12}$$

This shows that $P_T(x)$ splits as a product of linear factors. It also shows that $\lambda_1, \dots, \lambda_k$ are all the eigenvalues of T. It remains to show that $m_g(T; \lambda_i) = m_a(T; \lambda_i)$ for all i. Clearly, $v_1^{(i)}, \dots, v_{\ell_i}^{(i)} \in \mathrm{Eig}_T(\lambda_i)$ are linearly independent; hence $\ell_i \leq \dim \mathrm{Eig}_T(\lambda_i) = m_g(T; \lambda_i)$ by definition. But from (**??**), we also have $\ell_i = m_a(T; \lambda_i)$. So $m_a(T; \lambda_i) \leq m_g(T; \lambda_i)$ for all i. At the same time, we always have the inequality $m_g(T; \lambda_i) \leq m_a(T; \lambda_i)$. Therefore, $m_g(T; \lambda_i) = m_a(T; \lambda_i)$ for all i. (The proof also shows that $\ell_i = m_g(T; \lambda_i) = m_a(T; \lambda_i)$).

(c) $\implies$ **(d)**: We assume that $P_T(x) = (\lambda_1 - x)^{\ell_1} \dots (\lambda_k - x)^{\ell_k}$ and $m_g(T; \lambda_i) = m_a(T; \lambda_i)$. Then $\lambda_1, \dots, \lambda_k$ are exactly the eigenvalues of T. By definition, $\ell_i = m_a(T; \lambda_i)$. We have $n = \dim V = \sum_{i=1}^{k} m_a(T; \lambda_i)$ (because $\deg P_T(x) = n$). By assumption, this equals

$$\sum_{i=1}^{k} m_g(T; \lambda_i) = \sum_{i=1}^{k} \dim \mathrm{Eig}_T(\lambda_i).$$

(d) $\implies$ **(e)**: Recall that $\mathrm{Eig}_T(\lambda_1) + \dots + \mathrm{Eig}_T(\lambda_k)$ is a direct sum. Consequently,

$$\dim(\mathrm{Eig}_T(\lambda_1) + \dots + \mathrm{Eig}_T(\lambda_k)) = \dim \mathrm{Eig}_T(\lambda_1) + \dots + \dim \mathrm{Eig}_T(\lambda_k) = \dim V$$

by assumption. Since $\mathrm{Eig}_T(\lambda_1) + \dots + \mathrm{Eig}_T(\lambda_k) \subseteq V$ is a linear subspace, we have $\mathrm{Eig}_T(\lambda_1) + \dots + \mathrm{Eig}_T(\lambda_k) = V$. This implies that $\mathrm{Eig}_T(\lambda_1) + \dots + \mathrm{Eig}_T(\lambda_k) \cong \mathrm{Eig}_T(\lambda_1) \oplus \dots \oplus \mathrm{Eig}_T(\lambda_k)$.

(e) $\implies$ **(a)**: If $V \cong \mathrm{Eig}_T(\lambda_1) \oplus \dots \oplus \mathrm{Eig}_T(\lambda_k)$, then since $\mathrm{Eig}_T(\lambda_1) + \dots + \mathrm{Eig}_T(\lambda_k)$ is a direct sum and also a linear subspace of V, we must have $\mathrm{Eig}_T(\lambda_1) + \dots + \mathrm{Eig}_T(\lambda_k) = V$. If we choose a basis $v_1^{(i)}, \dots, v_{\ell_i}^{(i)}$ for $\mathrm{Eig}_T(\lambda_i)$ for all i (where $\ell_i = m_g(T; \lambda_i)$), then the union $(v_1^{(1)}, \dots, v_{\ell_1}^{(1)}, \dots, v_1^{(k)}, \dots, v_{\ell_k}^{(k)})$ is a basis of V and all the vectors in it are eigenvectors of T.

Q.E.D.

Corollary 21.c.2 (*Sufficient Condition for Diagonalizability*): If $P_T(x)$ splits as a product of linear factors and each eigenvalue λ of T has algebraic multiplicity 1, then T is diagonalizable.

Proof

If $m_a(T;\lambda) = 1$, this implies that $m_g(T;\lambda) = 1$. Now use point **(c)** of **??**. Q.E.D.

Trigonalization

(= bringing a matrix/endomorphism to upper triangular form)

Question: Let $T \in \mathrm{End}(V)$. We want to find a basis $\mathcal{B}$ for T such that $[T]^{\mathcal{B}}_{\mathcal{B}}$ has as simple a form as possible.

- If T is diagonalizable, we are okay. But we have seen that this is not always the case. For example, $A = \left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$ is not diagonalizable (we saw: $m_a(A;1) = 2, m_g(A;1) = 1$).
- A weaker (hence more general) condition is trigonalizability.

Definition 21.c.3 (*Trigonalizable Endomorphism*): We say that $T \in \mathrm{End}(V)$ is *"trigonalizable"* if there exists a basis $\mathcal{B}$ for V such that $[T]^{\mathcal{B}}_{\mathcal{B}}$ is upper triangular, i.e.,

$$[T]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix} \quad (n = \dim V). \tag{21.13}$$

Theorem 21.c.4 (*Trigonalizability Condition*): Let V be a finite-dimensional vector space over K and $T \in \mathrm{End}(V)$. Then T is trigonalizable if and only if $P_T(x)$ splits as a product of linear factors, i.e., $P_T(x) = (\lambda_1 - x)^{\ell_1} \dots (\lambda_k - x)^{\ell_k}$.

Short Digression

Theorem 21.c.5 (*Fundamental Theorem of Algebra*): Every polynomial $p(x) \in \mathbb{C}[x]$ with complex coefficients splits as a product of linear factors and a constant, i.e., $p(x) = C\cdot(x-a_1)^{n_1} \dots (x-a_r)^{n_r}$, with $C \in \mathbb{C}$, $r \geq 0$, and $a_i \in \mathbb{C}$.

Remark: The proof is typically covered in an analysis course or a complex analysis course in the third semester.

Corollary 21.c.6 (*Trigonalizability over* $\mathbb{C}$): Let V be a finite-dimensional vector space over $\mathbb{C}$. Then every $T \in \mathrm{End}(V)$ is trigonalizable.

Proof of the Theorem on Trigonalizability

Proof

$\Longrightarrow$: If there exists a basis $\mathcal{B}$ such that $[T]^{\mathcal{B}}_{\mathcal{B}}$ is upper triangular as in (**??**), then:

$$P_T(x) = \det([T]^{\mathcal{B}}_{\mathcal{B}} - x \cdot I_n) = \det \begin{pmatrix} \lambda_1 - x & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_n - x \end{pmatrix} = (\lambda_1 - x) \dots (\lambda_n - x).$$

$\Longleftarrow$: We will use induction on $n = \dim V$.

Base case $n = 1$: In this case, V is 1-dimensional and $[T]^{\mathcal{B}}_{\mathcal{B}} = (\,\lambda\,)$, so T is even diagonalizable.

Inductive step: Let $n \geq 1$. Assume that the theorem holds for all vector spaces of dimension $\leq n$ and every endomorphism of such a space. Let V be an $(n+1)$-dimensional vector space and $T \in \mathrm{End}(V)$ such that $P_T(x)$ splits as a product of linear factors.

Let λ_1 be a zero of $P_T(x)$. This implies that there exists an eigenvector $u_1 \in \ker(T - \lambda_1 \cdot \mathrm{id}_V)$. Extend u_1 to a basis $\mathcal{B}' = (u_1, u_2, \dots, u_{n+1})$ of V. Then the matrix of T in basis $\mathcal{B}'$ has the block form:

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = \left(\begin{array}{c|c} \lambda_1 & * \cdots * \\ \hline 0 & \\ \vdots & A \\ 0 & \end{array}\right) =: M \tag{21.14}$$

where A is an $n \times n$ matrix in $\mathcal{M}_{n\times n}(K)$. By block determinant properties, we have $P_T(x) = (\lambda_1 - x) \cdot P_A(x)$. Since $P_T(x)$ splits as a product of linear factors, so does $P_A(x)$.

Consider the linear map $T_A : K^n \to K^n$ defined by $T_A(u) = A \cdot u$. By the induction hypothesis, there exists a basis $\mathcal{C}$ of K^n such that:

$$[T_A]_{\mathcal{C}}^{\mathcal{C}} = \begin{pmatrix} \lambda_2 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_{n+1} \end{pmatrix}. \tag{21.15}$$

Denote by $\mathcal{E}$ the standard basis of K^n. We know that $[T_A]_{\mathcal{C}}^{\mathcal{C}} = [\mathrm{id}]_{\mathcal{C}}^{\mathcal{E}} \cdot [T_A]_{\mathcal{E}}^{\mathcal{E}} \cdot [\mathrm{id}]_{\mathcal{E}}^{\mathcal{C}} = Q^{-1}AQ$, where $Q = [\mathrm{id}]_{\mathcal{C}}^{\mathcal{E}}$. Consequently, $Q^{-1}AQ = \begin{pmatrix} \lambda_2 & \dots & * \\ 0 & \ddots & \vdots \\ 0 & \dots & \lambda_{n+1} \end{pmatrix}$.

Now, define the $(n+1) \times (n+1)$ block matrix:

$$\underline{P} := \left(\begin{array}{c|c} 1 & 0 \dots 0 \\ \hline 0 & \\ \vdots & Q \\ 0 & \end{array}\right) \in M_{(n+1)\times(n+1)}(K). \tag{21.16}$$

Claim: $\underline{P}$ is invertible with $\underline{P}^{-1} = \left(\begin{array}{c|c} 1 & 0 \dots 0 \\ \hline 0 & \\ \vdots & Q^{-1} \\ 0 & \end{array}\right)$, and $\underline{P}^{-1} M \underline{P} = \left(\begin{array}{c|c} \lambda_1 & * \cdots * \\ \hline 0 & \\ \vdots & Q^{-1}AQ \\ 0 & \end{array}\right)$.

Proof of Claim

This follows from direct block matrix multiplication. First, we verify the product $\underline{P} \cdot \underline{P}^{-1}$ to confirm the inverse:

$$\underline{P} \cdot \underline{P}^{-1} = \left(\begin{array}{c|c} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 0 \cdot Q^{-1} \\ \hline 0 \cdot 1 + Q \cdot 0 & 0 \cdot 0 + Q \cdot Q^{-1} \end{array}\right) = \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & I_n \end{array}\right).$$

Next, we compute the similarity transformation $\underline{P}^{-1} M \underline{P}$ in two steps. Let us first multiply $\underline{P}^{-1}$ and M:

$$\begin{aligned} \underline{P}^{-1} M &= \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & Q^{-1} \end{array}\right) \left(\begin{array}{c|c} \lambda_1 & * \\ \hline 0 & A \end{array}\right) \\ &= \left(\begin{array}{c|c} 1 \cdot \lambda_1 + 0 \cdot 0 & 1 \cdot * + 0 \cdot A \\ \hline 0 \cdot \lambda_1 + Q^{-1} \cdot 0 & 0 \cdot * + Q^{-1} \cdot A \end{array}\right) = \left(\begin{array}{c|c} \lambda_1 & * \\ \hline 0 & Q^{-1}A \end{array}\right). \end{aligned}$$

Finally, we multiply the resulting matrix by $\underline{P}$ from the right:

$$\begin{aligned}(\underline{P}^{-1}M)\underline{P} &= \left(\begin{array}{c|c}\lambda_1 & * \\ \hline 0 & Q^{-1}A\end{array}\right)\left(\begin{array}{c|c}1 & 0 \\ \hline 0 & Q\end{array}\right)\\ &= \left(\begin{array}{c|c}\lambda_1\cdot 1 + *\cdot 0 & \lambda_1\cdot 0 + *\cdot Q \\ \hline 0\cdot 1 + Q^{-1}A\cdot 0 & 0\cdot 0 + Q^{-1}A\cdot Q\end{array}\right)\\ &= \left(\begin{array}{c|c}\lambda_1 & *' \\ \hline 0 & Q^{-1}AQ\end{array}\right),\end{aligned}$$

where $*'$ denotes the new row vector $*Q$. Since $Q^{-1}AQ$ is upper triangular with diagonal entries $\lambda_2,\dots,\lambda_{n+1}$ (as guaranteed by the induction hypothesis), the block structure ensures the entire matrix $\underline{P}^{-1}M\underline{P}$ is upper triangular. This proves the claim. **Q.E.D.**

We continue with the proof of the Theorem. Recall we have the basis $\mathcal{B}' = (u_1, u_2, \dots, u_{n+1})$, where $[T]^{\mathcal{B}'}_{\mathcal{B}'} = M$.

Claim: There exists a basis $\mathcal{B}$ for V such that $[\mathrm{id}]^{\mathcal{B}}_{\mathcal{B}'} = \underline{P}$.

Proof of Claim

Write the entries of $\underline{P}$ as P_{ij}. Define $m = n+1$ and set vectors $v_1,\dots,v_m$ as linear combinations of $\mathcal{B}'$ using the columns of $\underline{P}$:

$$v_k := \sum_{j=1}^{m} P_{jk}u_j \quad \text{for } k = 1,\dots,m.$$

The set $\mathcal{B} := (v_1,\dots,v_m)$ is a basis for V. To see this, assume $c_1v_1+\dots+c_mv_m = 0$. Substituting the definitions of v_k and using the linear independence of $(u_1,\dots,u_m)$ yields the matrix equation $\underline{P}\cdot(c_1,\dots,c_m)^T = 0$. Because $\underline{P}$ is invertible, $(c_1,\dots,c_m)^T = 0$, proving linear independence.

Q.E.D.

We now have the basis $\mathcal{B}$. Computing the matrix of T in this new basis yields:

$$[T]^{\mathcal{B}}_{\mathcal{B}} = [\mathrm{id}]^{\mathcal{B}'}_{\mathcal{B}}\cdot[T]^{\mathcal{B}'}_{\mathcal{B}'}\cdot[\mathrm{id}]^{\mathcal{B}}_{\mathcal{B}'} = \underline{P}^{-1}\cdot M\cdot\underline{P}. \tag{21.17}$$

As proven above, $\underline{P}^{-1}M\underline{P}$ is strictly upper triangular. Therefore, T is trigonalizable. The induction is now complete.

Q.E.D.

An Alternative (And More Elegant) Proof

Proof

$\Longleftarrow$: We will use induction on $n = \dim V$.

Define the 1-dimensional subspace $U := \mathrm{Sp}(v_1)$ and consider the quotient space V/U. Since v_1 is an eigenvector, $T(U)\subseteq U$. This invariance property allows us to obtain a well-defined induced endomorphism on the quotient space, $\overline{T}: V/U \to V/U$, defined by $\overline{T}([v]) := [Tv]$.

Extend v_1 to a basis $\mathcal{B}_1 = (v_1, w_2, \dots, w_{n+1})$ of V. The equivalence classes $[w_2],\dots,[w_{n+1}]$ form a basis for V/U. In this basis, $\overline{T}$ is represented by an $n\times n$ matrix A, and the full transformation T in basis $\mathcal{B}_1$ takes the block form:

$$[T]^{\mathcal{B}_1}_{\mathcal{B}_1} = \left(\begin{array}{c|ccc}\lambda_1 & * & \dots & * \\ \hline 0 & & & \\ \vdots & & A & \\ 0 & & & \end{array}\right). \tag{21.18}$$

By the properties of block-triangular determinants, we have $P_T(x) = (\lambda_1 - x) \cdot P_A(x)$. Because $\overline{T}$ has the matrix representation A, we know $P_{\overline{T}}(x) = P_A(x)$, which gives us:

$$P_T(x) = (\lambda_1 - x) \cdot P_{\overline{T}}(x).$$

Since $P_T(x)$ splits into a product of linear factors by our initial assumption, $P_{\overline{T}}(x)$ must also split into linear factors.

Because $\dim(V/U) = n$, we can apply the induction hypothesis to $\overline{T}$. There exists a basis $\mathcal{C} = (z_2, \ldots, z_{n+1})$ for V/U such that its matrix is upper triangular:

$$[\overline{T}]_{\mathcal{C}}^{\mathcal{C}} = \begin{pmatrix} \lambda_2 & \ldots & * \\ 0 & \ddots & \vdots \\ 0 & \ldots & \lambda_{n+1} \end{pmatrix}. \tag{21.19}$$

Now, each element in $\mathcal{C}$ is represented by some vector $v_i \in V$, meaning $z_i = [v_i]$. (Such a v_i is not unique, but we pick one specific representative for every i). Define the list of vectors $\mathcal{B} = (v_1, v_2, \ldots, v_{n+1})$. We claim that $\mathcal{B}$ is a basis for V.

To show this, we check for linear independence. Assume:

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_{n+1} v_{n+1} = 0.$$

Applying the quotient map to this equation (where $[v_1] = 0$), we get:

$$\alpha_2 [v_2] + \cdots + \alpha_{n+1} [v_{n+1}] = 0.$$

This implies that $\alpha_2 z_2 + \cdots + \alpha_{n+1} z_{n+1} = 0$. Since the z_i vectors form a basis for V/U, they are linearly independent, which forces $\alpha_2 = \cdots = \alpha_{n+1} = 0$. Plugging this back into our original sum leaves $\alpha_1 v_1 = 0$. Since $v_1 \neq 0$, we must have $\alpha_1 = 0$. Because the $n+1$ vectors are linearly independent and $\dim V = n+1$, $\mathcal{B}$ is indeed a basis for V.

Finally, we claim that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular. For the first column, we already know $T(v_1) = \lambda_1 v_1$. For the remaining columns ($2 \leq i \leq n+1$), we read from the upper-triangular matrix (**??**) that:

$$[\overline{T} z_i]_{\mathcal{C}} = \alpha_{2i} z_2 + \cdots + \alpha_{i-1,i} z_{i-1} + \lambda_i z_i$$

for some scalars $\alpha \in K$. Translating this back to the equivalence classes:

$$[\overline{T} v_i] = \alpha_{2i} [v_2] + \cdots + \alpha_{i-1,i} [v_{i-1}] + \lambda_i [v_i].$$

This implies that the difference between $T v_i$ and the linear combination of the v's must belong to the kernel of the quotient map, which is U:

$$T v_i = \alpha_{2i} v_2 + \cdots + \alpha_{i-1,i} v_{i-1} + \lambda_i v_i + u \quad \text{for some } u \in U.$$

Since $U = \mathrm{Sp}(v_1)$, we can write $u = \beta_i v_1$ for some scalar β_i. Thus,

$$T v_i = \beta_i v_1 + \alpha_{2i} v_2 + \cdots + \alpha_{i-1,i} v_{i-1} + \lambda_i v_i.$$

Because $T v_i$ only requires basis vectors up to index i, $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular: Because $T v_i$ can be expressed as a linear combination of $v_1, \ldots, v_i$ (and not $v_{i+1}, \ldots, v_{n+1}$), $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \beta_2 & \ldots & \beta_{n+1} \\ 0 & \lambda_2 & \ldots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \lambda_{n+1} \end{pmatrix}.$$

This completes the inductive step, showing that T is trigonalizable. **Q.E.D.**

The Minimal Polynomial & The Cayley-Hamilton Theorem (Section 21.d)

A powerful tool in linear algebra is the ability to evaluate polynomials not just at scalar numbers, but at matrices and endomorphisms. Let $A \in M_{n\times n}(K)$. For any integer $k \geq 1$, the matrix power $A^k = A \cdot \cdots \cdot A$ is well-defined. Therefore, for any formal polynomial $g(x) = a_d x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0 \in K[x]$, we can define the matrix evaluation $g(A) \in M_{n\times n}(K)$ naturally as:

$$g(A) := a_d A^d + a_{d-1}A^{d-1} + \cdots + a_1 A + a_0 I_n.$$

A natural question arises: can we find a polynomial that completely "kills" or annihilates a given matrix?

Claim (*Existence of an Annihilating Polynomial*): For every matrix $A \in \mathcal{M}_{n\times n}(K)$, there exists a non-zero polynomial $g \in K[x]$ such that $g(A) = 0$.

Proof

Consider the sequence of successive matrix powers. Since the vector space $\mathcal{M}_{n\times n}(K)$ has dimension n^2, we can choose an integer $r \geq n^2$ to construct a sequence of length $r+1$:

$$\underbrace{I_n,\ A,\ A^2,\ A^3,\ \ldots,\ A^r}_{r+1 \text{ elements in a space of dimension } n^2} \tag{21.20}$$

Because the number of elements ($r+1$) strictly exceeds the dimension of the ambient space (n^2), the matrices in (**??**) cannot be linearly independent. This implies there exist scalars $a_0, a_1, \ldots, a_r \in K$, not all zero, such that:

$$a_0 I_n + a_1 A + a_2 A^2 + \cdots + a_r A^r = 0.$$

By setting $g(x) := a_r x^r + \cdots + a_1 x + a_0$, we obtain the desired non-zero annihilating polynomial.

Q.E.D.

A similar logic applies seamlessly to any abstract operator $T \in \mathrm{End}(V)$, where V is a finite-dimensional vector space over K, defining $g(T) := a_d T^d + \cdots + a_1 T + a_0 \,\mathrm{id}_V$. Since annihilating polynomials exist, it is mathematically fruitful to look for the "simplest" one, which acts as a sort of algebraic DNA for the operator.

Definition 21.d.1 (*The Minimal Polynomial*): Let V be a vector space over K and $T \in \mathrm{End}(V)$. A *"minimal polynomial"* for T is a non-zero, monic polynomial $m_T(x) \in K[x]$ of the minimal possible degree such that $m_T(T) = 0$.

While the minimal polynomial guarantees that an operator satisfies *some* polynomial equation, one of the most celebrated theorems in mathematics provides a canonical polynomial that always works: the characteristic polynomial.

Theorem 21.d.2 (*Cayley-Hamilton*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then

(a) $P_A(A) = 0$.

(b) Let V be a finite-dimensional vector space over K, and $T \in \mathrm{End}(V)$. Then $P_T(T) = 0$.

Exercise: Let $A \in \mathcal{M}_{n\times n}(K)$ be an invertible matrix. Use **??** to prove that the inverse matrix A^{-1} can be expressed as a polynomial in A of degree at most $n-1$.

Proof of Cayley-Hamilton Theorem (Analytic Approach)

Before we dive into a purely algebraic mechanism, it is highly instructive to view this theorem through the lens of topology. The core idea is elegant: the theorem is trivially true for diagonal matrices. If we can show that diagonalizable matrices are "dense" (meaning almost all matrices are diagonalizable), we can use the continuity of polynomials to force the theorem to hold for *every* matrix.

First, we note that it is enough to prove part **(a)** (for matrices). Suppose we know that **(a)** is true, and let $T \in \mathrm{End}(V)$. Pick a basis $\mathcal{B}$ for V, and assign $A := [T]_{\mathcal{B}}^{\mathcal{B}} \in \mathcal{M}_{n\times n}(K)$.

Claim (*Matrix Representation of Polynomials*): $[P_T(T)]_{\mathcal{B}}^{\mathcal{B}} = P_A(A)$.

Proof of Claim

The map assigning an endomorphism to its matrix representation is an algebra isomorphism. Thus, $[R+S]_{\mathcal{B}}^{\mathcal{B}} = [R]_{\mathcal{B}}^{\mathcal{B}} + [S]_{\mathcal{B}}^{\mathcal{B}}$, and $[R\circ S]_{\mathcal{B}}^{\mathcal{B}} = [R]_{\mathcal{B}}^{\mathcal{B}} \cdot [S]_{\mathcal{B}}^{\mathcal{B}}$. Since $P_T(x) = P_A(x)$ by definition, we write $P_A(x) = c_n x^n + \cdots + c_0$. Then:

$$\begin{aligned}
[P_T(T)]_{\mathcal{B}}^{\mathcal{B}} &= [c_n T^n + \cdots + c_1 T + c_0 \,\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} \\
&= c_n [T^n]_{\mathcal{B}}^{\mathcal{B}} + \cdots + c_1 [T]_{\mathcal{B}}^{\mathcal{B}} + c_0 I_n \\
&= c_n A^n + \cdots + c_1 A + c_0 I_n = P_A(A).
\end{aligned}$$

Q.E.D.

If Cayley-Hamilton holds for matrices, then $P_A(A) = 0$, meaning $[P_T(T)]_{\mathcal{B}}^{\mathcal{B}} = 0$. Since the zero matrix uniquely represents the zero endomorphism, we conclude $P_T(T) = 0$. We now concentrate entirely on proving the theorem for matrices, beginning over the complex field $\mathbb{C}$.

Lemma 21.d.3 (*Cayley-Hamilton for Diagonalizable Matrices*): Suppose $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is diagonalizable. Then $P_A(A) = 0$.

Proof

By assumption, there exists an invertible matrix Q such that

$$Q^{-1}AQ = \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

Since the roots of the characteristic polynomial are exactly the eigenvalues, we know $P_A(\lambda_i) = 0$ for all i. Writing $P_A(x) = \sum c_j x^j$, we substitute A:

$$Q^{-1}P_A(A)Q = \sum c_j Q^{-1}A^j Q = \sum c_j (Q^{-1}AQ)^j = \sum c_j \Lambda^j = P_A(\Lambda).$$

Because Λ is diagonal, $P_A(\Lambda)$ is simply the diagonal matrix with entries $P_A(\lambda_1), \ldots, P_A(\lambda_n)$. Since every entry is zero, $Q^{-1}P_A(A)Q = 0$, which immediately implies $P_A(A) = 0$. Q.E.D.

Lemma 21.d.4 (*Density of Diagonalizable Matrices*): Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be any arbitrary matrix. Then there exists a sequence of matrices $(A_k)_{k=1}^{\infty}$ such that:

(a) Each A_k is diagonalizable.

(b) $\lim_{k\to\infty} A_k = A$.

Proof

Because $\mathbb{C}$ is algebraically closed, every matrix is trigonalizable. Thus, there exists an invertible matrix P such that $U = P^{-1}AP$ is an upper triangular matrix with the eigenvalues $\lambda_1, \ldots, \lambda_n$ on its diagonal.

To ensure diagonalizability, we must prevent repeated eigenvalues. We can fix this by introducing a tiny perturbation. For each integer $k \geq 1$, we define a diagonal matrix of small shifts:

$$E_k = \mathrm{diag}\left(\frac{\epsilon_1}{k}, \frac{\epsilon_2}{k}, \ldots, \frac{\epsilon_n}{k}\right)$$

We carefully choose the constants ϵ_i such that the new diagonal entries $\lambda_i + \frac{\epsilon_i}{k}$ are strictly distinct from one another. We now define our sequence:

$$A_k := P(U + E_k)P^{-1}$$

Because $U + E_k$ is still upper triangular but now possesses exactly n distinct eigenvalues, it is guaranteed to be diagonalizable. Finally, as $k \to \infty$, the perturbation matrix $E_k \to 0$, meaning $\lim_{k\to\infty} A_k = PUP^{-1} = A$. **Q.E.D.**

Lemma 21.d.5 (*Cayley-Hamilton over $\mathbb{C}$*): Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be any complex matrix. The Cayley-Hamilton theorem holds for A.

Proof

By our construction in **??**, we can find a sequence of diagonalizable matrices $A_k \to A$. By **??**, the theorem holds for all of them: $P_{A_k}(A_k) = 0$.

Notice that the mapping $\Phi : \mathcal{M}_{n\times n}(\mathbb{C}) \to \mathcal{M}_{n\times n}(\mathbb{C})$ defined by $\Phi(M) = P_M(M)$ is built entirely from taking determinants, adding matrices, and multiplying matrices. Therefore, Φ is a continuous polynomial function of the matrix entries. Because continuous functions commute with limits, we have:

$$P_A(A) = \Phi(A) = \Phi\left(\lim_{k\to\infty} A_k\right) = \lim_{k\to\infty} \Phi(A_k) = \lim_{k\to\infty} 0 = 0.$$

Q.E.D.

To generalize this result from $\mathbb{C}$ to any arbitrary field K, we rely on a formal algebraic identity regarding multi-variable polynomials.

Lemma 21.d.6 (*Formal Zero of Multi-Variable Polynomials*):
Suppose $f(x_1, \dots, x_r) \in \mathbb{R}\ [x_1, \dots, x_r]$ satisfies $f(a_1, \dots, a_r) = 0$ for all $(a_1, \dots, a_r) \in \mathbb{R}^r$. Then $f \equiv 0$ as a formal polynomial (all coefficients are 0).

Proof

Write $f(x_1, \dots, x_r) = \sum_{\mathbf{i}\in\mathcal{I}} c_{\mathbf{i}} x_1^{i_1} \dots x_r^{i_r}$. Because f is zero everywhere on $\mathbb{R}^r$, all of its partial derivatives evaluated at the origin $(0, \dots, 0)$ must be zero. Taking the appropriate mixed partial derivatives isolates the coefficient $c_{\mathbf{i}}$. Thus, $c_{\mathbf{i}} = 0$ for all $\mathbf{i}$. **Q.E.D.**

Analytic Proof of Cayley-Hamilton over every field K

For an $n \times n$ matrix A, the entries of $P_A(A)$ are given by multi-variable polynomials in the n^2 variables $A_{11}, A_{12}, \dots, A_{nn}$. These polynomials have integer (and hence real) coefficients. Let the polynomial representing the (i, j)-th entry be $q_{ij}(A_{11}, \dots, A_{nn})$.

Since Cayley-Hamilton holds for $\mathbb{C}$ (and thus for any matrix in $\mathbb{R}$), we know $q_{ij}(A_{11}, \dots) = 0$ for all evaluations in $\mathbb{R}^{n^2}$. By **??**, $q_{ij} \equiv 0$ as formal polynomials. Because they are formally the zero polynomial, evaluating them over *any* field K will always yield 0. Thus, $P_A(A) = 0$ for any matrix over any field. **Q.E.D.**

A Purely Algebraic Proof of Cayley-Hamilton Thm

While the analytic proof is mathematically beautiful, it relies on topological machinery (limits and continuity) that feels slightly alien in pure algebra. Let us now construct a purely structural, algebraic proof using invariant subspaces.

Definition 21.d.7 (*T-Cyclic Subspace*): Let V be a vector space over K and $T \in \mathrm{End}(V)$. Let $v \in V$. We define the *"T-cyclic subspace"* generated by v as the span of the orbit of v under T:

$$\langle v\rangle_T := \mathrm{Sp}\{T^k v \mid k \in \mathbb{Z}_{\geq 0}\} = \mathrm{Sp}(v, Tv, T^2 v, \dots).$$

Lemma 21.d.8 (*Basis and Companion Matrix*): Suppose that $v, Tv, \dots, T^{d-1}v$ $(d \geq 1)$ are linearly independent, and $T^d v = c_0 v + c_1 Tv + \dots + c_{d-1}T^{d-1}v$. Then:

(a) $\mathcal{B} = (v, Tv, \ldots, T^{d-1}v)$ forms a basis for $\langle v \rangle_T$.

(b) Let $S = T|_{\langle v \rangle_T}$. Then $[S]^{\mathcal{B}}_{\mathcal{B}}$ is a companion matrix:

$$[S]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & \ldots & 0 & c_0 \\ 1 & 0 & \ldots & 0 & c_1 \\ 0 & 1 & \ldots & 0 & c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \ldots & 1 & c_{d-1} \end{pmatrix}.$$

Proof

(a) Let $U = \mathrm{Sp}(\mathcal{B})$. Applying T to any basis vector $T^k v$ yields either the next basis element $T^{k+1}v \in U$, or (in the case of $k = d-1$) it yields $T^d v \in U$ due to the given linear combination. Thus $T(U) \subseteq U$. Since $\langle v \rangle_T$ is by definition the smallest T-invariant subspace containing v, we must have $U = \langle v \rangle_T$. Since the set is linearly independent and spans the space, it is a basis.

(b) This follows directly from the definition of the matrix representation; each vector maps to the next standard basis vector, creating the 1s on the subdiagonal, and the last vector maps to the prescribed linear combination, populating the final column. **Q.E.D.**

Lemma 21.d.9 (*Characteristic Polynomial of a Companion Matrix*)**:** Under the assumptions of **??**, the characteristic polynomial of the restriction S is given by:

$$P_S(x) = (-1)^d(x^d - c_{d-1}x^{d-1} - \cdots - c_1 x - c_0).$$

Proof

We compute $\det([S]^{\mathcal{B}}_{\mathcal{B}} - xI_d)$ using cofactor expansion along the first row:

$$P_S(x) = \begin{vmatrix} -x & 0 & \ldots & 0 & c_0 \\ 1 & -x & \ldots & 0 & c_1 \\ 0 & 1 & \ldots & 0 & c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \ldots & 1 & c_{d-1} - x \end{vmatrix}$$

Expanding across the first row, we get two non-zero terms:

$$P_S(x) = (-x)\det(M_{11}) + (-1)^{1+d}c_0 \det(M_{1d})$$

where M_{11} is the submatrix obtained by deleting the first row and first column, and M_{1d} is the submatrix obtained by deleting the first row and d-th column. Notice that M_{1d} is a lower triangular matrix with entirely 1s on its main diagonal. Thus, $\det(M_{1d}) = 1$. The matrix M_{11} is exactly the $(d-1)\times(d-1)$ matrix representing the characteristic polynomial of the smaller companion matrix. By induction on d, assuming the formula holds for $\det(M_{11})$, multiplying it by $-x$ shifts the degree of all terms up by one. Adding the $(-1)^{d+1}c_0$ term gives the exact formula claimed. **Q.E.D.**

Algebraic Proof of Cayley-Hamilton

Let V be finite-dimensional, and $T \in \mathrm{End}(V)$. We want to show $P_T(T) = 0$, meaning $P_T(T)v = 0$ for all $v \in V$.

Let $v \in V$, $v \neq 0$. Consider the cyclic subspace $\langle v \rangle_T$. Take the maximal integer $d \geq 1$ such that $v, Tv, \ldots, T^{d-1}v$ are linearly independent. Then $T^d v = c_0 v + \cdots + c_{d-1}T^{d-1}v$. Extend this basis $\mathcal{B}$ to a full basis $\mathcal{C} = (v, \ldots, T^{d-1}v, w_1, \ldots, w_{n-d})$ for V. The matrix representation of T with respect to $\mathcal{C}$ is block upper-triangular:

$$[T]_{\mathcal{C}}^{\mathcal{C}} = \begin{pmatrix} [S]_{\mathcal{B}}^{\mathcal{B}} & * \\ 0 & [C] \end{pmatrix}$$

Because the determinant of a block-triangular matrix is the product of the determinants of its diagonal blocks, we have $P_T(x) = P_S(x) \cdot P_C(x)$. Consider the endomorphism $R := P_T(T) \in \operatorname{End}(V)$. Since polynomials evaluated at T commute with one another, we can factor R: Because the determinant of a block-triangular matrix is the product of the determinants of its diagonal blocks (as shown in Proposition 20.40), we have $P_T(x) = P_S(x) \cdot P_C(x)$. Consider the endomorphism $R := P_T(T) \in \operatorname{End}(V)$. Since polynomials evaluated at T commute with one another, we can factor R:

$$R = P_C(T) \circ P_S(T).$$

Therefore, $Rv = P_C(T)\Big(P_S(T)v\Big)$. However, applying **??** directly to v:

$$P_S(T)v = (-1)^d (T^d v - c_{d-1}T^{d-1}v - \cdots - c_0 v).$$

By the very definition of our constants c_i, this evaluates exactly to 0. Hence $Rv = P_C(T)(0) = 0$. Since $v \neq 0$ was chosen completely arbitrarily, $P_T(T)$ evaluates to zero on every vector in V, meaning $P_T(T) = 0$. **Q.E.D.**

Remark (*The Lazy Proof*): A completely incorrect "proof" often presented by students is the following: $P_A(A) = \det(A - A \cdot I) = \det(0) = 0$. This is fundamentally invalid. The expression $P_A(x) = \det(A - xI)$ evaluates to a polynomial with scalar coefficients. Substituting matrices into a scalar expression *before* expanding the determinant leads to a severe category error. One must compute the scalar polynomial first, and only then evaluate it at the matrix A.

Column Rank and Row Rank (Section 21.e)

Theorem 21.e.1 (*Equality of Column and Row Rank*): Let $A \in \mathcal{M}_{m\times n}(K)$. Then, $\operatorname{rank}_{\text{col}}(A) = \operatorname{rank}_{\text{row}}(A)$.

Remark (*Notation and Properties of Rank*):

(a) Because of this theorem, we will from now on denote $\operatorname{rank}(A) := \operatorname{rank}_{\text{col}}(A) = \operatorname{rank}_{\text{row}}(A)$.

(b) Suppose A' is a row-reduced echelon matrix that is row-equivalent to A. Then, we have:

$$\operatorname{rank}(A) = \operatorname{rank}_{\text{row}}(A) = \operatorname{rank}_{\text{row}}(A') = \text{number of pivots in } A'$$

(c) Recall that for a linear map $T : V \to W$, we defined $\operatorname{rank}(T) := \dim \operatorname{Im}(T)$. Moreover, if $A \in \mathcal{M}_{m\times n}(K)$, then $\operatorname{rank}(T_A) = \dim \operatorname{Im}(T_A) = \operatorname{rank}_{\text{col}}(A)$.

Preparation for the Proof

Lemma 21.e.2 (*Rank Invariance under Composition with Isomorphisms*): Let $T : V \to W$ be a linear map, and let $S : W \to W'$ and $L : P \to V$ be isomorphisms. Then:

(a) $\operatorname{rank}(S \circ T) = \operatorname{rank}(T)$.

(b) $\operatorname{rank}(T \circ L) = \operatorname{rank}(T)$.

Proof

Proof of (a): By definition, $\operatorname{rank}(S \circ T) = \dim(S(T(V)))$. Since S is an isomorphism, its restriction to the subspace $T(V) \subseteq W$ is injective. An injective linear map preserves the dimension of a vector space, and therefore:

$$\dim(S(T(V))) = \dim(T(V)) = \operatorname{rank}(T)$$

Proof of (b): By definition, $\operatorname{rank}(T \circ L) = \dim(T(L(P)))$. Since L is an isomorphism, it is surjective, which implies that $L(P) = V$. Consequently:

$$\dim(T(L(P))) = \dim(T(V)) = \operatorname{rank}(T)$$

This completes the proof. **Q.E.D.**

Lemma 21.e.3 (*Invertibility of the Transpose*): Let $A \in \mathcal{M}_{n\times n}(K)$. Then, $A \in \mathrm{GL}_n(K)$ if and only if $A^T \in \mathrm{GL}_n(K)$.

Proof

We know that A is invertible if and only if the associated linear map $T_A : K^n \to K^n$ is an isomorphism. This is true if and only if $\operatorname{rank}(T_A) = n$, which is equivalent to $\operatorname{rank}_{\mathrm{col}}(A) = n$.

We also know that A is row-equivalent to some row-reduced echelon matrix A'. Since row-equivalence corresponds to multiplying from the left by invertible matrices, we have that $A \in \mathrm{GL}_n(K)$ if and only if $A' \in \mathrm{GL}_n(K)$. Because A' is an $n \times n$ row-reduced echelon matrix, A' is invertible if and only if it has no zero rows. This holds if and only if A' has n pivots, which is equivalent to $\operatorname{rank}_{\mathrm{row}}(A') = n$. Since row operations preserve the row space, this means $\operatorname{rank}_{\mathrm{row}}(A) = n$.

In summary, we have established that $A \in \mathrm{GL}_n(K)$ if and only if $\operatorname{rank}_{\mathrm{row}}(A) = n$. But by definition, the row rank of A is exactly the column rank of its transpose, so $\operatorname{rank}_{\mathrm{row}}(A) = \operatorname{rank}_{\mathrm{col}}(A^T)$. Therefore, $A \in \mathrm{GL}_n(K)$ if and only if $\operatorname{rank}_{\mathrm{col}}(A^T) = n$, which means A^T is invertible. **Q.E.D.**

Lemma 21.e.4 (*Rank Invariance under Matrix Equivalence*): Let $A, B \in \mathcal{M}_{m\times n}(K)$ such that $B = PAQ$, where $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$. Then:

(a) $\operatorname{rank}_{\mathrm{col}}(A) = \operatorname{rank}_{\mathrm{col}}(B)$.

(b) $\operatorname{rank}_{\mathrm{row}}(A) = \operatorname{rank}_{\mathrm{row}}(B)$.

Proof

(a) Transitioning to the associated linear maps, we have $T_B = T_P \circ T_A \circ T_Q$. Since P and Q are invertible matrices, both T_P and T_Q are isomorphisms. Applying **??**, we deduce that $\operatorname{rank}(T_B) = \operatorname{rank}(T_A)$. Recalling that the rank of a linear map given by a matrix matches the column rank of that matrix, we obtain $\operatorname{rank}_{\mathrm{col}}(B) = \operatorname{rank}_{\mathrm{col}}(A)$.

(b) Taking the transpose of the equation $B = PAQ$ yields $B^T = Q^T A^T P^T$. Since P and Q are invertible matrices, their transposes P^T and Q^T are also invertible (we have previously seen that $(P^T)^{-1} = (P^{-1})^T$). We can now apply part (a) of this Lemma, which we have already proven, to the matrices A^T and B^T. This gives us $\operatorname{rank}_{\mathrm{col}}(B^T) = \operatorname{rank}_{\mathrm{col}}(A^T)$. By the definition of the transpose, this immediately translates to $\operatorname{rank}_{\mathrm{row}}(B) = \operatorname{rank}_{\mathrm{row}}(A)$. **Q.E.D.**

Proof of the Main Theorem

We are now ready for the proof of **??**.

Proof of ??

Let $r := \mathrm{rank}_{\mathrm{col}}(A)$. By a previous proposition regarding the equivalence of matrices, we know that there exist invertible matrices $P \in \mathrm{GL}_m(K)$ and $Q \in \mathrm{GL}_n(K)$ such that:

$$PAQ = \begin{pmatrix} I_r & O \\ O & O \end{pmatrix} =: B$$

By **??**, we know that $\mathrm{rank}_{\mathrm{row}}(A) = \mathrm{rank}_{\mathrm{row}}(B)$. Thus, it suffices to determine the row rank of B.

The matrix B is of the block form $\left(\begin{smallmatrix} I_r & O \\ O & O \end{smallmatrix}\right)$. The non-zero rows of B are precisely the first r standard basis vectors $e_1^T, e_2^T, \ldots, e_r^T \in K^n_{\mathrm{row}}$. Since these vectors form a linearly independent set, the dimension of the row space of B is exactly r. Consequently, $\mathrm{rank}_{\mathrm{row}}(B) = r$.

Therefore, we conclude that $\mathrm{rank}_{\mathrm{row}}(A) = r = \mathrm{rank}_{\mathrm{col}}(A)$, completing the proof. Q.E.D.

Euclidean and Hermitian Spaces (Chapter 22)

Euclidean and Hermitian (Unitary) Spaces (Section 22.a)

At last (!) we endow our spaces with a geometric structure. We work here with vector spaces V over $K = \mathbb{R}$ or $K = \mathbb{C}$. Over $\mathbb{R}$, we will have *"Euclidean spaces"*. Over $\mathbb{C}$, we will have *"Hermitian"* (a.k.a. unitary) spaces.

Definition 22.a.1 (*Scalar Product over $\mathbb{R}$*): Let V be a vector space over $\mathbb{R}$. A *"scalar product"* (a.k.a. *"inner product"*) is a function $V \times V \longrightarrow \mathbb{R}$, denoted by $v, w \mapsto \langle v, w\rangle$ (or (v, w)), that has the following properties:

(a) Linearity in the first variable:

$$\begin{gathered}\langle v_1 + v_2, w\rangle = \langle v_1, w\rangle + \langle v_2, w\rangle \quad \forall v_1, v_2, w \in V \\ \langle \alpha v, w\rangle = \alpha\langle v, w\rangle \quad \forall \alpha \in \mathbb{R},\ \forall v, w \in V\end{gathered}$$

(b) Linearity in the second variable:

$$\begin{gathered}\langle v, w_1 + w_2\rangle = \langle v, w_1\rangle + \langle v, w_2\rangle \quad \forall v, w_1, w_2 \in V \\ \langle v, \alpha w\rangle = \alpha\langle v, w\rangle \quad \forall \alpha \in \mathbb{R},\ \forall v, w \in V\end{gathered}$$

(c) Symmetry:

$$\langle v, w\rangle = \langle w, v\rangle \quad \forall v, w \in V$$

(d) Positivity:

$$0 \neq v \in V \implies \langle v, v\rangle > 0$$

We call $(V, \langle\cdot,\cdot\rangle)$ a *"Euclidean space"*, or a space endowed with a scalar product.

Definition 22.a.2 (*Hermitian Product over $\mathbb{C}$*): Let V be a vector space over $\mathbb{C}$. A *"Hermitian product"* (or *"inner product"*) on V is a function $V \times V \longrightarrow \mathbb{C}$, denoted by $v, w \mapsto \langle v, w\rangle$, with the following properties:

(a) Linearity in the first variable:

$$\begin{gathered}\langle v_1 + v_2, w\rangle = \langle v_1, w\rangle + \langle v_2, w\rangle \quad \forall v_1, v_2, w \in V \\ \langle \alpha v, w\rangle = \alpha\langle v, w\rangle \quad \forall \alpha \in \mathbb{C},\ \forall v, w \in V\end{gathered}$$

(b) Sesquilinearity/Conjugate-linearity (complex anti-linearity) in the second variable:

$$\begin{gathered}\langle v, w_1 + w_2\rangle = \langle v, w_1\rangle + \langle v, w_2\rangle \quad \forall v, w_1, w_2 \in V \\ \langle v, \alpha w\rangle = \overline{\alpha}\langle v, w\rangle \quad \forall \alpha \in \mathbb{C},\ \forall v, w \in V\end{gathered}$$

(c) Hermitian property:

$$\langle v, w\rangle = \overline{\langle w, v\rangle} \quad \forall v, w \in V$$

(d) Positivity:

$$0 \neq v \in V \implies \langle v, v\rangle > 0$$

We call $(V, \langle\cdot,\cdot\rangle)$ a *"Hermitian"* (or unitary) space.

Remark (*Complex Conjugate and Inner Products*): Recall that if $\alpha = a + ib \in \mathbb{C}$, where $a, b \in \mathbb{R}$ and $i = \sqrt{-1}$, then $\overline{\alpha} := a - ib$ is the complex conjugate of α. Recall also, that $\alpha \cdot \overline{\alpha} = a^2 + b^2 = |\alpha|^2 \geq 0$, with equality if and only if $\alpha = 0$.

Note that the definition of a Hermitian product coincides with a scalar product if V is over $\mathbb{R}$, and we allow α only in $\mathbb{R}$.

Lemma 22.a.3 (*Basic Properties of the Inner Product*): Let V be a Euclidean or Hermitian vector space. Then:

(a) $\forall v \in V$, $\langle 0, v\rangle = \langle v, 0\rangle = 0$.

(b) Let $w \in V$. If $\langle v, w\rangle = 0$ for all $v \in V$, then $w = 0$. (Exercise)

(c) Let $w_1, w_2 \in V$. If $\langle v, w_1\rangle = \langle v, w_2\rangle$ for all $v \in V$, then $w_1 = w_2$.

Definition 22.a.4 (*Norm, Distance, and Orthogonality*): Let V be a Euclidean or Hermitian vector space.

(a) For all $v \in V$, we define $\|v\| := \sqrt{\langle v, v\rangle} \in \mathbb{R}_{\geq 0}$. The function $\|\cdot\|$ is called the *"norm"* on V induced by $\langle \cdot, \cdot \rangle$.

(b) A vector $u \in V$ is called a *"unit vector"* if $\|u\| = 1$. Note that if $0 \neq v \in V$ is any vector, then $\frac{1}{\|v\|}v$ is a unit vector (which is positively proportional to v, so it points in the same direction as v). We call $\frac{v}{\|v\|}$ the *"normalization"* of v.

(c) For $v, w \in V$, we define the *"distance"* between v and w by $d(v, w) := \|v - w\|$.

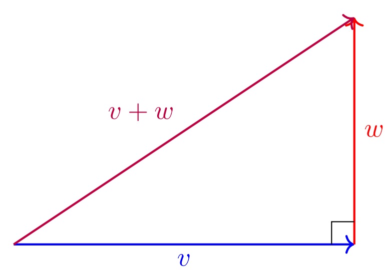
(d) Two vectors $v, w \in V$ are called *"orthogonal"* (or perpendicular to one another) if $\langle v, w\rangle = 0$. If this is the case, we sometimes write $v \perp w$.

(e) A subset $S \subseteq V$ is called an *"orthogonal system"* if $0 \notin S$, and for every two distinct elements $v, w \in S$, we have $v \perp w$.

(f) An orthogonal system $S \subseteq V$ is called *"orthonormal"* if for all $u \in S$, we have $\|u\| = 1$.

Theorem 22.a.5 (*Pythagoras' Theorem*): Let V be a Euclidean or Hermitian vector space, and let $v, w \in V$ with $v \perp w$. Then:

$$\|v + w\|^2 = \|v\|^2 + \|w\|^2$$



Proof

By definition of the norm and the properties of the inner product, we have:

$$\begin{aligned} \|v + w\|^2 &= \langle v + w, v + w\rangle \\ &= \langle v, v\rangle + \langle v, w\rangle + \langle w, v\rangle + \langle w, w\rangle \end{aligned}$$

Since $v \perp w$, we know that $\langle v, w\rangle = 0$. By Hermitian symmetry, this also implies $\langle w, v\rangle = \overline{\langle v, w\rangle} = \overline{0} = 0$. Therefore, the cross terms vanish, leaving us with:

$$\begin{aligned} \|v + w\|^2 &= \|v\|^2 + 0 + 0 + \|w\|^2 \\ &= \|v\|^2 + \|w\|^2 \end{aligned}$$

Q.E.D.

Exercise (*Generalized Pythagoras' Theorem*): Let $v_1, \ldots, v_k \in V$ (where V is a Euclidean or Hermitian space) be mutually orthogonal (i.e., $\langle v_i, v_j\rangle = 0$ for $i \neq j$). Then:

$$\|v_1 + \cdots + v_k\|^2 = \sum_{i=1}^{k} \|v_i\|^2$$

Exercise (*Linear Independence of Orthogonal Systems*): Let V be a Euclidean or Hermitian space. Show that any orthogonal system $S \subseteq V$ is linearly independent.

Exercise (*Reconstructing the Scalar Product from the Norm*): Let V be a Euclidean space. Show that for all $u, v \in V$, we have:

$$\langle u, v\rangle = \frac{1}{2}\left(\|u+v\|^2 - \|u\|^2 - \|v\|^2\right) \quad (*)$$

This shows that the norm $\|\cdot\|$ completely determines the scalar product $\langle\cdot,\cdot\rangle$.

Exercise (*Polarization Identity*): Let V be a Hermitian space. Then:

$$\langle u, v\rangle = \frac{1}{4}\left(\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2\right) \quad (**)$$

Important remark (*The Parallelogram Law*): Warning: For general norms $\|\cdot\|$, one cannot necessarily find a scalar product that induces it. A norm $\|\cdot\|$ comes from a scalar product if and only if it satisfies the *"Parallelogram law"*:

$$\|u+v\|^2 + \|u-v\|^2 = 2\left(\|u\|^2 + \|v\|^2\right)$$

Examples

Example (*Standard and Matrix-Induced Inner Products*):

(a) Standard inner products on $\mathbb{R}^n$ and $\mathbb{C}^n$:

$$\langle x, y\rangle_{\mathbb{R}^n} = \sum_{i=1}^n x_i y_i$$

$$\langle z, w\rangle_{\mathbb{C}^n} = \sum_{i=1}^n z_i \overline{w}_i$$

(b) Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ be a symmetric matrix. Define a function $V \times V \longrightarrow \mathbb{R}$ by:

$$\langle v, w\rangle_A = v^T A w$$

Check that properties (1)–(3) of a scalar product hold for *any* symmetric matrix A. Specifically, for symmetry, we have:

$$\langle w, v\rangle_A = w^T A v = (w^T A v)^T = v^T A^T w = v^T A w = \langle v, w\rangle_A$$

(Note that $w^T A v$ is a 1×1 matrix, and is therefore equal to its transpose).

What about property (4), positivity? For example, let $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, which is symmetric. Then:

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle_A = x_1 y_1 - x_2 y_2$$

But taking $v = \binom{0}{1}$, we get:

$$\left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle_A = -1 < 0$$

So, there is a lack of positivity.

Matrices and Positive Definiteness

Definition 22.a.6 (*Symmetric Matrix*): A matrix $A \in \mathcal{M}_{n\times n}(K)$ (where K is any field) is called *"symmetric"* if $A^T = A$ (i.e., $a_{ij} = a_{ji}$ for all i, j).

Definition 22.a.7 (*Adjoint and Hermitian Matrices*):

(a) Let $A \in \mathcal{M}_{n\times m}(\mathbb{C})$. We write $\overline{A}$ for the matrix whose entry at index (i,j) is $\overline{A_{ij}}$, representing complex conjugation. Thus, $(\overline{A})_{ij} := \overline{A_{ij}}$.

(b) Let $A \in \mathcal{M}_{n\times m}(\mathbb{C})$. We define $A^* := \overline{A}^T \in \mathcal{M}_{m\times n}(\mathbb{C})$. The matrix A^* is called the *"adjoint matrix"* of A (this has nothing to do with the classical adjoint, $\operatorname{adj}(A)$).

(c) A matrix $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is called *"Hermitian"* if $A^* = A$ (which is equivalent to $A = \overline{A}^T$).

Example (*Important example*): Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$. Define $\langle \cdot,\cdot\rangle_A : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ by

$$\langle v, w\rangle_A = v^T \cdot A \cdot w.$$

Exercise: Show that $\langle \cdot,\cdot\rangle_A$ is bilinear (i.e., linear in both variables).

Assume now that A is symmetric. Then, $\langle v,w\rangle_A = \langle w,v\rangle_A$ for all v,w. Indeed,

$$\langle w, v\rangle_A = w^T \cdot A \cdot v = w^T A^T v = (v^T A w)^T = (\langle v,w\rangle_A)^T = \langle v,w\rangle_A.$$

(Note that $v^T A w$ is just a scalar, which can be viewed as a 1×1 matrix, hence it equals its own transpose).

Is $\langle \cdot,\cdot\rangle_A$ a scalar product? In general, it is not. For example, consider $A = \left(\begin{smallmatrix} -1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. Then, $\langle \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)\rangle_A = -1$. However, for a scalar product, we need $\langle v,v\rangle > 0$ for all $v \neq 0$. Therefore, we lack positivity here.

Definition 22.a.8 (*Positive Definite Matrix (Real Case)*): A symmetric matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is called *"positive definite"* if $v^T A v > 0$ for all $0 \neq v \in \mathbb{R}^n$.

Claim: If A is positive definite, then $\langle \cdot,\cdot\rangle_A$ is a scalar product.

Example:

(a) Let $a_1, \ldots, a_n > 0$ (with $a_i \in \mathbb{R}$). Then, the diagonal matrix

$$D = \begin{pmatrix} a_1 & & 0 \\ & \ddots & \\ 0 & & a_n \end{pmatrix}$$

is positive definite. Indeed, if $v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$, then

$$\langle v, v\rangle_D = v^T \cdot D \cdot v = a_1 v_1^2 + \cdots + a_n v_n^2 \geq 0,$$

with equality if and only if $v_i = 0$ for all i.

(b) Let $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix} \in \mathcal{M}_{2\times 2}(\mathbb{R})$. Then, A is positive definite if and only if $a > 0$ and $\det(A) > 0$ (Exercise).

The situation for Hermitian products is similar.

Definition 22.a.9 (*Positive Definite Matrix (Complex Case)*): Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be a Hermitian matrix. We say that A is *"positive definite"* if for all $0 \neq v \in \mathbb{C}^n$, we have $v^T A \overline{v} > 0$.

Remark: Note that $v^T A \overline{v} \in \mathbb{R}$, because

$$\overline{(v^T A \overline{v})} = \overline{v}^T \overline{A} v = (\overline{v}^T \overline{A} v)^T = v^T \overline{A}^T \overline{v} = v^T A \overline{v}.$$

Here, we used the facts that for a 1×1 matrix (a scalar), $(\cdot)^T = (\cdot)$, and that A is Hermitian (so $\overline{A}^T = A$). Also, recall that if $v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{C}^n$, then $\overline{v} := \begin{pmatrix} \overline{v}_1 \\ \vdots \\ \overline{v}_n \end{pmatrix}$.

Given a positive definite complex matrix A, we can define a Hermitian product on $\mathbb{C}^n$ by

$$\langle v, w\rangle_A = v^T A \overline{w}.$$

Claim: The operation $\langle \cdot, \cdot \rangle_A$ is a Hermitian product if and only if A is positive definite.

Exercise: A matrix $A = \begin{pmatrix} a & b \\ \overline{b} & d \end{pmatrix} \in \mathcal{M}_{2\times 2}(\mathbb{C})$ is positive definite if and only if $a > 0$, $\det A \in \mathbb{R}$, and $\det A > 0$.

Proof of the Claim

We will prove it for complex matrices and Hermitian products (the case of symmetric matrices and scalar products is very similar and even simpler, which is left as an exercise).

Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ and suppose that $\langle v, w\rangle_A := v^T A \overline{w}$ is a Hermitian product. For all v, w, we have

$$\langle v, w\rangle_A = \overline{\langle w, v\rangle}_A, \tag{22.1}$$

which implies

$$v^T A \overline{w} = \overline{w^T A \overline{v}}.$$

This simplifies as follows:

$$v^T A \overline{w} = \overline{w}^T \overline{A} v \implies v^T A \overline{w} = (\overline{w}^T \overline{A} v)^T \implies v^T A \overline{w} = v^T \overline{A}^T \overline{w}.$$

Take $v = e_i$ and $w = e_j$ (the standard basis vectors). Write A as

$$A = \begin{pmatrix} a_{11} & \dots & a_{1j} & \dots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \dots & a_{nj} & \dots & a_{nn} \end{pmatrix}.$$

Then,

$$e_i^T \cdot A \cdot e_j = \begin{pmatrix} 0 & \dots & 1 & \dots & 0 \end{pmatrix} A \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & \dots & 1 & \dots & 0 \end{pmatrix} \begin{pmatrix} a_{1j} \\ \vdots \\ a_{nj} \end{pmatrix} = a_{ij}.$$

Similarly, $e_i^T \overline{A}^T e_j = \overline{a}_{ji}$. Therefore, $a_{ij} = \overline{a}_{ji}$ for all i, j, which means $\overline{A}^T = A$, proving that A is Hermitian.

The fact that A is positive definite is precisely the condition that $\langle v, v\rangle_A > 0$ for all $v \neq 0$, which translates to $v^T A \overline{v} > 0$.

The opposite statement, that if A is positive definite then $\langle \cdot, \cdot \rangle_A$ is a Hermitian product, is left as an exercise. **Q.E.D.**

The geometry on V and the choice of inner product

The geometry on V depends on the choice of inner product.

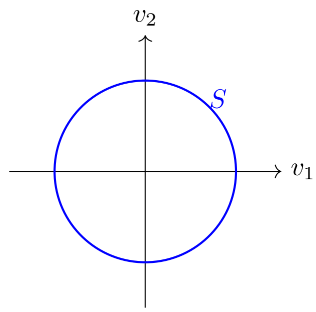
Example: Let $V = \mathbb{R}^2$. Let $\langle v, w \rangle = v_1w_1 + v_2w_2$ be the standard scalar product on $\mathbb{R}^2$. We define another product $\langle v, w \rangle' = a_1v_1w_1 + a_2v_2w_2$, where $a_1, a_2 > 0$. Notice that this is exactly $\langle v, w \rangle_A$ for the matrix $A = \begin{pmatrix} a_1 & 0 \\ 0 & a_2 \end{pmatrix}$.

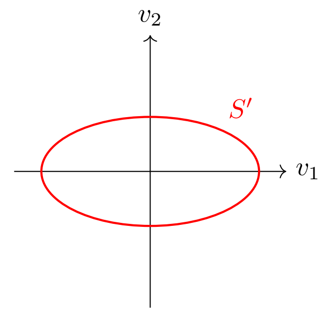
The unit sphere for the standard product $\langle \cdot, \cdot \rangle$ is:

$$S = \{v \in \mathbb{R}^2 \mid \|v\| = 1\} = \{(v_1, v_2) \mid v_1^2 + v_2^2 = 1\}.$$

The unit sphere for $\langle \cdot, \cdot \rangle'$ is:

$$S' = \{v \in \mathbb{R}^2 \mid \|v\|' = 1\} = \{(v_1, v_2) \mid a_1v_1^2 + a_2v_2^2 = 1\}.$$





Example: Let $A = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$. This matrix is positive definite (why?). Let $\langle \cdot, \cdot \rangle$ be the standard scalar product, and define $\langle \cdot, \cdot \rangle' := \langle \cdot, \cdot \rangle_A$.

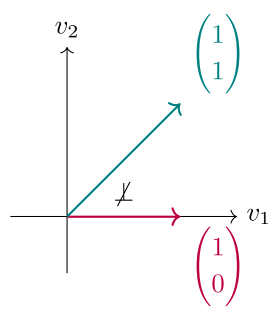
We compute:

$$\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\rangle = 1.$$

However, using the modified product:

$$\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\rangle' = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0.$$

So, $\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right) \perp' \left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$, but $\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right) \not\perp \left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$.



A different type of example

Example (*A different type of example*): Let $C[a, b]$ be the space of continuous functions $f : [a, b] \to \mathbb{R}$. It is a vector space over $\mathbb{R}$ with respect to the standard operations. We define a scalar product:

$$\langle f, g \rangle := \int_a^b f(x) \cdot g(x) \, dx.$$

This is well-defined and finite. (Why?)

Exercise: Show that $\langle\cdot,\cdot\rangle$ is linear in both variables and symmetric.

Let us check positivity: let $0 \neq f \in C[a,b]$.

$$\langle f,f\rangle = \int_a^b (f(x))^2\,dx.$$

Since $f \neq 0$, there exists an $x_0 \in (a,b)$ such that $f(x_0) \neq 0$. Since f is continuous, so is f^2. Hence, there exists a $\delta > 0$ such that $[x_0-\delta, x_0+\delta] \subseteq [a,b]$ and $f(x) \neq 0$ for all $x \in [x_0-\delta, x_0+\delta]$. This implies that $(f(x))^2 > 0$ for all $x \in [x_0-\delta, x_0+\delta]$. Therefore,

$$\langle f,f\rangle = \int_a^b (f(x))^2\,dx \geq \int_{x_0-\delta}^{x_0+\delta} (f(x))^2\,dx \geq 2\delta \cdot \min_{x\in[x_0-\delta,x_0+\delta]} (f(x))^2 > 0.$$

Remark: By this scalar product on $C[-\pi,\pi]$, the functions $f(x) = \sin x$ and $g(x) \equiv 1$ are orthogonal.

Norms and Angles

Definition 22.a.10 (*Norm*): Let V be a vector space over K (where $K = \mathbb{R}$ or $\mathbb{C}$). A *"norm"* on V is a function $\|\cdot\| : V \to \mathbb{R}_{\geq 0}$ such that:

(a) $\|u+v\| \leq \|u\| + \|v\|$ for all $u,v \in V$ (Triangle inequality).

(b) $\|\alpha \cdot v\| = |\alpha| \cdot \|v\|$ for all $\alpha \in K$ and $v \in V$ (Homogeneity).

(c) If $\|v\| = 0$, then $v = 0$ (Non-degeneracy).

Claim: If $\langle\cdot,\cdot\rangle$ is an inner product on V, then $\|v\| := \sqrt{\langle v,v\rangle}$ is a norm on V. We will prove this soon.

Exercise: For $\binom{a}{b} \in \mathbb{R}^2$, define $\|\binom{a}{b}\|' := \max\{|a|,|b|\}$, and $\|\binom{a}{b}\|'' := |a| + |b|$. Show that $\|\cdot\|'$ and $\|\cdot\|''$ are norms, but they are not induced by any scalar product.

Hint: Recall that if the scalar product $\langle\cdot,\cdot\rangle$ induces $\|\cdot\|$, then $\langle u,w\rangle = \frac{1}{2}(\|u+w\|^2 - \|u\|^2 - \|w\|^2)$.

Theorem 22.a.11 (*Cauchy-Schwarz Inequality*): Let V be a Euclidean or Hermitian space. For all $u,v \in V$, we have

$$|\langle u,v\rangle| \leq \|u\| \cdot \|v\|.$$

Furthermore, equality holds if and only if u and v are linearly dependent.

Proof

If $v = 0$, then both sides are 0, and the inequality holds (with equality). Since $v = 0$, the vectors u and v are linearly dependent, and therefore, the second statement also holds.

Assume now that $v \neq 0$. For any scalar $\lambda \in K$ (where $K = \mathbb{R}$ or $\mathbb{C}$), we have, by the positivity of the inner product,

$$0 \leq \langle u - \lambda v, u - \lambda v\rangle.$$

Using the linearity in the first variable and conjugate-linearity (or linearity) in the second variable, we expand this to obtain

$$0 \leq \langle u,u\rangle - \lambda\langle v,u\rangle - \overline{\lambda}\langle u,v\rangle + \lambda\overline{\lambda}\langle v,v\rangle.$$

Choose $\lambda = \frac{\langle u,v\rangle}{\|v\|^2}$. Substituting this into the inequality yields

$$\begin{aligned} 0 &\leq \|u\|^2 - \frac{\langle u,v\rangle}{\|v\|^2}\langle v,u\rangle - \frac{\overline{\langle u,v\rangle}}{\|v\|^2}\langle u,v\rangle + \frac{\langle u,v\rangle\overline{\langle u,v\rangle}}{\|v\|^4}\|v\|^2 \\ &= \|u\|^2 - \frac{\langle u,v\rangle\overline{\langle u,v\rangle}}{\|v\|^2} - \frac{\overline{\langle u,v\rangle}\langle u,v\rangle}{\|v\|^2} + \frac{\langle u,v\rangle\overline{\langle u,v\rangle}}{\|v\|^2} \\ &= \|u\|^2 - \frac{|\langle u,v\rangle|^2}{\|v\|^2}. \end{aligned}$$

Multiplying by $\|v\|^2$ (which is strictly positive), we obtain

$$0 \leq \|u\|^2\|v\|^2 - |\langle u,v\rangle|^2 \implies |\langle u,v\rangle|^2 \leq \|u\|^2\|v\|^2.$$

Taking the square root of both sides gives the Cauchy-Schwarz inequality.

For the equality condition, notice that equality holds if and only if $\langle u-\lambda v, u-\lambda v\rangle = 0$. By the non-degeneracy of the inner product, this is equivalent to $u - \lambda v = 0$, meaning $u = \lambda v$. Therefore, equality holds if and only if u and v are linearly dependent. **Q.E.D.**

Corollary 22.a.12: If $\langle\cdot,\cdot\rangle$ is an inner product on V, then $\|v\| := \sqrt{\langle v,v\rangle}$ is a norm on V.

Proof

The positivity and homogeneity properties are straightforward to verify. We will show the triangle inequality, namely $\|u+v\| \leq \|u\| + \|v\|$. We compute

$$\begin{aligned} \|u+v\|^2 &= \langle u+v, u+v\rangle \\ &= \langle u,u\rangle + \langle u,v\rangle + \langle v,u\rangle + \langle v,v\rangle \\ &= \|u\|^2 + \langle u,v\rangle + \overline{\langle u,v\rangle} + \|v\|^2 \\ &= \|u\|^2 + 2\,\mathrm{Re}(\langle u,v\rangle) + \|v\|^2. \end{aligned}$$

Since $\mathrm{Re}(z) \leq |z|$ for any complex number z, and applying the Cauchy-Schwarz inequality, we get

$$\|u+v\|^2 \leq \|u\|^2 + 2|\langle u,v\rangle| + \|v\|^2 \leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2.$$

Taking the square root yields the desired result. **Q.E.D.**

Definition 22.a.13 (*Angle*): Let V be a Euclidean space. For any two non-zero vectors $u, v \in V$, we define the *"angle"* α between them by the equation

$$\cos\alpha = \frac{\langle u,v\rangle}{\|u\|\cdot\|v\|},$$

where $\alpha \in [0,\pi]$. The Cauchy-Schwarz inequality guarantees that $-1 \leq \frac{\langle u,v\rangle}{\|u\|\cdot\|v\|} \leq 1$, so this definition is well-posed.

Definition (*Orthogonality*): Let V be a Euclidean or Hermitian space. We say that $u, v \in V$ are *"orthogonal"* if $\langle u,v\rangle = 0$. In this case, we write $u \perp v$.

Example: In $\mathbb{R}^2$ with the standard scalar product, the vectors $e_1 = \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$ and $e_2 = \left(\begin{smallmatrix}0\\1\end{smallmatrix}\right)$ are orthogonal.

Remark: The relationship between orthogonality and the norm is captured by the Pythagorean Theorem (**??**), which states that for orthogonal vectors u and v, the squared norm of their sum is the sum of their squared norms: $\|u+v\|^2 = \|u\|^2 + \|v\|^2$.

The Gram-Schmidt Orthogonalization Process

Theorem 22.a.14 (*Properties of Orthogonal Systems*): Let V be a Euclidean or Hermitian space, and let $(v_1, \dots, v_k)$ be an orthogonal system of non-zero vectors in V. Then:

- The vectors $v_1, \dots, v_k$ are linearly independent.
- If $v \in \mathrm{Sp}(v_1, \dots, v_k)$, meaning $v = \sum_{i=1}^{k} \alpha_i v_i$, then the coefficients are given by

$$\alpha_i = \frac{\langle v, v_i \rangle}{\|v_i\|^2} \quad \text{for all } 1 \le i \le k.$$

Proof

We prove the second property first. Suppose $v = \sum_{j=1}^{k} \alpha_j v_j$. Taking the inner product with v_i yields

$$\langle v, v_i \rangle = \left\langle \sum_{j=1}^{k} \alpha_j v_j, v_i \right\rangle = \sum_{j=1}^{k} \alpha_j \langle v_j, v_i \rangle.$$

Since the system is orthogonal, $\langle v_j, v_i \rangle = 0$ for all $j \ne i$. Thus, the sum collapses to a single term:

$$\langle v, v_i \rangle = \alpha_i \langle v_i, v_i \rangle = \alpha_i \|v_i\|^2.$$

Since $v_i \ne 0$, we have $\|v_i\|^2 > 0$. Dividing by $\|v_i\|^2$, we obtain

$$\alpha_i = \frac{\langle v, v_i \rangle}{\|v_i\|^2}.$$

To prove the first property, assume that there exist scalars $\alpha_1, \dots, \alpha_k \in K$ such that $\sum_{i=1}^{k} \alpha_i v_i = 0$. By the formula we just established, the coefficients are given by

$$\alpha_i = \frac{\langle 0, v_i \rangle}{\|v_i\|^2} = 0.$$

Since $\alpha_i = 0$ for all $1 \le i \le k$, the vectors $v_1, \dots, v_k$ are linearly independent. Q.E.D.

Proposition 22.a.15 (*Matrix Representation in Orthonormal Bases*): Let V and W be Euclidean or Hermitian spaces. Let $\mathcal{B} = (v_1, \dots, v_n)$ be an orthonormal basis of V, and let $\mathcal{C} = (w_1, \dots, w_m)$ be an orthonormal basis of W. Let $T : V \to W$ be a linear transformation. Then the entries of the representation matrix $A = [T]_{\mathcal{C}}^{\mathcal{B}} \in \mathcal{M}_{m \times n}(K)$ are given by:

$$A_{ij} = \langle T(v_j), w_i \rangle,$$

for $1 \le i \le m$ and $1 \le j \le n$.

Proof

By definition of the representation matrix $[T]_{\mathcal{C}}^{\mathcal{B}}$, the j-th column of A consists of the coordinates of $T(v_j)$ with respect to the basis $\mathcal{C}$. That is,

$$T(v_j) = \sum_{i=1}^{m} A_{ij} w_i.$$

Taking the inner product of both sides with w_k (for some $1 \le k \le m$), we obtain

$$\langle T(v_j), w_k \rangle = \left\langle \sum_{i=1}^{m} A_{ij} w_i, w_k \right\rangle = \sum_{i=1}^{m} A_{ij} \langle w_i, w_k \rangle.$$

Since $\mathcal{C}$ is an orthonormal basis, $\langle w_i, w_k \rangle = \delta_{ik}$ (which is 1 if $i = k$ and 0 otherwise). Thus, the sum collapses to the single term where $i = k$, giving:

$$\langle T(v_j), w_k \rangle = A_{kj}.$$

Relabeling the index k to i yields the desired formula $A_{ij} = \langle T(v_j), w_i \rangle$. Q.E.D.

Theorem 22.a.16 (*Gram-Schmidt Orthogonalization Process*): Let V be a Euclidean or Hermitian vector space. Let $v_1, \dots, v_n \in V$ be linearly independent vectors. Then, there exists an orthogonal system of non-zero vectors $w_1, \dots, w_n \in V$ such that for all $1 \le j \le n$,

$$\mathrm{Sp}(w_1, \dots, w_j) = \mathrm{Sp}(v_1, \dots, v_j).$$

Proof

We construct the vectors $w_1, \dots, w_n$ by induction on j.

For $j = 1$, we set $w_1 := v_1$. Since $v_1, \dots, v_n$ are linearly independent, we know that $v_1 \neq 0$, and thus $w_1 \neq 0$. Clearly, $\mathrm{Sp}(w_1) = \mathrm{Sp}(v_1)$.

Assume by induction that we have constructed an orthogonal system $w_1, \dots, w_{j-1}$ of non-zero vectors such that $\mathrm{Sp}(w_1, \dots, w_k) = \mathrm{Sp}(v_1, \dots, v_k)$ for all $1 \le k \le j-1$. We define the next vector as

$$w_j := v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i \rangle}{\langle w_i, w_i \rangle} w_i.$$

First, we show that $w_j \neq 0$. If $w_j = 0$, then $v_j = \sum_{i=1}^{j-1} \frac{\langle v_j, w_i \rangle}{\langle w_i, w_i \rangle} w_i$. This implies that $v_j \in \mathrm{Sp}(w_1, \dots, w_{j-1})$. By the inductive hypothesis, $\mathrm{Sp}(w_1, \dots, w_{j-1}) = \mathrm{Sp}(v_1, \dots, v_{j-1})$, so $v_j \in \mathrm{Sp}(v_1, \dots, v_{j-1})$. This means v_j is a linear combination of $v_1, \dots, v_{j-1}$, which contradicts the linear independence of the set $\{v_1, \dots, v_j\}$. This shows that $w_j \neq 0$.

By the inductive hypothesis, $w_1, \dots, w_{j-1}$ is an orthogonal system. So, we have to prove that $w_j \perp w_1, \dots, w_{j-1}$. Indeed, let $k \le j-1$. Using the linearity of the inner product in the first variable, we compute

$$\langle w_j, w_k \rangle = \left\langle v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i \rangle}{\langle w_i, w_i \rangle} w_i, w_k \right\rangle = \langle v_j, w_k \rangle - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i \rangle}{\langle w_i, w_i \rangle} \langle w_i, w_k \rangle.$$

Since $w_1, \dots, w_{j-1}$ are mutually orthogonal, $\langle w_i, w_k \rangle = 0$ for all $i \neq k$. Thus, the sum collapses to the single term where $i = k$:

$$\langle w_j, w_k \rangle = \langle v_j, w_k \rangle - \frac{\langle v_j, w_k \rangle}{\langle w_k, w_k \rangle} \langle w_k, w_k \rangle = \langle v_j, w_k \rangle - \langle v_j, w_k \rangle = 0.$$

This shows that $(w_1, \dots, w_j)$ is an orthogonal system.

To complete the induction, it remains to show that $\mathrm{Sp}(w_1, \dots, w_j) = \mathrm{Sp}(v_1, \dots, v_j)$. By the definition of w_j, we have

$$w_j \in \mathrm{Sp}(w_1, \dots, w_{j-1}, v_j) = \mathrm{Sp}(v_1, \dots, v_{j-1}, v_j),$$

where the equality follows from the inductive hypothesis. Therefore,

$$\underbrace{\mathrm{Sp}(w_1, \dots, w_{j-1}, w_j)}_{\text{dimension } j} \subseteq \underbrace{\mathrm{Sp}(v_1, \dots, v_{j-1}, v_j)}_{\text{dimension } j}.$$

But both of these vector spaces have dimension j (because $v_1, \dots, v_j$ are linearly independent, and $w_1, \dots, w_j$ are linearly independent, being an orthogonal system of non-zero vectors). Since one is a subspace of the other and their dimensions are equal, they must be equal. This completes the induction and the proof. **Q.E.D.**

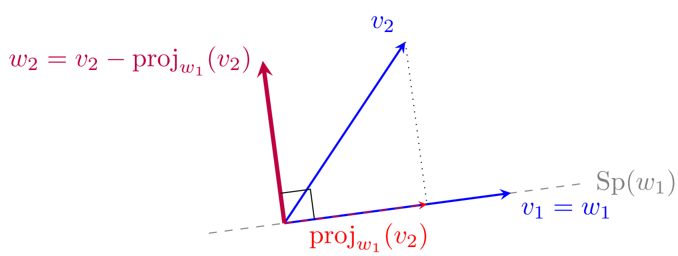
Corollary 22.a.17: Every finite-dimensional Euclidean or Hermitian space V possesses an orthonormal basis $\mathcal{B}$.

Proof

Let $n = \dim V$ and let $(v_1, \dots, v_n)$ be any basis of V. By the Gram-Schmidt process (Theorem 22.a.16), there exists an orthogonal system $(w_1, \dots, w_n)$ of non-zero vectors such that $\mathrm{Sp}(w_1, \dots, w_n) = \mathrm{Sp}(v_1, \dots, v_n) = V$. Since $(w_1, \dots, w_n)$ is an orthogonal system of n non-zero vectors in a space of dimension n, it forms an orthogonal basis for V. Finally, we define $u_i := \frac{w_i}{\|w_i\|}$ for each $1 \leq i \leq n$. Then $\mathcal{B} = (u_1, \ldots, u_n)$ is the required orthonormal basis. Q.E.D.

Geometric Interpretation of Gram-Schmidt

The Gram-Schmidt process can be viewed as a recursive method of "straightening out" a basis. At each step, we take a new vector v_j and subtract its projection onto the subspace spanned by the previously orthogonalized vectors $w_1, \ldots, w_{j-1}$. This leaves only the component of v_j that is orthogonal to that subspace.



Step 1: Set $w_1 = v_1$.
Step 2: Subtract the projection of v_2 onto w_1 to get $w_2 \perp w_1$.

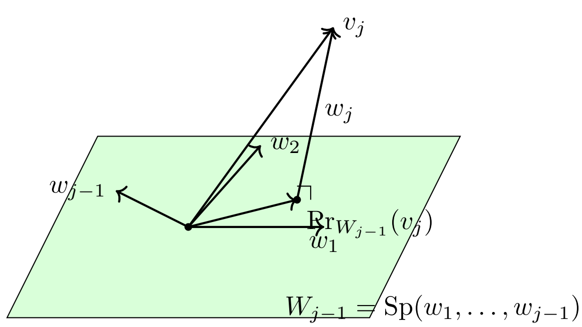


Figure 22.1: Geometric idea of Gram–Schmidt: the vectors $w_1, \ldots, w_{j-1}$ span W_{j-1}, and v_j decomposes as $v_j = w_j + \mathrm{Pr}_{W_{j-1}}(v_j)$ with $w_j \perp W_{j-1}$.

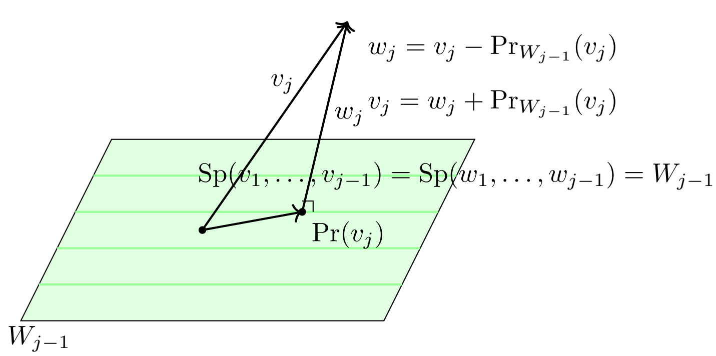


Figure 22.2: Geometric idea behind the Gram–Schmidt process.

Remark: In practice, the Gram-Schmidt process is often followed by a normalization step to produce an orthonormal basis. This two-stage process is essential for many numerical algorithms and theoretical results in linear algebra, such as the QR decomposition.

The Orthogonal Complement (Section 22.b)

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over K ($\mathbb{R}$ or $\mathbb{C}$).

Definition 22.b.1 (*The Orthogonal Complement*): Let $\emptyset \neq S \subseteq V$ be a subset. We define the orthogonal complement of S as:

$$S^\perp := \{v \in V \mid v \perp s \ \forall s \in S\} = \{v \in V \mid \langle v, s\rangle = 0\}.$$

When $S = \{v\}$ we write $v^\perp := \{v\}^\perp$.

Lemma 22.b.2 (*Properties of the Orthogonal Complement*): Let $\emptyset \neq S, T \subseteq V$. Then:

(a) $S^\perp \subseteq V$ is a linear subspace.

(b) $0^\perp = V$, $V^\perp = \{0\}$.

(c) $S \cap S^\perp$ is either $\emptyset$ or equal to $\{0\}$.

(d) If $S \subseteq T$ then $S^\perp \supseteq T^\perp$.

(e) $(\mathrm{Sp}(S))^\perp = S^\perp$.

(f) $S \subseteq (S^\perp)^\perp$.

Proof

(a) We have $0 \in S^\perp$ because $\langle 0, s\rangle = 0$ for all $s \in S$. If $\alpha, \beta \in K$ and $v, w \in S^\perp$, then for all $s \in S$ we have:

$$\langle \alpha v + \beta w, s\rangle = \alpha\langle v, s\rangle + \beta\langle w, s\rangle = 0.$$

This implies $\alpha v + \beta w \in S^\perp$. Thus, $S^\perp$ is a linear subspace.

(b) Showing $0^\perp = V$ is left as an exercise. Let $v \in V^\perp$. Then $\langle v, w\rangle = 0$ for all $w \in V$. In particular, $\langle v, v\rangle = 0$, which implies $v = 0$. This shows $V^\perp \subseteq \{0\}$. Since clearly $\{0\} \subseteq V^\perp$, we conclude $V^\perp = \{0\}$.

(c) If $S \cap S^\perp = \emptyset$, we are done. Otherwise, assume that $s \in S \cap S^\perp$. This implies $\langle s, s\rangle = 0$, which means $s = 0$.

(d) (Exercise).

(e) We have $S \subseteq \mathrm{Sp}(S)$. By statement **(d)**, $(\mathrm{Sp}(S))^\perp \subseteq S^\perp$. We will show that $S^\perp \subseteq (\mathrm{Sp}(S))^\perp$. Indeed, let $v \in S^\perp$, and let $u \in \mathrm{Sp}(S)$. Write $u = \sum_{i=1}^k \alpha_i u_i$ with $\alpha_i \in K$ and $u_i \in S$ for all $1 \leq i \leq k$. We compute:

$$\langle v, u\rangle = \sum_{i=1}^k \overline{\alpha_i}\langle v, u_i\rangle = 0,$$

where we used the fact that $\langle v, u_i\rangle = 0$ because $v \in S^\perp$ and $u_i \in S$. Thus $v \perp u$. This holds for all $u \in \mathrm{Sp}(S)$, hence $v \in (\mathrm{Sp}(S))^\perp$. This shows $S^\perp \subseteq (\mathrm{Sp}(S))^\perp$.

(f) Let $s \in S$. Then for all $v \in S^\perp$ we have $\langle v, s\rangle = 0$, which implies $\langle s, v\rangle = 0$. Therefore, $s \perp v$ for all $v \in S^\perp$. This implies $s \in (S^\perp)^\perp$.

Q.E.D.

Theorem 22.b.3 (*Orthogonal Complement in Finite Dimensions*): Let V be an inner product space (possibly of infinite dimension), and $U \subseteq V$ a finite-dimensional subspace. Then $U^\perp$ is a complement of U in V (i.e., $U + U^\perp = V$ and $U \cap U^\perp = \{0\}$).

Proof

By **??**, we already have $U \cap U^\perp = \{0\}$. So it is enough to show that $U + U^\perp = V$. Put $r := \dim U$ and pick an orthonormal basis $(e_1, \ldots, e_r)$ for U. Let $v \in V$. Define $\widetilde{P}_U(v) := \langle v, e_1\rangle e_1 + \cdots + \langle v, e_r\rangle e_r \in U$. (This is the orthogonal projection of v to U.) We have

$$v = \underbrace{\widetilde{P}_U(v)}_{\in U} + (v - \widetilde{P}_U(v)). \quad (*)$$

We claim that $v - \widetilde{P}_U(v) \in U^\perp$. Indeed, for any $1 \le i \le r$,

$$\langle v - \widetilde{P}_U(v), e_i\rangle = \langle v, e_i\rangle - \sum_{k=1}^{r} \langle v, e_k\rangle\langle e_k, e_i\rangle = \langle v, e_i\rangle - \langle v, e_i\rangle = 0.$$

This holds for all $1 \le i \le r$, hence $v - \widetilde{P}_U(v) \perp \mathrm{Sp}(e_1, \ldots, e_r) = U$, i.e., $v - \widetilde{P}_U(v) \in U^\perp$. Going back to $(*)$, we see that

$$v = \underbrace{\widetilde{P}_U(v)}_{\in U} + \underbrace{(v - \widetilde{P}_U(v))}_{\in U^\perp}.$$

This holds for all $v \in V$, which implies $V = U + U^\perp$. **Q.E.D.**

Remark: If $U \subseteq V$ is a finite-dimensional subspace, then since $U^\perp$ is a complement of U in V, we have a canonical isomorphism $V \cong U \oplus U^\perp$.

Definition 22.b.4 (*Orthogonal Projection*): Let V be an inner product space and $U \subseteq V$ a finite-dimensional subspace. We have seen that $U^\perp$ is a complement of U in V (which implies that for all $v \in V$, there exist unique $u \in U$ and $w \in U^\perp$ such that $v = u + w$). Define a map $P_U \colon V \to U$ as follows: for all $v \in V$, write $v = u + w$ with $u \in U$ and $w \in U^\perp$ in a unique way. We define $P_U(v) := u$. The map P_U is called the *"orthogonal projection"* on U.

Proposition 22.b.5 (*Properties of Orthogonal Projection*): P_U has the following properties:

(a) P_U is a linear map.

(b) $\mathrm{Im}\, P_U = U$, $\ker P_U = U^\perp$.

(c) For all $v \in V$, $v - P_U(v) \in U^\perp$.

(d) For all $u \in U$, $P_U(u) = u$.

(e) If $(e_1, \ldots, e_r)$ is any orthonormal basis for U, then $P_U(v) = \sum_{i=1}^r \langle v, e_i\rangle \cdot e_i$ (which is equal to $\widetilde{P}_U(v)$ from the previous proof).

Proof

Properties **(a)–(d)** follow from the general fact that if W_1, W_2 are vector subspaces of W and $W_1 \oplus W_2 = W$, then the map $P_{W_1} \colon W \to W_1$, defined by $P_{W_1}(w) = w_1$, where we write $w = w_1 + w_2$ in a unique way with $w_1 \in W_1, w_2 \in W_2$, is a linear map, $\mathrm{Im}\, P_{W_1} = W_1$, $\ker P_{W_1} = W_2$, and for all $w \in W$ we have $w - P_{W_1}(w) \in W_2$. (So $w = P_{W_1}(w) + (w - P_{W_1}(w))$ is the decomposition of w as $w = w_1 + w_2$.) We also have $P_{W_1}(w) = w$ for all $w \in W_1$.

For **(e)**, we can write

$$v = \underbrace{\left(\sum_{i=1}^{r} \langle v, e_i\rangle \cdot e_i\right)}_{(1)} + \underbrace{\left(v - \sum_{i=1}^{r} \langle v, e_i\rangle e_i\right)}_{(2)} \tag{22.2}$$

Now, $(1) = \widetilde{P}_U(v)$ from the previous proof, and we have also seen in that proof that $(2) \in U^\perp$. This implies that (**??**) is the unique decomposition of v with respect to $V \cong U \oplus U^\perp$. Hence, $P_U(v) = (1)$. **Q.E.D.**

Corollary 22.b.6 (*Bidual Orthogonal Complement*): Let V be an inner product space and $U \subseteq V$ a finite-dimensional subspace. Then $(U^\perp)^\perp = U$.

Proof

By a previous Lemma, $U \subseteq (U^\perp)^\perp$. We will show that we also have $(U^\perp)^\perp \subseteq U$. Indeed, let $v \in (U^\perp)^\perp$. Recall that $U^\perp$ is a complement of U in V. Write $v = u + w$ with $u \in U$ and $w \in U^\perp$. Since $u \in U \subseteq (U^\perp)^\perp$ and also $v \in (U^\perp)^\perp$, we have that $w = v - u \in (U^\perp)^\perp$. But $w \in U^\perp$, hence $w \in U^\perp \cap (U^\perp)^\perp = \{0\}$. This shows that $w = 0$. Consequently, $v = u \in U$. We have proven that $(U^\perp)^\perp \subseteq U$. Q.E.D.

Corollary 22.b.7 (*Dimension Formula for the Orthogonal Complement*): If V is a finite-dimensional inner product space and $U \subseteq V$ is a subspace, then $\dim U + \dim U^\perp = \dim V$.

Orthogonal & Unitary Matrices

Definition 22.b.8 (*Orthogonal and Unitary Matrices*):

(a) A matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is called *"orthogonal"* if its columns form an orthonormal system in $\mathbb{R}^n$, with respect to the standard scalar product of $\mathbb{R}^n$. We denote the set of orthogonal matrices $n \times n$ by $\mathrm{O}(n) \subseteq \mathcal{M}_{n\times n}(\mathbb{R})$.

(b) A matrix $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is called *"unitary"* if its columns form an orthonormal system in $\mathbb{C}^n$, with respect to the standard Hermitian product on $\mathbb{C}^n$. We denote the set of all unitary matrices $n \times n$ by $\mathrm{U}(n) \subseteq \mathcal{M}_{n\times n}(\mathbb{C})$.

Lemma 22.b.9 (*Characterization of Orthogonal Matrices*): For a matrix $A \in \mathcal{M}_{n\times n}(\mathbb{R})$, the following are equivalent:

$$A \text{ is orthogonal} \iff A^T A = I_n \iff AA^T = I_n \iff A^{-1} = A^T.$$

Lemma 22.b.10 (*Characterization of Unitary Matrices*): For a matrix $A \in \mathcal{M}_{n\times n}(\mathbb{C})$, the following are equivalent:

$$A \text{ is unitary} \iff A^* A = I_n \iff AA^* = I_n \iff A^{-1} = A^*.$$

Proof of ??

(The proof of **??** is similar). Write $A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$, where $v_k \in \mathbb{C}^n_{\text{col}}$ for $1 \le k \le n$. We have:

$$A^*A = \begin{pmatrix} - & \overline{v_1}^\top & - \\ & \vdots & \\ - & \overline{v_n}^\top & - \end{pmatrix} \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix} = \left(\overline{v_i}^\top \cdot v_j\right)_{1\le i\le n, 1\le j\le n} = \left(\langle v_j, v_i\rangle\right)_{1\le i\le n, 1\le j\le n}. \quad (22.3)$$

Now, A is unitary if and only if $v_1, \dots, v_n$ are an orthonormal system, which is equivalent to saying that (**??**) is exactly I_n. The equivalence of these to $AA^* = I_n$ is a general property of invertible matrices. Q.E.D.

Corollary 22.b.11 (*Rows of Orthogonal and Unitary Matrices*):

(a) Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$. Then $A \in \mathrm{O}(n) \iff A^T \in \mathrm{O}(n) \iff$ the rows of A form an orthonormal basis for $\mathbb{R}^n$.

(b) Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$. Then $A \in \mathrm{U}(n) \iff A^* \in \mathrm{U}(n) \iff$ the rows of A form an orthonormal basis for $\mathbb{C}^n$.

QR Decomposition

Theorem 22.b.12 (*QR Decomposition for Invertible Matrices*):

(a) Let $A \in \mathrm{GL}_n(\mathbb{R})$. Then there exists $Q \in \mathrm{O}(n)$ and an upper triangular matrix $R \in \mathcal{M}_{n\times n}(\mathbb{R})$ such that $A = Q \cdot R$.

(b) Let $A \in \mathrm{GL}_n(\mathbb{C})$. Then there exists $Q \in \mathrm{U}(n)$ and an upper triangular matrix $R \in \mathcal{M}_{n\times n}(\mathbb{C})$ such that $A = Q \cdot R$.

Proof (for (a) & (b) together)

Write $A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$ with $v_k \in K^n_{\mathrm{col}}$ (where $K = \mathbb{R}$ or $\mathbb{C}$). Since $A \in \mathrm{GL}_n(K)$, the vectors $v_1, \dots, v_n$ form a basis for K^n. Apply the Gram-Schmidt orthogonalization process to $v_1, \dots, v_n$ and obtain an orthonormal basis $(e_1, \dots, e_n)$ for K^n. Recall that for all $1 \le j \le n$, $\mathrm{Sp}(v_1, \dots, v_j) = \mathrm{Sp}(e_1, \dots, e_j)$. And we also have:

$$\begin{aligned} v_1 &= \langle v_1, e_1\rangle e_1 \\ v_2 &= \langle v_2, e_1\rangle e_1 + \langle v_2, e_2\rangle e_2 \\ &\ \vdots \\ v_j &= \langle v_j, e_1\rangle e_1 + \dots + \langle v_j, e_j\rangle e_j \\ &\ \vdots \\ v_n &= \langle v_n, e_1\rangle e_1 + \dots + \langle v_n, e_n\rangle e_n. \end{aligned}$$

This implies the matrix equation:

$$\begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix} = \begin{pmatrix} | & & | \\ e_1 & \dots & e_n \\ | & & | \end{pmatrix} \begin{pmatrix} \langle v_1, e_1\rangle & \langle v_2, e_1\rangle & \dots & \langle v_j, e_1\rangle & \dots & \langle v_n, e_1\rangle \\ 0 & \langle v_2, e_2\rangle & \dots & \langle v_j, e_2\rangle & \dots & \langle v_n, e_2\rangle \\ \vdots & 0 & \ddots & \vdots & & \vdots \\ \vdots & \vdots & & \langle v_j, e_j\rangle & \dots & \vdots \\ 0 & 0 & \dots & 0 & \dots & \langle v_n, e_n\rangle \end{pmatrix}$$

Which is precisely $A = Q \cdot R$, where $Q \in \mathrm{O}(n)$ (or $\mathrm{U}(n)$), and R is an upper triangular matrix.

Q.E.D.

Theorem 22.b.13 (*Extension of QR Decomposition for All Matrices*): Let $A \in \mathcal{M}_{n\times n}(K)$, and let $r := \mathrm{rank}(A)$. Then there exist matrices Q, R with $Q \in \mathrm{O}(n)$ if $K = \mathbb{R}$ and $Q \in \mathrm{U}(n)$ if $K = \mathbb{C}$, and $R = \left(\begin{smallmatrix} C & * \\ 0 & 0 \end{smallmatrix}\right) \in \mathcal{M}_{n\times n}(K)$, where $C \in \mathcal{M}_{r\times r}(K)$ is an upper triangular matrix, such that $A = Q \cdot R$.

Proof

Note that $r = \dim \mathrm{Sp}(v_1, \dots, v_n)$. We may assume $r \ge 1$, otherwise $r = 0$, hence $A = 0$ and we can take $R = 0$ and Q arbitrary. Note that there exist indices $1 \le i_1 < \dots < i_r \le n$ such that:

(a) $v_{i_1}, \dots, v_{i_r}$ are linearly independent.

(b) $(v_{i_1}, \dots, v_{i_k})$ is a basis for $\mathrm{Sp}(v_1, v_2, \dots, v_{i_k})$ (and $k = \dim \mathrm{Sp}(v_1, v_2, \dots, v_{i_k}) \le i_k$).

To see this, let $1 \le i_1$ be the minimal index j such that $v_j \ne 0$. (Since $r \ge 1$, there exists such an index.) Clearly $\mathrm{Sp}(v_1, \dots, v_{i_1}) = \mathrm{Sp}(v_{i_1})$. Suppose we have defined $1 \le i_1 < \dots < i_k < n$ as above. Consider now the minimal index ℓ, with $i_k < \ell \le n$ and such that $v_\ell \notin \mathrm{Sp}(v_{i_1}, \dots, v_{i_k})$. If there is no such ℓ, then $k = r$ and we are done. Otherwise take $i_{k+1} := \ell$. Continue the construction by induction.

Now, apply the Gram-Schmidt process to the list of vectors $v_{i_1}, \dots, v_{i_r}$ and obtain an orthonormal system $e_1, \dots, e_r$ such that $(e_1, \dots, e_k)$ is a basis for $\mathrm{Sp}(v_{i_1}, \dots, v_{i_k}) = \mathrm{Sp}(v_1, v_2, \dots, v_{i_k})$. Extend $e_1, \dots, e_r$ to an orthonormal basis $(e_1, \dots, e_r, e_{r+1}, \dots, e_n)$ of K^n (how?). Take $Q = \begin{pmatrix} | & & | \\ e_1 & \cdots & e_n \\ | & & | \end{pmatrix}$, and let R be constructed analogously as in the previous proof, which will have the block form $R = \left(\begin{smallmatrix} C & * \\ 0 & 0 \end{smallmatrix}\right)$. **Q.E.D.**

Exercise: Let V be an inner product space. Suppose $v_1, \dots, v_m \in V$ are not necessarily linearly independent, but not all of them are 0. What happens if we apply the Gram-Schmidt orthogonalization process? Suppose that at stage j we already have an orthogonal system $e_1, \dots, e_k$ that spans $\mathrm{Sp}(v_1, \dots, v_{i_k})$ and $v_{i_{k+1}} \in \mathrm{Sp}(v_1, v_2, \dots, v_{i_k})$. How will the next vector in the Gram-Schmidt process look like?

Dual Spaces and Inner Products (Chapter 23)

Dual Spaces and Inner Products (Section 23.a)

Let V be a vector space over K (any field). Recall that we have the dual space $V^* := \operatorname{Hom}(V, K)$, which is the space of linear maps $\varphi\colon V \to K$ (also called "*functionals*"). The dual space V^* is also a vector space over K.

Warm up: Assume $V = K^n$ and let $\varphi\colon K^n \to K$ be a functional. Then there exists a unique matrix $A = (a_1, \dots, a_n) \in \mathcal{M}_{1\times n}(K)$ such that $\varphi(v) = A \cdot v$ for all $v \in K^n$.

Indeed, let $e_1, \dots, e_n$ be the standard basis for K^n, and define $a_1 = \varphi(e_1), \dots, a_n = \varphi(e_n)$. Let $v = (v_1, \dots, v_n)^\top \in K^n$. Then

$$\varphi(v) = \varphi(v_1 e_1 + \cdots + v_n e_n) = v_1 \varphi(e_1) + \cdots + v_n \varphi(e_n) = v_1 a_1 + \cdots + v_n a_n = A \cdot v.$$

Consider now an inner product space $(V, \langle \cdot, \cdot \rangle)$ over K (where $K = \mathbb{R}$ or $\mathbb{C}$). Let $u \in V$. We define a map $\varphi_u\colon V \to K$ by mapping $v \mapsto \langle v, u \rangle \in K$.

Claim: The map φ_u is linear.

(This is left as an exercise).

Theorem 23.a.1: Let $(V, \langle \cdot, \cdot \rangle)$ be a finite-dimensional inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$). Let $\varphi \in V^*$. Then there exists a unique vector $u \in V$ such that $\varphi = \varphi_u$, that is, $\varphi(v) = \langle v, u \rangle$ for all $v \in V$.

In fact, for $K = \mathbb{R}$, the map $\Phi\colon V \to V^*$, given by $\Phi(u) := \varphi_u$, is an isomorphism. For $K = \mathbb{C}$, this map is injective and surjective, but it is only $\mathbb{C}$-antilinear, meaning that $\Phi(u_1 + u_2) = \Phi(u_1) + \Phi(u_2)$ and $\Phi(c \cdot u) = \overline{c} \cdot \Phi(u)$ for all $c \in \mathbb{C}$ and $u \in V$.

So, at least for $K = \mathbb{R}$, a scalar product on V gives us a canonical isomorphism $V \cong V^*$. (However, this isomorphism depends on the choice of the scalar product!)

Proof

First, we prove that for all $\varphi \in V^*$ there exists a unique $u \in V$ such that $\varphi = \varphi_u$. Fix an orthonormal basis $\mathcal{B} = (e_1, \dots, e_n)$ for V, where $n = \dim V$. Let $\varphi \in V^*$, and let $v \in V$. We have $v = \sum_{i=1}^n \langle v, e_i \rangle \cdot e_i$. This implies:

$$\varphi(v) = \varphi\left(\sum_{i=1}^n \langle v, e_i \rangle \cdot e_i\right) = \sum_{i=1}^n \langle v, e_i \rangle \cdot \varphi(e_i) = \left\langle v, \sum_{i=1}^n \overline{\varphi(e_i)} \cdot e_i \right\rangle.$$

(Note that the conjugation is needed here only if $K = \mathbb{C}$). Define $u := \sum_{i=1}^n \overline{\varphi(e_i)} \cdot e_i$. It follows that $\varphi = \varphi_u$.

Uniqueness of u: Suppose that $\varphi_{u_1} = \varphi_{u_2}$ for $u_1, u_2 \in V$. This implies that $\langle v, u_1 \rangle = \langle v, u_2 \rangle$ for all $v \in V$. By a previous result, this implies that $u_1 = u_2$.

The statements about the linearity of Φ are left as an exercise. **Q.E.D.**

The Adjoint Map (Section 23.b)

Let V, W be vector spaces over K. Let $T\colon V \to W$ be a linear map. We have seen in Linear Algebra I that T induces a linear map $T^*\colon W^* \to V^*$ defined by $T^*(\varphi) := \varphi \circ T$.

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[dr, "T^*(\varphi)=\varphi\circ T"'] & W \arrow[d, "\varphi"] \\ & K \end{tikzcd}$$

Assume now that $(V, \langle\cdot,\cdot\rangle_V)$ and $(W, \langle\cdot,\cdot\rangle_W)$ are inner product spaces over K (where $K = \mathbb{R}$ or $K = \mathbb{C}$). Assume that $\dim V < \infty$. Given T as above, we can define a map $T' \colon W \to V$ using our *"canonical"* identifications $W \cong W^*$ and $V \cong V^*$. (This is straightforward for $K = \mathbb{R}$, but we have to be careful with $K = \mathbb{C}$.)

$$\begin{tikzcd} W^* \arrow[r, "T^*"] & V^* \arrow[d, bend left, "\Phi_V^{-1}"] \\ W \arrow[u, "\Phi_W"] \arrow[r, dashed, "T'"'] & V \arrow[u, "\Phi_V"'] \end{tikzcd}$$

Here, we define $T' := \Phi_V^{-1} \circ T^* \circ \Phi_W$. (Note that we are using the fact that $\dim V < \infty$ so that Φ_V is invertible).

What properties does T' have? Note that T' is uniquely defined by $\Phi_V \circ T' = T^* \circ \Phi_W$.

$$\begin{aligned} &\Phi_V \circ T' = T^* \circ \Phi_W \\ &\qquad \iff \forall w \in W, \quad (\Phi_V \circ T')(w) = (T^* \circ \Phi_W)(w) \\ &\qquad \iff \varphi_{T'(w)} = T^*(\varphi_w) \quad (\text{in } V^*) \\ &\qquad \iff \forall v \in V, \quad \varphi_{T'(w)}(v) = (T^*(\varphi_w))(v) \\ &\qquad \iff \langle v, T'(w)\rangle_V = \varphi_w(T(v)) = \langle T(v), w\rangle_W. \end{aligned}$$

In other words,

$$\langle T(v), w\rangle_W = \langle v, T'(w)\rangle_V \quad \forall v \in V, \forall w \in W. \tag{23.1}$$

Claim: T' is a linear map.

Proof

For $K = \mathbb{R}$, the maps Φ_W, Φ_V, and T^* are linear, and therefore T' is also linear. For $K = \mathbb{C}$, the maps Φ_W and Φ_V are additive but only $\mathbb{C}$-antilinear. This implies that Φ_V^{-1} is also additive and $\mathbb{C}$-antilinear (this is an exercise to check!). And T^* is linear. Consequently, $T'(w_1 + w_2) = T'(w_1) + T'(w_2)$ (exercise). Now, let $c \in \mathbb{C}$ and $w \in W$. We compute:

$$\begin{aligned} T'(cw) &= (\Phi_V^{-1} \circ T^* \circ \Phi_W)(cw) \\ &= \Phi_V^{-1}(T^*(\bar{c} \cdot \Phi_W(w))) \\ &= \Phi_V^{-1}(\bar{c} \cdot T^*(\Phi_W(w))) \\ &= c \cdot \Phi_V^{-1}(T^*(\Phi_W(w))) \\ &= c \cdot T'(w). \end{aligned}$$

This proves that T' is linear. **Q.E.D.**

Notation: The linear map $T' \colon W \to V$ is often denoted by $T^* \colon W \to V$. This makes sense because of the *"canonical"* identifications:

$$T \colon V \to W \quad \rightsquigarrow \quad \left(\begin{array}{ccc} W^* & \xrightarrow{T^*} & V^* \\ \wr\!\wr & & \wr\!\wr \\ W & \xrightarrow{T'} & V \end{array}\right)$$

(Here we also assume $\dim W < \infty$ to have the canonical identification on the left). The map $T^* \colon W \to V$ is called the "*adjoint*" of T.

Basic Properties of Adjoint Maps

Lemma 23.a.1 (*Properties of the Adjoint Map*)**:** Let V, W, U be finite-dimensional inner product spaces over K (where $K = \mathbb{R}$ or $\mathbb{C}$). Let $S, T\colon V \to W$ and $R\colon W \to U$ be linear maps. Then:

(a) T^* is a linear map.

(b) $(S+T)^* = S^* + T^*$.

(c) $(\lambda T)^* = \overline{\lambda} T^*$ for all $\lambda \in K$.

(d) $(T^*)^* = T$.

(e) $(\mathrm{id}_V)^* = \mathrm{id}_V$.

(f) $(R \circ T)^* = T^* \circ R^*$.

Proof

(a) This was proved already. (We will use repeatedly that if $\langle x, y_1\rangle = \langle x, y_2\rangle$ for all x, then $y_1 = y_2$).

(b) Let $w \in W$ and $v \in V$. We have:

$$\begin{aligned}\langle v, (S+T)^*(w)\rangle &= \langle (S+T)(v), w\rangle \\ &= \langle S(v), w\rangle + \langle T(v), w\rangle \\ &= \langle v, S^*(w)\rangle + \langle v, T^*(w)\rangle \\ &= \langle v, (S^* + T^*)(w)\rangle.\end{aligned}$$

This holds for all $v \in V$, which implies $(S+T)^*(w) = (S^*+T^*)(w)$. Since this holds for all $w \in W$, we conclude that $(S+T)^* = S^* + T^*$.

(c) This is left as an exercise.

(d) Let $v \in V$ and $w \in W$. We have:

$$\begin{aligned}\langle w, (T^*)^*(v)\rangle &= \langle T^*(w), v\rangle \\ &= \overline{\langle v, T^*(w)\rangle} \\ &= \overline{\langle T(v), w\rangle} \\ &= \langle w, T(v)\rangle.\end{aligned}$$

This holds for all $w \in W$, which implies $(T^*)^*(v) = T(v)$. Since this holds for all $v \in V$, we conclude that $(T^*)^* = T$.

(e) This is left as an exercise.

(f) Let $v \in V$ and $u \in U$. We have:

$$\begin{aligned}\langle v, (R\circ T)^*(u)\rangle &= \langle (R\circ T)(v), u\rangle \\ &= \langle R(T(v)), u\rangle \\ &= \langle T(v), R^*(u)\rangle \\ &= \langle v, T^*(R^*(u))\rangle \\ &= \langle v, (T^*\circ R^*)(u)\rangle.\end{aligned}$$

This holds for all $v \in V$, which implies $(R\circ T)^*(u) = (T^*\circ R^*)(u)$. Since this holds for all $u \in U$, we conclude that $(R \circ T)^* = T^* \circ R^*$.

Q.E.D.

Lemma 23.a.2 (*Orthogonal Relations of Adjoint Maps*)**:** Let V, W be finite-dimensional inner product spaces. Let $T\colon V \to W$ be a linear map, and let $T^*\colon W \to V$ be the adjoint of T. Then:

(a) $\ker(T^*) = (\operatorname{Im} T)^\perp$.

(b) $\operatorname{Im} T^* = (\ker T)^\perp$.

(c) $\ker T = (\operatorname{Im} T^*)^\perp$.

(d) $\operatorname{Im} T = (\ker T^*)^\perp$.

Proof

Let $w \in W$. Then:

$$\begin{aligned} w \in \ker(T^*) &\iff T^*(w) = 0 \\ &\iff \forall v \in V, \quad \langle v, T^*(w)\rangle = 0 \\ &\iff \forall v \in V, \quad \langle T(v), w\rangle = 0 \\ &\iff w \perp T(v) \quad \forall v \in V \\ &\iff w \perp \operatorname{Im}(T) \\ &\iff w \in (\operatorname{Im} T)^\perp. \end{aligned}$$

This proves statement **(a)**.

Apply the orthogonal complement $(-)^\perp$ to both sides of **(a)**:

$$(\ker T^*)^\perp = ((\operatorname{Im} T)^\perp)^\perp = \operatorname{Im} T.$$

(Recall that $\operatorname{Im} T$ is finite-dimensional). This proves statement **(d)**.

Now, use statement **(d)** with T^* instead of T:

$$\operatorname{Im} T^* = (\ker(T^*)^*)^\perp = (\ker T)^\perp,$$

which proves statement **(b)**.

To prove statement **(c)**, apply the orthogonal complement $(-)^\perp$ to statement **(b)**. **Q.E.D.**

Matrix Representation of Adjoint Maps

Proposition 23.a.3 (*Matrix Representation of the Adjoint*)**:** Let V, W be finite-dimensional inner product spaces and $T\colon V \to W$ a linear map. Let $\mathcal{B} = (e_1, \dots, e_n)$ be an orthonormal basis for V, and $\mathcal{C} = (f_1, \dots, f_m)$ an orthonormal basis for W. Then:

$$[T^*]^{\mathcal{C}}_{\mathcal{B}} = \left(\overline{[T]^{\mathcal{B}}_{\mathcal{C}}}\right)^\top = \left([T]^{\mathcal{B}}_{\mathcal{C}}\right)^*$$

(where the latter denotes the adjoint matrix).

Proof

Write $A = (a_{ij})_{\substack{1\le i\le m\\1\le j\le n}} = [T]^{\mathcal{B}}_{\mathcal{C}}$ and $B = (b_{ij})_{\substack{1\le i\le n\\1\le j\le m}} = [T^*]^{\mathcal{C}}_{\mathcal{B}}$. We have:

$$b_{ij} = \langle T^*(f_j), e_i\rangle = \overline{\langle e_i, T^*(f_j)\rangle} = \overline{\langle T(e_i), f_j\rangle} = \overline{a_{ji}}.$$

Therefore, $B = (\overline{A})^\top = A^*$. **Q.E.D.**

Corollary 23.a.4 (*Adjoint of a Matrix Map*)**:** Let $A \in \mathcal{M}_{m\times n}(K)$ (where $K = \mathbb{R}$ or $\mathbb{C}$), and let $T_A\colon K^n \to K^m$ be the linear map defined by $T_A(v) = A \cdot v$. Endow K^n and K^m with their standard inner products. Then $(T_A)^* = T_{A^*}$.

Proposition 23.a.5 (*Properties of the Adjoint Matrix*)**:** Let $K = \mathbb{R}$ or $\mathbb{C}$. Let $A, B \in \mathcal{M}_{m\times n}(K)$ and $C \in \mathcal{M}_{n\times p}(K)$. Then:

(a) $\overline{(A+B)}^\top = \overline{A}^\top + \overline{B}^\top$.

(b) $\overline{(\lambda A)}^\top = \overline{\lambda A}^\top$ for all $\lambda \in K$.

(c) $\overline{(\overline{A}^\top)}^\top = A$.

(d) $\overline{I_n}^\top = I_n$.

(e) $\overline{(A \cdot C)}^\top = \overline{C}^\top \cdot \overline{A}^\top$.

Proof

This is left as an exercise. **Q.E.D.**

Exercise Solutions

Proof of Linearity of φ_u (on ??)

Let $v_1, v_2 \in V$ and $c \in K$. We have:

$$\varphi_u(v_1 + v_2) = \langle v_1 + v_2, u\rangle = \langle v_1, u\rangle + \langle v_2, u\rangle = \varphi_u(v_1) + \varphi_u(v_2),$$

and

$$\varphi_u(cv_1) = \langle cv_1, u\rangle = c\langle v_1, u\rangle = c\varphi_u(v_1).$$

Thus, φ_u is a linear map. **Q.E.D.**

Proof of ??

For $K = \mathbb{R}$, we have $\Phi(u) = \varphi_u$. Then:

$$\Phi(u_1 + u_2)(v) = \langle v, u_1 + u_2\rangle = \langle v, u_1\rangle + \langle v, u_2\rangle = \Phi(u_1)(v) + \Phi(u_2)(v),$$

and

$$\Phi(cu)(v) = \langle v, cu\rangle = c\langle v, u\rangle = c\Phi(u)(v).$$

(Here we used the fact that for $K = \mathbb{R}$, $c = \overline{c}$). Thus Φ is linear.

For $K = \mathbb{C}$, the additivity follows identically. For scalar multiplication:

$$\Phi(cu)(v) = \langle v, cu\rangle = \overline{c}\langle v, u\rangle = \overline{c}\Phi(u)(v).$$

Thus $\Phi(cu) = \overline{c}\Phi(u)$, making it $\mathbb{C}$-antilinear.

For the inverse Φ^{-1}: since Φ is additive, taking $\varphi_1 = \Phi(u_1)$ and $\varphi_2 = \Phi(u_2)$ gives $\Phi^{-1}(\varphi_1 + \varphi_2) = \Phi^{-1}(\varphi_1) + \Phi^{-1}(\varphi_2)$, so Φ^{-1} is additive. For scalar multiplication, let $\varphi = \Phi(u)$ and $\lambda \in \mathbb{C}$. Then $\overline{\lambda}\varphi = \overline{\lambda}\Phi(u) = \Phi(\lambda u)$. Applying Φ^{-1} to both sides gives $\Phi^{-1}(\overline{\lambda}\varphi) = \lambda u = \lambda\Phi^{-1}(\varphi)$. Setting $c = \overline{\lambda}$ (so $\lambda = \overline{c}$) gives $\Phi^{-1}(c\varphi) = \overline{c}\Phi^{-1}(\varphi)$, proving Φ^{-1} is $\mathbb{C}$-antilinear. **Q.E.D.**

Proof of Additivity of T' (on ??)

We have $T'(w_1 + w_2) = (\Phi_V^{-1} \circ T^* \circ \Phi_W)(w_1 + w_2)$. Since Φ_W is additive, this equals $(\Phi_V^{-1} \circ T^*)(\Phi_W(w_1) + \Phi_W(w_2))$. Since T^* is linear (and therefore additive), this becomes $\Phi_V^{-1}(T^*(\Phi_W(w_1)) + T^*(\Phi_W(w_2)))$. Finally, since Φ_V^{-1} is additive, this splits into $\Phi_V^{-1}(T^*(\Phi_W(w_1))) + \Phi_V^{-1}(T^*(\Phi_W(w_2))) = T'(w_1) + T'(w_2)$. **Q.E.D.**

Proof of ?? (c)

Let $v \in V$ and $w \in W$. We have:

$$\langle v, (\lambda T)^*(w)\rangle = \langle (\lambda T)(v), w\rangle = \langle \lambda T(v), w\rangle = \lambda\langle T(v), w\rangle = \lambda\langle v, T^*(w)\rangle = \langle v, \overline{\lambda}T^*(w)\rangle.$$

This holds for all $v \in V$, so $(\lambda T)^*(w) = \overline{\lambda}T^*(w)$. Thus $(\lambda T)^* = \overline{\lambda}T^*$. **Q.E.D.**

Proof of ?? (e)

Let $v, w \in V$. We have:

$$\langle v, (\mathrm{id}_V)^*(w)\rangle = \langle \mathrm{id}_V(v), w\rangle = \langle v, w\rangle.$$

This holds for all $v \in V$, so $(\mathrm{id}_V)^*(w) = w = \mathrm{id}_V(w)$. Thus $(\mathrm{id}_V)^* = \mathrm{id}_V$. Q.E.D.

Solution

. These follow directly from the properties of matrix transpose and complex conjugation:

(a) $\overline{(A+B)}^\top = (\overline{A}+\overline{B})^\top = \overline{A}^\top + \overline{B}^\top$.

(b) $\overline{(\lambda A)}^\top = (\overline{\lambda A})^\top = \overline{\lambda A}^\top$.

(c) $\overline{(\overline{A}^\top)}^\top = (\overline{\overline{A}}^\top)^\top = (A^\top)^\top = A$.

(d) $\overline{I_n}^\top = I_n^\top = I_n$ (since I_n is real and symmetric).

(e) $\overline{(A \cdot C)}^\top = (\overline{A} \cdot \overline{C})^\top = \overline{C}^\top \cdot \overline{A}^\top$.

Q.E.D.

Spectral Theorems (Chapter 24)

AI-Note: The professor left a note to *"explain the name"* of the Spectral Theorems. In linear algebra, the *"spectrum"* of a linear operator is its set of eigenvalues. The Spectral Theorems provide conditions under which an operator can be completely described by its spectrum—specifically, when there exists a basis for the entire space consisting solely of the operator's eigenvectors (making the operator diagonalizable).

Definition 24.a.1: Let V be an inner product space and $T\colon V \to V$ an endomorphism. We say that T is "*orthogonally diagonalizable*" if there exists an orthonormal basis for V consisting of eigenvectors of T.

Lemma 24.a.2: If T is orthogonally diagonalizable, then $T \circ T^* = T^* \circ T$.

Proof

Let $\mathcal{B}$ be an orthonormal basis for V such that all the vectors in $\mathcal{B}$ are eigenvectors of T. This implies that the representation matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ is a diagonal matrix.

Since $\mathcal{B}$ is an orthonormal basis, we also know that the matrix of the adjoint map is the conjugate transpose of the matrix of the map, meaning $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^*$. Since the conjugate transpose of a diagonal matrix is still a diagonal matrix, $[T^*]_{\mathcal{B}}^{\mathcal{B}}$ is also a diagonal matrix.

If M and N are diagonal matrices, then they must commute, meaning $M \cdot N = N \cdot M$. Therefore, we have:

$$[T]_{\mathcal{B}}^{\mathcal{B}} \cdot [T^*]_{\mathcal{B}}^{\mathcal{B}} = [T^*]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}}.$$

By the properties of representation matrices, this implies:

$$[T \circ T^*]_{\mathcal{B}}^{\mathcal{B}} = [T^* \circ T]_{\mathcal{B}}^{\mathcal{B}} \implies T \circ T^* = T^* \circ T.$$

Q.E.D.

Definition 24.a.3: Let V be an inner product space. An endomorphism $T\colon V \to V$ is called "*normal*" if $T \circ T^* = T^* \circ T$.

Similarly, a matrix $A \in \mathcal{M}_{n\times n}(K)$ (where $K = \mathbb{R}$ or $\mathbb{C}$) is called "*normal*" if $A \cdot \overline{A}^{\top} = \overline{A}^{\top} \cdot A$. (Note that A is a normal matrix if and only if the map $T_A\colon K^n \to K^n$ is a normal endomorphism, provided we endow K^n with its standard inner product).

We have seen in the previous lemma that if $T\colon V \to V$ is orthogonally diagonalizable, then T is normal.

Question: Is the converse statement true?

Over $K = \mathbb{R}$, the answer is NO! For example, consider the matrix $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. The linear map $T_A\colon \mathbb{R}^2 \to \mathbb{R}^2$ is a clockwise rotation by 90°, so there are no eigenvalues (in $\mathbb{R}$) and no eigenvectors. Thus, it cannot be diagonalizable. However, $\overline{A}^{\top} = A^{\top} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and we can easily check that $A \cdot A^{\top} = A^{\top} \cdot A = I_2$, meaning A is indeed normal.

What about over $\mathbb{C}$?

Theorem 24.a.4 (*Spectral Theorem over $\mathbb{C}$*): Let V be a finite-dimensional inner product space over $\mathbb{C}$, and let $T\colon V \to V$ be a linear map. Then T is orthogonally diagonalizable if and only if T is normal.

For the proof, we need a lemma.

Lemma 24.a.5 (*Properties of Normal Endomorphisms*): Let V be a finite-dimensional inner product space over K (where $K=\mathbb{R}$ or $\mathbb{C}$). Let $T\colon V\to V$ be a normal endomorphism. Then:

(a) **(a)** For all $v\in V$, $\|Tv\|=\|T^*v\|$.

(b) **(b)** For all $\lambda\in K$, the endomorphism $T-\lambda\,\mathrm{id}_V$ is also normal.

(c) **(c)** If v is an eigenvector of T with eigenvalue λ, then v is also an eigenvector of T^* with eigenvalue $\overline{\lambda}$.

(d) **(d)** Let v_1,v_2 be eigenvectors of T with eigenvalues λ_1,λ_2 respectively. If $\lambda_1\neq\lambda_2$, then $\langle v_1,v_2\rangle=0$ (i.e., $v_1\perp v_2$).

Proof

Proof of (a): We compute the squared norm:

$$\|Tv\|^2=\langle Tv,Tv\rangle=\langle v,T^*Tv\rangle=\langle v,TT^*v\rangle=\langle T^*v,T^*v\rangle=\|T^*v\|^2,$$

where we used the definition of the adjoint and the fact that T is normal ($T\circ T^*=T^*\circ T$).

Proof of (b): We expand the compositions:

$$\begin{aligned}(T-\lambda\,\mathrm{id}_V)\circ(T-\lambda\,\mathrm{id}_V)^*&=(T-\lambda\,\mathrm{id}_V)\circ(T^*-\overline{\lambda}\,\mathrm{id}_V)\\&=T\circ T^*-\overline{\lambda}T-\lambda T^*+\lambda\overline{\lambda}\,\mathrm{id}_V\,.\quad(*)\end{aligned}$$

On the other hand:

$$\begin{aligned}(T-\lambda\,\mathrm{id}_V)^*\circ(T-\lambda\,\mathrm{id}_V)&=(T^*-\overline{\lambda}\,\mathrm{id}_V)\circ(T-\lambda\,\mathrm{id}_V)\\&=T^*\circ T-\lambda T^*-\overline{\lambda}T+\overline{\lambda}\lambda\,\mathrm{id}_V\,.\quad(**)\end{aligned}$$

Because $T\circ T^*=T^*\circ T$, the expressions $(*)$ and $(**)$ are equal.

Proof of (c): If v is an eigenvector of T with eigenvalue λ, then $(T-\lambda\,\mathrm{id}_V)v=0$. By statement **(b)**, $T-\lambda\,\mathrm{id}_V$ is normal, hence by statement **(a)**, we have:

$$\|(T^*-\overline{\lambda}\,\mathrm{id}_V)v\|=\|(T-\lambda\,\mathrm{id}_V)v\|=\|0\|=0.$$

This implies that $(T^*-\overline{\lambda}\,\mathrm{id}_V)v=0$, and therefore $T^*v=\overline{\lambda}v$.

Proof of (d): We compute the inner product in two ways:

$$\lambda_1\langle v_1,v_2\rangle=\langle\lambda_1v_1,v_2\rangle=\langle Tv_1,v_2\rangle=\langle v_1,T^*v_2\rangle.$$

By statement **(c)**, $T^*v_2=\overline{\lambda_2}v_2$. Thus:

$$\langle v_1,T^*v_2\rangle=\langle v_1,\overline{\lambda_2}v_2\rangle=\lambda_2\langle v_1,v_2\rangle.$$

Equating the two results gives $\lambda_1\langle v_1,v_2\rangle=\lambda_2\langle v_1,v_2\rangle$, which implies $(\lambda_1-\lambda_2)\langle v_1,v_2\rangle=0$. Since $\lambda_1\neq\lambda_2$, we must have $\langle v_1,v_2\rangle=0$.

Q.E.D.

Proof of the Spectral Theorem over $\mathbb{C}$ (??)

We only need to prove the direction *"T is normal $\implies$ T is orthogonally diagonalizable"*, as the reverse direction was already established in **??**.

By a previous theorem, T is trigonalizable (here we explicitly use the fact that $K=\mathbb{C}$ and the Fundamental Theorem of Algebra). That is, there exists a basis $\mathcal{C}$ for V such that $[T]_{\mathcal{C}}^{\mathcal{C}}$ is an upper triangular matrix. By applying the Gram-Schmidt orthogonalization process to the basis $\mathcal{C}$, we obtain a new basis $\mathcal{B}=(v_1,\dots,v_n)$ which is orthonormal, and such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is still upper triangular. (This is left as an exercise to check. The main point is that if $\mathcal{C}=(u_1,\dots,u_n)$, then Gram-Schmidt gives an orthonormal basis $\mathcal{B}=(v_1,\dots,v_n)$ with $\mathrm{Sp}(v_1,\dots,v_k)=\mathrm{Sp}(u_1,\dots,u_k)$ for all $1\le k\le n$).

We claim that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is actually diagonal. To see this, let us write the matrix explicitly as $[T]_{\mathcal{B}}^{\mathcal{B}} = (a_{ij})_{1\le i\le n, 1\le j\le n}$:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}.$$

Since $\mathcal{B}$ is orthonormal, the norm of Tv_1 is simply the norm of the first column of the matrix, which means $\|Tv_1\|^2 = |a_{11}|^2$. Since $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^*$, the first column of $[T^*]_{\mathcal{B}}^{\mathcal{B}}$ is the conjugate transpose of the first row of $[T]_{\mathcal{B}}^{\mathcal{B}}$. Therefore, we also have:

$$\|T^*v_1\|^2 = \sum_{k=1}^{n} |a_{1k}|^2 = |a_{11}|^2 + |a_{12}|^2 + \dots + |a_{1n}|^2.$$

As T is normal, **??** part **(a)** tells us that $\|Tv_1\| = \|T^*v_1\|$. This implies:

$$|a_{11}|^2 = |a_{11}|^2 + |a_{12}|^2 + \dots + |a_{1n}|^2,$$

hence $a_{12} = \dots = a_{1n} = 0$. This shows that the matrix now looks like:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & 0 & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}.$$

Now, consider v_2. We have $\|Tv_2\|^2 = |a_{22}|^2$ and $\|T^*v_2\|^2 = |a_{22}|^2 + |a_{23}|^2 + \dots + |a_{2n}|^2$. As $\|Tv_2\| = \|T^*v_2\|$, we similarly obtain $a_{23} = \dots = a_{2n} = 0$.

We can continue this process by induction, and show that for all $1 \le k < n$ we have $a_{k,k+1} = \dots = a_{k,n} = 0$. Hence, $[T]_{\mathcal{B}}^{\mathcal{B}}$ is diagonal. Since $\mathcal{B}$ is an orthonormal basis that diagonalizes T, it consists of eigenvectors, proving that T is orthogonally diagonalizable. **Q.E.D.**

Remark: In the proof above, we used $K = \mathbb{C}$ only at one place: when we stated that T is trigonalizable. Therefore, we immediately obtain the following corollary for the real case.

Corollary 24.a.6: Let V be a finite-dimensional inner product space over $\mathbb{R}$, and $T\colon V \to V$ an endomorphism which is normal. If T is trigonalizable (which is true if and only if the characteristic polynomial $P_T(x)$ factors as a product of linear factors over $\mathbb{R}$), then T is orthogonally diagonalizable.

The Spectral Theorem over $\mathbb{R}$

We've seen that over $\mathbb{R}$, the statement *"$T \in \mathrm{End}(V)$ is normal $\implies$ T is orthogonally diagonalizable"* does not hold in general. We need a stronger condition.

Lemma 24.a.7: Let V be a finite-dimensional inner product space over $\mathbb{R}$, and $T \in \mathrm{End}(V)$ which is orthogonally diagonalizable. Then $T^* = T$.

Proof

Let $\mathcal{B}$ be an orthonormal basis such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is a diagonal matrix. Since we are over $\mathbb{R}$, the conjugate transpose is simply the standard transpose. Then:

$$[T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^{\top} = [T]_{\mathcal{B}}^{\mathcal{B}}.$$

(The last equality holds because $[T]_{\mathcal{B}}^{\mathcal{B}}$ is diagonal). Because their representation matrices with respect to the same basis are equal, it follows that $T^* = T$. **Q.E.D.**

Definition 24.a.8: Let V be an inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$), and $T \in \operatorname{End}(V)$. T is called "*self adjoint*" if $T^* = T$.

Similarly, a matrix $A \in \mathcal{M}_{n\times n}(K)$ is called "*self adjoint*" if $A^* = A$. (Note that over $K = \mathbb{R}$, a self adjoint matrix is precisely a symmetric matrix, $A^\top = A$).

Exercise: Show that $T\colon V \to V$ is self adjoint if and only if for every orthonormal basis $\mathcal{B}$ of V, the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ is self adjoint.

Theorem 24.a.9 (*Spectral Theorem over* $\mathbb{R}$): Let V be a Euclidean space (i.e., an inner product space over $\mathbb{R}$) of finite dimension. Then T is orthogonally diagonalizable if and only if T is self adjoint.

For the proof, we need the following lemma:

Lemma 24.a.10: Let V be an inner product space over K (where $K = \mathbb{R}$ or $\mathbb{C}$), and let $T \in \operatorname{End}(V)$ be self adjoint. Then:

(a) **(a)** All eigenvalues of T are real.

(b) **(b)** The characteristic polynomial $P_T(x)$ factors into a product of linear factors over K.

Proof

Proof of (a): If $K = \mathbb{R}$, there is nothing to show. So assume $K = \mathbb{C}$. Let $\lambda \in \mathbb{C}$ be an eigenvalue and $v \in V$ a corresponding non-zero eigenvector. We've seen in **??** **(c)** that v is also an eigenvector of T^* with eigenvalue $\overline{\lambda}$. But $T^* = T$. This implies:

$$\lambda v = Tv = T^* v = \overline{\lambda} v.$$

Since $v \neq 0$, we must have $\lambda = \overline{\lambda}$, which implies $\lambda \in \mathbb{R}$.

Proof of (b): For $K = \mathbb{C}$, this follows immediately from the Fundamental Theorem of Algebra. So assume $K = \mathbb{R}$. Let $\mathcal{B}$ be an orthonormal basis for V and consider the matrix $A := [T]_{\mathcal{B}}^{\mathcal{B}} \in \mathcal{M}_{n\times n}(\mathbb{R})$. The statement would follow if we show that the characteristic polynomial $P_A(x)$ factors as a product of linear terms with real coefficients.

Now, $T^* = T$, hence $A = [T]_{\mathcal{B}}^{\mathcal{B}} = [T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^* = A^* = A^\top$. Consider $\mathbb{C}^n$ endowed with its standard inner product, and view A as a matrix in $\mathcal{M}_{n\times n}(\mathbb{C})$. Let $T_A\colon \mathbb{C}^n \to \mathbb{C}^n$ be the corresponding linear map. The standard basis $\mathcal{E}_n = (e_1, \dots, e_n)$ is orthonormal, and $[T_A]_{\mathcal{E}_n}^{\mathcal{E}_n} = A$.

We know that $P_{T_A}(x) = P_A(x)$. When viewed as a polynomial over $\mathbb{C}$ (i.e., in $\mathbb{C}[x]$), it factors as a product of linear terms:

$$P_A(x) = \prod_{k=1}^{n} (x - \lambda_k) \in \mathbb{C}[x], \quad (*) \tag{24.1}$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of T_A. But T_A is self adjoint (even when viewed over $\mathbb{C}$ since $A^\top = A = A^*$), so by part **(a)**, all the eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbb{R}$. Therefore, the factorization $(*)$ actually holds in $\mathbb{R}[x]$.

Q.E.D.

Proof of the Spectral Theorem over $\mathbb{R}$ (??)

The forward direction was proven in **??**. For the reverse direction, if T is self adjoint, then by **??**, its characteristic polynomial splits over $\mathbb{R}$. This implies T is trigonalizable over $\mathbb{R}$. Furthermore, since $T = T^*$, T trivially commutes with its adjoint, making it normal. By **??**, a normal map that is trigonalizable over $\mathbb{R}$ is orthogonally diagonalizable. **Q.E.D.**

Spectral Theorems for Matrices

Let $K = \mathbb{R}$ or $\mathbb{C}$. Recall that a matrix $A \in \mathcal{M}_{n\times n}(K)$ is called normal if $A \cdot A^* = A^* \cdot A$.

Example:

(a) For $K = \mathbb{R}$: Symmetric matrices and orthogonal matrices $\mathrm{O}(n)$ are normal.

(b) For $K = \mathbb{C}$: Self adjoint matrices and unitary matrices $\mathrm{U}(n)$ are normal.

Remark: There exist symmetric matrices in $\mathcal{M}_{n\times n}(\mathbb{C})$ that are not normal. For example, let $A = \left(\begin{smallmatrix} i & i \\ i & 1 \end{smallmatrix}\right)$. We have $A^\top = A$, so it is symmetric. However, $A^* = \left(\begin{smallmatrix} -i & -i \\ -i & 1 \end{smallmatrix}\right)$. A quick calculation shows that $A \cdot A^* \neq A^* \cdot A$, so A is not normal.

Theorem 24.a.11 (*Matrix Spectral Theorems*):

(a) If $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is normal, then there exists a unitary matrix $U \in \mathrm{U}(n)$ such that $U^{-1}AU$ is diagonal. (Note that $U^{-1} = U^*$).

(b) If $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is a symmetric matrix, then there exists an orthogonal matrix $O \in \mathrm{O}(n)$ such that $O^{-1}AO$ is diagonal. (Note that $O^{-1} = O^\top$).

Proof Sketch

The proof is left as an exercise, based on the following key points translating between maps and matrices:

(a) $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is normal if and only if $T_A \colon \mathbb{C}^n \to \mathbb{C}^n$ is normal (when we endow $\mathbb{C}^n$ with its standard inner product).

(b) If $\mathcal{B}$ is an orthonormal basis for $\mathbb{C}^n$, then the transition matrix $U := [\mathrm{id}_{\mathbb{C}^n}]^{\mathcal{B}}_{\mathcal{E}_n}$ is unitary.

(c) $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is symmetric if and only if $T_A \colon \mathbb{R}^n \to \mathbb{R}^n$ is self adjoint, when we endow $\mathbb{R}^n$ with its standard inner product.

(d) If $\mathcal{B}$ is an orthonormal basis for $\mathbb{R}^n$, then the transition matrix $O := [\mathrm{id}_{\mathbb{R}^n}]^{\mathcal{B}}_{\mathcal{E}_n}$ is orthogonal.

Q.E.D.

Exercise Solutions

Proof of Gram-Schmidt Preserving Upper Triangularity (on ??)

Let $\mathcal{C} = (u_1, \ldots, u_n)$ be a basis for V such that the representation matrix $[T]^{\mathcal{C}}_{\mathcal{C}}$ is upper triangular. This means that for any $1 \leq k \leq n$, the vector $T(u_k)$ is a linear combination of $u_1, \ldots, u_k$. In other words, $T(u_k) \in \mathrm{Sp}(u_1, \ldots, u_k)$. Consequently, the subspace $W_k := \mathrm{Sp}(u_1, \ldots, u_k)$ is T-invariant, meaning $T(W_k) \subseteq W_k$.

Applying the Gram-Schmidt orthogonalization process to the basis $\mathcal{C}$ produces an orthonormal basis $\mathcal{B} = (v_1, \ldots, v_n)$ with the property that for every $1 \leq k \leq n$, the spans are preserved: $\mathrm{Sp}(v_1, \ldots, v_k) = \mathrm{Sp}(u_1, \ldots, u_k) = W_k$.

Now, consider the matrix $[T]^{\mathcal{B}}_{\mathcal{B}}$. The k-th column of this matrix consists of the coordinates of $T(v_k)$ with respect to the basis $\mathcal{B}$. Since $v_k \in W_k$ and W_k is T-invariant, we must have $T(v_k) \in W_k = \mathrm{Sp}(v_1, \ldots, v_k)$. This means $T(v_k)$ can be written strictly as a linear combination of $v_1, \ldots, v_k$, meaning all coefficients for vectors v_j with $j > k$ are exactly zero. Therefore, $[T]^{\mathcal{B}}_{\mathcal{B}}$ is an upper triangular matrix. Q.E.D.

Proof of Matrix Self Adjointness Equivalence (on ??)

Let $T \in \mathrm{End}(V)$. First, suppose T is self adjoint, so $T = T^*$. Let $\mathcal{B}$ be an arbitrary orthonormal basis for V. By the properties of representation matrices for adjoint maps in orthonormal bases, we have $[T^*]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^*$. Since $T = T^*$, we substitute to get $[T]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^*$, which means the matrix $[T]^{\mathcal{B}}_{\mathcal{B}}$ is self adjoint.

Conversely, suppose that for any orthonormal basis $\mathcal{B}$ of V, the matrix $[T]^{\mathcal{B}}_{\mathcal{B}}$ is self adjoint, i.e., $[T]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^*$. Again, using the identity $[T^*]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^*$, we see that $[T]^{\mathcal{B}}_{\mathcal{B}} = [T^*]^{\mathcal{B}}_{\mathcal{B}}$. Because the map sending an endomorphism to its representation matrix with respect to a fixed basis is an isomorphism, this equality of matrices implies the equality of the maps, hence $T = T^*$, meaning T is self adjoint. **Q.E.D.**

Proof of ??

Proof of (a): Let $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ be a normal matrix. By point (a), the linear map $T_A\colon \mathbb{C}^n \to \mathbb{C}^n$ is a normal endomorphism with respect to the standard inner product. By the Spectral Theorem over $\mathbb{C}$ (**??**), T_A is orthogonally diagonalizable. Thus, there exists an orthonormal basis $\mathcal{B} = (v_1, \ldots, v_n)$ for $\mathbb{C}^n$ consisting of eigenvectors of T_A.

The representation matrix of T_A with respect to $\mathcal{B}$, which is $[T_A]^{\mathcal{B}}_{\mathcal{B}}$, is a diagonal matrix D. Let $\mathcal{E}_n$ be the standard basis for $\mathbb{C}^n$. The transition matrix from $\mathcal{B}$ to $\mathcal{E}_n$ is $U := [\mathrm{id}_{\mathbb{C}^n}]^{\mathcal{B}}_{\mathcal{E}_n}$, whose columns are precisely the vectors $v_1, \ldots, v_n$. Since $\mathcal{B}$ is an orthonormal basis, point (b) guarantees that U is a unitary matrix.

Using the change of basis formula, we have:

$$D = [T_A]^{\mathcal{B}}_{\mathcal{B}} = [\mathrm{id}_{\mathbb{C}^n}]^{\mathcal{E}_n}_{\mathcal{B}} \cdot [T_A]^{\mathcal{E}_n}_{\mathcal{E}_n} \cdot [\mathrm{id}_{\mathbb{C}^n}]^{\mathcal{B}}_{\mathcal{E}_n} = U^{-1} \cdot A \cdot U.$$

Thus, $U^{-1}AU$ is diagonal.

Proof of (b): Let $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ be a symmetric matrix. By point (c), the linear map $T_A\colon \mathbb{R}^n \to \mathbb{R}^n$ is self adjoint. By the Spectral Theorem over $\mathbb{R}$ (**??**), T_A is orthogonally diagonalizable. Thus, there exists an orthonormal basis $\mathcal{B}$ for $\mathbb{R}^n$ consisting of eigenvectors of T_A, making $[T_A]^{\mathcal{B}}_{\mathcal{B}}$ a diagonal matrix D.

Let $O := [\mathrm{id}_{\mathbb{R}^n}]^{\mathcal{B}}_{\mathcal{E}_n}$ be the transition matrix. Since $\mathcal{B}$ is an orthonormal basis in $\mathbb{R}^n$, point (d) guarantees that O is an orthogonal matrix. By the change of basis formula:

$$D = [T_A]^{\mathcal{B}}_{\mathcal{B}} = [\mathrm{id}_{\mathbb{R}^n}]^{\mathcal{E}_n}_{\mathcal{B}} \cdot [T_A]^{\mathcal{E}_n}_{\mathcal{E}_n} \cdot [\mathrm{id}_{\mathbb{R}^n}]^{\mathcal{B}}_{\mathcal{E}_n} = O^{-1} \cdot A \cdot O.$$

Thus, $O^{-1}AO$ is diagonal.

Q.E.D.

Orthogonal and Unitary Maps / Isometries (Chapter 25)

Linear Isometries (Section 25.a)

Definition 25.a.1 (*Linear Isometry*): Let V, W be inner product spaces over K (where $K = \mathbb{R}$ or $\mathbb{C}$). A map $T\colon V \to W$ is called a "*linear isometry*" if T is linear and for all $v_1, v_2 \in V$,

$$\langle Tv_1, Tv_2 \rangle_W = \langle v_1, v_2 \rangle_V .$$

(In other words, T preserves the inner product).

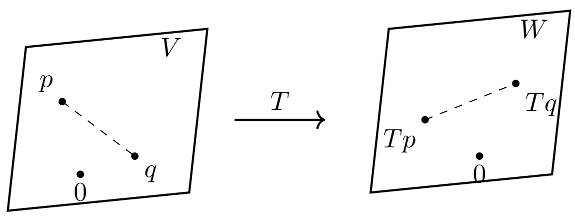
Remark: If $T\colon V \to W$ is a linear isometry, then:

(a) For all $v \in V$, $\|Tv\| = \|v\|$.

(b) For all $p, q \in V$, the distance is preserved:

$$\operatorname{dist}_W(Tp, Tq) = \|Tp - Tq\|_W = \|p - q\|_V = \operatorname{dist}_V(p, q).$$

(c) T is injective. Indeed, if $v \in \ker T$, then $0 = \|Tv\| = \|v\|$, which implies $v = 0$. So, $\ker T = \{0\}$.



From now on, we will concentrate on the case where $W = V$, meaning $T \in \operatorname{End}(V)$. If such a map T is an isometry, we also call it an "*orthogonal endomorphism*" if $K = \mathbb{R}$, or a "*unitary endomorphism*" if $K = \mathbb{C}$.

Theorem 25.a.2 (*Characterization of Isometries*): Let V be a finite-dimensional inner product space over $\mathbb{R}$ or $\mathbb{C}$, and let $T \in \operatorname{End}(V)$. The following statements are equivalent:

(a) T is an isometry.

(b) For all $v \in V$, $\|Tv\| = \|v\|$.

(c) For every orthonormal system $e_1, \ldots, e_k$ of V, the list of vectors $Te_1, \ldots, Te_k$ is also an orthonormal system.

(d) There exists an orthonormal basis $\mathcal{B} = (e_1, \ldots, e_n)$ of V such that $Te_1, \ldots, Te_n$ is also an orthonormal basis.

(e) $T \circ T^* = \operatorname{id}_V$.

(f) $T^* \circ T = \operatorname{id}_V$.

(g) $T^* = T^{-1}$.

(h) T^* is an isometry.

Proof

We will prove some of the equivalences here; the rest are left as exercises (and for the exercise sheet).

(b) $\implies$ (a): Using the polarization identity (for the real case), we have:

$$\begin{aligned}\langle Tv, Tw\rangle &= \frac{1}{2}\left(\|Tv+Tw\|^2 - \|Tv\|^2 - \|Tw\|^2\right)\\ &= \frac{1}{2}\left(\|T(v+w)\|^2 - \|Tv\|^2 - \|Tw\|^2\right)\\ &= \frac{1}{2}\left(\|v+w\|^2 - \|v\|^2 - \|w\|^2\right)\\ &= \langle v, w\rangle.\end{aligned}$$

(a) $\implies$ (c): Let $e_1, \dots, e_k$ be an orthonormal system. Then, for all $1 \le i, j \le k$, we have:

$$\langle Te_i, Te_j\rangle = \langle e_i, e_j\rangle = \delta_{ij}.$$

This implies that $Te_1, \dots, Te_k$ is also an orthonormal system.

AI-Note: In the handwritten notes, the following proof is labeled as **(d) $\implies$ (e)**. However, it actually proves **(d) $\implies$ (f)**, which is $T^* \circ T = \mathrm{id}_V$. We have corrected the labeling here.

(d) $\implies$ (f): Let $\mathcal{B} = (e_1, \dots, e_n)$ be an orthonormal basis such that $Te_1, \dots, Te_n$ is also an orthonormal system. Let $1 \le j \le n$. We have, for all $1 \le i \le n$:

$$\langle e_i, e_j\rangle = \delta_{ij} = \langle Te_i, Te_j\rangle = \langle e_i, T^*Te_j\rangle. \quad (*) \tag{25.1}$$

Since $(e_1, \dots, e_n)$ is a basis for V, the equality $(*)$ implies that by linearity in the first argument,

$$\langle v, e_j\rangle = \langle v, T^*Te_j\rangle \quad \forall v \in V.$$

This implies that $T^*Te_j = e_j$. This last equality holds for all $1 \le j \le n$, which implies that $T^* \circ T = \mathrm{id}_V$. **Q.E.D.**

Orthogonal and Unitary Matrices (Section 25.b)

Back to matrices. Recall that:

- $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ is orthogonal if $A^TA = I \iff AA^T = I \iff A^{-1} = A^T$. The set of such matrices is $\mathrm{O}(n)$.
- $A \in \mathcal{M}_{n\times n}(\mathbb{C})$ is unitary if $A^*A = I \iff AA^* = I \iff A^{-1} = A^*$. The set of such matrices is $\mathrm{U}(n)$.

Proposition 25.a.1: Let V be an n-dimensional inner product space over $\mathbb{R}$ (resp. $\mathbb{C}$). Let $\mathcal{B}$ be an orthonormal basis for V, and let $T \in \mathrm{End}(V)$. Then, T is orthogonal (resp. unitary) if and only if $[T]_{\mathcal{B}}^{\mathcal{B}} \in \mathrm{O}(n)$ (resp. $[T]_{\mathcal{B}}^{\mathcal{B}} \in \mathrm{U}(n)$).

Proof

This is left as an exercise. **Q.E.D.**

Note that for all $A \in \mathrm{O}(n)$, we have $\det(A) = \pm 1$. Indeed:

$$1 = \det(I) = \det(AA^T) = \det(A)\cdot\det(A^T) = \det(A)\cdot\det(A) = \det(A)^2.$$

Similarly, for all $A \in \mathrm{U}(n)$, we have $|\det(A)| = 1$. Indeed:

$$1 = \det(AA^*) = \det(A)\cdot\det(\overline{A}^T) = \det(A)\cdot\det(\overline{A}) = \det(A)\cdot\overline{\det(A)} = |\det(A)|^2.$$

(Note that $\det(A) \in \mathbb{C}$).

Definition 25.a.2 (*Special Orthogonal and Unitary Groups*): We define the following subsets:

$$\begin{aligned}\mathrm{SO}(n) &:= \{A \in \mathrm{O}(n) \mid \det(A) = 1\} \subseteq \mathrm{O}(n) \subseteq \mathrm{GL}_n(\mathbb{R}),\\ \mathrm{SU}(n) &:= \{A \in \mathrm{U}(n) \mid \det(A) = 1\} \subseteq \mathrm{U}(n) \subseteq \mathrm{GL}_n(\mathbb{C}).\end{aligned}$$

Proposition 25.a.3 (*Group Properties*): Let G be any of the sets $\mathrm{O}(n), \mathrm{SO}(n), \mathrm{U}(n)$, or $\mathrm{SU}(n)$. Then:

(a) $I_n \in G$.

(b) For all $A, B \in G$, $A \cdot B \in G$.

(c) For all $A \in G$, $A^{-1} \in G$.

Proof

This is left as an exercise. **Q.E.D.**

Classification of the Elements of O(2), O(3), etc. (Section 25.c)

Lemma 25.a.1: Let $A \in \mathrm{O}(n)$. If λ is a real eigenvalue of A, then $\lambda = \pm 1$.

Proof

Suppose $Av = \lambda v$, where $0 \neq v \in \mathbb{R}^n$. Then:

$$\langle v, v\rangle = \langle Av, Av\rangle = \langle \lambda v, \lambda v\rangle = \lambda^2 \langle v, v\rangle.$$

As $v \neq 0$, we have $\langle v, v\rangle \neq 0$, which implies $\lambda^2 = 1$, and therefore $\lambda = \pm 1$. **Q.E.D.**

Proposition 25.a.2 (*Classification of* $\mathrm{O}(2)$): Let $A \in \mathrm{O}(2)$.

(a) If $\det A = 1$, then there exists $\theta \in [0, 2\pi)$ such that

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} := R_\theta.$$

The map $T_A \colon \mathbb{R}^2 \to \mathbb{R}^2$ is an anti-clockwise rotation by an angle θ.

(b) If $\det A = -1$, then there exists a basis $\mathcal{B} = (v_1, v_2)$ for $\mathbb{R}^2$ such that $T_A \colon \mathbb{R}^2 \to \mathbb{R}^2$ has the following representation matrix in the basis $\mathcal{B}$:

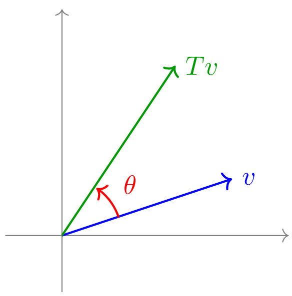
$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{25.2}$$

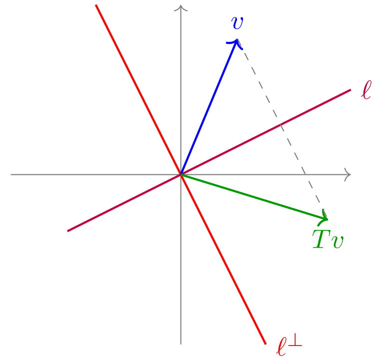
In other words, T_A performs a reflection with respect to the line $\ell := \mathrm{Sp}(v_1)$. Moreover, there exists $\theta \in [0, 2\pi)$ such that

$$A = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix} = R_\theta \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The basis $\mathcal{B}$ for which (**??**) holds is given by

$$v_1 = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) \end{pmatrix}, \quad v_2 = \begin{pmatrix} \cos\left(\frac{\theta}{2} + \frac{\pi}{2}\right) \\ \sin\left(\frac{\theta}{2} + \frac{\pi}{2}\right) \end{pmatrix}.$$





Proof

Note that if $v \in \mathbb{R}^2$ has $\|v\| = 1$, then there exists $\theta \in [0, 2\pi)$ such that $v = \left(\begin{smallmatrix}\cos\theta \\ \sin\theta\end{smallmatrix}\right)$.

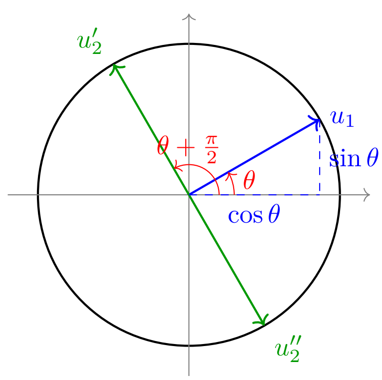
Now, let $A = \begin{pmatrix} | & | \\ u_1 & u_2 \\ | & | \end{pmatrix} \in \mathrm{O}(2)$. Since A is orthogonal, its columns form an orthonormal basis. Thus, $\|u_1\| = 1$, which implies $u_1 = \left(\begin{smallmatrix}\cos\theta \\ \sin\theta\end{smallmatrix}\right)$ for some $\theta \in [0, 2\pi)$. We also have $u_1 \perp u_2$ and $\|u_2\| = 1$.

The orthogonal complement to u_1 is:

$$\begin{pmatrix}\cos\theta \\ \sin\theta\end{pmatrix}^\perp = \mathrm{Sp}\begin{pmatrix}\cos(\theta + \frac{\pi}{2}) \\ \sin(\theta + \frac{\pi}{2})\end{pmatrix} = \mathrm{Sp}\begin{pmatrix}-\sin\theta \\ \cos\theta\end{pmatrix}.$$

Within the space $\mathrm{Sp}\left(\begin{smallmatrix}-\sin\theta \\ \cos\theta\end{smallmatrix}\right)$, there are precisely two elements with norm 1, namely:

$$u_2' = \begin{pmatrix}-\sin\theta \\ \cos\theta\end{pmatrix}, \quad u_2'' = \begin{pmatrix}\sin\theta \\ -\cos\theta\end{pmatrix}.$$



If $u_2 = u_2'$, then $A = \begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix} = R_\theta$, and $\det A = 1$.

If $u_2 = u_2''$, then $A = \begin{pmatrix}\cos\theta & \sin\theta \\ \sin\theta & -\cos\theta\end{pmatrix} = R_\theta \cdot \begin{pmatrix}1 & 0 \\ 0 & -1\end{pmatrix}$, and $\det A = -1$.

In the second case, a direct calculation shows that $v_1 = \left(\begin{smallmatrix}\cos(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2})\end{smallmatrix}\right)$ and $v_2 = \left(\begin{smallmatrix}\cos(\frac{\theta}{2}+\frac{\pi}{2}) \\ \sin(\frac{\theta}{2}+\frac{\pi}{2})\end{smallmatrix}\right)$ are eigenvectors of A for the eigenvalues $\lambda_1 = 1$ and $\lambda_2 = -1$, respectively. **Q.E.D.**

Summary: $\mathrm{SO}(2) =$ rotations.
$\mathrm{O}(2) \setminus \mathrm{SO}(2) =$ reflections.

Corollary 25.a.3: Let V be a 2-dimensional Euclidean space and $T\colon V \to V$ a linear isometry. Then $\det(T) = \pm 1$. Moreover, there exists an orthonormal basis $\mathcal{B}$ for V such that:

(a) If $\det(T) = 1$, then $[T]^{\mathcal{B}}_{\mathcal{B}} = R_\theta$.

(b) If $\det(T) = -1$, then $[T]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Proposition 25.a.4 (*Classification of* $\mathrm{SO}(3)$): Let $A \in \mathrm{SO}(3)$. Then 1 is an eigenvalue of A. Moreover, there exists an orthonormal basis $\mathcal{B} = (v_1, v_2, v_3)$ and $\theta \in [0, 2\pi)$ such that

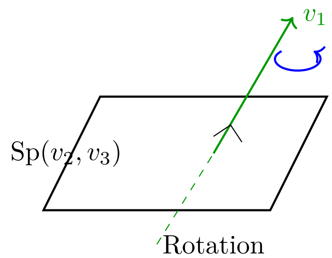
$$[T_A]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & R_\theta \end{pmatrix}.$$

So, T_A is a rotation with angle θ around the axis $\mathrm{Sp}(v_1)$. Moreover:

(a) If -1 is an eigenvalue of A, then $\theta = \pi$, hence

$$[T_A]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

(b) If -1 is <u>not</u> an eigenvalue of A (which is true if and only if 1 is the only real eigenvalue), then $\theta \neq \pi$.



Proof

The characteristic polynomial $P_A(x)$ of A is a polynomial of degree 3 with real coefficients. By the Intermediate Value Theorem from analysis, any odd-degree polynomial with real coefficients has at least one real zero $\lambda \in \mathbb{R}$ of $P_A(x)$ (i.e., $P_A(\lambda) = 0$). This implies that A has at least one real eigenvalue. By a previous theorem (**??**), $\lambda = \pm 1$.

Let $v_1 \in \mathbb{R}^3$ be an eigenvector of λ. By normalizing it if necessary, we may assume $\|v_1\| = 1$. Consider the subspace

$$L := v_1^\perp = \{v \in \mathbb{R}^3 \mid v \perp v_1\}.$$

Since A is orthogonal, we have for all $v \in L$, $A \cdot v \in L$. (i.e., $T_A(L) \subseteq L$, meaning L is invariant under T_A). Indeed, if $v \in L$, this implies $0 = \langle v, v_1 \rangle$. Because $A \in \mathrm{O}(3)$, it preserves inner products:

$$0 = \langle v, v_1 \rangle = \langle Av, Av_1 \rangle = \langle Av, \pm v_1 \rangle = \pm \langle Av, v_1 \rangle \implies Av \in L.$$

We also have $\mathbb{R}^3 = \mathrm{Sp}(v_1) \oplus L$, hence $\dim L = 2$. Now, $T_A|_L\colon L \to L$ is a linear isometry. Let $\mathcal{L} = (w_1, w_2)$ be an orthonormal basis for L. By a previous result, the representation matrix is $Q := [T_A|_L]^{\mathcal{L}}_{\mathcal{L}} \in \mathrm{O}(2)$.

Case (a): $\lambda = 1$. Take the basis $\mathcal{B} := (v_1, w_1, w_2)$. Then

$$[T_A]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \multicolumn{2}{c}{\multirow{2}{*}{Q}} \\ 0 & & \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & Q \end{pmatrix}.$$

Now, $1 = \det A = \det(T_A) = 1 \cdot \det(Q)$, which implies $\det Q = 1$. So, $Q \in \mathrm{SO}(2)$. By the previous proposition, there exists $\theta \in [0, 2\pi)$ such that $Q = R_\theta$. Thus,

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & R_\theta \end{pmatrix}.$$

Case (b): If $\lambda = -1$. By the same argument as for case (a), we get that $1 = \det A = (-1) \cdot \det(T_A|_L)$, which implies $\det(T_A|_L) = -1$. By **??**, there exists an orthonormal basis $\mathcal{L} = (w_1, w_2)$ for L such that $[T_A|_L]_{\mathcal{L}}^{\mathcal{L}} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. Take $\mathcal{B}' = (v_1, w_1, w_2)$. Then

$$[T_A]_{\mathcal{B}'}^{\mathcal{B}'} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

And for the reordered basis $\mathcal{B} = (w_1, v_1, w_2)$, we have

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & R_\pi \end{pmatrix}.$$

Note that in case (b) too, 1 is an eigenvalue of A (corresponding to the eigenvector w_1).

Q.E.D.

Proposition 25.a.5 (*Classification of* $\mathrm{O}(3) \setminus \mathrm{SO}(3)$): Let $A \in \mathrm{O}(3) \setminus \mathrm{SO}(3)$. Then there exists an orthonormal basis $\mathcal{B} = (v_1, v_2, v_3)$ for $\mathbb{R}^3$ such that

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & R_\theta \end{pmatrix}$$

for some $\theta \in [0, 2\pi)$. (So T_A does a rotation by θ around $\mathrm{Sp}(v_1)$, followed by a reflection with respect to $\mathrm{Sp}(v_2, v_3)$).

Proof

Let $A' := -A = (-I) \cdot A$. Then $A' \in \mathrm{O}(3)$ and $\det A' = (-1)^3 \cdot \det A = (-1) \cdot (-1) = 1$. This implies $A' \in \mathrm{SO}(3)$. By the previous proposition, there exists an orthonormal basis $\mathcal{B}$ for $\mathbb{R}^3$, and $\theta' \in [0, 2\pi)$, such that $[T_{A'}]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & R_{\theta'} \end{smallmatrix}\right)$. This implies:

$$\begin{aligned} [T_A]_{\mathcal{B}}^{\mathcal{B}} &= [-\mathrm{id}_{\mathbb{R}^3}]_{\mathcal{B}}^{\mathcal{B}} \cdot [T_{A'}]_{\mathcal{B}}^{\mathcal{B}} \\ &= -I \cdot \begin{pmatrix} 1 & 0 \\ 0 & R_{\theta'} \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ 0 & -R_{\theta'} \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ 0 & R_\theta \end{pmatrix}, \end{aligned}$$

where $\theta = (\theta' + \pi) \pmod{2\pi}$. (Note that $-R_{\theta'} = R_\pi \circ R_{\theta'} = R_{\theta' + \pi}$).

Q.E.D.

Exercise Solutions

Proof of ?? (d) $\implies$ (e)

Assume there exists an orthonormal basis $\mathcal{B} = (e_1, \ldots, e_n)$ of V such that $Te_1, \ldots, Te_n$ is also an orthonormal basis. Since it is an orthonormal basis, we have $\langle Te_i, Te_j \rangle = \delta_{ij}$. By the definition of the adjoint map, we can rewrite the inner product as:

$$\delta_{ij} = \langle Te_i, Te_j \rangle = \langle e_i, T^*Te_j \rangle.$$

Because the vectors e_i form a basis, the fact that the inner product of T^*Te_j with every basis vector e_i yields δ_{ij} (which are precisely the coordinates of e_j) implies that $T^*Te_j = e_j$ for all $1 \leq j \leq n$. Since the linear map $T^* \circ T$ acts as the identity on a basis, it must be the identity map on the entire space V. Thus, $T^* \circ T = \mathrm{id}_V$. (This proves **(d)** $\implies$ **(f)**). Since V is finite-dimensional, a left inverse is also a right inverse, so $T \circ T^* = \mathrm{id}_V$, proving **(e)**. **Q.E.D.**

Proof of ??

Let $\mathcal{B} = (e_1, \ldots, e_n)$ be an orthonormal basis for V. First, suppose T is an orthogonal (resp. unitary) endomorphism. By **??**, $T \circ T^* = \mathrm{id}_V$ and $T^* \circ T = \mathrm{id}_V$. Taking the representation matrix with respect to $\mathcal{B}$, we have:

$$[T^* \circ T]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}} = I_n.$$

By the properties of representation matrices, this splits into $[T^*]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} = I_n$. Since $\mathcal{B}$ is an orthonormal basis, $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^*$. Thus, $\left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^* \cdot [T]_{\mathcal{B}}^{\mathcal{B}} = I_n$, which means the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ is orthogonal (resp. unitary).

Conversely, assume that $A := [T]_{\mathcal{B}}^{\mathcal{B}}$ is orthogonal (resp. unitary). This means $A^*A = I_n$. Because $\mathcal{B}$ is an orthonormal basis, $A^* = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^* = [T^*]_{\mathcal{B}}^{\mathcal{B}}$. Substituting this back yields $[T^*]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} = I_n$, and therefore $[T^* \circ T]_{\mathcal{B}}^{\mathcal{B}} = [\mathrm{id}_V]_{\mathcal{B}}^{\mathcal{B}}$. This implies the maps are equal, so $T^* \circ T = \mathrm{id}_V$. By **??**, this guarantees T is an isometry. **Q.E.D.**

Proof of ??

We will prove this generically for $G = \mathrm{U}(n)$ and $G = \mathrm{SU}(n)$ (the real cases are identical, replacing the conjugate transpose * with the transpose $^\top$).

Proof of (a): The identity matrix I_n satisfies $I_n^* \cdot I_n = I_n \cdot I_n = I_n$, so $I_n \in \mathrm{U}(n)$. Furthermore, $\det(I_n) = 1$, so $I_n \in \mathrm{SU}(n)$.

Proof of (b): Let $A, B \in \mathrm{U}(n)$. We evaluate $(AB)^*(AB) = (B^*A^*)(AB) = B^*(A^*A)B$. Since $A \in \mathrm{U}(n)$, $A^*A = I_n$, so this reduces to $B^*I_nB = B^*B$. Since $B \in \mathrm{U}(n)$, $B^*B = I_n$. Thus, $AB \in \mathrm{U}(n)$. If additionally $A, B \in \mathrm{SU}(n)$, then $\det(AB) = \det(A)\det(B) = 1 \cdot 1 = 1$, so $AB \in \mathrm{SU}(n)$.

Proof of (c): Let $A \in \mathrm{U}(n)$. This implies $A^*A = I_n$, meaning $A^{-1} = A^*$. We check if A^{-1} is unitary: $(A^{-1})^*A^{-1} = (A^*)^*A^* = AA^* = I_n$. Thus, $A^{-1} \in \mathrm{U}(n)$. If additionally $A \in \mathrm{SU}(n)$, then $\det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{1} = 1$, so $A^{-1} \in \mathrm{SU}(n)$.

Q.E.D.

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Singular Value Decomposition (Chapter 26)

Theorem 26.a.1 (*Singular Value Decomposition*): Let $K=\mathbb{R}$ or $\mathbb{C}$, and let $A\in\mathcal{M}_{m\times n}(K)$. Then there exist $P\in \mathrm{O}(m)$ (if $K=\mathbb{R}$) or $P\in \mathrm{U}(m)$ (if $K=\mathbb{C}$), and $Q\in\mathrm{O}(n)$ (if $K=\mathbb{R}$) or $Q\in\mathrm{U}(n)$ (if $K=\mathbb{C}$), and a *"diagonal"* matrix

$$D=\left(\begin{array}{ccc}\lambda_1 & & 0\\ & \ddots & \\ 0 & & \lambda_k\\ \hline & 0\dots 0 & \end{array}\right)\in\mathcal{M}_{m\times n}(\mathbb{R})$$

with $\lambda_1,\dots,\lambda_k>0$, where $k=\operatorname{rank}A$, such that $A=PDQ^{-1}$.

Remark: Recall that we previously defined the equivalence of matrices. Here, the factorization $A=PDQ^{-1}$ demonstrates that we have an orthogonal (or unitary) equivalence.

Proof

We will prove only a special case: we assume $\operatorname{rank}A=n$ (which is true if and only if T_A is injective. Hence, we also assume $n\le m$).

Define $B:=A^*\cdot A\in\mathcal{M}_{n\times n}(K)$. We claim that the linear map $T_B\colon K^n\to K^n$ is injective. Indeed, for all $0\neq v\in K^n$, we have:

$$\langle T_Bv,v\rangle=\langle Bv,v\rangle=\langle A^*Av,v\rangle=\langle Av,Av\rangle>0. \tag{26.1}$$

This implies that $T_Bv\neq 0$. Therefore, $\ker T_B=\{0\}$, hence T_B is injective.

Note also that if $\eta\in K$ is an eigenvalue of B, then $\eta\in\mathbb{R}$ (even if $K=\mathbb{C}$) and $\eta>0$. Indeed, B is self adjoint ($B^*=(A^*A)^*=A^*(A^*)^*=A^*A=B$), hence $\eta\in\mathbb{R}$. To see that $\eta>0$, let v be an eigenvector of η. From (**??**), we have:

$$\langle Bv,v\rangle>0\implies\langle\eta v,v\rangle>0\implies\eta\langle v,v\rangle>0\implies\eta>0.$$

By a previous theorem, B is orthogonally diagonalizable, hence there exists $Q\in\mathrm{O}(n)$ (or $\mathrm{U}(n)$) such that

$$Q^{-1}A^*AQ=\begin{pmatrix}\eta_1 & & 0\\ & \ddots & \\ 0 & & \eta_n\end{pmatrix},$$

where $\eta_j\in\mathbb{R}$, and $\eta_j>0$ for all j.

Let $\lambda_j:=+\sqrt{\eta_j}>0$ for $j=1,\dots,n$. Define

$$C:=AQ\cdot\begin{pmatrix}\lambda_1^{-1} & & 0\\ & \ddots & \\ 0 & & \lambda_n^{-1}\end{pmatrix}\in\mathcal{M}_{m\times n}(K).$$

We have:

$$C^*C=\begin{pmatrix}\lambda_1^{-1} & & 0\\ & \ddots & \\ 0 & & \lambda_n^{-1}\end{pmatrix}\underbrace{Q^{-1}A^*AQ}_{=\begin{pmatrix}\eta_1 & & 0\\ & \ddots & \\ 0 & & \eta_n\end{pmatrix}}\begin{pmatrix}\lambda_1^{-1} & & 0\\ & \ddots & \\ 0 & & \lambda_n^{-1}\end{pmatrix}$$

$$=I_n.$$

It follows that the columns of C form an orthonormal system of vectors in K^m. To see this explicitly, let $c_1, \dots, c_n$ be the columns of C. The identity $C^*C = I_n$ means:

$$C^*C = \begin{pmatrix} - & \overline{c_1}^\top & - \\ & \vdots & \\ - & \overline{c_n}^\top & - \end{pmatrix} \begin{pmatrix} | & & | \\ c_1 & \dots & c_n \\ | & & | \end{pmatrix} = I_n.$$

Extend $c_1, \dots, c_n$ to an orthonormal basis $c_1, \dots, c_n, f_{n+1}, \dots, f_m$ of K^m and define:

$$P := \begin{pmatrix} | & & | & | & & | \\ c_1 & \dots & c_n & f_{n+1} & \dots & f_m \\ | & & | & | & & | \end{pmatrix} \in \mathcal{M}_{m \times m}(K).$$

Clearly, $P \in \mathrm{O}(m)$ (or $\mathrm{U}(m)$).

We have:

$$AQ \begin{pmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix} = C = P \cdot \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \\ \hline & 0 & \end{pmatrix} \begin{array}{l} \left.\vphantom{\begin{matrix} 1 \\ \ddots \\ 1 \end{matrix}}\right\} n \\ \left.\vphantom{0}\right\} m-n \end{array}$$

Multiplying from the right by the diagonal matrix of λ_j's and then by Q^{-1}, we obtain:

$$A = P \cdot \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \\ \hline & 0 & \end{pmatrix} \cdot Q^{-1}.$$

This proves the theorem (under the assumption that $\operatorname{rank} A = n$).

For the general case, see the script, chapter 7.5. Q.E.D.

Bilinear and Quadratic Forms (Chapter 27)

Definition 27.a.1 (*Bilinear Form*): Let V be a vector space over K (a general field). A function $B\colon V\times V\to K$ is called a "*bilinear form*" if the following two conditions are satisfied:

(a) For all $w\in V$, the map $B(-,w)\colon V\to K$, $v\mapsto B(v,w)$ is linear.

(b) For all $v\in V$, the map $B(v,-)\colon V\to K$, $w\mapsto B(v,w)$ is linear.

Lemma 27.a.2 (*Matrix Representation of a Bilinear Form*): Let V be a vector space over K and $\mathcal{C}=(e_1,\ldots,e_n)$ a basis for V. Let $B\colon V\times V\to K$ be a bilinear form. Then there exists a matrix $[B]_{\mathcal{C}}\in\mathcal{M}_{n\times n}(K)$ such that

$$B\left(\sum_{i=1}^n c_ie_i,\sum_{j=1}^n d_je_j\right)=(c_1,\ldots,c_n)\cdot[B]_{\mathcal{C}}\cdot(d_1,\ldots,d_n)^\top\quad\forall c_i,d_j\in K.$$

The matrix $[B]_{\mathcal{C}}$ is called the "*matrix representation*" of B in the basis $\mathcal{C}$.

Proof

Define $[B]_{\mathcal{C}}=(a_{ij})$, where $a_{ij}:=B(e_i,e_j)$. Then we compute:

$$\begin{aligned}B\left(\sum_{i=1}^n c_ie_i,\sum_{j=1}^n d_je_j\right)&=\sum_{i=1}^n c_iB\left(e_i,\sum_{j=1}^n d_je_j\right)\\&=\sum_{i=1}^n c_i\left(\sum_{j=1}^n d_jB(e_i,e_j)\right)\\&=\sum_{i=1}^n\sum_{j=1}^n c_id_jB(e_i,e_j)\\&=(c_1,\ldots,c_n)\cdot[B]_{\mathcal{C}}\cdot\begin{pmatrix}d_1\\\vdots\\d_n\end{pmatrix}.\end{aligned}$$

Q.E.D.

We saw that also for $T\in\operatorname{End}(V)$ and $\mathcal{C}$ a basis for V, we have a matrix representation $[T]_{\mathcal{C}}^{\mathcal{C}}\in\mathcal{M}_{n\times n}(K)$, and that if $\mathcal{D}$ is another basis for V, then

$$[T]_{\mathcal{D}}^{\mathcal{D}}=[\mathrm{id}_V]_{\mathcal{D}}^{\mathcal{C}}\cdot[T]_{\mathcal{C}}^{\mathcal{C}}\cdot[\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}},\tag{27.1}$$

where $[\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}}=([\mathrm{id}_V]_{\mathcal{D}}^{\mathcal{C}})^{-1}$ is the transition matrix from basis $\mathcal{D}$ to $\mathcal{C}$.

Question: What happens for bilinear forms?

Lemma 27.a.3 (*Change of Basis for Bilinear Forms*): Let V be a finite-dimensional vector space over K and $\mathcal{C},\mathcal{D}$ two bases of V. Let $B\colon V\times V\to K$ be a bilinear form. Then

$$[B]_{\mathcal{D}}=\left([\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}}\right)^\top\cdot[B]_{\mathcal{C}}\cdot[\mathrm{id}_V]_{\mathcal{C}}^{\mathcal{D}}.$$

(Note that this is quite different from (**??**)).

Proof

Let $\mathcal{C} = (v_1, \dots, v_n)$ and $\mathcal{D} = (w_1, \dots, w_n)$. We have $([B]_\mathcal{C})_{j,k} = B(v_j, v_k)$, and $([B]_\mathcal{D})_{j,k} = B(w_j, w_k)$. Recall that column j in $[\mathrm{id}_V]_\mathcal{C}^\mathcal{D}$ contains the coordinates of w_j in the basis $\mathcal{C}$:

$$w_j = \sum_{p=1}^{n} \left([\mathrm{id}_V]_\mathcal{C}^\mathcal{D}\right)_{p,j} \cdot v_p.$$

This implies:

$$\begin{aligned}
([B]_\mathcal{D})_{j,k} &= B(w_j, w_k) \\
&= B\left(\sum_{p=1}^{n} \left([\mathrm{id}_V]_\mathcal{C}^\mathcal{D}\right)_{p,j} v_p, \sum_{q=1}^{n} \left([\mathrm{id}_V]_\mathcal{C}^\mathcal{D}\right)_{q,k} v_q\right) \\
&= \sum_{p,q=1}^{n} \left([\mathrm{id}_V]_\mathcal{C}^\mathcal{D}\right)_{p,j} \cdot B(v_p, v_q) \cdot \left([\mathrm{id}_V]_\mathcal{C}^\mathcal{D}\right)_{q,k} \\
&= \left(\left([\mathrm{id}_V]_\mathcal{C}^\mathcal{D}\right)^\top \cdot [B]_\mathcal{C} \cdot [\mathrm{id}_V]_\mathcal{C}^\mathcal{D}\right)_{j,k}.
\end{aligned}$$

Q.E.D.

Definition 27.a.4 (*Symmetric, Positive Definite, and Quadratic Forms*):

(a) A bilinear form $B\colon V \times V \to K$ is called "*symmetric*" if $B(v,w) = B(w,v)$ for all $v, w \in V$.

(b) Assume $K = \mathbb{R}$. A bilinear form $B\colon V \times V \to \mathbb{R}$ is called "*positive definite*" if $B(v,v) > 0$ for all $0 \neq v \in V$.

(c) For a bilinear form $B\colon V \times V \to K$, we define $q_B\colon V \to K$ by $q_B(v) := B(v,v)$. We call q_B the "*quadratic form*" associated to B. Sometimes we just denote it by q.

Example:

(a) Let $(V, \langle \cdot, \cdot \rangle)$ be a Euclidean vector space. Then $B(v,w) := \langle v, w \rangle$ is a symmetric, positive definite bilinear form. Its associated quadratic form is $q(v) = \|v\|^2$.

(b) Let $B \in \mathcal{M}_{n\times n}(K)$. Then $B\colon K^n \times K^n \to K$ defined by $B(v,w) := v^\top B w$ is a bilinear form. If $B^\top = B$ (i.e., B is a symmetric matrix), then B is a symmetric bilinear form (left as an exercise). The associated quadratic form is $q(v) = v^\top B v$.

Remark: Note that $B \in \mathcal{M}_{n\times n}(K)$ and $B^\top \in \mathcal{M}_{n\times n}(K)$ give the same quadratic form $q\colon K^n \to K$. Indeed, for all $v \in K^n$, we have:

$$q_B(v) = v^\top B v = (v^\top B v)^\top = v^\top B^\top v = q_{B^\top}(v),$$

where we used the fact that $v^\top B v$ is a scalar, and thus equals its own transpose.

Assume now that $2 \neq 0$ in K (i.e., $1 + 1 \neq 0$). Then $\frac{1}{2}(B + B^\top)$ also gives the same quadratic form as B.

Conclusion: If $2 \neq 0$ in K, then for the discussion on quadratic forms $q\colon K^n \to K$, we can assume they all come from symmetric matrices (or symmetric bilinear forms).

Theorem (*Polarization Identity*): Assume $2 \neq 0$ in K. (Note that $2 \neq 0 \implies 4 \neq 0$. Why? This is left as an exercise). If B is symmetric and defines $q := q_B$, then for all $v, w \in V$ we have:

(a) $B(v,w) = \frac{1}{4}q(v+w) - \frac{1}{4}q(v-w)$.

(b) $B(v,w) = \frac{1}{2}(q(v+w) - q(v) - q(w))$.

Theorem 27.a.5 (*Sylvester's Law of Inertia*): Let $B\colon V\times V\to\mathbb{R}$ be a symmetric bilinear form on a vector space V over $\mathbb{R}$, of dimension n. Then there exists a basis $\mathcal{C}$ for V such that

$$[B]_{\mathcal{C}} = \begin{pmatrix} I_k & 0 & 0 \\ 0 & -I_\ell & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

for some $k,\ell\in\mathbb{Z}_{\geq 0}$ with $k+\ell\leq n$. (The zero block in the bottom right has size $n-(k+\ell)$).

Moreover, k and ℓ do not depend on the basis $\mathcal{C}$ (they depend only on B). In fact:

$$\begin{aligned} k &= \max\{\dim W \mid W\subseteq V \text{ is a subspace and } B|_{W\times W} \text{ is positive definite}\}, \\ \ell &= \max\{\dim U \mid U\subseteq V \text{ is a subspace and } -B|_{U\times U} \text{ is positive definite}\}, \\ n-(k+\ell) &= \max\{\dim Z \mid Z\subseteq V \text{ is a subspace and } B|_{Z\times Z}\equiv 0\}. \end{aligned}$$

Remark:

(a) In the basis $\mathcal{C}$, the quadratic form q associated to B looks like

$$q(x_1,\dots,x_n) = x_1^2+\cdots+x_k^2 - x_{k+1}^2-\cdots-x_{k+\ell}^2.$$

(b) k is called the "*positive index*" of B, ℓ is the "*negative index*", and $n-(k+\ell)$ is the "*nullity*". The value $\sigma(B):=\operatorname{sgn}(B):=k-\ell$ is called the "*signature*" of B. (This is sometimes referred to as type $(k,\ell,0)$).

Proof

Choose any basis $\mathcal{C}'$ for V and let $A:=[B]_{\mathcal{C}'}\in\mathcal{M}_{n\times n}(\mathbb{R})$. Note that A is a symmetric matrix (since B is symmetric). This implies that A is orthogonally diagonalizable, i.e., there exists $P\in\mathrm{O}(n)$ such that $P^{-1}AP$ is a diagonal matrix. But $P^{-1}=P^\top$, so $P^\top AP$ is diagonal.

Let $\mathcal{C}''$ be the (unique) basis of V such that $[\mathrm{id}_V]_{\mathcal{C}''}^{\mathcal{C}'}=P$. (That is, if $\mathcal{C}'=(v_1',\dots,v_n')$ and $\mathcal{C}''=(v_1'',\dots,v_n'')$, then $v_j''=\sum_{i=1}^n P_{ij}v_i'$ for all j). We have

$$[B]_{\mathcal{C}''} = P^\top AP = \begin{pmatrix} \eta_1 & & 0 \\ & \ddots & \\ 0 & & \eta_n \end{pmatrix},$$

where $\eta_j\in\mathbb{R}$ for all j.

By changing the order of the elements in the basis $\mathcal{C}''$, we may assume $\eta_1,\dots,\eta_k>0$, $\eta_{k+1},\dots,\eta_{k+\ell}<0$, and $\eta_{k+\ell+1}=\cdots=\eta_n=0$ for some $k,\ell\in\mathbb{Z}_{\geq 0}$ with $k+\ell\leq n$.

We can write:

$$\begin{aligned} &\eta_i=\lambda_i^2, \text{ with } \lambda_i>0, \quad \forall 1\leq i\leq k \quad (\text{so } B(v_i'',v_i'')=\lambda_i^2), \\ &\eta_i=-\lambda_i^2, \text{ with } \lambda_i>0, \quad \forall k+1\leq i\leq k+\ell \quad (\text{so } B(v_i'',v_i'')=-\lambda_i^2), \\ &\eta_i=0, \quad \forall k+\ell+1\leq i\leq n \quad (\text{so } B(v_i'',v_i'')=0). \end{aligned}$$

Define $\mathcal{C}=(v_1,\dots,v_k,v_{k+1},\dots,v_{k+\ell},v_{k+\ell+1},\dots,v_n)$, where

$$v_i=\frac{1}{\lambda_i}v_i'' \quad \forall 1\leq i\leq k+\ell, \qquad v_i=v_i'' \quad \forall k+\ell+1\leq i\leq n.$$

In other words:

$$\mathcal{C}=\left(\frac{1}{\lambda_1}v_1'',\dots,\frac{1}{\lambda_k}v_k'',\frac{1}{\lambda_{k+1}}v_{k+1}'',\dots,\frac{1}{\lambda_{k+\ell}}v_{k+\ell}'',v_{k+\ell+1}'',\dots,v_n''\right).$$

Note that $v_i=c_iv_i''$ for some scalars $c_i\neq 0$, for all i. It is left as an exercise to show that $\mathcal{C}$ is indeed a basis.

We have $([B]_{\mathcal{C}})_{i,j}=B(v_i,v_j)=B(c_iv_i'',c_jv_j'')=0$ for all $i\neq j$. And for $i=j$:

- if $1\leq i\leq k$, then $([B]_{\mathcal{C}})_{i,i}=B(v_i,v_i)=B\left(\frac{1}{\lambda_i}v_i'',\frac{1}{\lambda_i}v_i''\right)=\frac{1}{\lambda_i^2}\lambda_i^2=1$,

- if $k+1 \le i \le k+\ell$, then $([B]_{\mathcal{C}})_{i,i} = B(v_i, v_i) = B\left(\frac{1}{\lambda_i} v_i'', \frac{1}{\lambda_i} v_i''\right) = \frac{1}{\lambda_i^2}(-\lambda_i^2) = -1$,
- if $k+\ell+1 \le i \le n$, then $([B]_{\mathcal{C}})_{i,i} = B(v_i, v_i) = B(v_i'', v_i'') = 0$.

This proves the existence of the basis $\mathcal{C}$ claimed in the theorem.

We will prove now that k is independent of $\mathcal{C}$. Put $k' := \max\{\dim W \mid W \subseteq V$ is a subspace such that $B|_{W\times W}$ is positive definite$\}$. We will show $k = k'$. Indeed, $k \le k'$, because $B|_{U\times U}$ is positive definite on $U = \operatorname{Sp}(v_1, \dots, v_k)$. This is because:

$$B(a_1v_1+\cdots+a_kv_k, a_1v_1+\cdots+a_kv_k) = (a_1, \dots, a_k, 0, \dots, 0)\cdot\begin{pmatrix} I_k & 0 & 0 \\ 0 & -I_\ell & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} a_1 \\ \vdots \\ a_k \\ 0 \\ \vdots \\ 0 \end{pmatrix} = a_1^2+\cdots+a_k^2 > 0,$$

if not all $a_i = 0$.

Suppose by contradiction that $k' \ge k+1$. This implies there exists $W \subseteq V$ of dimension $\ge k+1$ such that $B|_{W\times W}$ is positive definite. Take $X := \operatorname{Sp}(v_{k+1}, \dots, v_n)$. Then, by the dimension formula:

$$\dim(W\cap X) = \dim W + \dim X - \dim(W+X) \ge (k+1)+(n-k) - \dim(W+X) \ge n+1-n = 1.$$

This implies $W \cap X \ne \{0\}$. Take $0 \ne w \in W \cap X$. We can write $w = a_{k+1}v_{k+1} + \cdots + a_n v_n$. Then $B(w,w) = -a_{k+1}^2 - \cdots - a_{k+\ell}^2 \le 0$. But, by the assumption that W is positive definite, $B(w,w) > 0$. This is a contradiction, which implies $k' \le k$. This proves $k = k'$.

The proofs for ℓ and $n-(k+\ell)$ are similar. **Q.E.D.**

Theorem 27.a.6 (*Sylvester's Law for Euclidean Spaces*): Let V be a finite-dimensional Euclidean space of dimension n. Let $B\colon V \times V \to \mathbb{R}$ be a symmetric bilinear form. Then there exists an orthonormal basis $\mathcal{C}$ for V in which

$$[B]_{\mathcal{C}} = \begin{pmatrix} \eta_1 & & & & & 0 \\ & \ddots & & & & \\ & & \eta_k & & & \\ & & & \eta_{k+1} & & \\ & & & & \ddots & \\ 0 & & & & & \eta_{k+\ell} \end{pmatrix},$$

where $k+\ell \le n$, with $\eta_1, \dots, \eta_k > 0$ and $\eta_{k+1}, \dots, \eta_{k+\ell} < 0$. As before, k, ℓ are the positive and negative indices of B. Moreover, the scalars $\eta_1, \dots, \eta_k, \eta_{k+1}, \dots, \eta_{k+\ell}, 0, \dots, 0$ are independent of $\mathcal{C}$. They depend only on B (and on the scalar product on V). They are the eigenvalues of $[B]_{\mathcal{D}}$ for every orthonormal basis $\mathcal{D}$ of V (independent of $\mathcal{D}$ and on whether or not $[B]_{\mathcal{D}}$ is diagonal).

Remark: This theorem has a geometric meaning. If $q\colon \mathbb{R}^n \to \mathbb{R}$ is a quadratic form $q(x_1, \dots, x_n) = \sum a_{ij}x_ix_j$ for some $a_{ij} \in \mathbb{R}$, then the set

$$Q = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid q(x_1, \dots, x_n) = C\} \quad \text{(for some constant } C \in \mathbb{R})$$

is called a "*quadric*". (Examples include an ellipse/ellipsoid, parabola/paraboloid, hyperbola/hyperboloid). The theorem tells us that there exists a linear isometry $T\colon \mathbb{R}^n \to \mathbb{R}^n$ such that

$$T(Q) = \left\{ (y_1, \dots, y_n) \in \mathbb{R}^n \;\middle|\; \underbrace{\eta_1 y_1^2 + \cdots + \eta_k y_k^2}_{\text{positive coeffs.}} + \underbrace{\eta_{k+1}y_{k+1}^2 + \cdots + \eta_{k+\ell}y_{k+\ell}^2}_{\text{negative coeffs.}} = C \right\}.$$

Proof Outline

Pick any orthonormal basis $\mathcal{C}'$ for V (similar to the previous proof, only now we take $\mathcal{C}'$ to be orthonormal). Define $A = [B]_{\mathcal{C}'}$, and proceed exactly as in the previous proof, but instead of defining $\mathcal{C}$ with the scaling factors λ_i, simply take $\mathcal{C} = \mathcal{C}''$.

To see why $\eta_1, \dots, \eta_{k+\ell}, 0, \dots, 0$ depend only on B etc., note that if $\mathcal{D}$ and $\mathcal{D}'$ are two orthonormal bases for V, then $[\mathrm{id}_V]^{\mathcal{D}}_{\mathcal{D}'} \in \mathrm{O}(n)$ (left as an exercise). Put $Q := [\mathrm{id}_V]^{\mathcal{D}}_{\mathcal{D}'}$. We have

$$[B]_{\mathcal{D}} = Q^\top \cdot [B]_{\mathcal{D}'} \cdot Q = Q^{-1}[B]_{\mathcal{D}'}Q,$$

since $Q \in \mathrm{O}(n)$. Hence, the matrices $[B]_{\mathcal{D}}$ and $[B]_{\mathcal{D}'}$ are similar, which means their eigenvalues are exactly the same. **Q.E.D.**

Theorem 27.a.7 (*Diagonalization of Symmetric Bilinear Forms*): Let V be a finite-dimensional vector space over a field K of characteristic 0. Let $B \colon V \times V \to K$ be a symmetric bilinear form. Then there exists a basis $\mathcal{C}$ of V such that $[B]_{\mathcal{C}}$ is a diagonal matrix.

Proof

We proceed by induction on $n := \dim V$. We need to show there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ such that $B(v_i, v_j) = 0$ for all $i \neq j$.

If $B = 0$, or if $n = 1$, there is nothing to prove.

Let $n \geq 2$. Assume the statement of the theorem holds for all vector spaces of dimension $\leq n-1$. Let V be an n-dimensional vector space. If $B(v,v) = 0$ for all $v \in V$, then the polarization formula shows $B = 0$ and we are done. So assume that there exists $v_1 \in V$ such that $B(v_1, v_1) \neq 0$.

Put $W = \mathrm{Sp}(v_1)$. The subspace W is 1-dimensional (because $v_1 \neq 0$). Define the orthogonal complement with respect to B as:

$$W^{\perp_B} := \{w \in V \mid B(v_1, w) = 0\}.$$

It is left as an exercise to show that $W^{\perp_B} \subseteq V$ is a subspace.

We claim that $W \cap W^{\perp_B} = \{0\}$. Indeed, if $w \in W \cap W^{\perp_B}$, then $w = c \cdot v_1$ and $B(v_1, w) = 0$. But $B(v_1, w) = B(v_1, c \cdot v_1) = c \cdot B(v_1, v_1)$. Since $B(v_1, v_1) \neq 0$, this implies $c = 0$, which means $w = 0$.

We claim that $V = W + W^{\perp_B}$. Indeed, if $v \in V$, then the vector

$$u := v - \frac{B(v_1, v)}{B(v_1, v_1)} \cdot v_1$$

belongs to $W^{\perp_B}$. To see this, we calculate:

$$B(v_1, u) = B(v_1, v) - \frac{B(v_1, v)}{B(v_1, v_1)} \cdot B(v_1, v_1) = 0.$$

This implies $u \in W^{\perp_B}$. So we can write v as:

$$v = \underbrace{\frac{B(v_1, v)}{B(v_1, v_1)} \cdot v_1}_{\in W} + \underbrace{u}_{\in W^{\perp_B}} \in W + W^{\perp_B}.$$

It follows that $V = W \oplus W^{\perp_B}$. This implies $\dim W^{\perp_B} = n - 1$.

Now, $B|_{W^{\perp_B} \times W^{\perp_B}}$ is a symmetric bilinear form on an $(n-1)$-dimensional vector space ($W^{\perp_B}$). By the induction hypothesis, there exists a basis $\mathcal{C}' = (v_2, \dots, v_n)$ for $W^{\perp_B}$ such that $B(v_i, v_j) = 0$ for all $i \neq j$ with $i, j \geq 2$.

Take $\mathcal{C} = (v_1, \dots, v_n)$. Since $v_1 \in W$ and $v_2, \dots, v_n \in W^{\perp_B}$, we obtain that $B(v_i, v_j) = 0$ for all $i \neq j$. **Q.E.D.**

Question: What happens over $\mathbb{C}$?

Theorem 27.a.8 (*Complex Symmetric Bilinear Forms*): Let $B\colon V\times V\to\mathbb{C}$ be a symmetric bilinear form on a vector space V over $\mathbb{C}$, of dimension n. Then there exists a basis $\mathcal{C}$ of V such that

$$[B]_{\mathcal{C}} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix},$$

for some $0\le r\le n$. (The zero block in the bottom right has size $n-r$).

The number r is independent of the basis $\mathcal{C}$ — it depends only on B. In fact,

$$n-r=\max\{\dim W \mid W\subseteq V \text{ is a subspace such that } B|_{W\times W}\equiv 0\}.$$

The number r is sometimes called the "*rank*" of B.

Proof

By the previous theorem, there exists a basis $\mathcal{C}'=(v'_1,\dots,v'_n)$ of V such that

$$[B]_{\mathcal{C}'} = \begin{pmatrix} \eta_1 & & 0 \\ & \ddots & \\ 0 & & \eta_n \end{pmatrix},$$

where $0\le r\le n$ and $\eta_j\in\mathbb{C}$ for all $1\le j\le r$, and $\eta_j=0$ for $r<j\le n$.

Choose $\lambda_j\in\mathbb{C}$ such that $\lambda_j^2=\eta_j$ for $j=1,\dots,r$. (This is always possible over $\mathbb{C}$). Define $v_j=\frac{1}{\lambda_j}v'_j$ for $j=1,\dots,r$, and $v_q=v'_q$ for all $r<q\le n$.

Take $\mathcal{C}=(v_1,\dots,v_n)$. Then $B(v_i,v_j)=0$ for all $i\neq j$. Furthermore, for $1\le j\le r$, we have:

$$B(v_j,v_j)=B\left(\frac{1}{\lambda_j}v'_j,\frac{1}{\lambda_j}v'_j\right)=\frac{1}{\lambda_j^2}\cdot\eta_j=1,$$

and for all $r+1\le j\le n$, we have $B(v_j,v_j)=0$. **Q.E.D.**

More on Positive Definite Bilinear Forms

Recall: A symmetric bilinear form $B\colon V\times V\to\mathbb{R}$ is called "*positive definite*" if $B(v,v)>0$ for all $0\neq v\in V$.

Theorem 27.a.9 (*Sylvester's Criterion*): Let V be a finite-dimensional vector space over $\mathbb{R}$ and $B\colon V\times V\to\mathbb{R}$ a symmetric bilinear form. Then the following conditions are equivalent:

(a) B is positive definite.

(b) There exists a basis $\mathcal{C}=(e_1,\dots,e_n)$ for V such that the principal minors $\det(B(e_j,e_k))_{j,k=1,\dots,\ell}$ for all $1\le\ell\le n$ are positive.

(c) Like **(b)**, but instead of "there exists a basis...", it holds for "all bases...".

Proof

(b) $\implies$ **(a)**: We proceed by induction on $n=\dim V$.

For $n=1$, the assumption in **(b)** says $B(e_1,e_1)>0$. This implies that for all $0\neq c\in\mathbb{R}$, $B(ce_1,ce_1)=c^2B(e_1,e_1)>0$.

Suppose the implication holds for all spaces of dimension $\le n$. Let V be an $(n+1)$-dimensional vector space, and $B\colon V\times V\to\mathbb{R}$ a symmetric bilinear form. Let $\mathcal{C}=(e_1,\dots,e_{n+1})$ be a basis for V such that all the principal minors of $[B]_{\mathcal{C}}$ are positive.

Let $W:=\mathrm{Sp}(e_1,\dots,e_n)$ with the basis $\mathcal{C}_W=(e_1,\dots,e_n)$. The principal minors of $[B|_{W\times W}]_{\mathcal{C}_W}$ are exactly the same as the first n principal minors of $[B]_{\mathcal{C}}$. Since these are positive, the induction hypothesis implies that $B|_{W\times W}$ is positive definite.

This implies that if we define $\langle w,w'\rangle_B:=B(w,w')$, then $\langle\cdot,\cdot\rangle_B$ is a valid scalar product on W.

Consider the linear functional $L\colon W \to \mathbb{R}$ defined by $L(w) = B(w, e_{n+1})$. By a previous result (Riesz Representation Theorem, **??**), there exists $w' \in W$ such that $L(w) = \langle w, w' \rangle_B$ for all $w \in W$. This implies:

$$B(w, w') = B(w, e_{n+1}) \quad \forall w \in W. \tag{27.2}$$

Define a new basis for V, $\mathcal{C}' = (e_1, \ldots, e_n, e_{n+1} - w')$. (It is left as an exercise to show this is a basis. Hint: $w' \in W$). By (**??**), we have $B(e_j, e_{n+1} - w') = 0$ for all $1 \leq j \leq n$.

This implies that the matrix representation in the new basis is block diagonal:

$$[B]_{\mathcal{C}'} = \begin{pmatrix} C & 0 \\ 0 & d \end{pmatrix} = S^\top [B]_{\mathcal{C}} S,$$

where $d = B(e_{n+1} - w', e_{n+1} - w')$, $S = [\mathrm{id}_V]^{\mathcal{C}}_{\mathcal{C}'}$, and $C = [B|_{W \times W}]_{\mathcal{C}_W}$.

Taking the determinant, we get:

$$\det([B]_{\mathcal{C}'}) = \det(C) \cdot d = \det(S)^2 \cdot \det([B]_{\mathcal{C}}).$$

We know $\det(C) > 0$ (it is a principal minor), $\det(S)^2 > 0$ (since S is invertible), and $\det([B]_{\mathcal{C}}) > 0$ (by assumption). This implies that $d > 0$.

Let $v \in V$. We can write $v = w + \alpha \cdot (e_{n+1} - w')$, with $w \in W$ and $\alpha \in \mathbb{R}$. Recall that $\langle w, e_{n+1} - w' \rangle_B = 0$. This implies:

$$B(v, v) = B(w, w) + \alpha^2 B(e_{n+1} - w', e_{n+1} - w') = B(w, w) + \alpha^2 d > 0,$$

whenever either $w \neq 0$ or $\alpha \neq 0$. This implies that B is positive definite. The proof of "**(b)** $\Longrightarrow$ **(a)**" follows now by induction.

(a) $\Longrightarrow$ **(b)**: Left as an exercise. Similar ideas apply. **Q.E.D.**

Exercise Solutions

Proof that $2 \neq 0 \implies 4 \neq 0$

Assume $2 \neq 0$ in a field K. This means $1+1 \neq 0$. If $4 = 0$, then $2+2 = 0$, which means $2 \cdot 2 = 0$. Since K is a field, it has no zero divisors. Thus, $2 \cdot 2 = 0$ implies $2 = 0$, which contradicts our assumption. Therefore, $4 \neq 0$. **Q.E.D.**

Proof that $W^{\perp_B}$ is a subspace (on ??)

Let $w_1, w_2 \in W^{\perp_B}$ and $\alpha, \beta \in K$. We need to show that $\alpha w_1 + \beta w_2 \in W^{\perp_B}$. By definition of $W^{\perp_B}$, we have $B(v_1, w_1) = 0$ and $B(v_1, w_2) = 0$. Using the linearity of B in the second argument, we compute:

$$B(v_1, \alpha w_1 + \beta w_2) = \alpha B(v_1, w_1) + \beta B(v_1, w_2) = \alpha \cdot 0 + \beta \cdot 0 = 0.$$

Thus, $\alpha w_1 + \beta w_2 \in W^{\perp_B}$, proving it is a subspace. **Q.E.D.**

Proof that $\mathcal{C}$ is a basis (on ??)

We are given a basis $\mathcal{C}'' = (v''_1, \ldots, v''_n)$. The new set $\mathcal{C} = (v_1, \ldots, v_n)$ is formed by scaling each vector in $\mathcal{C}''$ by a non-zero scalar c_i (where $c_i = 1/\lambda_i$ or $c_i = 1$). Since scaling basis vectors by non-zero scalars preserves linear independence and the span, $\mathcal{C}$ remains a basis for V. **Q.E.D.**

Proof that $\mathcal{C}'$ is a basis (on ??)

We are given the basis $\mathcal{C} = (e_1, \ldots, e_n, e_{n+1})$. We define $\mathcal{C}' = (e_1, \ldots, e_n, e_{n+1} - w')$, where $w' \in W = \mathrm{Sp}(e_1, \ldots, e_n)$. We can write $w' = \sum_{i=1}^n \alpha_i e_i$. Suppose a linear combination of $\mathcal{C}'$

equals zero:

$$\sum_{i=1}^{n} c_i e_i + c_{n+1}(e_{n+1} - w') = 0 \implies \sum_{i=1}^{n} (c_i - c_{n+1}\alpha_i)e_i + c_{n+1}e_{n+1} = 0.$$

Because $\mathcal{C}$ is a basis, all coefficients must be zero. In particular, $c_{n+1} = 0$. Substituting this back into the other coefficients yields $c_i = 0$ for all $1 \le i \le n$. Thus, the vectors in $\mathcal{C}'$ are linearly independent and form a basis. **Q.E.D.**

Proof of ?? (a) $\implies$ (b)

Assume B is positive definite. Let $\mathcal{C} = (e_1, \ldots, e_n)$ be any basis for V. For any $1 \le \ell \le n$, consider the subspace $W_\ell = \mathrm{Sp}(e_1, \ldots, e_\ell)$. Since B is positive definite on V, its restriction $B|_{W_\ell \times W_\ell}$ is also positive definite. The matrix representation of this restriction in the basis $(e_1, \ldots, e_\ell)$ is exactly the $\ell \times \ell$ principal submatrix of $[B]_{\mathcal{C}}$. A positive definite matrix has strictly positive eigenvalues (as shown in the proof of Sylvester's Law of Inertia), and its determinant is the product of these eigenvalues. Therefore, the determinant of every principal submatrix (the principal minors) must be strictly positive. **Q.E.D.**

The Jordan Normal Form (Chapter 28)

Introduction and Motivation (Section 28.a)

Let V be a finite-dimensional vector space over K, and let $T\colon V \to V$ be a linear map.

The easiest situation to understand is when T is diagonalizable. That is, there exists a basis $\mathcal{B}$ for V such that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$

where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of T.

Using the basis $\mathcal{B}$, it is also easy to calculate powers of T. For example,

$$[T^k]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1^k & & 0 \\ & \ddots & \\ 0 & & \lambda_n^k \end{pmatrix} \quad \forall k \in \mathbb{Z}_{\geq 1}.$$

But <u>not all</u> $T \in \mathrm{End}(V)$ are diagonalizable! Why? There are two possible reasons:

(a) The characteristic polynomial $P_T(x)$ does not split as a product of linear factors. This is equivalent to saying that T is not (even) trigonalizable (hence, all the more so not diagonalizable!). In other words, $P_T(x)$ doesn't have enough zeroes in K (or even no zeroes at all).

For example, consider $T_A\colon \mathbb{R}^2 \to \mathbb{R}^2$, where $A = R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, with $\theta \neq \pi k$ for $k \in \mathbb{Z}$. The map T_A has <u>no</u> eigenvalues (in $\mathbb{R}$). It is a rotation, but not by 0° or 180°.

Sometimes we can overcome this "problem" by enlarging (extending) K. For example, $K = \mathbb{R}$ can be extended to $K = \mathbb{C}$. Then the above matrix A has two eigenvalues when viewed as $A \in \mathcal{M}_{2\times 2}(\mathbb{C})$, namely $\lambda_1 = e^{i\theta}$ and $\lambda_2 = e^{-i\theta}$ (Exercise: check this).

(b) T is trigonalizable, or equivalently $P_T(x)$ splits as a product of linear factors, but for some eigenvalues λ of T, the geometric multiplicity is strictly less than the algebraic multiplicity:

$$m_g(\lambda) < m_a(\lambda).$$

Let's try to understand case **(b)** better.

Definition 28.a.1 (*Jordan Matrix*): Let $\lambda \in K$ and $n \in \mathbb{Z}_{\geq 1}$. Define:

$$J_{\lambda,n} = \begin{pmatrix} \lambda & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{pmatrix} \in \mathcal{M}_{n\times n}(K).$$

This matrix has λ on the main diagonal, 1's on the diagonal immediately above the main diagonal, and 0's everywhere else.

For $n = 1$, $J_{\lambda,1} = (\lambda)$. For $n = 2$, $J_{\lambda,2} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$. For $n = 3$, $J_{\lambda,3} = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$, etc.

The matrix $J_{\lambda,n}$ is called the "*n-dimensional Jordan matrix with eigenvalue λ*".

Lemma 28.a.2 (*Properties of the Jordan Matrix*):

(a) **(a)** λ is the only eigenvalue of $J_{\lambda,n}$.
(b) **(b)** $P_{J_{\lambda,n}}(x) = (\lambda - x)^n$, and $m_a(J_{\lambda,n};\lambda) = n$.
(c) **(c)** $m_g(J_{\lambda,n};\lambda) = 1$. In fact, $\operatorname{Eig}_{J_{\lambda,n}}(\lambda) = \operatorname{Sp}(e_1)$, where $e_1 = (1,0,\dots,0)^\top$.

Proof

Proof of (a) and (b): The characteristic polynomial is the determinant of an upper triangular matrix:

$$P_{J_{\lambda,n}}(x) = \det \begin{pmatrix} \lambda - x & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda - x \end{pmatrix} = (\lambda - x)^n.$$

This proves both **(a)** and **(b)**.

Proof of (c): To find $\operatorname{Eig}_{J_{\lambda,n}}(\lambda)$, we need to solve the equation $(J_{\lambda,n} - \lambda I_n)x = 0$:

$$(J_{\lambda,n} - \lambda I_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \implies \begin{pmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

This immediately implies the system of equations:

$$\begin{cases} x_2 = 0 \\ x_3 = 0 \\ \vdots \\ x_n = 0 \\ 0 = 0 \end{cases}$$

The only free variable is x_1. Therefore, $\operatorname{Eig}_{J_{\lambda,n}}(\lambda) = \operatorname{Sp}\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \operatorname{Sp}(e_1)$, which means the geometric multiplicity is 1.

Q.E.D.

We've seen that the statement *"$P_T(x)$ splits as a product of linear factors"* does <u>not</u> imply that *"T is diagonalizable"*, and $J_{\lambda,n}$ is a clear counterexample.

Question: Is $J_{\lambda,n}$ the *"only reason"* for this non-implication?

Theorem 28.a.3 (*Jordan Normal Form*): Let V be an n-dimensional vector space over K and $T \in \operatorname{End}(V)$ such that $P_T(x)$ splits as a product of linear factors in $K[x]$ (e.g., if $K = \mathbb{C}$, this always happens). Then there exists a basis $\mathcal{B}$ for V such that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} J_{\lambda_1,n_1} & & & \\ & J_{\lambda_2,n_2} & & \\ & & \ddots & \\ & & & J_{\lambda_k,n_k} \end{pmatrix},$$

where $k \geq 1$, $\lambda_1, \dots, \lambda_k \in K$, $n_1, \dots, n_k \geq 1$, and $n_1 + \cdots + n_k = n$.

Moreover, up to permutation, the blocks $J_{\lambda_1,n_1}, \dots, J_{\lambda_k,n_k}$ are unique.

Remark: The basis $\mathcal{B}$ is called a "*Jordan basis*" for T. The resulting block diagonal matrix is called the "*Jordan Normal Form*" (JNF) of T. A similar theorem holds for matrices $A \in \mathcal{M}_{n\times n}(K)$: there exists an invertible matrix P such that $P^{-1}AP$ is in Jordan Normal Form.

Example: Consider a matrix $A \in \mathcal{M}_{13\times 13}(K)$ in Jordan Normal Form:

$$A = \begin{pmatrix}
\lambda_1 & 1 & 0 & & & & & & & \\
0 & \lambda_1 & 1 & & & & & & & \\
0 & 0 & \lambda_1 & & & & & & & \\
 & & & \lambda_1 & 1 & 0 & & & & \\
 & & & 0 & \lambda_1 & 1 & & & & \\
 & & & 0 & 0 & \lambda_1 & & & & \\
 & & & & & & \lambda_2 & 1 & & \\
 & & & & & & 0 & \lambda_2 & & \\
 & & & & & & & & \lambda_2 & \\
 & & & & & & & & & \lambda_2 \\
\lambda_3 & 1 & 0 & & & & & & & \\
0 & \lambda_3 & 1 & & & & & & & \\
0 & 0 & \lambda_3 & & & & & & &
\end{pmatrix}$$

Here, $n = 13$. Let's assume the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ are distinct.

- **Characteristic polynomial:** $P_A(x) = (\lambda_1 - x)^6 \cdot (\lambda_2 - x)^4 \cdot (\lambda_3 - x)^3$.
- **Algebraic multiplicity:** $m_a(\lambda_1) = 6$, $m_a(\lambda_2) = 4$, $m_a(\lambda_3) = 3$.
- **Geometric multiplicity:** $m_g(\lambda)$ is exactly the number of blocks with eigenvalue λ. So, $m_g(\lambda_1) = 2$, $m_g(\lambda_2) = 3$, and $m_g(\lambda_3) = 1$.
- **Minimal polynomial:** $\mu_A(x) = (x-\lambda_1)^{\ell_1} \cdot (x-\lambda_2)^{\ell_2} \cdot (x-\lambda_3)^{\ell_3}$, where $1 \le \ell_i \le m_a(\lambda_i)$. We will see that ℓ_i is precisely the size of the <u>biggest</u> block with eigenvalue λ_i. Thus,

$$\mu_A(x) = (x-\lambda_1)^3 \cdot (x-\lambda_2)^2 \cdot (x-\lambda_3)^3.$$

Powers of Jordan Matrices and Minimal Polynomials (Section 28.b)

Let's go back to the Jordan matrices $J_{\lambda,n}$. Recall that $J_{\lambda,n} \in \mathcal{M}_{n\times n}(K)$ for $n \in \mathbb{Z}_{\geq 1}$ and $\lambda \in K$. We can write $J_{\lambda,n} = \lambda \cdot I_n + N$, where

$$N = \begin{pmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{pmatrix} = J_{0,n}.$$

Lemma 28.a.1 (*Powers of the Nilpotent Matrix N and Jordan Blocks*):

(a) **(a)** The powers of N shift the diagonal of 1's further up:

$$N^2 = \begin{pmatrix} 0 & 0 & 1 & & 0 \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & \ddots & 0 \\ 0 & & & & 0 \end{pmatrix}, \quad N^3 = \begin{pmatrix} 0 & 0 & 0 & 1 & \dots \\ & \ddots & \ddots & \ddots & \ddots \\ & & \ddots & \ddots & \ddots \end{pmatrix}, \quad \dots, \quad N^{n-1} = \begin{pmatrix} 0 & \dots & 0 & 1 \\ \vdots & & & 0 \\ \vdots & & & \vdots \\ 0 & \dots & \dots & 0 \end{pmatrix}.$$

Furthermore, $N^n = 0$, and $N^k = 0$ for all $k \geq n$. In other words, N^k has 1's in the k-th *"diagonal"* above the *"central diagonal"* and 0's everywhere else.

(b) **(b)** For all $k \geq 1$, we have:

$$J_{\lambda,n}^k = \begin{pmatrix} \lambda^k & \binom{k}{1}\lambda^{k-1} & \binom{k}{2}\lambda^{k-2} & \dots \\ & \lambda^k & \binom{k}{1}\lambda^{k-1} & \dots \\ & & \ddots & \dots \\ & & & \binom{k}{1}\lambda^{k-1} \\ 0 & & & \lambda^k \end{pmatrix}.$$

Written differently, we have $J_{\lambda,n}^k = \sum_{j=0}^k \binom{k}{j}\lambda^{k-j}N^j$ for all $k \geq 1$.

Proof

Proof of (a): This can be shown by direct calculation (exercise). Alternatively, denote by $e_1, \dots, e_n$ the standard basis for K^n. Then the linear map $T_N \colon K^n \to K^n$ acts as a shift operator:

$$T_N e_i = e_{i-1} \quad \forall 2 \leq i \leq n, \quad \text{and} \quad T_N e_1 = 0.$$

In other words:

$$e_n \xrightarrow{T_N} e_{n-1} \xrightarrow{T_N} \dots \xrightarrow{T_N} e_1 \xrightarrow{T_N} 0.$$

The map $T_{N^2} (= T_N \circ T_N = T_N^2)$ applies the shift twice:

$$e_n \xrightarrow{T_N^2} e_{n-2} \xrightarrow{T_N^2} e_{n-4} \dots \to 0$$
$$e_{n-1} \xrightarrow{T_N^2} e_{n-3} \xrightarrow{T_N^2} \dots \to 0$$

By induction on k, it follows that for all $k \geq 1$:

$$T_{N^k}(e_j) = \begin{cases} e_{j-k} & \text{if } k+1 \leq j \leq n \\ 0 & \text{if } j \leq k \end{cases}.$$

Since $N^k = [T_{N^k}]_{\mathcal{E}_n}^{\mathcal{E}_n}$, the matrix form follows immediately.

Proof of (b): Since the scalar matrix λI_n and the matrix N commute, we can use the standard binomial theorem:

$$J_{\lambda,n}^k = (\lambda I_n + N)^k = \sum_{j=0}^{k} \binom{k}{j} (\lambda I_n)^{k-j} N^j = \sum_{j=0}^{k} \binom{k}{j} \lambda^{k-j} N^j.$$

Using the structure of N^j from part **(a)**, this yields the upper triangular matrix with binomial coefficients along the diagonals.

Q.E.D.

Corollary 28.a.2 (*Multiplicities and the Minimal Polynomial from JNF*): Suppose $T \in \operatorname{End}(V)$ admits a matrix representation in Jordan Normal Form in some basis $\mathcal{B}$:

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} J_{\lambda_1,n_1} & & \\ & \ddots & \\ & & J_{\lambda_k,n_k} \end{pmatrix}. \tag{28.1}$$

Then for every eigenvalue λ of T, the geometric multiplicity $m_g(\lambda)$ equals the number of blocks in (**??**) labeled with λ (i.e., blocks of the type $J_{\lambda,\dots}$). Note that blocks of size 1×1 also count. In other words,

$$m_g(\lambda) = \#\{j \mid 1 \le j \le k, \lambda_j = \lambda\}.$$

Furthermore, the minimal polynomial $\mu_T(x)$ is given by:

$$\mu_T(x) = \prod_{\lambda \in \operatorname{Spec}(T)} (x-\lambda)^{s(\lambda)},$$

where $s(\lambda)$ is the size of the largest block labeled by λ, i.e., $s(\lambda) = \max\{n_j \mid 1 \le j \le k, \lambda_j = \lambda\}$.

Proof

Geometric Multiplicity: Suppose that $A \in \mathcal{M}_{n\times n}(K)$ has the block diagonal shape $A = \begin{pmatrix} B & 0 \\ 0 & C \end{pmatrix}$ with $B \in \mathcal{M}_{\ell\times\ell}(K)$ and $C \in \mathcal{M}_{m\times m}(K)$. Let $0 \neq u = \begin{pmatrix} v \\ w \end{pmatrix}$ with $v \in K^\ell$ and $w \in K^m$. Then u is an eigenvector of A for the eigenvalue λ if and only if $Bv = \lambda v$ and $Cw = \lambda w$.

Moreover, the eigenspace decomposes as $\operatorname{Eig}_A(\lambda) = \operatorname{Eig}_B(\lambda) \oplus \operatorname{Eig}_C(\lambda)$. In particular, $m_g(A,\lambda) = m_g(B,\lambda) + m_g(C,\lambda)$. (We use the convention here that if λ is not an eigenvalue of a matrix D, then $m_g(D,\lambda) = 0$).

Now recall from **??** that for a single Jordan block, $m_g(J_{\lambda,r};\lambda') = \begin{cases} 1 & \text{if } \lambda' = \lambda \\ 0 & \text{if } \lambda' \neq \lambda \end{cases}$. Applying this additively to $A := [T]_{\mathcal{B}}^{\mathcal{B}}$, we obtain:

$$m_g(T,\lambda) = m_g(A,\lambda) = \sum_{i=1}^{k} m_g(J_{\lambda_i,n_i};\lambda) = \#\{i \mid \lambda_i = \lambda\}.$$

Minimal Polynomial: Let $A = \begin{pmatrix} B & 0 \\ 0 & C \end{pmatrix}$ as above. We have $A^k = \begin{pmatrix} B^k & 0 \\ 0 & C^k \end{pmatrix}$ (check!). This implies that for every polynomial $q(x) \in K[x]$, we have $q(A) = \begin{pmatrix} q(B) & 0 \\ 0 & q(C) \end{pmatrix}$.

Consider a single Jordan matrix $J_{\lambda,n}$. For every $q(x) \in K[x]$, we have:

$$q(J_{\lambda,n}) = \begin{pmatrix} q(\lambda) & & * \\ & \ddots & \\ 0 & & q(\lambda) \end{pmatrix} \quad \text{(exercise)}.$$

Note the following key properties:

(a) If $q(\lambda) \neq 0$, then $q(J_{\lambda,n})$ is an upper triangular matrix with non-zero diagonal entries, hence it is invertible.

(b) If $q(x) = (x-\lambda)^s$, then $q(J_{\lambda,n}) = (J_{\lambda,n} - \lambda I_n)^s = N^s$. So, for $q(x) = (x-\lambda)^s$, $q(J_{\lambda,n}) = 0$ if and only if $s \geq n$.

(c) Now, take a general polynomial $q(x) \in K[x]$ of positive degree (i.e., $q(x) \neq$ const.) and let λ be a zero of $q(x)$. We can write $q(x) = (x-\lambda)^s \cdot p(x)$, where $s \geq 1$ and $p(\lambda) \neq 0$. This implies $q(J_{\lambda,n}) = N^s \cdot p(J_{\lambda,n})$. Since $p(\lambda) \neq 0$, $p(J_{\lambda,n})$ is invertible by **(a)**. Therefore, $q(J_{\lambda,n}) = 0$ if and only if $N^s = 0$, which occurs if and only if $s \geq n$.

Let A be a matrix in Jordan Normal Form, and let $q(x) \in K[x]$ be a polynomial of positive degree such that $q(A) = 0$. We must have $q(\lambda_j) = 0$ for all $1 \leq j \leq k$, because if $q(\lambda_j) \neq 0$, then $q(J_{\lambda_j,n_j})$ would be invertible (and thus non-zero) by **(a)**, which would imply that $q(A) \neq 0$. This proves that $\lambda_1, \ldots, \lambda_k$ are all zeroes of $q(x)$.

Let $\lambda \in \{\lambda_1, \ldots, \lambda_k\}$. Write $q(x) = (x-\lambda)^s \cdot p(x)$ where $p(\lambda) \neq 0$. By **(c)**, we must have $s \geq n_j$ for all j with $\lambda_j = \lambda$. That is, $s \geq \max\{n_j \mid \lambda_j = \lambda\} = s(\lambda)$. This implies that any annihilating polynomial $q(x)$ must be of the form:

$$q(x) = \prod_{\lambda \in \{\lambda_1,\ldots,\lambda_k\}} (x-\lambda)^{s(\lambda)} \cdot r(x) \quad \text{for some } r(x) \in K[x]. \tag{28.2}$$

On the other hand, we claim that the specific polynomial $\tilde{q}(x) := \prod_{\lambda \in \{\lambda_1,\ldots,\lambda_k\}} (x-\lambda)^{s(\lambda)}$ satisfies $\tilde{q}(A) = 0$. If we prove this, then together with (**??**), it establishes that $\tilde{q}(x)$ is the monic polynomial of least degree annihilating A, meaning $\mu_T(x) = \tilde{q}(x)$.

Indeed, let $A := \operatorname{blockdiag}(J_{\lambda_1,n_1}, \ldots, J_{\lambda_k,n_k})$. We have:

$$\tilde{q}(A) = \operatorname{blockdiag}(\tilde{q}(J_{\lambda_1,n_1}), \ldots, \tilde{q}(J_{\lambda_k,n_k})). \tag{28.3}$$

Let $1 \leq j \leq k$, and consider the block $\tilde{q}(J_{\lambda_j,n_j})$. We can write $\tilde{q}(x) = (x-\lambda_j)^{s(\lambda_j)} \cdot p_j(x)$ where $p_j(x) \in K[x]$. By the definition of $s(\lambda_j)$, we have $s(\lambda_j) \geq n_j$. By property **(c)** above, $\tilde{q}(J_{\lambda_j,n_j}) = 0$. This holds for all $1 \leq j \leq k$, hence the entire block diagonal matrix in (**??**) is zero. This proves the last claim, and hence concludes the proof of the corollary. **Q.E.D.**